

Future Convergence Theory

A Unified Axiomatic Theory of
Manifestation Constraints, the Future Causal Information Field,
Gravity, Quantum Theory, Black Holes, and Cosmology

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June 29, 2026

Abstract

This paper reconstructs Future Convergence Theory (FCT) as a unified axiomatic framework founded upon the concepts of Manifestation Constraints, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the Reprojection Closed Loop. The purpose of this work is not to present FCT merely as a philosophical worldview. Rather, its objective is to establish a quantitatively testable theoretical framework that explicitly predicts measurable deviations from existing theories of General Relativity, Quantum Theory, Black Hole Physics, and Cosmology, thereby producing observationally falsifiable consequences. Within the present framework, gravity is not regarded as an interaction between particles. Instead, gravity is defined as the Manifestation Constraint that maintains the mapping from the Future Causal Information Field onto the Three-Dimensional Manifestation Layer. Quantum Theory is likewise redefined. Rather than representing the ultimate description of physical reality, quantum theory is interpreted as the Quantum Permission Layer situated between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. Accordingly, the graviton is not regarded as the fundamental origin of gravity. Instead, it is interpreted as a perturbative manifestation of the Manifestation Constraint appearing within the intermediate quantum layer. Within this framework, Dark Matter is reinterpreted as the residual structure of the Future Causal Information Field that possesses no optical manifestation, Dark Energy is interpreted as the cosmological average Manifestation Constraint, and black holes are redefined as information-recovery boundaries through which information is transferred from the Three-Dimensional Manifestation Layer back into the Future Causal Information Field. The principal observational target investigated in this paper is the spectrum of black-hole quasinormal modes. In particular, we derive the conditions under which

$$\omega_{\ell mn}^{\text{FCT}} \neq \omega_{\ell mn}^{\text{GR}}$$

holds, and introduce the measurable deviation

$$\Delta\omega_{\ell mn} = \omega_{\ell mn}^{\text{obs}} - \omega_{\ell mn}^{\text{GR}}.$$

The primary objective is to determine whether this observable deviation correlates with

$$\Lambda_{\text{BH}}, \quad \Sigma_{\mu\nu}, \quad S_F, \quad C_{\mu\nu},$$

thereby providing a direct observational test of the gravitational predictions of Future Convergence Theory. Accordingly, the role of FCT in this paper is not merely conceptual. Instead, it is formulated as a predictive physical theory whose measurable observables are defined explicitly through deviations from General Relativity.

Keywords: Future Convergence Theory; Future Causal Information Field; Manifestation Constraint; Quantum Permission Layer; Black Hole Ringdown; Quasinormal Modes; Dark Matter; Dark Energy; Cosmology; Future Entropy.

Contents

I	Axiomatic Foundations of Unified Ontology	7
1	Fundamental Axioms of Future Convergence Theory	8
1.1	Fundamental Axioms of Future Convergence Theory	8
1.1.1	1.1 Meaning Generation Template and the Future Causal Information Field	8
1.1.2	1.2 Definition of the Quantum Permission Layer	10
1.1.3	1.3 Definition of the Three-Dimensional Manifestation Layer	12
1.1.4	1.4 Reprojection Closed Loop	15
1.1.5	1.5 The FCT Closure Equation	17
II	Gravity as a Manifestation Constraint	20
2	Gravity as a Manifestation Constraint	21
2.0.1	2.1 Gravity Is Not an Interparticle Force but a Manifestation Constraint .	22
2.0.2	2.2 Definition of the Manifestation Constraint Tensor	23
2.0.3	2.3 The Gravity Universality Theorem	25
2.0.4	2.4 Derivation of the Non-Screenability of Gravity	27
2.0.5	2.5 Why Gravity Acts on Light	29
2.0.6	2.6 Why Gravity Is Weak	32
2.0.7	2.7 The Relationship Between Gravitational Time Dilation and the Double Fixed Point Theory	34
2.0.8	2.8 The General Relativistic Limit	36
2.0.9	2.9 Falsifiable Predictions	39
III	Why Quantum Theory Must Be an Intermediate Layer	42
3	Why Quantum Theory Must Be an Intermediate Layer	43
3.0.1	3.1 The Layer Non-Identity Axiom	44
3.0.2	3.2 The FCT Definition of the Quantum State	44
3.0.3	3.3 Reformulation of the Measurement Problem	47
3.0.4	3.4 Redefinition of Decoherence	49
3.0.5	3.5 Vacuum Fluctuation Corrections	51
3.0.6	3.6 Corrections to the Casimir Effect	53
3.0.7	3.7 Predictions for the Delayed Choice Experiment	56
3.0.8	3.8 Predictions for the Quantum Eraser	58
3.0.9	3.9 Quantum Theory Limit and Falsifiability	60

IV	The Graviton Is Not the Essence of Gravity	63
4	The Graviton Is Not the Essence of Gravity	64
4.0.1	4.1 Metric Perturbations and the Graviton	65
4.0.2	4.2 The Layer Position of the Graviton	66
4.0.3	4.3 Non-Identity of the Manifestation Constraint and Quantum Perturbations	68
4.0.4	4.4 What the Detection of the Graviton Would Mean	70
4.0.5	4.5 Corrections to High-Energy Gravitational Scattering	72
4.0.6	4.6 Connection to Gravitational-Wave Dispersion	74
4.0.7	4.7 Falsifiability Conditions	77
V	The Position of String Theory within Future Convergence Theory	80
5	The Position of String Theory within Future Convergence Theory	81
5.0.1	5.1 String Theory Is Not the Final Layer but a Quantum Intermediate Layer	82
5.0.2	5.2 String Actions with FCT Corrections	83
5.0.3	5.3 Effective Background Metric	85
5.0.4	5.4 Corrections to Vibrational Modes	88
5.0.5	5.5 Corrections to the Mass Spectrum	90
5.0.6	5.6 Compactification Conditions	92
5.0.7	5.7 The Vacuum Selection Problem	95
5.0.8	5.8 Stability Conditions for Calabi–Yau Manifolds	97
5.0.9	5.9 Observability and Theoretical Limits	100
VI	Redefinition of Dark Matter and Observational Predictions	103
6	Redefinition of Dark Matter and Observational Predictions	104
6.0.1	6.1 Dark Matter in Earlier Future Convergence Theory	105
6.0.2	6.2 Redefinition of Dark Matter after Paper 92	105
6.0.3	6.3 Non-Optical Manifestation Residue	107
6.0.4	6.4 Effective Energy–Momentum Tensor	110
6.0.5	6.5 Prediction for Galactic Rotation Curves	112
6.0.6	6.6 Prediction for Halo Structure	115
6.0.7	6.7 Prediction for Galaxy-Cluster Collisions	117
6.0.8	6.8 Prediction for Gravitational Lensing	119
6.0.9	6.9 Predictions for the Cosmic Microwave Background	122
6.0.10	6.10 Criteria for Distinguishing Future Convergence Theory from Standard CDM	124
VII	Black-Hole Quasi-Normal Modes and Ringdown Predictions	128
7	FCT Reconstruction of Black-Hole Perturbation Theory	129
7.1	FCT Reconstruction of Black-Hole Perturbation Theory	129
7.1.1	Problem Setting	129
7.1.2	Modification of the Background Spacetime by the Future Information Field	131
7.1.3	FCT Reconstruction of the Regge–Wheeler Equation	134
7.1.4	Perturbation Theory of Quasi-Normal-Mode Eigenvalues	136
7.1.5	Derivation of the Effective Potential from the FCT Action Principle . . .	140
7.1.6	FCT Correction-Coefficient Constraint Theorem	144

7.1.7	Observational Correspondence Axiom for Future Convergence Entropy . .	147
7.1.8	Correction Formula for the Ringdown Frequency	153
VIII	Black-Hole Boundary Theory and Future Boundary Theory	157
7.2	Redefinition of the Event Horizon by the Future Information Field and Prediction of Gravitational-Wave Echoes	158
7.2.1	8.1 Introduction	158
7.2.2	8.2 Mathematical Definition of the Future Boundary	159
7.2.3	8.3 Derivation of the Future Boundary Reflection Coefficient	162
7.2.4	8.4 Derivation of the Gravitational-Wave Echo Delay Time at the Future Boundary	166
7.2.5	8.5 Amplitude Structure and Decay Law of the Gravitational-Wave Echo Train	168
7.2.6	8.6 Spectral Structure and Frequency Response of the Future Boundary .	171
7.2.7	8.7 Gravitational-Wave Detection Templates Based on Future Boundary Theory	174
7.2.8	8.8 Detectability and Observational Strategy of Future Boundary Theory	177
7.2.9	8.9 Summary of the Chapter and Verification Roadmap for Future Boundary Theory	180
IX	Predictions for Gravitational Waves as a Whole	183
7.3	Predictions for Gravitational Waves as a Whole	184
7.3.1	9.1 Introduction: From Local Black-Hole Theory to Cosmological Propagation Theory	184
7.3.2	9.2 Derivation of the Gravitational-Wave Propagation Equation Including the Future Information Field	185
7.3.3	9.3 Derivation of the Gravitational-Wave Dispersion Relation from the Future Information Field	188
7.3.4	9.4 Group Velocity, Phase Velocity, and Gravitational-Wave Arrival-Time Correction	190
7.3.5	9.5 Redefinition of Gravitational-Wave Polarization by the Future Information Field	193
7.3.6	9.6 Cosmological Phase Accumulation and the Integrated Effect of the Future Information Field	196
7.3.7	9.7 Future Information Field Corrections to Standard Sirens	198
7.3.8	9.8 Detectability of the Future Information Field with LIGO, Virgo, KAGRA, and LISA	201
7.3.9	9.9 Summary of the Chapter and Roadmap for Future Observational Tests	204
X	Dark Matter Theory Based on the Future Information Field	207
7.3.10	10.1 Introduction: Redefining the Dark Matter Problem	208
7.3.11	10.2 The Mathematical Structure of Dark Matter in General Relativity and Λ CDM Cosmology, and Its Reformulation by the Future Information Field	211
7.3.12	10.3 Derivation of the Effective Gravitational Density from the Future Information Field	213
7.3.13	10.4 Modification of Galactic Rotation Curves and Observational Predictions of the Future Information Field	216
7.3.14	10.5 Gravitational Lensing Theory Based on the Future Information Field	219

7.3.15	10.6 Predictions of the Future Information Field in the Bullet Cluster . .	221
7.3.16	10.7 Formation of the Large-Scale Structure of the Universe by the Future Information Field	224
7.3.17	10.8 Predictions of the Future Information Field for the Cosmic Microwave Background	227
7.3.18	10.9 Comparison with Existing Cosmological Models and the Distinguishability of the Future Information Field	229
7.3.19	10.10 Cosmological Observational Predictions Based on the Future Information Field	232
7.3.20	10.11 Summary of This Chapter and the Position of the Dark Matter Theory Based on the Future Information Field	234
XI	Dark Energy Theory Based on the Future Information Field	237
7.3.21	11.1 Introduction: Ontological Redefinition of the Dark Energy Problem .	238
7.3.22	11.2 The Mathematical Structure of Dark Energy in Standard Cosmology and Its Reformulation by the Future Information Field	239
7.3.23	11.3 Derivation of the Modified Friedmann Equation by the Future Information Field	242
7.3.24	11.4 Derivation of the Equation of State Based on the Future Information Field and the Origin of Negative Pressure	244
7.3.25	11.5 Onset Condition of Cosmic Accelerated Expansion and the Time Evolution of the Future Information Field	247
7.3.26	11.6 Reformulation of the Cosmological Constant Problem and the Interpretation of Vacuum Energy Based on the Future Information Field . . .	250
7.3.27	11.7 Reinterpretation of Vacuum Fluctuations, Zero-Point Energy, and the Casimir Effect Based on the Future Information Field	253
7.3.28	11.8 Unified Theory of Cosmic Termination, Heat Death, and Child-Universe Formation Based on the Future Information Field	256
7.3.29	11.9 Summary of This Chapter and the Position of the Future Information Field Theory of Cosmic Accelerated Expansion	259
XII	Early-Universe Theory Based on the Future Information Field	262
7.3.30	12.1 Introduction: Ontological Redefinition of the Early-Universe Problem	263
7.3.31	12.2 Mathematical Structure of Standard Early-Universe Cosmology and Its Reformulation by the Future Information Field	265
7.3.32	12.3 Derivation of the Inflationary Mechanism from the Future Information Field	268
7.3.33	12.4 Quantum Fluctuations of the Future Projection and the Origin of Primordial Density Perturbations and Primordial Gravitational Waves . .	271
7.3.34	12.5 Unified Reformulation of the Horizon Problem, Flatness Problem, and Monopole Problem	274
7.3.35	12.6 Derivation of the Cosmic Microwave Background, Spectral Index, and Non-Gaussianity	277
7.3.36	12.7 Unified Conditions for Cosmic Birth Based on the Child Universe Theory and the Future Information Field	280
7.3.37	12.8 Summary and Achievements of the Early-Universe Theory Based on the Future Information Field	284

XIII	Unified Action Principle of Future Convergence Theory	287
7.3.38	13.1 Introduction: The Necessity of a Unified Action Principle in Future Convergence Theory	288
7.3.39	13.2 Unified Lagrangian Density of Future Convergence Theory	290
7.3.40	13.3 Fundamental Field Equations of Future Convergence Theory Derived from the Unified Action	293
7.3.41	13.4 Unique Derivation of Black Holes, Dark Matter, Dark Energy, and Cosmology from the Unified Field Equations	295
7.3.42	13.5 Noether's Theorem and the Future Entropy Conservation Law	299
7.3.43	13.6 Consistency with General Relativity, the Standard Model, Λ CDM, and Paper 92 in the Appropriate Limits	301
7.3.44	13.7 Quantitative Predictions of Future Convergence Theory and Its Distinctions from Existing Theories	304
7.3.45	13.8 Summary of This Chapter and the Mathematical Closure of Future Convergence Theory	307
XIV	List of Observables and Falsification Conditions	311
8	List of Observables and Falsification Conditions	312
8.0.1	14.1 Ringdown Residuals	313
8.0.2	14.2 Echo Detection Conditions	314
8.0.3	14.3 Gravitational-Wave Dispersion	316
8.0.4	14.4 Dark Matter Halo Differences	318
8.0.5	14.5 Gravitational Lensing Differences	320
8.0.6	14.6 CMB Differences	322
8.0.7	14.7 Page Curve Differences	324
8.0.8	14.8 Casimir Differences	326
8.0.9	14.9 Equivalence Principle Limit Corrections	328
8.0.10	14.10 Comprehensive Falsification Conditions	330
XV	Conclusion	333
	Conclusion	334

Part I

Axiomatic Foundations of Unified Ontology

Chapter 1

Fundamental Axioms of Future Convergence Theory

This chapter establishes the minimal axiomatic system that supports the entire framework of Future Convergence Theory (FCT). All subsequent developments presented throughout this work—including gravity, quantum theory, gravitons, string theory, dark matter, black-hole quasinormal modes, gravitational waves, the Page curve, the early universe, dark energy, child universes, the unified field equation, and the complete set of observational predictions—are derived exclusively from the ontological and mathematical structures introduced in this chapter. The purpose of this chapter is not to provide a philosophical exposition of FCT. Rather, its objective is to establish the ontological entities and mathematical structures required for deriving quantitative deviations from General Relativity in the subsequent chapters.

1.1 Fundamental Axioms of Future Convergence Theory

In this chapter, we define the minimal axiomatic system that constitutes the entire theoretical framework of Future Convergence Theory (hereafter, FCT). Every result derived throughout this paper—including gravity, quantum theory, black holes, gravitational waves, dark matter, dark energy, the early universe, child universes, the unified field equation, and all observational predictions—is obtained solely from the mathematical structures introduced in this chapter. Accordingly, this chapter is not intended as a review or reformulation of existing physical theories. Nor does it merely modify General Relativity or Quantum Theory. Instead, its purpose is to define the ontological entities adopted by Future Convergence Theory, to specify the mathematical relationships among those entities, and to establish the complete set of axioms upon which all subsequent chapters are constructed.

Part I

Axiomatic Foundations of Unified Ontology

1.1.1 1.1 Meaning Generation Template and the Future Causal Information Field

Modern physics constructs its theoretical framework upon spacetime, matter, energy, and quantum states as its fundamental objects. In General Relativity, the spacetime metric

$$g_{\mu\nu}$$

constitutes the most fundamental geometrical object. In Quantum Field Theory, a field

$$\phi(x)$$

or an operator algebra represents the fundamental degree of freedom. However, all of these frameworks presuppose that an observable physical state already exists. In other words, existing physical theories describe “interactions among entities that already exist,” but do not address, within their own theoretical foundations, the ontological question of “why a particular state becomes manifest as an existing physical reality.” Future Convergence Theory adopts a different starting point. The present theory assumes that every physical state appearing within the three-dimensional world is the result of manifestation from a more fundamental informational structure. This primordial informational structure is referred to as the **Meaning Generation Template**. The Meaning Generation Template belongs neither to three-dimensional space nor to time. It is neither matter, energy, nor a quantum state. Rather, it is the highest ontological structure that contains the potential possibility of every physical state capable of future manifestation. The Meaning Generation Template is denoted by

$$\boxed{\mathcal{M}}$$

where

$$\mathcal{M}$$

does not represent observable physical states themselves. Instead, it represents an abstract ontological structure that defines the entire set of potentially manifestable physical states. Within Future Convergence Theory, the three-dimensional world is not generated directly from the Meaning Generation Template. Instead, the Meaning Generation Template first gives rise to the **Future Causal Information Field**. The Future Causal Information Field is defined by

$$\boxed{\mathcal{F}}$$

where

$$\mathcal{F}$$

denotes an informational field in which the information contained in the Meaning Generation Template is continuously organized under causal constraints directed toward the future. Throughout this paper, the terms Future Information Field, Future Causal Information Field, and the corresponding Japanese term “Future Causal Information Field” are treated as referring to the identical ontological object. The Future Causal Information Field is fundamentally different from ordinary physical fields. Electromagnetic fields, gravitational fields, and quantum fields are all degrees of freedom that have already manifested within the three-dimensional physical world. In contrast, the Future Causal Information Field describes informational constraints that exist prior to physical manifestation. Consequently, the Future Causal Information Field is neither identical to the metric field of General Relativity nor to the operator fields of Quantum Field Theory. Within Future Convergence Theory, the highest ontological hierarchy is defined as

$$\boxed{\mathcal{M} \rightarrow \mathcal{F}}$$

That is, the Meaning Generation Template

$$\mathcal{M}$$

generates the Future Causal Information Field

$$\mathcal{F}.$$

This mapping is not a physical law. Rather, it is an ontological rule of generation. Accordingly, the first axiom of the theory is formulated as follows.

Axiom 1.1 (Meaning Generation Axiom). *Every observable physical state can become manifest only through the Future Causal Information Field*

$$\mathcal{F},$$

which is generated from the Meaning Generation Template

$$\mathcal{M}.$$

No observable physical state can manifest independently of this ontological generation process.

We next introduce the second axiom concerning the Future Causal Information Field.

Axiom 1.2 (Future Causal Information Field Axiom). *The Future Causal Information Field*

$$\mathcal{F}$$

is an informational constraint field that exists independently of three-dimensional spacetime and preserves the complete causal structure of all potentially manifestable physical states.

Consequently, the highest ontological structure adopted by Future Convergence Theory is summarized by the following ontological sequence:

$$\boxed{\mathcal{M} \rightarrow \mathcal{F}}$$

In the following sections, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, gravity, black holes, and cosmology will be derived successively from the Future Causal Information Field.

1.1.2 1.2 Definition of the Quantum Permission Layer

In the previous section, the Meaning Generation Template

$$\mathcal{M}$$

and the Future Causal Information Field

$$\mathcal{F}$$

were introduced as the highest ontological structures of Future Convergence Theory. However, not all information contained within the Future Causal Information Field is manifested directly into the three-dimensional world. If every informational state contained in the Future Causal Information Field were allowed to manifest without restriction, the three-dimensional world would necessarily contain mutually contradictory physical states simultaneously. For example, states in which the same particle manifests simultaneously at different spatial locations, mutually incompatible physical laws coexist, or conservation laws are violated would all become physically realized. Such states are not observed in nature. Therefore, an intermediate layer must exist between the Future Causal Information Field and the Three-Dimensional Manifestation Layer that restricts which informational states are permitted to become physically manifest. Future Convergence Theory defines this intermediate layer as the **Quantum Permission Layer**. The Quantum Permission Layer is denoted by

$$\boxed{\mathcal{Q}}$$

where

$$\mathcal{Q}$$

represents the constraint layer that selects, from among all informational states contained in the Future Causal Information Field, only those states that are permitted to manifest within the

three-dimensional physical world. Accordingly, within Future Convergence Theory, Quantum Theory is not regarded as a theory governing the motion of particles. Instead, Quantum Theory is redefined as the complete set of permission rules that determine which informational states may transition from the Future Causal Information Field into the Three-Dimensional Manifestation Layer. From this standpoint, the wave function does not represent a physical particle. Nor does it merely represent an observational probability distribution. Rather, the wave function is interpreted as the mathematical representation of the set of physically permissible manifestation states defined on the Quantum Permission Layer. Consequently, the conventional quantum state

$$|\psi\rangle$$

is reinterpreted in Future Convergence Theory as a state belonging to the Quantum Permission Layer:

$$|\psi\rangle \in \mathcal{Q}$$

It is important to emphasize that the Quantum Permission Layer is not identical to the Future Causal Information Field itself. The Future Causal Information Field contains the entirety of potentially manifestable informational states. The Quantum Permission Layer, by contrast, extracts only those states that are allowed to become physically manifest under the constraints imposed by the current physical laws. Accordingly, the ontological hierarchy of Future Convergence Theory is extended as follows:

$$\mathcal{M} \longrightarrow \mathcal{F} \longrightarrow \mathcal{Q}$$

The mapping

$$\mathcal{F} \rightarrow \mathcal{Q}$$

is therefore interpreted as a constraint mapping that imposes manifestation conditions upon the informational states contained within the Future Causal Information Field. This mapping is defined as the **Quantum Permission Operator**. The Quantum Permission Operator is denoted by

$$\mathcal{P}_Q : \mathcal{F} \rightarrow \mathcal{Q}$$

where

$$\mathcal{P}_Q$$

maps only those informational states contained within the Future Causal Information Field that satisfy conservation laws, causality, Manifestation Constraints, and the Future Convergence conditions introduced in subsequent chapters, into the Quantum Permission Layer. Accordingly, within Future Convergence Theory, quantum superposition does not imply that multiple physical states exist simultaneously. Instead, superposition represents the set of permissible manifestation states defined on the Quantum Permission Layer, for which a unique manifested state has not yet been determined. On the basis of this interpretation, we introduce the third axiom.

Axiom 1.3 (Quantum Permission Axiom). *Every informational state contained within the Future Causal Information Field is subjected to the constraint imposed by the Quantum Permission Operator*

$$\mathcal{P}_Q.$$

Only the states belonging to the resulting Quantum Permission Layer

$$\mathcal{Q}$$

constitute candidates for manifestation within the Three-Dimensional Manifestation Layer.

We further establish the following axiom concerning the ontological status of the Quantum Permission Layer.

Axiom 1.4 (Layer Non-Identity Axiom). *The Quantum Permission Layer*

$$\mathcal{Q}$$

is ontologically distinct from both the Future Causal Information Field

$$\mathcal{F}$$

and the Three-Dimensional Manifestation Layer. Quantum Theory exists solely as the intermediate constraint layer that connects these two ontologically distinct domains.

Consequently, the ontological hierarchy of Future Convergence Theory is defined as the following three-layer structure:

$$\boxed{\mathcal{M} \longrightarrow \mathcal{F} \longrightarrow \mathcal{Q}}$$

In the next section, we formulate the mechanism by which states passing through the Quantum Permission Layer become physically manifest within three-dimensional spacetime and provide the mathematical definition of the Three-Dimensional Manifestation Layer.

1.1.3 1.3 Definition of the Three-Dimensional Manifestation Layer

In the previous section, we established that the Future Causal Information Field

$$\mathcal{F},$$

generated from the Meaning Generation Template

$$\mathcal{M},$$

is constrained by the Quantum Permission Operator

$$\mathcal{P}_Q,$$

thereby forming the Quantum Permission Layer

$$\mathcal{Q}.$$

However, the states contained within the Quantum Permission Layer are still not observable physical states. The Quantum Permission Layer merely preserves those states that are *permitted to become manifest*; it does not itself constitute the physical reality that has already appeared in the observable universe. Accordingly, Future Convergence Theory requires the existence of a further, ontologically independent layer following the Quantum Permission Layer, namely, the layer in which actual physical reality is realized. This final layer is defined as the **Three-Dimensional Manifestation Layer**. The Three-Dimensional Manifestation Layer is denoted by

$$\boxed{\mathcal{X}}$$

where

$$\mathcal{X}$$

represents the ontological domain in which all observable physical quantities exist, including spacetime, matter, energy, gravity, electromagnetic fields, quantum particles, and every other object studied by contemporary physics. Accordingly, the spacetime metric

$$g_{\mu\nu},$$

in General Relativity, the quantum fields

$$\phi(x),$$

of Quantum Field Theory, every particle described by the Standard Model, and even the observer, all belong to the Three-Dimensional Manifestation Layer. Future Convergence Theory therefore does not reject either General Relativity or Quantum Theory. Rather, both theories are interpreted as physical theories governing the Three-Dimensional Manifestation Layer.

Manifestation Projection

The transition from the Quantum Permission Layer to the Three-Dimensional Manifestation Layer is not merely a state transition. Within Future Convergence Theory, it is the mapping that determines physical manifestation itself. This mapping is referred to as the **Manifestation Projection**, and is defined by

$$\boxed{\mathcal{P}_X : \mathcal{Q} \rightarrow \mathcal{X}}$$

where

$$\mathcal{P}_X$$

is the operator that determines which quantum-permitted states become physically realized within three-dimensional spacetime. Only through this projection do observable physical states come into existence.

Ontological Hierarchy

The ontological hierarchy of Future Convergence Theory is therefore extended to the following four-layer structure:

$$\boxed{\mathcal{M} \longrightarrow \mathcal{F} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{X}}$$

Each layer is ontologically independent and must not be identified with any other. The Meaning Generation Template is not the Future Causal Information Field. The Future Causal Information Field is not the Quantum Permission Layer. The Quantum Permission Layer is not the Three-Dimensional Manifestation Layer. This principle of ontological non-identity constitutes one of the fundamental principles of Future Convergence Theory.

Manifested States

Let an arbitrary observable physical state belonging to the Three-Dimensional Manifestation Layer be denoted by

$$x \in \mathcal{X}.$$

Within Future Convergence Theory, every manifested state is generated according to

$$\boxed{x = \mathcal{P}_X (\mathcal{P}_Q(F))}$$

where

$$F \in \mathcal{F}$$

represents an informational state belonging to the Future Causal Information Field. Accordingly, no observable physical state is generated directly from the Future Causal Information Field. Every manifested state necessarily passes through the Quantum Permission Layer prior to physical realization.

Interpretation of Quantum Superposition

Under this formulation, quantum superposition does not exist within the Three-Dimensional Manifestation Layer. Superposition exists exclusively within the Quantum Permission Layer

$$\mathcal{Q}.$$

By contrast, within the Three-Dimensional Manifestation Layer

$$\mathcal{X},$$

only the state selected by the Manifestation Projection

$$\mathcal{P}_X$$

is physically realized. Accordingly, within Future Convergence Theory, the conventional measurement problem is reformulated. Rather than being interpreted as the collapse of the wave function, it becomes the problem of determining the Manifestation Projection.

Fourth Axiom

On the basis of the preceding discussion, we introduce the following axiom.

Axiom 1.5 (Three-Dimensional Manifestation Axiom). *Every observable physical state is generated through the Manifestation Projection*

$$\mathcal{P}_X$$

from the Quantum Permission Layer

$$\mathcal{Q}.$$

Consequently, the Three-Dimensional Manifestation Layer

$$\mathcal{X}$$

is the unique ontological layer within Future Convergence Theory that is directly accessible to physical observation.

Ontological Chain

The fundamental ontological structure of Future Convergence Theory is therefore represented by the following chain:

$$\boxed{\mathcal{M} \xrightarrow{\mathcal{P}_F} \mathcal{F} \xrightarrow{\mathcal{P}_Q} \mathcal{Q} \xrightarrow{\mathcal{P}_X} \mathcal{X}}$$

where

$$\mathcal{P}_F$$

denotes the ontological generative mapping that produces the Future Causal Information Field from the Meaning Generation Template.

Conclusion of This Section

In this section, we introduced the Three-Dimensional Manifestation Layer

$$\mathcal{X},$$

and demonstrated that the observable physical universe is generated from the Future Causal Information Field through the intermediate Quantum Permission Layer. This establishes the four-layer ontological architecture consisting of the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, and the Three-Dimensional Manifestation Layer. In the next section, we demonstrate that this four-layer architecture is not a purely one-way generative process. Instead, it forms a closed ontological structure incorporating the recovery of information into the Future Causal Information Field, thereby introducing the concept of the *Reprojection Closed Loop*.

1.1.4 1.4 Reprojection Closed Loop

In the preceding sections, Future Convergence Theory has been formulated as a four-layer ontological architecture consisting of the Meaning Generation Template

$$\mathcal{M},$$

the Future Causal Information Field

$$\mathcal{F},$$

the Quantum Permission Layer

$$\mathcal{Q},$$

and the Three-Dimensional Manifestation Layer

$$\mathcal{X}.$$

However, if this architecture were merely a one-way process of ontological generation, Future Convergence Theory could explain only the emergence of the universe. It would not provide a unified account of black-hole information preservation, the thermodynamic heat death of the universe, child-universe generation, and Future Convergence. Within the present theory, information that has become manifest is never annihilated through a one-way process. Every piece of information existing within the Three-Dimensional Manifestation Layer is ultimately recovered into the Future Causal Information Field, where it is subsequently utilized as a new condition for future manifestation. This cyclic ontological structure is defined as the **Reprojection Closed Loop**.

Information Recovery

Let an arbitrary physical state belonging to the Three-Dimensional Manifestation Layer be denoted by

$$x \in \mathcal{X}.$$

Such a state does not simply disappear through temporal evolution. Instead, Future Convergence Theory postulates that every manifested state is preserved as information within the Future Causal Information Field. The mapping responsible for this informational recovery is defined by

$$\boxed{\mathcal{R} : \mathcal{X} \rightarrow \mathcal{F}}$$

where

$$\mathcal{R}$$

is referred to as the **Recovery Operator** or equivalently the **Reprojection Operator**.

Principle of Information Preservation

The introduction of the Information Recovery Mapping implies that information which appears to disappear from the Three-Dimensional Manifestation Layer is instead preserved within the Future Causal Information Field. Accordingly, Future Convergence Theory contains no concept of fundamental information loss. Information is not destroyed. Rather, it is preserved by changing its ontological mode of existence—from the Manifestation Layer into the Future Causal Information Field. Consequently, within Future Convergence Theory, information preservation is not merely a physical conservation law. It is an intrinsic property of ontology itself.

Closed-Loop Structure

Including the Information Recovery Mapping

$$\mathcal{R},$$

together with the mappings introduced previously,

$$\mathcal{P}_F, \quad \mathcal{P}_Q, \quad \mathcal{P}_X,$$

the complete ontological structure of Future Convergence Theory becomes the following closed loop:

$$\boxed{\mathcal{M} \xrightarrow{\mathcal{P}_F} \mathcal{F} \xrightarrow{\mathcal{P}_Q} \mathcal{Q} \xrightarrow{\mathcal{P}_X} \mathcal{X} \xrightarrow{\mathcal{R}} \mathcal{F}}$$

This ontological cycle is referred to as the **Future Convergence Loop**.

Relation to Black Holes

As demonstrated in Part IX, a black hole is not an object that destroys information. Within Future Convergence Theory, a black hole is interpreted as a region in which the Information Recovery Mapping

$$\mathcal{R}$$

dominates the local dynamics. Accordingly, the black-hole information preservation problem does not require the introduction of an additional conservation law. Instead, it is understood as the local realization of the Reprojection Closed Loop.

Relation to the Heat Death of the Universe

Within Future Convergence Theory, the thermodynamic heat death of the universe does not signify the termination of information. Even after the entire Three-Dimensional Manifestation Layer reaches thermal equilibrium, all manifested information remains preserved within the Future Causal Information Field. Accordingly, the heat death of the universe represents the cessation of manifestation rather than the destruction of information.

Relation to Child-Universe Generation

Information recovered into the Future Causal Information Field does not remain permanently static. Once the conditions required for Future Convergence are satisfied, the Future Causal Information Field once again generates the Quantum Permission Layer, which subsequently gives rise to a new Three-Dimensional Manifestation Layer. This process constitutes the mechanism of **Child Universe Generation**, which will be formulated in Part XII. Accordingly, a child universe is not created from nothing. Rather, it emerges as a consequence of the Reprojection Closed Loop.

Fifth Axiom

The preceding discussion leads to the following axiom.

Axiom 1.6 (Reprojection Closed Loop Axiom). *Every piece of information generated within the Three-Dimensional Manifestation Layer is preserved within the Future Causal Information Field through the Information Recovery Mapping*

$$\mathcal{R}.$$

The Future Causal Information Field subsequently utilizes the preserved information to generate a new Quantum Permission Layer and a new Three-Dimensional Manifestation Layer. Consequently, Future Convergence Theory forms an ontologically closed loop.

Mathematical Representation of the Closed Loop

The ontological closed-loop structure of Future Convergence Theory is summarized by

$$\boxed{\mathcal{M} \rightarrow \mathcal{F} \rightarrow \mathcal{Q} \rightarrow \mathcal{X} \rightarrow \mathcal{F}}$$

Within this closed loop, meaning generation, manifestation, information preservation, and child-universe generation are understood not as independent phenomena, but as different aspects of a single ontological cycle.

Conclusion of This Section

In this section, we demonstrated that Future Convergence Theory is not merely a one-way theory of ontological generation. Instead, it is formulated as a Reprojection Closed Loop incorporating information recovery. This closed-loop structure provides the ontological foundation for a unified interpretation of black-hole information preservation, the thermodynamic heat death of the universe, child-universe generation, and the conservation law of Future Entropy. In the next section, all axioms introduced thus far are unified into a single mathematical closure condition. This leads to the formulation of the fundamental equation of Future Convergence Theory, namely the **FCT Closure Equation**.

1.1.5 1.5 The FCT Closure Equation

In the preceding sections, we introduced the ontological hierarchy that constitutes Future Convergence Theory, namely, the Meaning Generation Template

$\mathcal{M}$,

the Future Causal Information Field

$\mathcal{F}$,

the Quantum Permission Layer

$\mathcal{Q}$,

the Three-Dimensional Manifestation Layer

$\mathcal{X}$,

together with the Information Recovery Mapping

$\mathcal{R}$.

These entities were not introduced as independent conceptual components. Rather, within Future Convergence Theory, they collectively constitute a single, closed ontological structure. In this section, we formulate this structure mathematically and introduce the **FCT Closure Equation** as the highest-level governing equation common to all subsequent chapters.

Necessary Condition for Ontological Closure

In general, a theoretical framework is said to be closed if every object introduced within the theory can be reconstructed exclusively through mappings defined within the theory itself. The same principle applies to Future Convergence Theory. Meaning generation, manifestation, information preservation, and reprojection must not be regarded as mutually independent processes. If the ontological cycle could be completed only by introducing information from outside the theory, Future Convergence Theory would fail to constitute a closed theoretical framework. Accordingly, Future Convergence Theory must satisfy the requirement of ontological self-closure.

Composition of the Ontological Mappings

The mappings introduced thus far are summarized as

$$\mathcal{P}_F : \mathcal{M} \rightarrow \mathcal{F},$$

$$\mathcal{P}_Q : \mathcal{F} \rightarrow \mathcal{Q},$$

$$\mathcal{P}_X : \mathcal{Q} \rightarrow \mathcal{X},$$

and

$$\mathcal{R} : \mathcal{X} \rightarrow \mathcal{F}.$$

The composition of these mappings defines the complete ontological process originating from the Meaning Generation Template:

$$\boxed{\mathcal{C} = \mathcal{R} \circ \mathcal{P}_X \circ \mathcal{P}_Q \circ \mathcal{P}_F}$$

The operator

$$\mathcal{C}$$

is referred to as the **Future Convergence Operator**.

Closure Condition

For ontological closure to hold within Future Convergence Theory, the Future Causal Information Field must remain preserved after one complete application of the Future Convergence Operator. Accordingly, the following closure condition is imposed:

$$\boxed{\mathcal{C}(\mathcal{M}) \subseteq \mathcal{F}}$$

That is, the information generated from the Meaning Generation Template, subsequently manifested, and finally recovered, always returns to the Future Causal Information Field.

Fixed-Point Condition

Within Future Convergence Theory, the Future Causal Information Field is not a structure generated only once. Instead, in the asymptotic limit of repeated information recovery, the Future Causal Information Field converges toward a stable ontological state. This limiting state is referred to as the **Future Fixed Point**. The fixed point

$$\mathcal{F}^*$$

is defined by

$$\boxed{\mathcal{C}(\mathcal{F}^*) = \mathcal{F}^*}$$

This fixed point constitutes the first fixed point of the **Double Fixed Point Theory**, which will be introduced in a subsequent part of this work.

The FCT Closure Equation

We therefore define the following equation as the highest governing equation of Future Convergence Theory:

$$\boxed{\mathcal{F} = \mathcal{R} \circ \mathcal{P}_X \circ \mathcal{P}_Q \circ \mathcal{P}_F(\mathcal{M})}$$

This equation is referred to as the **Future Convergence Theory Closure Equation** (or simply, the **FCT Closure Equation**). It constitutes the highest-level equation that unifies existence, manifestation, information preservation, and Future Convergence within a single mathematical framework.

Ontological Interpretation

The FCT Closure Equation does not imply that the Future Causal Information Field is generated only once from the Meaning Generation Template. Rather, the Future Causal Information Field continuously recovers all information generated within the Three-Dimensional Manifestation Layer. While preserving that accumulated information, it simultaneously forms the conditions for future manifestations. Accordingly, existence itself is not understood as a one-way process of generation. Instead, it is interpreted as a self-maintaining informational structure sustained within a closed ontological loop.

Summary of Part I

In this chapter, we established the following five fundamental components as the minimal axiomatic system of Future Convergence Theory.

1. The Meaning Generation Template

$\mathcal{M}$

2. The Future Causal Information Field

$\mathcal{F}$

3. The Quantum Permission Layer

$\mathcal{Q}$

4. The Three-Dimensional Manifestation Layer

$\mathcal{X}$

5. The FCT Closure Equation

$$\mathcal{F} = \mathcal{R} \circ \mathcal{P}_X \circ \mathcal{P}_Q \circ \mathcal{P}_F(\mathcal{M})$$

These five elements constitute the fundamental axioms upon which the entirety of Future Convergence Theory is constructed.

Conclusion of Part I

Part I has established the ontological foundation of Future Convergence Theory. Within this framework, gravity, quantum theory, black holes, and cosmology are not regarded as mutually independent theoretical systems. Instead, each of them is derived from a single ontological closed loop that originates with the Meaning Generation Template, proceeds through the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the Information Recovery Mapping, before converging once again upon the Future Causal Information Field. In the subsequent parts of this work, the FCT Closure Equation serves as the starting point for deriving gravity as a Manifestation Constraint, the reformulation of Quantum Theory, black-hole physics, cosmology, the unified field equation, and the observational predictions that distinguish Future Convergence Theory from General Relativity.

Part II

Gravity as a Manifestation Constraint

Chapter 2

Gravity as a Manifestation Constraint

In Part I, we defined the minimal axiomatic system constituting Future Convergence Theory, namely, the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the FCT Closure Equation. Through these axioms, the observable physical world was positioned as a manifestation process extending from the Future Causal Information Field to the Three-Dimensional Manifestation Layer. However, Part I has not yet defined how manifested states are stably maintained within the Three-Dimensional Manifestation Layer. If only the Manifestation Projection existed, without any subsequent constraint mechanism, the three-dimensional world would easily lose its manifested states through temporal evolution, and stable physical laws would not be established. Therefore, a constraint mechanism that maintains manifestation itself is required. Future Convergence Theory identifies precisely this constraint mechanism as the essence of the phenomenon traditionally called gravity. The present theory does not introduce gravity as a fundamental interaction acting between masses. Nor does it begin by postulating the graviton as a fundamental particle. Within this theory, gravity is defined as the **Manifestation Constraint** that maintains the mapping from the Future Causal Information Field to the Three-Dimensional Manifestation Layer so that it does not break down under temporal evolution. Thus, gravity is not a force. It is a constraint action that preserves the structure of manifestation. From this standpoint, spacetime curvature in General Relativity is understood as the geometrical representation obtained when the Manifestation Constraint is projected onto three-dimensional spacetime. Newtonian gravity, by contrast, becomes an approximate theory corresponding to the weak-gravity limit of the Manifestation Constraint. Furthermore, as will be derived in later chapters, black holes, gravitational waves, dark matter, and dark energy are all obtained as distinct boundary conditions of the Manifestation Constraint. Accordingly, within Future Convergence Theory, gravity is not a physical phenomenon placed alongside other phenomena. Rather, gravity is positioned as the most fundamental constraint principle that makes the Three-Dimensional Manifestation Layer itself possible. In this chapter, we first provide the mathematical definition of the Manifestation Constraint. We then introduce the Manifestation Constraint Tensor and derive, successively from the axiomatic system of Future Convergence Theory, its universality, its non-screenability, its action on light, the weakness of gravity, and its relation to time delay. Finally, we show that General Relativity is recovered as the low-energy limit of the theory of Manifestation Constraints, and we summarize the observational differences between Future Convergence Theory and General Relativity.

Part II

Gravity as a Manifestation Constraint

2.0.1 2.1 Gravity Is Not an Interparticle Force but a Manifestation Constraint

In Newtonian mechanics, gravity is defined as an attractive force acting between two masses. In General Relativity, gravity is described as spacetime curvature produced by mass and energy. Both theories have achieved extraordinary empirical success. However, neither theory explains why gravity itself exists. General Relativity takes the spacetime metric

$$g_{\mu\nu}$$

as its fundamental variable. Yet the questions of why the spacetime metric itself exists, and why matter always manifests into the same spacetime, remain assumptions external to the theory. Future Convergence Theory explains this problem from the manifestation structure defined in Part I. In Part I, we showed that every physical state is projected from the Future Causal Information Field, through the Quantum Permission Layer, into the Three-Dimensional Manifestation Layer. However, the Manifestation Projection

$$\mathcal{P}_X : \mathcal{Q} \rightarrow \mathcal{X}$$

alone does not guarantee that a state, once manifested, remains maintained under temporal evolution. Therefore, Future Convergence Theory requires a constraint action that preserves manifested states. This constraint action is defined as the **Manifestation Constraint**. The Manifestation Constraint is denoted by

$$\boxed{\mathcal{C}_M}$$

where

$$\mathcal{C}_M$$

is the constraint operator that guarantees that all states existing within the Three-Dimensional Manifestation Layer continue to exist stably while preserving consistency with the Future Causal Information Field. Under this definition, gravity is positioned not as an attractive force acting between particles, but as a constraint that preserves manifested states. Accordingly, within Future Convergence Theory, gravity is an ontological constraint established between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. The spacetime curvature observed in General Relativity is merely the geometrical representation of this constraint projected onto four-dimensional spacetime. On this basis, the present theory introduces the following fundamental axiom.

Axiom 2.1 (Manifestation Constraint Axiom). *Every physical state existing within the Three-Dimensional Manifestation Layer maintains consistency with the Future Causal Information Field through the Manifestation Constraint*

$$\mathcal{C}_M.$$

Gravity is the physical representation of this Manifestation Constraint as it appears within observable spacetime.

Thus, Future Convergence Theory redefines gravity not as a fundamental interaction, but as a constraint principle that maintains existence itself. In the next section, we introduce the **Manifestation Constraint Tensor** in order to treat this Manifestation Constraint on four-dimensional spacetime, and we construct its mathematical correspondence with General Relativity.

2.0.2 2.2 Definition of the Manifestation Constraint Tensor

In the previous section, we defined gravity in Future Convergence Theory not as an interparticle interaction, but as the **Manifestation Constraint** that maintains consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. However, in order to treat the Manifestation Constraint as a physical theory, a mathematical object is required that describes this constraint on four-dimensional spacetime. In General Relativity, spacetime geometry is described by the metric tensor

$$g_{\mu\nu},$$

from which the connection, curvature tensor, Ricci tensor, and Einstein tensor are derived. Future Convergence Theory does not reject these geometrical objects. However, it regards them not as gravity itself, but as secondary descriptions that appear when the Manifestation Constraint is projected onto the Three-Dimensional Manifestation Layer. Accordingly, the present theory introduces, as a quantity more fundamental than spacetime curvature, the **Manifestation Constraint Tensor**. The Manifestation Constraint Tensor is defined by

$$\boxed{\mathcal{C}_{\mu\nu}}$$

where

$$\mathcal{C}_{\mu\nu}$$

is a symmetric rank-two tensor representing the local degree to which a manifested state, mapped from the Future Causal Information Field into the Three-Dimensional Manifestation Layer, is constrained. Therefore, within Future Convergence Theory, the gravitational field is none other than the Manifestation Constraint Tensor field

$$\mathcal{C}_{\mu\nu}(x).$$

Properties of the Manifestation Constraint Tensor

The Manifestation Constraint Tensor satisfies the following properties.

1. It is a symmetric tensor:

$$\boxed{\mathcal{C}_{\mu\nu} = \mathcal{C}_{\nu\mu}}$$

2. It is defined on the Three-Dimensional Manifestation Layer.
3. It locally represents the consistency condition with the Future Causal Information Field.
4. In the General Relativity limit, it corresponds one-to-one with the Einstein tensor.

Manifestation Constraint Density

The strength of the Manifestation Constraint is represented by the scalar quantity

$$\Lambda_{\text{M}}.$$

We define this scalar as the **Manifestation Constraint Density**.

$$\boxed{\Lambda_{\text{M}} = g^{\mu\nu} \mathcal{C}_{\mu\nu}}$$

Here,

$$\Lambda_{\text{M}}$$

represents the total amount of constraint required to maintain the manifested state at a given spacetime point. In general, regions of stronger gravity correspond to larger values of

$$\Lambda_{\text{M}}.$$

Coupling to the Future Causal Information Field

The Future Causal Information Field

$$\mathcal{F},$$

defined in Part I, constitutes the informational source of the Manifestation Constraint Tensor. Accordingly, Future Convergence Theory assumes the following relation:

$$\boxed{\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}(\mathcal{F})}$$

That is, the Manifestation Constraint Tensor is not an independent field. Rather, it is an effective tensor derived from the Future Causal Information Field. In Part XIII, this relation will be derived from the unified field equation.

Relation to Spacetime Curvature

In General Relativity, the Ricci tensor

$$R_{\mu\nu}$$

and the Einstein tensor

$$G_{\mu\nu}$$

represent gravity. By contrast, in Future Convergence Theory, these quantities are interpreted as projections of the Manifestation Constraint Tensor. That is, in the General Relativity limit, there exists a projection operator

$$\Pi$$

such that

$$\boxed{G_{\mu\nu} = \Pi(\mathcal{C}_{\mu\nu})}$$

Therefore, the Einstein tensor is not gravity itself, but the geometrical representation of the Manifestation Constraint.

Physical Meaning

Under this definition, the conventional view that mass generates gravity is replaced by the following interpretation. Mass locally modifies the correspondence between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. As a result, the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

changes, and this change is observed as spacetime curvature. Thus, mass is not the cause of gravity. Rather, mass is a local boundary condition that modifies the Manifestation Constraint.

Second Axiom

On this basis, the present chapter adopts the following axiom.

Axiom 2.2 (Manifestation Constraint Tensor Axiom). *Gravity within the Three-Dimensional Manifestation Layer is completely described by the Manifestation Constraint Tensor*

$$\mathcal{C}_{\mu\nu}.$$

Spacetime curvature in General Relativity is understood as a projection of the Manifestation Constraint Tensor.

Conclusion of This Section

In this section, we introduced the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

as the fundamental field of Future Convergence Theory. Consequently, gravity has been mathematically formulated not as spacetime curvature, but as a Manifestation Constraint field derived from the Future Causal Information Field. In the next section, we derive from this Manifestation Constraint Tensor the reason why gravity acts universally on all matter and energy, and we prove the Gravity Universality Theorem.

2.0.3 2.3 The Gravity Universality Theorem

In the previous section, we introduced the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

as the fundamental gravitational field of Future Convergence Theory. However, if the Manifestation Constraint is adopted as the essence of gravity, Future Convergence Theory must itself explain one of the most fundamental facts accounted for by General Relativity: namely, why gravity acts universally upon all forms of matter, all forms of energy, and even upon light. In General Relativity, the universality of gravity is postulated through the Equivalence Principle. By contrast, in Future Convergence Theory, the universality of gravity is not an independent axiom but a theorem that follows necessarily from the definition of the Manifestation Constraint.

Objects Subject to the Manifestation Constraint

As established in Part I, every physical state belonging to the Three-Dimensional Manifestation Layer

$$\mathcal{X}$$

is generated from the Future Causal Information Field through the Manifestation Projection

$$\mathcal{P}_X.$$

Accordingly, every state

$$x \in \mathcal{X}$$

necessarily undergoes the Manifestation Constraint during its process of manifestation. The crucial point is that the Manifestation Constraint is not a mechanism that acts selectively according to particle species. Rather, it is the constraint that maintains the Manifestation Projection itself. Consequently, every manifested state is subjected to the Manifestation Constraint simply by virtue of existing.

Manifestation Constraint and Existence

For an arbitrary physical state

$$x,$$

Future Convergence Theory adopts the following proposition.

Proposition 2.1. *Every state existing within the Three-Dimensional Manifestation Layer satisfies the Manifestation Constraint*

$$\mathcal{C}_{\mu\nu}$$

as a necessary condition for its existence.

This proposition implies that the Manifestation Constraint is not an interaction applied to an already existing physical state. Rather, it is a prerequisite for the existence of that physical state.

Gravity Universality Theorem

The following theorem therefore holds.

Theorem 2.1 (Gravity Universality Theorem). *Every physical state existing within the Three-Dimensional Manifestation Layer shares the same Manifestation Constraint. Consequently, gravity acts upon every manifested state independently of particle species, internal degrees of freedom, electric charge, spin, or the presence or absence of rest mass.*

Proof. Consider an arbitrary physical state

$$x \in \mathcal{X}.$$

From Part I, every manifested state is generated through the Manifestation Projection

$$\mathcal{P}_X.$$

From Section 2.1, the Manifestation Projection is maintained by the Manifestation Constraint

$$\mathcal{C}_{\mu\nu}.$$

Accordingly, any state that is not subject to the Manifestation Constraint cannot exist within the Three-Dimensional Manifestation Layer. Therefore, every existing physical state necessarily shares the same Manifestation Constraint. It follows immediately that gravity acts universally upon all manifested physical states. $\square$

Comparison with General Relativity

In General Relativity, the universality of gravity is introduced through the Equivalence Principle. Future Convergence Theory does not adopt the Equivalence Principle as an independent axiom. Instead, gravitational universality is derived as a theorem from the existence of the Manifestation Constraint itself. The Equivalence Principle is therefore interpreted as a consequence of the underlying ontological constraint.

Application to Light

A photon possesses no rest mass. Nevertheless, a photon is also a state belonging to the Three-Dimensional Manifestation Layer. Therefore, by the Gravity Universality Theorem, photons are likewise subject to the Manifestation Constraint. As a consequence, light is gravitationally deflected despite having no rest mass. Although this conclusion agrees with General Relativity, the underlying reason is fundamentally different. In General Relativity, light follows geodesics of curved spacetime. In Future Convergence Theory, light propagates according to the Manifestation Structure because it shares the same Manifestation Constraint as every other manifested state.

Universality and the Conservation Law

The universality of gravity is also required by the conservation law of Future Entropy. If certain particles were exempt from the Manifestation Constraint, the Information Recovery Mapping

$$\mathcal{R}$$

would necessarily acquire different structures for different particle species. Under such circumstances, the FCT Closure Equation established in Part I could no longer hold. Therefore, the universality of the Manifestation Constraint is itself a necessary condition for maintaining ontological closure.

Observational Consequences

The Gravity Universality Theorem reproduces the presently verified Weak Equivalence Principle. However, Future Convergence Theory predicts that under extreme physical conditions, small corrections originating from the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

may appear. Accordingly, minute deviations from General Relativity may arise near black holes, during ultra-high-energy scattering processes, and in the extremely early universe. These corrections will be derived quantitatively in Part VII and the subsequent parts of this work.

Conclusion of This Section

In this section, we derived the reason why gravity acts upon every physical state from the existence of the Manifestation Constraint. Consequently, within Future Convergence Theory, gravitational universality is not assumed as an independent Equivalence Principle but is established as a necessary condition for the very existence of the Three-Dimensional Manifestation Layer. In the next section, we use this result to derive, from the standpoint of the Manifestation Constraint, why gravity cannot be screened in the same manner as electromagnetic interactions.

2.0.4 2.4 Derivation of the Non-Screenability of Gravity

In the previous section, we demonstrated that within Future Convergence Theory, the universality of gravity follows directly from the Manifestation Constraint without introducing the Equivalence Principle as an independent axiom. Gravity, however, possesses another remarkable property that distinguishes it from the other fundamental interactions. Gravity cannot be screened. Electromagnetic interactions may be weakened by conductors or dielectric materials. Likewise, both the strong and weak interactions exhibit effective screening through their finite interaction ranges or through the surrounding medium. By contrast, throughout all experiments performed to date, complete gravitational screening has never been observed. Although General Relativity is consistent with this fact, it does not derive the impossibility of gravitational screening from first principles. Within Future Convergence Theory, non-screenability follows directly from the definition of the Manifestation Constraint itself.

General Condition for Screening

For an interaction

$$\mathcal{I}$$

to be screenable, there must exist a mechanism capable of locally disconnecting the interaction from the field that mediates it. That is, for some spatial region

$$\Omega,$$

there must exist an operation satisfying

$$\boxed{\mathcal{I} \rightarrow 0}$$

within that region. Electromagnetic fields satisfy this condition because induced charges inside a conductor locally cancel the effective electric field. This is possible because the electromagnetic field exists independently of the matter upon which it acts.

The Manifestation Constraint Has No Screening Boundary

In Future Convergence Theory, however, gravity is identified with the Manifestation Constraint

$$\mathcal{C}_{\mu\nu}.$$

The Manifestation Constraint is not a force acting upon matter. It is the condition that allows matter itself to exist. Consequently, screening the Manifestation Constraint would mean screening the manifested state itself. Yet every state belonging to the Three-Dimensional Manifestation Layer is maintained by the Manifestation Constraint. Therefore, any region in which the Manifestation Constraint were locally removed could no longer support physical existence.

Ontological Contradiction

Assume, for the sake of argument, that the Manifestation Constraint were completely screened within a region

$$\Omega.$$

Then

$$\boxed{\mathcal{C}_{\mu\nu} = 0}$$

would hold. However, from Section 2.1, every physical state

$$x \in \mathcal{X}$$

requires the Manifestation Constraint as a necessary condition for existence. Therefore,

$$\mathcal{C}_{\mu\nu} = 0$$

implies

$$\boxed{x \notin \mathcal{X}}$$

within that region. In other words, neither the screening apparatus nor the object being screened could themselves exist there. The very agent required to perform the screening operation would disappear. Hence, the assumption of complete gravitational screening leads to an ontological self-contradiction.

Non-Screenability Theorem

The preceding argument immediately yields the following theorem.

Theorem 2.2 (Non-Screenability Theorem). *The Manifestation Constraint is a necessary condition for the existence of the Three-Dimensional Manifestation Layer. Accordingly, any operation intended to screen the Manifestation Constraint locally would simultaneously eliminate the manifested states themselves and is therefore physically impossible. Hence, gravity cannot be completely screened.*

Proof. Let gravity be identified with the Manifestation Constraint

$$\mathcal{C}_{\mu\nu}.$$

For gravity to be screenable, there must exist a finite region in which

$$\mathcal{C}_{\mu\nu} = 0.$$

However, the Manifestation Constraint constitutes the necessary condition for the existence of the Three-Dimensional Manifestation Layer. Consequently, no physical state can exist within such a region. Since the screening apparatus itself is also a physical state belonging to the Three-Dimensional Manifestation Layer, it likewise cannot exist there. Therefore, the screening operation itself cannot be defined. Hence, complete gravitational screening is physically impossible. $\square$

Comparison with General Relativity

Within General Relativity, the impossibility of gravitational screening is generally understood as a consequence of the positivity of mass-energy and the absence of negative gravitational charge capable of cancelling a gravitational source. Future Convergence Theory provides a more fundamental interpretation. Gravity is not an interaction but a necessary condition for existence itself. Accordingly, the impossibility of screening follows not from the sign of a physical charge but from the underlying ontological structure.

Relation to Black Holes

Near a black hole, the Manifestation Constraint Density

$$\Lambda_M$$

becomes extremely large. However, even as the constraint density increases, the Manifestation Constraint itself is never screened. Rather, the Manifestation Constraint increasingly governs the recovery of information into the Future Causal Information Field. Accordingly, a black hole is interpreted not as a region where gravity is screened, but as a region in which the Manifestation Constraint reaches an extreme intensity.

Observational Consequences

The present theory predicts that no physical device capable of gravitational screening can exist. Conversely, under extreme gravitational conditions, higher-order corrections to

$$\mathcal{C}_{\mu\nu}$$

become significant. Therefore, observable deviations from General Relativity are expected to appear not through gravitational screening itself, but through small corrections in black-hole physics, gravitational-wave observations, and cosmological phenomena.

Conclusion of This Section

In this section, we derived the impossibility of gravitational screening from the ontological nature of the Manifestation Constraint. Consequently, the non-screenability of gravity is understood not as a peculiar property of a particular interaction, but as a necessary condition for the existence of the Three-Dimensional Manifestation Layer itself. In the next section, we derive, from the standpoint of Future Convergence Theory, why photons, despite possessing no rest mass, are nevertheless subject to gravity.

2.0.5 2.5 Why Gravity Acts on Light

In the previous section, we demonstrated that gravity cannot be screened because the Manifestation Constraint constitutes the existential condition of the Three-Dimensional Manifestation Layer itself. However, another fundamental fact must be explained in any theory claiming the universality of gravity. Photons possess no rest mass, yet they are nevertheless influenced by gravity. In Newtonian mechanics, gravity is defined as a force proportional to mass. Accordingly, massless photons should not experience gravity. General Relativity resolves this problem by asserting that photons do not experience a gravitational force; instead, they propagate along geodesics of curved spacetime, thereby producing gravitational lensing and gravitational redshift. Future Convergence Theory adopts a fundamentally different interpretation. Within the present framework, light is influenced by gravity neither because it possesses energy nor because it possesses mass. Rather, photons are subject to gravity because they are manifested states belonging to the Three-Dimensional Manifestation Layer and therefore share the same Manifestation Constraint as every other manifested physical state.

The Ontological Status of the Photon

As established in Part I, every state belonging to the Three-Dimensional Manifestation Layer

$$\mathcal{X}$$

is generated from the Future Causal Information Field through the Manifestation Projection

$$\mathcal{P}_X.$$

Photons are no exception. Let the photon state be denoted by

$$\gamma \in \mathcal{X}.$$

Within Future Convergence Theory,

$$\boxed{\gamma = \mathcal{P}_X(\mathcal{P}_Q(F_\gamma))}$$

where

$$F_\gamma$$

denotes the informational state corresponding to the photon within the Future Causal Information Field. Accordingly, photons share the Manifestation Constraint in exactly the same manner as every other physical state.

Propagation Governed by the Manifestation Constraint

Within Future Convergence Theory, light is not regarded as a particle freely moving through empty space. Rather, light propagates within the Three-Dimensional Manifestation Structure maintained by the Manifestation Constraint. Consequently, whenever the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

changes, the propagation path of light changes simultaneously. To an observer, this appears as though gravity has bent the path of light. However, within the present theory, no external force acts directly upon the photon. Instead, the propagation path changes because the underlying Manifestation Constraint itself has changed.

Correspondence with General Relativity

In General Relativity, light propagates along geodesics determined by the spacetime metric

$$g_{\mu\nu}.$$

As established in the previous section, Future Convergence Theory interprets the spacetime metric as the geometrical projection of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Accordingly, the geodesic equation is understood, at a more fundamental level, as the geometrical representation of manifestation trajectories determined by the Manifestation Constraint. In other words, the geodesics of General Relativity correspond to the low-energy limit of the Manifestation Constraint in Future Convergence Theory.

Gravitational Lensing

Gravitational lensing is the phenomenon in which the trajectory of light is observed to bend in the vicinity of a massive object. Within Future Convergence Theory, this phenomenon is not interpreted as mass pulling upon light. Rather, the distribution of mass locally modifies the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

thereby deforming the set of physically permissible manifestation trajectories. Consequently, gravitational lensing is interpreted as a direct observation of the spatial distribution of the Manifestation Constraint itself.

Gravitational Redshift

Likewise, gravitational redshift is not interpreted as the consequence of a photon being decelerated by gravity. Within Future Convergence Theory, regions possessing different Manifestation Constraint Densities

$$\Lambda_M$$

correspond to different manifestation structures. As a consequence, differences arise in the observed proper time, frequency, and wavelength. General Relativity describes this phenomenon as gravitational time dilation. Future Convergence Theory instead interprets it as a change in the criteria of manifestation produced by differences in the Manifestation Constraint Density.

Universality of Light

From the Gravity Universality Theorem established in the previous section, the Manifestation Constraint is independent of particle species. Accordingly, photons, electrons, protons, neutrons, neutrinos, and composite particles all share the same Manifestation Constraint. Consequently, light does not obey a special law of gravity distinct from that governing other particles. The reason light is influenced by gravity is therefore not its mass, but the fact that its mode of existence is identical to that of every other manifested physical state.

Observational Consequences

Future Convergence Theory reproduces the General Relativistic limits for ordinary gravitational lensing and gravitational redshift. However, in regimes where higher-order corrections to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

become dominant, the theory predicts small deviations from General Relativity in light-path corrections, polarization rotation, phase delay, and the propagation of ultra-high-energy photons. These corrections will be derived quantitatively in Parts VIII and XIV.

Conclusion of This Section

In this section, we derived the reason that light is influenced by gravity not from mass or energy, but from the universality of the Manifestation Constraint. Consequently, within Future Convergence Theory, the gravitational deflection of light is reinterpreted not as a direct consequence of spacetime curvature itself, but as a phenomenon in which changes in the Manifestation Constraint Structure deform the manifestation trajectory of light. In the next section, we derive the reason gravity is extraordinarily weak compared with the other fundamental interactions from the ontological properties of the Manifestation Constraint.

2.0.6 2.6 Why Gravity Is Weak

In the previous section, we showed that within Future Convergence Theory the reason light is influenced by gravity is not that photons possess mass, but that every state belonging to the Three-Dimensional Manifestation Layer shares the same Manifestation Constraint. Gravity, however, possesses yet another essential characteristic. Compared with the other fundamental interactions, gravity is extraordinarily weak. For example, when the interaction between two protons is considered, gravitational attraction is approximately

$$10^{-36}$$

times weaker than the electromagnetic interaction. To date, General Relativity provides no explanation for this enormous hierarchy. Within General Relativity, the gravitational constant

$$G$$

is introduced as an empirical constant. Consequently, neither the origin of its numerical value nor the reason gravity is so much weaker than the other interactions is explained within the theory itself. Future Convergence Theory derives the weakness of gravity from the fact that the Manifestation Constraint plays a fundamentally different role from the other interactions.

The Difference Between Interactions and Constraints

The electromagnetic, strong, and weak interactions are all local interactions that modify the states of particles that have already become manifest. The Manifestation Constraint, by contrast, is not a mechanism that causes particles to interact. Rather, it is the foundational condition that makes the Three-Dimensional Manifestation Layer itself possible. Accordingly, the Manifestation Constraint does not actively alter particle motion. Instead, it acts as a background constraint extending throughout all space, ensuring that manifested states remain stable and do not collapse.

Dispersion of the Constraint Density

As established in Part I, the Manifestation Constraint does not act only upon individual localized particles. Within Future Convergence Theory, the Manifestation Constraint is distributed throughout the entire Three-Dimensional Manifestation Layer. Consequently, the effective strength of the constraint observed within any local region represents only a small fraction of the total distributed constraint. We refer to this principle as the **Constraint Dispersion Principle**. Let the **Constraint Dispersion Coefficient** be denoted by

$$\eta_C.$$

The locally effective Manifestation Constraint is then expressed as

$$\boxed{\mathcal{C}_{\mu\nu}^{\text{local}} = \eta_C \mathcal{C}_{\mu\nu}}$$

where

$$0 < \eta_C \ll 1.$$

Thus, the gravity observed locally corresponds only to a very small portion of the total Manifestation Constraint.

Reinterpretation of the Gravitational Constant

Accordingly, within Future Convergence Theory, the gravitational constant

$$G$$

is not regarded as a fundamental constant. Instead, it is interpreted as the proportionality constant representing the effective local coupling of the Manifestation Constraint. That is,

$$G = G(\eta_C)$$

In General Relativity, the gravitational constant is an empirically given constant of nature. In Future Convergence Theory, by contrast, it emerges effectively from the spatial dispersion of the Manifestation Constraint.

The Origin of the Weakness of Gravity

The preceding discussion leads to the following theorem.

Theorem 2.3 (Weak Gravitational Coupling Theorem). *The Manifestation Constraint is not a local interaction between particles but a background constraint maintaining the entire Three-Dimensional Manifestation Layer. Consequently, the locally observed constraint strength represents only a very small fraction of the total constraint. Therefore, gravity is observed to be extraordinarily weak compared with the other fundamental interactions.*

Proof. The Manifestation Constraint acts simultaneously upon every manifested state. Accordingly, the constraint is distributed throughout the entire Three-Dimensional Manifestation Layer rather than being concentrated between particular particles. Consequently, the effective constraint strength appearing between localized particles is extremely small compared with the total constraint density. Hence, gravity appears as a weakly coupled interaction in local observations. $\square$

Comparison with General Relativity

General Relativity does not explain the numerical value of the gravitational constant

$$G.$$

Within Future Convergence Theory, the gravitational constant is interpreted as an effective constant representing the fraction of the Manifestation Constraint projected locally. Its numerical value therefore originates from the underlying ontological structure. Accordingly, the weakness of gravity is not regarded as an accidental property of nature but as a direct consequence of the fact that the Manifestation Constraint functions as a background constraint rather than as a localized interaction.

Observational Consequences

Future Convergence Theory reproduces the same effective gravitational constant as General Relativity in ordinary gravitational experiments. However, under conditions involving extremely strong gravitational fields, the very early universe, the vicinity of black holes, or regions where the Manifestation Constraint Density changes rapidly, higher-order corrections to the Constraint Dispersion Coefficient

$$\eta_C$$

may become significant. These corrections will appear observationally in gravitational-wave signals, black-hole quasinormal modes, and cosmological observables, and will be derived quantitatively beginning in Part VII.

Conclusion of This Section

In this section, we derived the reason gravity is extraordinarily weak compared with the other fundamental interactions. The weakness of gravity follows from the fact that the Manifestation Constraint is not a localized interaction but a background constraint maintaining the entire Three-Dimensional Manifestation Layer. Consequently, within Future Convergence Theory, the gravitational constant is reinterpreted not as a fundamental constant but as the effective local coupling of the Manifestation Constraint. In the next section, we introduce the relationship between the Manifestation Constraint and the Double Fixed Point Theory, and formulate how gravitational time dilation emerges within Future Convergence Theory.

2.0.7 2.7 The Relationship Between Gravitational Time Dilation and the Double Fixed Point Theory

In the previous section, we showed that within Future Convergence Theory the weakness of gravity follows from the fact that the Manifestation Constraint is a background constraint maintaining the entire Three-Dimensional Manifestation Layer rather than a localized interaction. Gravity, however, exhibits another phenomenon requiring explanation. This phenomenon is gravitational time dilation. Within General Relativity, gravitational time dilation arises because the temporal component of the spacetime metric

$$g_{\mu\nu}$$

changes in a gravitational field. Future Convergence Theory adopts a different interpretation. Time itself is not regarded as changing. Instead, gravitational time dilation is interpreted as the difference in the rate of manifestation resulting from a change in the synchronization condition between the Future Causal Information Field and the Three-Dimensional Manifestation Layer.

The Double Fixed Point Theory

Part I established that Future Convergence Theory possesses a closed-loop structure converging toward the Future Fixed Point

$$\mathcal{F}^*.$$

However, manifestation within Future Convergence Theory is not determined by a single fixed point alone. The present theory proposes that the stability of existence requires two fixed points. The first fixed point is defined as the point of informational convergence within the Future Causal Information Field,

$$\boxed{\mathcal{F}^*}$$

The second fixed point is defined as the point of manifestation stability within the Three-Dimensional Manifestation Layer,

$$\boxed{\mathcal{X}^*}$$

These two fixed points together constitute the **Double Fixed Point**.

Synchronization Condition

For a manifested state to remain stable within Future Convergence Theory, the fixed point of the Future Causal Information Field and that of the Three-Dimensional Manifestation Layer must remain synchronized. This synchronization condition is defined by

$$\boxed{\mathcal{S}(\mathcal{F}^*, \mathcal{X}^*) = 0}$$

where

$$\mathcal{S}$$

denotes the **Fixed Point Synchronization Function**. When this synchronization condition is satisfied, manifested states remain stable and ordinary temporal evolution proceeds normally.

Origin of Gravitational Time Dilation

Within a gravitational field, the Manifestation Constraint Tensor

$$C_{\mu\nu}$$

increases in magnitude. As the Manifestation Constraint becomes stronger, the synchronization condition between the Future Causal Information Field and the Three-Dimensional Manifestation Layer is correspondingly modified. Consequently,

$$\boxed{\mathcal{S} \neq 0}$$

holds. Within Future Convergence Theory, this synchronization mismatch is observed as gravitational time dilation. Accordingly, time itself does not slow. Rather, the effective update rate at which information from the Future Causal Information Field is manifested within the Three-Dimensional Manifestation Layer changes.

Manifestation Velocity

We define the **Manifestation Velocity** by

$$v_M.$$

This quantity represents the effective rate at which information contained in the Future Causal Information Field is reflected within the Three-Dimensional Manifestation Layer. Within Future Convergence Theory, as the Manifestation Constraint Density

$$\Lambda_M$$

increases, the Manifestation Velocity becomes

$$\boxed{v_M = v_0 f(\Lambda_M)}$$

where

$$f(\Lambda_M)$$

is a monotonically decreasing function satisfying

$$f \rightarrow 1$$

in the weak-gravity limit. Thus, stronger gravitational fields correspond to lower Manifestation Velocities. This reduction constitutes the fundamental origin of gravitational time dilation within Future Convergence Theory.

Correspondence with General Relativity

In General Relativity, gravitational time dilation is represented by variations in the temporal component of the spacetime metric,

$$g_{00}.$$

Within Future Convergence Theory, however, the spacetime metric is interpreted as the geometrical projection of the Manifestation Constraint Tensor. Accordingly, gravitational time dilation is understood, at a more fundamental level, as a reduction in the Manifestation Update Rate caused by changes in the synchronization between the two fixed points. General Relativity therefore corresponds to the low-energy geometric description of this more fundamental change in Manifestation Velocity.

The Black-Hole Limit

In the vicinity of a black hole, the Manifestation Constraint Density

$$\Lambda_M$$

approaches its maximum value. Consequently, the Manifestation Velocity decreases toward an extreme limit. Within Future Convergence Theory, the apparent freezing of time near the event horizon is not interpreted as time itself coming to a halt. Rather, it is understood as the limiting slowdown of manifestation updates caused by the increasingly delayed synchronization between the Future Causal Information Field and the Three-Dimensional Manifestation Layer.

Observational Consequences

In ordinary gravitational fields, corrections arising from fixed-point synchronization are extremely small. Accordingly, Future Convergence Theory reproduces the same gravitational time dilation predicted by General Relativity. However, in the vicinity of black holes, during the very early universe, and in regions of extremely large spacetime curvature, higher-order corrections to the fixed-point synchronization become significant. These corrections are predicted to produce observable deviations in black-hole ringdown signals, gravitational-wave phase evolution, and black-hole information recovery times. These predictions will be derived quantitatively in Parts VII, IX, and XIII.

Conclusion of This Section

In this section, gravitational time dilation was redefined as a consequence of changes in the synchronization condition between the Future Causal Information Field and the Three-Dimensional Manifestation Layer within the framework of the Double Fixed Point Theory. Consequently, gravitational time dilation is interpreted not as an intrinsic property of time itself, but as a change in the Manifestation Update Rate. In the next section, we demonstrate that the Manifestation Constraint Theory reduces to General Relativity in the weak-constraint limit and derive the General Relativistic limit of Future Convergence Theory.

2.0.8 2.8 The General Relativistic Limit

In the previous section, we demonstrated that gravitational time dilation in Future Convergence Theory is derived from the synchronization condition between the two fixed points associated with the Future Causal Information Field and the Three-Dimensional Manifestation Layer. However, the introduction of new concepts alone is insufficient for a gravitational theory to qualify as a physical theory. General Relativity has been verified to extremely high precision over a broad range of observations, including the Solar System, binary pulsars, gravitational lensing, gravitational waves, and black-hole shadows. Accordingly, Future Convergence Theory is not intended to replace General Relativity outright. Rather, it must reproduce General Relativity in the appropriate physical limit. In this section, we derive the conditions under which the Manifestation Constraint Theory reduces to General Relativity.

The Weak-Constraint Limit

The Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

is the constraint field derived from the Future Causal Information Field. In sufficiently weak gravitational fields, the Manifestation Constraint Density

$$\Lambda_M$$

is small, and higher-order corrections arising from the Future Causal Information Field become negligible. We define this regime as the **Weak-Constraint Limit**, given by

$$\boxed{\Lambda_M \rightarrow 0}$$

In this limit, only the leading-order contribution from the Future Causal Information Field remains.

Projection of the Constraint Tensor

As introduced in Section 2.2, the Manifestation Constraint Tensor is the more fundamental quantity from which the Einstein tensor emerges. Accordingly, in the Weak-Constraint Limit, the Manifestation Constraint Tensor is projected into spacetime geometry. That is,

$$\boxed{\Pi(\mathcal{C}_{\mu\nu}) = G_{\mu\nu}}$$

where

$$\Pi$$

denotes the projection operator mapping the Manifestation Constraint onto the geometry of four-dimensional spacetime.

Limit of the Field Equations

The complete field equations of Future Convergence Theory will be derived in Part XIII. However, in the Weak-Constraint Limit, the correction terms associated with the Future Causal Information Field vanish, and the field equations reduce to

$$\boxed{G_{\mu\nu} = 8\pi G T_{\mu\nu}}$$

which is precisely the Einstein field equation of General Relativity. Accordingly, General Relativity is identified as the low-energy, weak-constraint limit of Future Convergence Theory.

The Newtonian Limit

In the additional limit of weak gravity, low velocities, and static fields, the spacetime metric becomes

$$g_{00} = - \left(1 + \frac{2\Phi}{c^2} \right).$$

In this limit, Future Convergence Theory reproduces the Poisson equation,

$$\boxed{\nabla^2 \Phi = 4\pi G \rho}$$

Thus, Newtonian gravity is likewise recovered as an approximation to Future Convergence Theory.

Correspondence with the Equivalence Principle

As demonstrated in Section 2.3, gravitational universality follows directly from the Manifestation Constraint. In the Weak-Constraint Limit, the higher-order structure of the Manifestation Constraint disappears. Consequently, the Equivalence Principle becomes fully identical to that of General Relativity. Future Convergence Theory is therefore consistent with all existing precision tests of the Equivalence Principle.

Correspondence of Time Dilation

For the Manifestation Velocity

$$v_M,$$

introduced in Section 2.7, the Weak-Constraint Limit yields

$$f(\Lambda_M) = 1 + \mathcal{O}(\Lambda_M).$$

Accordingly, gravitational time dilation becomes identical to that derived from the Schwarzschild solution of General Relativity.

Deviations from General Relativity

Future Convergence Theory differs from General Relativity only within high-constraint regimes. These include, in particular,

- the vicinity of black holes,
- black-hole ringdown following mergers,
- the extremely early universe,
- galactic scales dominated by dark matter,
- and cosmological scales dominated by dark energy.

Within these regimes, higher-order corrections originating from the Manifestation Constraint Tensor become significant and produce observable deviations from General Relativity.

Correspondence Principle

Future Convergence Theory therefore satisfies a correspondence principle analogous to Bohr's Correspondence Principle. Specifically,

$$\boxed{\lim_{\Lambda_M \rightarrow 0} \text{FCT} = \text{General Relativity}}$$

Hence, Future Convergence Theory is not a theory that rejects General Relativity, but rather a more fundamental theory of which General Relativity constitutes an appropriate limiting case.

Conclusion of This Section

In this section, we demonstrated that Future Convergence Theory reduces to General Relativity in the Weak-Constraint Limit and further reduces to Newtonian gravity in the weak-gravity limit. Accordingly, Future Convergence Theory remains consistent with every observational domain in which General Relativity has thus far been verified, while predicting genuinely new physics only in high-constraint regimes. In the next section, we summarize the quantitative deviations predicted by Future Convergence Theory relative to General Relativity and systematically identify the observables through which the theory may be falsified.

2.0.9 2.9 Falsifiable Predictions

In the previous section, we demonstrated that Future Convergence Theory reduces to General Relativity in the Weak-Constraint Limit. Accordingly, within the observational regimes where General Relativity has already been confirmed to high precision, Future Convergence Theory necessarily reproduces the same predictions. However, reproducing General Relativity alone is insufficient for establishing a new physical theory. A successful theory must also provide predictions that are distinguishable from those of General Relativity and that can, in principle, be supported or refuted by observation. In this section, we summarize the observables for which Future Convergence Theory predicts deviations from General Relativity and define the falsification criteria for the theory as a whole.

Principle of Falsifiability

Throughout this work, the value of a theory is not judged by the novelty of its conceptual framework. Future Convergence Theory is regarded as successful only if it predicts quantitative observational deviations that cannot be explained by General Relativity. Accordingly, the primary subject of experimental verification is not the existence of the fields themselves but the differences between observable quantities. Let the prediction of General Relativity be denoted by

$$\mathcal{O}_{\text{GR}},$$

and that of Future Convergence Theory by

$$\mathcal{O}_{\text{FCT}}.$$

The observable quantity predicted by the present theory is therefore

$$\boxed{\Delta\mathcal{O} = \mathcal{O}_{\text{FCT}} - \mathcal{O}_{\text{GR}}}$$

The experimental verification of Future Convergence Theory is thus reduced to the measurement of

$$\Delta\mathcal{O}.$$

Black-Hole Quasinormal Modes

The principal observational target of this work is the spectrum of black-hole quasinormal modes. Within General Relativity, the ringdown frequencies are determined solely by the black-hole mass, angular momentum, and electric charge. Future Convergence Theory predicts additional corrections arising from the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

and from the Future Causal Information Field. Accordingly,

$$\boxed{\omega_{\ell mn}^{\text{FCT}} = \omega_{\ell mn}^{\text{GR}} + \Delta\omega_{\ell mn}}$$

The quantity

$$\Delta\omega_{\ell mn}$$

constitutes the primary observational prediction of Future Convergence Theory.

Gravitational-Wave Waveforms

During black-hole mergers, corrections are expected not only in the ringdown frequencies but also in the gravitational-wave phase evolution. Future Convergence Theory predicts that higher-order corrections to the Manifestation Constraint Tensor modify the waveform according to

$$h_{\text{FCT}} = h_{\text{GR}} + \Delta h$$

Future high-precision gravitational-wave observations should therefore be capable of testing the predicted correction

$$\Delta h.$$

Gravitational Time Dilation

As introduced in Section 2.7 through the Manifestation Velocity

$$v_{\text{M}},$$

Future Convergence Theory predicts that additional corrections to gravitational time dilation arise in high-constraint regimes. These corrections may become observable near black holes or in future ultra-high-precision atomic clock experiments.

Gravitational Lensing

If the Manifestation Constraint Tensor possesses higher-order corrections beyond those represented by General Relativity, corresponding corrections should also appear in gravitational lensing. Accordingly, the present theory predicts that the deflection angle

$$\theta$$

satisfies

$$\theta_{\text{FCT}} = \theta_{\text{GR}} + \Delta\theta$$

where

$$\Delta\theta$$

represents the measurable deviation from the General Relativistic prediction.

Black-Hole Information Recovery

As will be derived in Part IX, the information recovery time, the Page curve, and the scrambling time all receive corrections determined by the Manifestation Constraint Density. These quantities constitute essential tests of whether Future Convergence Theory functions not only as a gravitational theory but also as a viable theory of quantum gravity.

Cosmology

Within Future Convergence Theory, dark matter, dark energy, and the early universe arise as distinct boundary conditions of the Future Causal Information Field. Consequently, deviations from General Relativity and from the standard cosmological model are predicted for galaxy rotation curves, gravitational lensing, the cosmic microwave background, the stochastic gravitational-wave background, and the tensor-to-scalar ratio. These predictions will be derived quantitatively in Parts VI, X, and XI.

Falsification Criteria for the Theory

Future Convergence Theory is regarded as falsified if any one of the following conditions is satisfied.

1. Future Convergence Theory fails to produce quantitative observational deviations from General Relativity.
2. The derived deviations are mutually inconsistent.
3. The predicted corrections to black-hole quasinormal modes, gravitational waves, cosmological observables, or information recovery remain completely undetected even after observational precision becomes sufficient to test the theory.
4. The introduction of the Future Convergence Theory corrections does not improve agreement with observations beyond that achieved by General Relativity.

If any of these conditions is fulfilled, Future Convergence Theory cannot be regarded as a valid physical theory.

Conclusion of Part II

In this chapter, gravity has been redefined not as an interaction between particles but as the Manifestation Constraint that preserves the consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. From this perspective, the universality of gravity, its non-screenability, its action upon light, its apparent weakness, and gravitational time dilation were all derived from the ontological properties of the Manifestation Constraint. Furthermore, we demonstrated that General Relativity is recovered as the Weak-Constraint Limit of Future Convergence Theory and organized the falsifiable observational predictions that distinguish the two theories. Accordingly, the fundamental structure of the gravitational theory within Future Convergence Theory has now been established. In the next chapter, we demonstrate mathematically that quantum theory is not an independent final theory but necessarily constitutes the intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer.

Part III

Why Quantum Theory Must Be an Intermediate Layer

Chapter 3

Why Quantum Theory Must Be an Intermediate Layer

In Part I, we defined the minimal axiomatic system constituting Future Convergence Theory, namely, the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the FCT Closure Equation. In Part II, gravity was redefined not as a fundamental interaction acting between particles, but as the **Manifestation Constraint** that maintains consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. As a result, the universality of gravity, its non-screenability, its action on light, gravitational time dilation, and even the General Relativistic limit were shown to follow from the axiomatic system of Future Convergence Theory. However, one fundamental problem remains. In Part I, we introduced the Quantum

$\mathcal{Q}$

between the Future Causal Information Field

$\mathcal{F}$

and the Three-Dimensional Manifestation Layer

$\mathcal{X}$.

This introduction was given axiomatically. It has not yet been proven why such a layer must necessarily exist. If the Three-Dimensional Manifestation Layer were generated directly from the Future Causal Information Field, quantum theory would become unnecessary. Conversely, if quantum theory were the final layer of physics, then the ontological foundation provided by the Future Causal Information Field would become unnecessary. Therefore, in order for Future Convergence Theory to constitute a self-contained theory, it must be shown mathematically that quantum theory is not merely a layer that may exist, but a layer that must exist as an intermediate structure. This chapter addresses that problem. First, we prove that the Future Causal Information Field and the Three-Dimensional Manifestation Layer are ontologically distinct layers. We then show the contradictions that arise if the two are connected directly, and prove that the Quantum Permission Layer must necessarily be introduced in order to avoid those contradictions. Furthermore, we reformulate the wave function, superposition, the measurement problem, decoherence, vacuum fluctuations, the Casimir effect, delayed-choice experiments, and the Quantum Eraser from the standpoint of Future Convergence Theory. Finally, we show that only when quantum theory functions as the intermediate layer of Future Convergence Theory does it remain consistent with existing quantum mechanics while simultaneously enabling unification with General Relativity.

Part III

Why Quantum Theory Must Be an Intermediate Layer

3.0.1 3.1 The Layer Non-Identity Axiom

In Part I, the ontological structure of Future Convergence Theory was defined as a four-layer architecture consisting of the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, and the Three-Dimensional Manifestation Layer. However, this four-layer architecture is not merely a conceptual classification. Each layer plays a distinct mathematical role, and no two layers can be identified with one another under any limit. Future Convergence Theory refers to this property as **Layer Non-Identity**. First, the Future Causal Information Field

$$\mathcal{F}$$

is an ontological information field that contains all informational states capable of manifestation. By contrast, the Three-Dimensional Manifestation Layer

$$\mathcal{X}$$

consists only of physical states that are actually observed. Therefore, these two layers do not coincide either in informational content or in ontological status. Furthermore, the Quantum Permission Layer

$$\mathcal{Q}$$

is neither a subset of the Future Causal Information Field nor a subset of the Three-Dimensional Manifestation Layer. The Quantum Permission Layer is the constraint layer that selects, from the Future Causal Information Field, only those states that are permitted to manifest, and maps them toward the Three-Dimensional Manifestation Layer. Accordingly, within Future Convergence Theory, quantum theory is neither information itself nor the observational result itself. Quantum theory is an intermediate constraint structure required to connect two ontologically distinct layers. On this basis, Future Convergence Theory adopts the following fundamental axiom.

Axiom 3.1 (Layer Non-Identity Axiom). *The Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, and the Three-Dimensional Manifestation Layer are mutually ontologically independent. They cannot be identified with one another through any mapping, limit, or approximation. Accordingly, the Quantum Permission Layer is neither the Future Causal Information Field nor the Three-Dimensional Manifestation Layer, but exists as the unique intermediate layer connecting the two.*

This axiom constitutes the starting point of the present chapter. In the next section, we redefine the quantum state itself from the standpoint of Future Convergence Theory and clarify what the wave function mathematically represents.

3.0.2 3.2 The FCT Definition of the Quantum State

In the previous section, we showed that within Future Convergence Theory the Future Causal Information Field, the Quantum Permission Layer, and the Three-Dimensional Manifestation Layer are mutually ontologically independent, and that quantum theory is positioned as the unique intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer. However, positioning quantum theory as an intermediate layer is not sufficient. It is also necessary to define precisely what the wave function, or the quantum state itself, the most fundamental object of present quantum mechanics, means within Future Convergence Theory. In this section, we redefine the quantum state not as an ontological object but as a mathematical object representing a manifestation-permitted state.

Quantum States in Standard Quantum Theory

In ordinary quantum mechanics, a quantum state is defined as a state vector

$$|\psi\rangle$$

belonging to a Hilbert space

$$\mathcal{H}.$$

The time evolution of the state is described by the Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle.$$

However, this definition alone does not determine whether

$$|\psi\rangle$$

is a physical reality, a probability amplitude, a representation of knowledge, or a pre-measurement state. This ambiguity gives rise to the foundational problems of quantum theory, including the measurement problem.

Quantum States in Future Convergence Theory

In Future Convergence Theory, a quantum state is not the particle itself. Nor is it the observational result itself. A quantum state is a **permitted state** that specifies whether an informational state existing in the Future Causal Information Field is eligible to be projected into the Three-Dimensional Manifestation Layer. Accordingly, the quantum state is defined as a mathematical object belonging to the Quantum Permission Layer

$$\mathcal{Q}.$$

$$\boxed{|\psi\rangle \in \mathcal{Q}}$$

Here,

$$|\psi\rangle$$

is a permission vector that represents information from the Future Causal Information Field as a manifestable state.

Correspondence with Future Information

Let an informational state belonging to the Future Causal Information Field be

$$F \in \mathcal{F}.$$

Using the Quantum Permission Operator defined in Part I,

$$\mathcal{P}_Q : \mathcal{F} \rightarrow \mathcal{Q},$$

the quantum state is given by

$$\boxed{|\psi\rangle = \mathcal{P}_Q(F)}$$

Accordingly, the wave function is not the Future Causal Information Field itself, but a manifestation-candidate state generated from the Future Causal Information Field.

Meaning of the Quantum State

Within Future Convergence Theory, the wave function is not a function that describes the position of a particle in an indeterminate way. Nor is it a probability distribution itself. Rather, the wave function is a **permission condition** that specifies which informational states contained in the Future Causal Information Field are permitted to be projected into the Three-Dimensional Manifestation Layer. Thus, the quantum state is not existence itself, but a mathematical object that describes manifestability.

Superposition States

Quantum superposition is likewise not interpreted in Future Convergence Theory as multiple states existing physically at the same time. Within the Quantum Permission Layer, several manifestation candidates may be permitted simultaneously for a single Future Causal Information state. Accordingly, a superposition state may be written as

$$|\psi\rangle = \sum_i c_i |\psi_i\rangle$$

However, each

$$|\psi_i\rangle$$

does not represent a simultaneously existing particle. Rather, the states represent a set of manifestation possibilities.

Reinterpretation of Probability Amplitudes

In standard quantum theory, Born's rule interprets

$$|c_i|^2$$

as the probability of observing the state

$$|\psi_i\rangle.$$

In Future Convergence Theory, this quantity is not a probability in the fundamental sense. Instead,

$$|c_i|^2$$

represents the **Manifestation Permission Weight**, namely the degree to which the corresponding manifestation state is selected from the Future Causal Information Field. Accordingly, Born's rule is reinterpreted not as a fundamental probability law, but as the statistical expression of the Manifestation Constraint.

Third Theorem

The preceding discussion gives the following theorem.

Theorem 3.1 (Quantum State Theorem). *The quantum state*

$$|\psi\rangle$$

is not the particle itself. Rather, it is the manifestation-permitted state by which an informational state contained in the Future Causal Information Field becomes eligible for projection into the Three-Dimensional Manifestation Layer.

Conclusion of This Section

In this section, the quantum state was redefined within Future Convergence Theory as a manifestation-permitted state generated from the Future Causal Information Field. Consequently, the wave function has been positioned not as existence itself, but as a mathematical object representing manifestability. In the next section, we use this definition to reformulate the measurement problem, the central unresolved problem of quantum mechanics, from the standpoint of Future Convergence Theory.

3.0.3 3.3 Reformulation of the Measurement Problem

In the previous section, we showed that within Future Convergence Theory the quantum state

$$|\psi\rangle$$

is a manifestation-permitted state generated from the Future Causal Information Field, rather than the particle itself or the observational result itself. Once this definition is adopted, the measurement problem—the most fundamental unsolved problem in quantum mechanics—is reformulated in a fundamentally different way.

The Measurement Problem in Standard Quantum Theory

In conventional quantum mechanics, a quantum state evolves continuously and unitarily according to the Schrödinger equation. However, at the instant a measurement is performed, the wave function is assumed to collapse into a single eigenstate. That is,

$$\boxed{|\psi\rangle = \sum_i c_i |\psi_i\rangle \longrightarrow |\psi_k\rangle}$$

is postulated. However, this collapse process cannot be derived from the Schrödinger equation itself. Consequently, the questions of what constitutes a measurement, when collapse occurs, and whether an observer is fundamentally required remain unresolved.

The Standpoint of Future Convergence Theory

Within Future Convergence Theory, the wave function is a manifestation-permitted state. Accordingly, this theory does not adopt the view that the wave function itself changes as a consequence of measurement. Instead, measurement is defined as the process in which one manifestation candidate contained in the Quantum Permission Layer

$$\mathcal{Q}$$

is mapped into the Three-Dimensional Manifestation Layer through the Manifestation Projection

$$\mathcal{P}_X.$$

That is,

$$\boxed{\mathcal{P}_X : \mathcal{Q} \rightarrow \mathcal{X}}$$

is executed.

The Wave Function Does Not Collapse

From this standpoint, the wave function is never destroyed by measurement. Neither the informational state

$$F$$

contained in the Future Causal Information Field nor the Quantum Permission Layer

$$\mathcal{Q}$$

is lost through observation. Only the state that becomes physically manifested within the Three-Dimensional Manifestation Layer changes. Accordingly, Future Convergence Theory replaces

$$\boxed{|\psi\rangle \nrightarrow |\psi_k\rangle}$$

with

$$\boxed{\mathcal{P}_X(|\psi\rangle) = x_k}$$

where

$$x_k \in \mathcal{X}$$

is the manifested state that is actually observed.

The Ontological Meaning of Measurement

Within Future Convergence Theory, measurement is neither an operation that creates information nor one that destroys information. Measurement is the event in which one manifestation-permitted state contained within the Future Causal Information Field is projected into the Three-Dimensional Manifestation Layer. Consequently, measurement is not the generation of existence but the determination of manifestation.

The Role of the Observer

Under this definition, the observer is not the entity that collapses the wave function. The observer is itself a manifested state belonging to the Three-Dimensional Manifestation Layer

$$\mathcal{X}.$$

The observer merely records the outcome of the Manifestation Projection. Accordingly, Future Convergence Theory does not require the observer to be introduced as a special physical entity.

Measurement Theorem

The preceding discussion yields the following theorem.

Theorem 3.2 (Measurement Theorem). *Measurement is the process by which a manifestation-permitted state contained within the Quantum Permission Layer is projected into the Three-Dimensional Manifestation Layer through the Manifestation Projection*

$$\mathcal{P}_X.$$

Accordingly, the wave function itself does not collapse during measurement.

Proof. From the previous section, the quantum state is a manifestation-permitted state generated from the Future Causal Information Field. Furthermore, Part I established that the Three-Dimensional Manifestation Layer is generated through the Manifestation Projection

$$\mathcal{P}_X.$$

Therefore, the object that changes during measurement is not the quantum state itself but the output of the Manifestation Projection. Hence, measurement is understood as the determination of the manifested state, rather than the collapse of the wave function. $\square$

Comparison with the Standard Interpretation

In standard quantum theory, measurement is introduced as an operation external to the formalism. Within Future Convergence Theory, measurement is defined ontologically as one stage of the Future Convergence Loop. Accordingly, the measurement problem is reformulated not as a problem of quantum theory itself but as a problem concerning the Manifestation Projection.

Conclusion of This Section

In this section, the measurement problem has been reformulated as the determination of a manifested state through the Manifestation Projection, rather than as wave-function collapse. This formulation explains the measurement process consistently within the axiomatic system of Future Convergence Theory without introducing the observer as a privileged physical entity. The next section redefines decoherence from the standpoint of Future Convergence Theory on the basis of this measurement process.

3.0.4 3.4 Redefinition of Decoherence

In the previous section, we showed that within Future Convergence Theory, measurement is not the collapse of the wave function but the determination of a manifested state through the Manifestation Projection

$$\mathcal{P}_X.$$

Consequently, the measurement problem is reformulated as a problem concerning the manifestation process rather than the dynamics of the wave function. However, a phenomenon closely related to measurement is decoherence. In contemporary quantum theory, decoherence is understood as the rapid loss of quantum interference caused by the interaction between a quantum system and its environment. This interpretation has been remarkably successful in explaining the classical limit. Nevertheless, decoherence alone does not explain why only a single observational outcome ultimately appears. Future Convergence Theory therefore redefines decoherence not as the collapse of a quantum state but as the reduction of manifestation candidates.

Decoherence in Standard Quantum Theory

In ordinary quantum mechanics, a superposition consisting of multiple quantum states

$$|\psi_i\rangle$$

interacts with the environment so that the off-diagonal elements of the density operator

$$\rho$$

vanish. That is,

$$\boxed{\rho_{ij} \rightarrow 0 \quad (i \neq j)}$$

As a consequence, quantum interference disappears and the system approaches a classical statistical mixture.

The Standpoint of Future Convergence Theory

Within Future Convergence Theory, the quantum state is a manifestation-permitted state. Therefore, decoherence does not destroy the wave function itself. Rather, what disappears is the consistency among multiple manifestation candidates. All information continues to be preserved within the Future Causal Information Field. However, within the Quantum Permission Layer, changes in the consistency conditions imposed by the Manifestation Constraint gradually reduce the set of permitted manifestation candidates.

Manifestation Reduction

Future Convergence Theory defines decoherence as **Manifestation Reduction**. The Manifestation Reduction Operator is introduced as

$$\boxed{\mathcal{D} : \mathcal{Q} \rightarrow \mathcal{Q}}$$

where

$$\mathcal{D}$$

reduces the set of manifestation candidates contained within the Quantum Permission Layer according to environmental consistency conditions. Accordingly, decoherence is not an operation that destroys quantum states but one that gradually contracts the set of manifestation candidates.

Manifestation Candidate Sets

For a given quantum state, define the manifestation candidate set as

$$\boxed{\mathcal{A} = \{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle\}}$$

As decoherence proceeds, the Manifestation Reduction Operator produces

$$\boxed{\mathcal{D}(\mathcal{A}) = \mathcal{A}'}$$

where

$$\mathcal{A}' \subset \mathcal{A}.$$

Thus, the set of manifestable states decreases monotonically during time evolution.

Difference Between Decoherence and Measurement

Within Future Convergence Theory, decoherence and measurement are clearly distinguished. During decoherence, the manifestation candidate set shrinks while multiple candidates may still remain. During measurement, however, the Manifestation Projection

$$\mathcal{P}_X$$

acts to project only one candidate from the remaining set into the Three-Dimensional Manifestation Layer. Accordingly, decoherence constitutes the stage preceding measurement rather than measurement itself.

Fourth Theorem

The preceding discussion gives the following theorem.

Theorem 3.3 (Decoherence Theorem). *Decoherence is the process by which the manifestation candidate set contained within the Quantum Permission Layer is reduced according to environmental consistency conditions. Accordingly, decoherence never destroys the information contained within the Future Causal Information Field.*

Proof. Part I established that the Future Causal Information Field is the ontological information field containing all information. The quantum state is generated from that information as a manifestation-permitted state. Therefore, environmental interactions modify only the manifestation candidate set contained within the Quantum Permission Layer, while the Future Causal Information Field itself remains unchanged. Hence, decoherence is interpreted as the reduction of manifestation candidates rather than information loss. $\square$

Comparison with the Standard Theory

In standard quantum theory, decoherence is understood as an environmental effect that suppresses quantum interference. Future Convergence Theory preserves this description while additionally giving decoherence an ontological interpretation as the reduction of manifestation candidate sets. Accordingly, decoherence alone does not determine the observational outcome. Final manifestation occurs only when the Manifestation Projection

$$\mathcal{P}_X$$

acts.

Conclusion of This Section

In this section, decoherence has been redefined not as the collapse of quantum states but as the reduction of manifestation candidate sets according to environmental consistency conditions. This reformulation establishes decoherence and measurement as distinct processes with different roles within Future Convergence Theory. The next section redefines the quantum vacuum from the standpoint of Future Convergence Theory and derives the origin of vacuum fluctuations together with their correction terms.

3.0.5 3.5 Vacuum Fluctuation Corrections

In the previous section, we showed that within Future Convergence Theory, decoherence is not the collapse of quantum states but the reduction of manifestation candidate sets according to environmental consistency conditions. However, quantum theory contains another important phenomenon in which observable physical effects remain even when no observable particles are present. This phenomenon is known as the **Vacuum Fluctuation**. In standard quantum field theory, the vacuum is not defined as a state containing zero particles, but rather as the ground state possessing zero-point energy. Accordingly, even in apparently empty space, virtual particle pairs are assumed to be continuously created and annihilated. Future Convergence Theory adopts a different standpoint. The vacuum is not defined as a region without particles. Instead, the vacuum is defined as a region in which only manifestation-permitted states exist that have not yet been projected into the Three-Dimensional Manifestation Layer from the Future Causal Information Field.

Definition of the Vacuum in Future Convergence Theory

As established in Part I, the Future Causal Information Field

$$\mathcal{F}$$

contains every informational state that is capable of manifestation. Only a subset of these states passes through the Quantum Permission Layer

$$\mathcal{Q}$$

and ultimately manifests within the Three-Dimensional Manifestation Layer

$$\mathcal{X}.$$

Accordingly, within Future Convergence Theory, the vacuum is defined as the region containing manifestation-permitted states that have not yet been projected through the Manifestation Projection

$$\mathcal{P}_X.$$

This region is called the **Vacuum Information State** and is defined by

$$\boxed{\mathcal{V} \subset \mathcal{Q}}$$

Origin of Vacuum Fluctuations

Within Future Convergence Theory, vacuum fluctuations do not imply the actual creation and annihilation of virtual particles. Rather, vacuum fluctuations arise because multiple manifestation candidates contained within the Future Causal Information Field continuously vary in accordance with the consistency conditions imposed by the Manifestation Constraint. Accordingly, the manifestation candidate set contained within the Vacuum Information State

$$\mathcal{V}$$

evolves as

$$\mathcal{V}(t) = \mathcal{V}_0 + \delta\mathcal{V}(t)$$

where

$$\delta\mathcal{V}(t)$$

denotes the small fluctuation produced by changes in synchronization with the Future Causal Information Field.

Relation to the Manifestation Constraint

Vacuum fluctuations are not independent of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Whenever the Manifestation Constraint Density

$$\Lambda_M$$

changes, the set of manifestation candidates contained within the Quantum Permission Layer also changes. Consequently,

$$\delta\mathcal{V} = \delta\mathcal{V}(\Lambda_M)$$

This implies that the structure of vacuum fluctuations changes as the gravitational field becomes stronger.

Redefinition of Vacuum Energy

Standard quantum field theory defines vacuum energy as the sum of all zero-point oscillations. Future Convergence Theory instead defines vacuum energy as the **Manifestation Surplus Energy** stored within manifestation-permitted states that have not yet become manifest. Let the vacuum-energy density be

$$\rho_{\text{vac}}^{\text{FCT}}.$$

Then

$$\rho_{\text{vac}}^{\text{FCT}} = \rho_{\text{vac}}^{\text{QFT}} + \Delta\rho_{\text{FCT}}$$

where

$$\Delta\rho_{\text{FCT}}$$

is the correction arising from the Future Causal Information Field and the Manifestation Constraint Tensor.

Fifth Theorem

The preceding discussion yields the following theorem.

Theorem 3.4 (Vacuum Fluctuation Theorem). *Vacuum fluctuations do not represent the creation and annihilation of virtual particles. Rather, they are continuous variations in the manifestation candidate set contained within the Future Causal Information Field according to the consistency conditions imposed by the Manifestation Constraint.*

Proof. Within Future Convergence Theory, particles exist only as manifested states in the Three-Dimensional Manifestation Layer. Therefore, there is no need to assume the creation or annihilation of particles in the vacuum. On the other hand, the Quantum Permission Layer always contains manifestation-permitted candidate states. These candidate sets vary according to synchronization with the Future Causal Information Field and changes in the Manifestation Constraint Density. Hence, vacuum fluctuations are interpreted as the dynamical evolution of manifestation candidate sets. $\square$

Observational Consequences

Under ordinary experimental conditions, Future Convergence Theory reproduces the same vacuum-fluctuation limit as standard quantum field theory. However, near black holes, during the very early universe, and within regions of extremely high curvature, the correction

$$\Delta\rho_{\text{FCT}}$$

can no longer be neglected. Accordingly, the Casimir effect, Hawking radiation, the cosmological constant, and dark energy acquire small corrections beyond conventional quantum field theory. These corrections will be derived quantitatively in Parts V, IX, and XI.

Conclusion of This Section

In this section, vacuum fluctuations have been redefined not as the creation and annihilation of virtual particles but as the dynamical evolution of manifestation candidate sets contained within the Future Causal Information Field. Consequently, the vacuum is interpreted not as empty space but as an ontological region containing informational states prior to manifestation. The next section derives the Future Convergence Theory correction to the Casimir effect on the basis of this reinterpretation of vacuum fluctuations.

3.0.6 3.6 Corrections to the Casimir Effect

In the previous section, we showed that within Future Convergence Theory, vacuum fluctuations do not arise from the creation and annihilation of virtual particle pairs but from variations in manifestation candidate sets according to the consistency conditions imposed by the Manifestation Constraint. This reinterpretation also provides a new understanding of the Casimir effect, one of the most representative phenomena originating from the quantum vacuum. Standard quantum field theory explains the Casimir effect as arising from the difference in zero-point energy caused by the restriction of allowed vacuum modes between two conducting plates. This explanation has been confirmed experimentally with high precision. Future Convergence Theory accepts the experimental validity of the effect while assigning it a different ontological origin.

Manifestation Permission Boundary

Consider two conducting plates separated by a distance

$$a.$$

In Future Convergence Theory, the conducting plates do more than impose electromagnetic boundary conditions. They also modify the manifestation conditions within the Quantum Permission Layer. Consequently, the manifestation candidate set

$$\mathcal{A}$$

between the plates differs from that outside the plates. This boundary is called the **Manifestation Permission Boundary**.

Manifestation Candidate Density

Define the Manifestation Candidate Density by

$$\mu_M,$$

which denotes the density of manifestation-permitted states per unit volume. Outside the conducting plates,

$$\mu_M = \mu_0.$$

Inside the plates, however,

$$\mu_M = \mu_0 - \delta\mu$$

where

$$\delta\mu > 0$$

represents the reduction in manifestation candidates produced by the boundary conditions.

Manifestation Constraint Energy

Within Future Convergence Theory, the origin of the Casimir effect is not the zero-point energy itself. Changes in the manifestation candidate density modify the local Manifestation Constraint Density

$$\Lambda_M.$$

Accordingly, define the Manifestation Constraint Energy

$$E_M$$

by

$$E_M = E_{\text{vac}} + \Delta E_{\text{FCT}}$$

where

$$\Delta E_{\text{FCT}} = f(\delta\mu, \Lambda_M)$$

is the correction unique to Future Convergence Theory.

Casimir Force

The Casimir force is defined as the derivative of the Manifestation Constraint Energy with respect to the plate separation. Thus,

$$F_{\text{FCT}} = -\frac{\partial E_M}{\partial a}$$

which may be written as

$$F_{\text{FCT}} = F_{\text{Casimir}} + \Delta F.$$

Here,

$$F_{\text{Casimir}}$$

is the standard Casimir force predicted by quantum field theory, while

$$\Delta F$$

is the correction produced by the Manifestation Constraint.

Weak-Constraint Limit

Under ordinary laboratory conditions, the Manifestation Constraint Density is extremely small. Therefore,

$$\Lambda_M \rightarrow 0$$

implies

$$\boxed{\Delta F \rightarrow 0}$$

and Future Convergence Theory exactly reproduces the standard Casimir effect.

Corrections in Strong-Constraint Regions

Near black holes, in the early universe, or within regions of extremely high curvature, the Manifestation Constraint Density increases substantially. Future Convergence Theory therefore predicts higher-order corrections to the Casimir force. In general,

$$\boxed{\Delta F \propto \Lambda_M + \mathcal{O}(\Lambda_M^2)}$$

Hence, under extreme conditions, the observed Casimir force may deviate slightly from the prediction of standard quantum field theory.

Sixth Theorem

The preceding discussion yields the following theorem.

Theorem 3.5 (Casimir Correction Theorem). *The Casimir effect arises from changes in the boundary conditions imposed on the manifestation candidate set, thereby producing a difference in Manifestation Constraint Energy. In the weak-constraint limit, Future Convergence Theory reproduces the standard Casimir effect exactly. In strong-constraint regions, corrections proportional to the Manifestation Constraint Density appear.*

Observational Consequences

Under ordinary Casimir experiments, Future Convergence Theory is experimentally indistinguishable from standard quantum field theory. However, under extreme gravitational fields, ultra-high-precision nanoscale Casimir measurements, or cosmological boundary conditions, the correction

$$\Delta F$$

may become measurable. Furthermore, this correction shares a common origin with Hawking radiation, black-hole evaporation, and the cosmological constant problem.

Conclusion of This Section

In this section, the Casimir effect has been redefined not as a consequence of differences in zero-point energy but as the result of changes in Manifestation Constraint Energy caused by boundary conditions imposed on the manifestation candidate set. Consequently, Future Convergence Theory reproduces the standard Casimir effect under ordinary conditions while predicting small deviations within strong-constraint regimes. The next section analyzes the Delayed Choice Experiment from the standpoint of Future Convergence Theory and clarifies the relationship between the Future Causal Information Field and the Manifestation Projection.

3.0.7 3.7 Predictions for the Delayed Choice Experiment

In the previous section, we showed that within Future Convergence Theory the Casimir effect is understood as a consequence of changes in Manifestation Constraint Energy produced by boundary conditions imposed upon manifestation candidate sets. However, the quantum phenomenon that most directly tests the conceptual foundation of Future Convergence Theory is not vacuum fluctuations. Rather, it is the **Delayed Choice Experiment**, in which future measurement settings appear to influence present quantum interference. In standard quantum mechanics, the Delayed Choice Experiment is interpreted as a phenomenon in which the final measurement configuration appears to determine the past behavior of the particle. Nevertheless, this does not imply that the future physically influences the past. Within Future Convergence Theory, the phenomenon follows naturally once the Future Causal Information Field is introduced.

Standard Interpretation

In a delayed-choice experiment, the measurement apparatus is modified after the particle has already entered the interferometer. Nevertheless, the observed statistics always correspond to the final measurement configuration. Standard quantum mechanics explains this by allowing the wave function to evolve unitarily until the measurement is completed. However, it remains conceptually difficult to explain why the final measurement setting appears to be reflected in the quantum state from the very beginning.

The Standpoint of Future Convergence Theory

Within Future Convergence Theory, the quantum state is a manifestation- permitted state generated from the Future Causal Information Field. The Future Causal Information Field does not arise after manifestation. Instead, it already contains the informational structure from which manifestation becomes possible. Accordingly, measurement conditions are incorporated into the Manifestation Projection as consistency conditions relative to the Future Causal Information Field. The essential point is that no information is transmitted from the future to the past. Rather, the Manifestation Projection itself operates so as to satisfy the consistency conditions defined by the Future Causal Information Field. Consequently, only manifestation states compatible with the final measurement configuration are projected into the Three-Dimensional Manifestation Layer.

Manifestation Consistency Condition

Let the complete boundary condition of the measurement apparatus be denoted by

$$B.$$

Within Future Convergence Theory, the Manifestation Projection satisfies

$$\boxed{\mathcal{P}_X = \mathcal{P}_X(F, B)}$$

where

$$F$$

represents the informational state contained in the Future Causal Information Field. Accordingly, the manifested state is determined not only by the Future Causal Information Field itself but also by the consistency of the entire measurement configuration.

Synchronization with the Future Causal Information Field

The Future Causal Information Field is not generated sequentially according to ordinary temporal order. Instead, it is an ontological information field converging toward the Future Fixed Point

$$\mathcal{F}^*.$$

Consequently, the Manifestation Projection is determined not by local time alone but by consistency with the global fixed-point structure. Therefore, even when the manifested state agrees with the final measurement configuration in a delayed-choice experiment, ordinary local causality is not violated.

Delayed Choice Theorem

The preceding discussion yields the following theorem.

Theorem 3.6 (Delayed Choice Theorem). *The statistical distributions observed in delayed-choice experiments are not determined by causal influences propagating from the future into the past. Instead, they arise because the Manifestation Projection maps into the Three-Dimensional Manifestation Layer only those manifestation states that are consistent with both the Future Causal Information Field and the complete measurement boundary conditions.*

Proof. Within Future Convergence Theory, the quantum state is a manifestation-permitted state generated from the Future Causal Information Field. Furthermore, the Manifestation Projection allows projection into the Three-Dimensional Manifestation Layer only when consistency with the measurement boundary conditions is satisfied. Accordingly, the statistical distributions observed in delayed-choice experiments are determined by global manifestation consistency rather than by information propagating from the future into the past. Therefore, local temporal causality remains preserved. $\square$

Comparison with Standard Quantum Theory

Standard quantum mechanics explains delayed-choice experiments through the unitary evolution of the wave function. Future Convergence Theory instead explains the phenomenon through the consistency relation between the Future Causal Information Field and the Manifestation Projection. Accordingly, the theory requires no statement that "the future changes the past."

Observational Consequences

Under ordinary delayed-choice experiments, Future Convergence Theory reproduces exactly the same statistical distributions as standard quantum mechanics. However, when measurement boundary conditions change extremely rapidly or when delayed-choice experiments are performed under large gravitational potential differences, higher-order corrections associated with the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

may become observable. Future Convergence Theory predicts that such corrections may eventually be detectable in ultra-high-precision quantum interference experiments or in experiments performed near strong gravitational fields.

Conclusion of This Section

In this section, delayed-choice experiments were reinterpreted in terms of the consistency relation between the Future Causal Information Field and the Manifestation Projection. Consequently,

Future Convergence Theory explains the observed statistical distributions while preserving ordinary temporal causality. The next section analyzes the Quantum Eraser experiment and formulates the relationship between information erasure and the Manifestation Projection.

3.0.8 3.8 Predictions for the Quantum Eraser

In the previous section, we showed that within Future Convergence Theory the Delayed Choice Experiment is explained not through causal influence from the future but through consistency between the Future Causal Information Field and the measurement boundary conditions. A closely related quantum phenomenon is the **Quantum Eraser Experiment**. In a Quantum Eraser experiment, path information that initially appears to have been obtained is later erased through an appropriate measurement procedure, allowing interference fringes to reappear. Standard quantum mechanics explains this by stating that interference depends not on the existence of path information itself but on whether that information remains fundamentally accessible. Future Convergence Theory instead interprets the phenomenon through the structure of manifestation candidate sets and the Manifestation Projection.

Meaning of Path Information

Within standard quantum theory, path information identifies which path a particle has taken. Within Future Convergence Theory, however, path information is not regarded as the historical trajectory of a particle. Instead, path information represents an additional boundary condition imposed by the Manifestation Constraint upon the manifestation candidate set. Define the path information by

$$\boxed{I_{\text{path}}}$$

Action on the Manifestation Candidate Set

Using the manifestation candidate set introduced in the previous section,

$$\mathcal{A} = \{|\psi_1\rangle, |\psi_2\rangle, \dots\},$$

the presence of path information partitions the candidate set according to path-identification conditions. Thus,

$$\boxed{\mathcal{A} \rightarrow \mathcal{A}_1 \cup \mathcal{A}_2}$$

Consequently, interference cannot occur between candidates belonging to different subsets.

The Essence of the Quantum Eraser

Within Future Convergence Theory, erasing path information does not erase the history of the particle. Instead, it removes the boundary condition associated with path identification. Accordingly, the manifestation candidate subsets become unified once again. That is,

$$\boxed{\mathcal{A}_1 \cup \mathcal{A}_2 \rightarrow \mathcal{A}}$$

Interference therefore reappears not because the past has changed but because the manifestation candidate set has become unified once more.

Manifestation Consistency Condition

Within the Quantum Eraser experiment, the Manifestation Projection is determined by consistency with the complete boundary conditions, including the path information. Accordingly,

$$\boxed{\mathcal{P}_X = \mathcal{P}_X(F, B, I_{\text{path}})}$$

where

$$B$$

denotes the measurement boundary condition. When the path information is erased,

$$I_{\text{path}} \rightarrow 0,$$

the Manifestation Projection again selects manifestation states from the unified candidate set capable of producing interference.

Quantum Eraser Theorem

The preceding discussion yields the following theorem.

Theorem 3.7 (Quantum Eraser Theorem). *The interference fringes recovered in a Quantum Eraser experiment do not arise because the past state of the particle has changed. Rather, they result from the disappearance of the boundary conditions that had partitioned the manifestation candidate set, allowing the Manifestation Projection once again to select states from the unified candidate set.*

Proof. Within Future Convergence Theory, the quantum state is a manifestation- permitted state generated from the Future Causal Information Field. Path information acts as a boundary condition partitioning the manifestation candidate set. When the Quantum Eraser removes this boundary condition, the candidate sets become unified. Consequently, the Manifestation Projection can once again select manifested states from the unified candidate set, producing interference fringes. Throughout this process, neither the Future Causal Information Field nor the past physical state undergoes any modification. Therefore, the Quantum Eraser is understood as a change in manifestation structure produced by altered boundary conditions. $\square$

Comparison with Standard Quantum Theory

Standard quantum mechanics explains the Quantum Eraser through the accessibility of path information. Future Convergence Theory instead explains the phenomenon through the partitioning and reunification of manifestation candidate sets. Accordingly, information erasure is redefined not as the destruction of information itself but as a modification of manifestation boundary conditions.

Observational Consequences

Under ordinary Quantum Eraser experiments, Future Convergence Theory reproduces the same interference patterns predicted by standard quantum mechanics. However, when the measurement boundary conditions change extremely rapidly or when the experiment is performed within a strong gravitational field, higher-order corrections associated with the Manifestation Constraint Tensor

$$C_{\mu\nu}$$

may appear. The theory predicts that these corrections would manifest themselves as small differences in fringe visibility, phase, and correlation coefficients.

Conclusion of This Section

In this section, the Quantum Eraser experiment has been reinterpreted not as a modification of the past but as a change in the boundary conditions governing manifestation candidate sets. Future Convergence Theory therefore explains the Quantum Eraser while preserving ordinary temporal causality. The next section introduces the quantum-theory limit of Future Convergence Theory and formulates both the conditions under which ordinary quantum mechanics is recovered and the criteria by which the theory may be falsified.

3.0.9 3.9 Quantum Theory Limit and Falsifiability

In the preceding sections, Future Convergence Theory has reformulated quantum states, measurement, decoherence, vacuum fluctuations, the Casimir effect, the Delayed Choice Experiment, and the Quantum Eraser within a unified framework consisting of the Future Causal Information Field, the Quantum Permission Layer, and the Manifestation Projection. However, for any new physical theory to be scientifically meaningful, it must both reproduce the established theory within its domain of validity and make testable predictions beyond it. Accordingly, this section clarifies the limit in which Future Convergence Theory reduces to ordinary quantum mechanics and formulates explicit criteria under which the theory would be regarded as falsified.

Quantum Mechanical Limit

Consider the limit in which the corrections originating from the Future Causal Information Field

$$\mathcal{F}$$

and the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

become negligible. That is,

$$\Lambda_M \rightarrow 0$$

together with

$$\delta\mathcal{V} \rightarrow 0$$

are assumed. In this limit, higher-order corrections associated with the Future Causal Information Field disappear, and the mathematical structure of the Quantum Permission Layer becomes equivalent to the ordinary Hilbert space of standard quantum mechanics.

Reduction to the Schrödinger Equation

Within Future Convergence Theory, the quantum state

$$|\psi\rangle$$

is defined as a manifestation-permitted state. When corrections arising from the Future Causal Information Field become negligible, however, its temporal evolution reduces to the ordinary Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle$$

Accordingly, Future Convergence Theory does not replace quantum mechanics but rather contains it as its weak-constraint limit.

Born's Rule

Previous sections reinterpreted Born's rule as the manifestation-permission weight associated with each candidate state. In the weak-constraint limit, this manifestation-permission weight becomes identical to the ordinary observational probability. Therefore,

$$P_i = |c_i|^2$$

is experimentally indistinguishable from the prediction of standard quantum mechanics.

Consistency with Quantum Information Theory

Future Convergence Theory is likewise consistent with quantum information theory. The operational rules governing qubits, quantum entanglement, quantum communication, and quantum computation are all recovered in the weak- constraint limit. Accordingly, all currently established experimental results in quantum information science remain compatible with Future Convergence Theory.

Predictions Beyond Standard Quantum Theory

Observable deviations from standard quantum mechanics arise only when the coupling to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

or to the Future Causal Information Field can no longer be neglected. Typical situations include

- quantum interference near black holes,
- quantum entanglement across large gravitational potential differences,
- ultra-high-precision Casimir experiments,
- quantum fluctuations in the early Universe,
- ultra-sensitive atom interferometry,
- and quantum interference in gravitational-wave backgrounds.

In these regimes, Future Convergence Theory predicts higher-order corrections to the standard quantum mechanical description.

Falsifiability Criteria

Future Convergence Theory is regarded as experimentally falsified if any of the following conditions is satisfied.

1. The ordinary quantum mechanical limit cannot be reproduced.
2. The theory is inconsistent with Born's rule.
3. Existing experimental results concerning quantum interference, Casimir measurements, delayed-choice experiments, or Quantum Eraser experiments cannot be reproduced.
4. The higher-order corrections predicted by Future Convergence Theory remain absent despite experimental precision sufficient to detect them.
5. The introduction of Future Convergence Theory provides no improvement in explanatory power beyond standard quantum mechanics.

If any of these conditions is met, Future Convergence Theory cannot be regarded as a valid extension of quantum theory.

Conclusion of Part III

This chapter has established that quantum theory occupies the ontological intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer. Consequently, the wave function, measurement, decoherence, vacuum fluctuations, the Casimir effect, the Delayed Choice Experiment, and the Quantum Eraser have all been reformulated within a unified framework based on manifestation-permitted states, manifestation candidate sets, and the Manifestation Projection. Furthermore, Future Convergence Theory has been shown to reproduce ordinary quantum mechanics completely in the weak-constraint limit while predicting new observable corrections only in regimes where manifestation constraints become significant. The ontological role of quantum theory within Future Convergence Theory is thereby established. The next chapter demonstrates that the graviton should be understood not as gravity itself but as a quantum perturbation of the Manifestation Constraint, thereby deriving the basic structure of quantum gravity within Future Convergence Theory.

Part IV

The Graviton Is Not the Essence of Gravity

Chapter 4

The Graviton Is Not the Essence of Gravity

In Part I, we introduced the ontological foundation constituting Future Convergence Theory, namely, the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the FCT Closure Equation. In Part II, gravity was redefined not as an interparticle interaction, but as the **Manifestation Constraint** that maintains consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. As a result, gravitational universality, non-screenability, the action of gravity on light, the weakness of gravity, gravitational time dilation, and the General Relativistic limit were derived from the ontological properties of the Manifestation Constraint. Furthermore, in Part III, quantum theory was shown to be the intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer, and the wave function, measurement, decoherence, vacuum fluctuations, the Casimir effect, the Delayed Choice Experiment, and the Quantum Eraser were all explained in a unified manner through manifestation-permitted states and the Manifestation Projection. At this point, one of the most important problems in quantum gravity appears. That problem is whether the **graviton** is gravity itself. In perturbative quantum gravity, gravity is usually treated by expanding the spacetime metric

$$g_{\mu\nu}$$

around a background metric

$$\eta_{\mu\nu}.$$

Quantizing the resulting perturbation leads to a massless spin-2 boson, namely the graviton, which is taken to mediate the gravitational interaction. However, from this standpoint, the relationship between gravity in General Relativity and the graviton in quantum field theory is still not completely unified. Moreover, as shown in Part II, within Future Convergence Theory, gravity is not an interaction between particles. Rather, it is the Manifestation Constraint

$$\mathcal{C}_{\mu\nu}$$

that maintains the Three-Dimensional Manifestation Layer itself. Therefore, if gravity itself is the Manifestation Constraint, then the graviton cannot be identical with that constraint. In the present theory, the graviton is defined as a **Quantum Perturbation** that arises when the Manifestation Constraint is projected into the Quantum Permission Layer. That is, the graviton is not the essence of gravity. It is merely an effective excitation state that appears when gravity is observed as a quantum degree of freedom. Adopting this standpoint eliminates the need to quantize gravity itself. What is quantized is not the Manifestation Constraint, but the local perturbation of the Manifestation Constraint. As a result, the division of roles between General Relativity and Quantum Theory becomes clear, and the apparent contradiction between

the ontology of gravity and quantum field theory is resolved. In this chapter, we first review the standard framework of perturbative gravity and then redefine the graviton as a quantum perturbation of the Manifestation Constraint. We then prove that the Manifestation Constraint and its quantum perturbation are ontologically distinct, and clarify what would actually be demonstrated if the graviton were detected. Furthermore, we derive corrections to high-energy gravitational scattering and to gravitational-wave dispersion, and finally formulate falsifiability conditions predicted by Future Convergence Theory.

Part IV

The Graviton Is Not the Essence of Gravity

4.0.1 4.1 Metric Perturbations and the Graviton

In General Relativity, the gravitational field is described by the spacetime metric

$$g_{\mu\nu}.$$

In quantum gravity, this metric is expanded perturbatively around a flat background metric

$$\eta_{\mu\nu}.$$

That is, one writes

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

where

$$h_{\mu\nu}$$

represents a small perturbation away from the background spacetime. When this perturbation field is quantized, one obtains a massless spin-2 bosonic particle. This particle is commonly referred to as the **graviton**. However, this derivation only quantizes a perturbation of the metric. It does not define gravity itself. Within Future Convergence Theory, as shown in Part II, gravity is the Manifestation Constraint Tensor

$$C_{\mu\nu}$$

itself. Therefore, the metric perturbation

$$h_{\mu\nu}$$

is not the Manifestation Constraint itself. Rather, it is the geometrical representation of a local fluctuation of the Manifestation Constraint appearing within the Quantum Permission Layer. From this standpoint, Future Convergence Theory defines the graviton as follows.

Definition 4.1 (Graviton). *The graviton is the effective excitation state that appears when the Manifestation Constraint Tensor*

$$C_{\mu\nu}$$

arises as a local perturbation within the Quantum Permission Layer

$$\mathcal{Q}.$$

Accordingly, the graviton is not gravity itself, but a quantum perturbation of the Manifestation Constraint.

This definition establishes a clear division of roles. Gravity is an ontological object, whereas the graviton is a quantum excitation. In the next section, we formulate this division mathematically and prove that the layer in which the graviton exists is not identical to the layer in which the Manifestation Constraint exists. This is the principle of **non-identity of layer position**.

4.0.2 4.2 The Layer Position of the Graviton

In the previous section, we showed that within Future Convergence Theory, gravity is the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

whereas the graviton is defined as its quantum perturbation. However, this definition alone does not yet clarify where the graviton belongs within the ontological hierarchy of Future Convergence Theory. In this section, we prove that the Manifestation Constraint and the graviton belong to ontologically distinct layers.

The Ontological Layer of the Manifestation Constraint

The Manifestation Constraint Tensor introduced in Part II,

$$\mathcal{C}_{\mu\nu},$$

is the ontological constraint that maintains consistency between the Future Causal Information Field

$$\mathcal{F}$$

and the Three-Dimensional Manifestation Layer

$$\mathcal{X}.$$

Accordingly, the Manifestation Constraint does not exist as a particle. Rather, it is the foundational structure that makes the Three-Dimensional Manifestation Layer itself possible. Thus, the Manifestation Constraint governs the entire Three-Dimensional Manifestation Layer prior to the existence of individual particles.

The Ontological Layer of the Graviton

By contrast, as defined in the previous section, the graviton is a local perturbation of the Manifestation Constraint appearing within the Quantum Permission Layer

$$\mathcal{Q}.$$

Therefore, the graviton is not the Manifestation Constraint itself. Nor is it an entity that constrains the entire Three-Dimensional Manifestation Layer. The graviton is a quantum degree of freedom described within the Quantum Permission Layer. That is,

$$\boxed{g \in \mathcal{Q}}$$

where

$$g$$

denotes the graviton state.

Ontological Separation

It follows that the Manifestation Constraint and the graviton belong to different layers. Thus,

$$\boxed{\mathcal{C}_{\mu\nu} \notin \mathcal{Q}}$$

and also

$$\boxed{g \notin \mathcal{X}}$$

The Manifestation Constraint is not a quantum state. The graviton, likewise, is not gravity itself. The two have fundamentally different ontological roles.

Layer Mapping

Within Future Convergence Theory, the process by which the Manifestation Constraint appears as a quantum degree of freedom is represented by the mapping

$$\boxed{\mathcal{G} : \mathcal{C}_{\mu\nu} \longrightarrow g}$$

The operator

$$\mathcal{G}$$

is called the **Graviton Projection Operator**. It is crucial to note that this operator does not generate gravity. Gravity already exists as the Manifestation Constraint. The projection operator merely expresses that constraint as a local excitation within the Quantum Permission Layer.

Ontological Hierarchy

The relationship between gravity and the graviton within Future Convergence Theory is therefore summarized by the following hierarchy:

$$\boxed{\mathcal{F} \longrightarrow \mathcal{C}_{\mu\nu} \longrightarrow \mathcal{Q} \longrightarrow g \longrightarrow \mathcal{X}}$$

Here, the Manifestation Constraint is the ontological constraint derived from the Future Causal Information Field, whereas the graviton is the effective degree of freedom shown by that constraint within the Quantum Permission Layer.

Layer Non-Identity Theorem for the Graviton

The preceding considerations yield the following theorem.

Theorem 4.1 (Layer Non-Identity Theorem for the Graviton). *The Manifestation Constraint Tensor*

$$\mathcal{C}_{\mu\nu}$$

and the graviton

$$g$$

belong to distinct ontological layers within Future Convergence Theory. Therefore, even if the graviton were detected, such a detection would not constitute the direct observation of gravity itself.

Proof. The Manifestation Constraint is the ontological constraint that makes the Three-Dimensional Manifestation Layer possible. As defined in Part II, it is not a particle. By contrast, the graviton is a local perturbation belonging to the Quantum Permission Layer and is described as a quantum degree of freedom. The two differ in their domains of definition, ontological layers, and physical roles. Therefore, they cannot be identified. $\square$

Theoretical Consequence

As a result, even if the graviton were experimentally detected in the future, Future Convergence Theory would not interpret this as evidence that gravity is a particle. What would be demonstrated is only that the Manifestation Constraint possesses local excitations within the Quantum Permission Layer. Accordingly, graviton detection and the ontology of gravity must be kept logically distinct.

Conclusion of This Section

In this section, the ontological layer to which the graviton belongs was clearly defined, and it was proven that the Manifestation Constraint Tensor and the graviton are not identical. Future Convergence Theory has thereby formulated, on the basis of its ontological hierarchy, the position that gravity is a constraint, whereas the graviton is its quantum perturbation. In the next section, we use this result to show that the Manifestation Constraint and the quantum perturbation are non-identical not only ontologically, but also physically and mathematically, and derive why the two cannot be conflated.

4.0.3 4.3 Non-Identity of the Manifestation Constraint and Quantum Perturbations

In the previous section, we showed that within Future Convergence Theory, the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

and the graviton

$$g$$

belong to different ontological layers. However, the mere difference in ontological hierarchy is not sufficient to establish that they are also physically and mathematically distinct objects. In this section, we prove that the Manifestation Constraint and quantum perturbations are non-identical in their mathematical structures, temporal evolution, and domains of action.

Difference in the Objects of Action

The Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

provides the existence condition for the entire Three-Dimensional Manifestation Layer. Its object of action is not an individual particle. Electrons, protons, photons, neutrinos, and even the vacuum state are all subject to the Manifestation Constraint once they become manifested. By contrast, the graviton is a local excitation within the Quantum Permission Layer. Accordingly, the graviton is not a constituent of the Manifestation Constraint itself. Rather, it is merely a degree of freedom describing local fluctuations of that constraint.

Difference in Temporal Evolution

The Manifestation Constraint is preserved throughout the entire Future Convergence Loop introduced in Part I. Its temporal evolution is determined as a consistency condition with the Future Causal Information Field. Accordingly,

$$\boxed{\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}(\mathcal{F})}$$

holds. The graviton, on the other hand, evolves as a quantum state within the Quantum Permission Layer. Its dynamics are therefore described by an effective quantum theory. Consequently, the graviton obeys temporal evolution laws that differ from those of the Manifestation Constraint itself.

Difference in Mathematical Structure

The Manifestation Constraint Tensor is an ontological constraint field. Mathematically, it is a rank-two tensor field constraining the entire four-dimensional spacetime. The graviton, by

contrast, is a quantized excitation of a small fluctuation of that constraint field and is represented as a quantum state in quantum field theory. Accordingly,

$$\boxed{\mathcal{C}_{\mu\nu} \neq g}$$

Furthermore,

$$\boxed{\dim(\mathcal{C}_{\mu\nu}) \neq \dim(g)}$$

Thus, the two differ in their domains of definition, codomains, and physical interpretations.

Quantizability

In conventional quantum gravity, the objective is to quantize gravity itself. Within Future Convergence Theory, however, the object to be quantized is not the Manifestation Constraint itself. Rather, only the local perturbations derived from the Manifestation Constraint are quantized. Accordingly, Future Convergence Theory adopts the proposition

$$\boxed{Q(\mathcal{C}_{\mu\nu}) \neq g}$$

where

$$Q$$

denotes the quantization operator. That is, directly quantizing the Manifestation Constraint does not produce the graviton itself. The graviton appears only when local perturbations of the Manifestation Constraint are quantized as effective degrees of freedom.

Non-Identity Theorem

The preceding discussion establishes the following theorem.

Theorem 4.2 (Manifestation Constraint–Quantum Perturbation Non-Identity Theorem). *The Manifestation Constraint Tensor*

$$\mathcal{C}_{\mu\nu}$$

and the graviton

$$g$$

differ in their objects of action, temporal evolution, mathematical structures, and quantization properties. Accordingly, the graviton is not identical to the Manifestation Constraint itself.

Proof. The Manifestation Constraint is an ontological constraint field derived from the Future Causal Information Field and governs the entire Three-Dimensional Manifestation Layer. The graviton, by contrast, is defined as a local excitation within the Quantum Permission Layer. Since the two differ in their domains of definition, temporal evolution, mathematical structures, and objects of action, they cannot be identified. $\square$

Theoretical Consequence

It follows that, within Future Convergence Theory, the expression "quantizing gravity" is not strictly accurate. What is quantized is not gravity itself, but the local perturbations generated when gravity is projected into the Quantum Permission Layer. Quantum gravity therefore appears as an effective approximation to the theory of the Manifestation Constraint.

Conclusion of This Section

In this section, we demonstrated that the Manifestation Constraint Tensor and the graviton are physically as well as mathematically non-identical. Future Convergence Theory thereby establishes a clear division of roles between the ontology of gravity and quantum gravity. In the next section, we formulate what the experimental detection of the graviton would—and would not—mean from the standpoint of Future Convergence Theory.

4.0.4 4.4 What the Detection of the Graviton Would Mean

In the previous section, we showed that within Future Convergence Theory, the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

and the graviton

$$g$$

are distinct with respect to their domains of action, temporal evolution, mathematical structures, and ontological hierarchy. Accordingly, even if the graviton were experimentally detected in the future, that fact alone would not imply that gravity itself is a particle. In this section, we clarify precisely what graviton detection would—and would not—establish from the standpoint of Future Convergence Theory.

Meaning within the Standard Framework

In perturbative quantum gravity, the detection of the graviton would provide strong evidence that gravity is quantizable and that gravitational interactions are mediated by a massless spin-2 particle. Nevertheless, even within that framework, the observation concerns only a quantum excitation. It does not directly establish the ontological origin of gravity itself.

The Standpoint of Future Convergence Theory

Within Future Convergence Theory, the graviton is defined as the local excitation state exhibited by the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

within the Quantum Permission Layer. Accordingly, if the graviton is observed, what is directly verified is only that the Manifestation Constraint possesses quantum degrees of freedom. It does not follow that gravity itself exists as a particle.

Observable Quantity

Let the quantity measured in a graviton detection experiment be denoted by

$$\mathcal{G}_{\text{obs}}.$$

Within Future Convergence Theory, this observable is written as

$$\boxed{\mathcal{G}_{\text{obs}} = \Pi_Q(\delta\mathcal{C}_{\mu\nu})}$$

where

$$\Pi_Q$$

is the operator projecting local perturbations of the Manifestation Constraint into the Quantum Permission Layer, and

$$\delta\mathcal{C}_{\mu\nu}$$

denotes the local perturbation of the Manifestation Constraint Tensor. Accordingly, experiments observe not the Manifestation Constraint itself but its local perturbations.

Detectability

As shown in Part II, the Manifestation Constraint is the background constraint maintaining the entire Three-Dimensional Manifestation Layer. The background constraint itself cannot be directly observed by a local particle detector. Local perturbations, however, possess finite excitation energy and are therefore observable in principle. Accordingly, Future Convergence Theory yields

$$\boxed{\text{Detectable} \implies \delta\mathcal{C}_{\mu\nu}}$$

whereas

$$\boxed{\text{Undetectable} \implies \mathcal{C}_{\mu\nu}}$$

Thus, what is experimentally detectable is the excitation of the constraint, not the background constraint itself.

Interpretation of Detection Results

Within Future Convergence Theory, graviton detection supports only the following three conclusions.

1. The Manifestation Constraint possesses local quantum excitations.
2. Those excitations behave as spin-2 degrees of freedom.
3. A quantum limit of gravitational waves exists.

By contrast, graviton detection alone does not establish any of the following.

1. That gravity itself is a particle.
2. That spacetime itself is quantized.
3. That the ontological foundation of reality consists solely of quantum fields.

Graviton Detection Theorem

The foregoing considerations establish the following theorem.

Theorem 4.3 (Graviton Detection Theorem). *If the graviton is observed, what is experimentally established is the existence of local quantum excitations of the Manifestation Constraint. That observation alone does not justify the conclusion that gravity itself is a particle.*

Proof. From Sections 4.2 and 4.3, the Manifestation Constraint and the graviton are ontologically and mathematically distinct objects. Experimental apparatuses detect only local exchanges of finite energy. Therefore, the observed quantity is only the local perturbation

$$\delta\mathcal{C}_{\mu\nu},$$

not the background Manifestation Constraint itself. Consequently, graviton detection indirectly supports the existence of the Manifestation Constraint but does not establish that gravity itself is a particle. $\square$

Theoretical Consequence

This conclusion implies that even if the graviton is discovered in the future, Future Convergence Theory does not require any revision of the ontological principle established in Part II, namely,

$$\text{Gravity} = \text{Manifestation Constraint}.$$

Rather, the graviton naturally appears as the effective quantum degree of freedom predicted by the Manifestation Constraint Theory. Accordingly, the discovery of the graviton would not refute Future Convergence Theory but instead support its quantum limit.

Conclusion of This Section

In this section, we clarified the physical meaning of graviton detection on the basis of the ontology of Future Convergence Theory. As a result, we showed that the experimental detection of the graviton would verify only the local quantum excitations of the Manifestation Constraint, and would not imply that gravity itself is a particle. In the next section, this conclusion is applied to high-energy gravitational scattering, and the corrections to scattering amplitudes predicted by Future Convergence Theory are derived.

4.0.5 4.5 Corrections to High-Energy Gravitational Scattering

In the previous section, we showed that, within Future Convergence Theory, even if the graviton were detected, such an observation would only indicate the local quantum excitation of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

and would not imply that gravity itself is a particle. Once this standpoint is adopted, the understanding of gravitational scattering at high energies also differs from that of conventional perturbative quantum gravity. In standard quantum gravity, scattering amplitudes are fundamentally derived from graviton exchange processes. Within Future Convergence Theory, however, the graviton is merely the local excitation of the Manifestation Constraint. Consequently, high-energy scattering receives contributions not only from graviton exchange but also from the background structure of the Manifestation Constraint itself.

Standard Scattering Amplitude

In perturbative quantum gravity, the two-body scattering amplitude is generally expanded as

$$\mathcal{M}_{\text{GR}} = \sum_n \mathcal{M}_n$$

where each contribution

$$\mathcal{M}_n$$

represents graviton exchange, loop corrections, or higher-order perturbative effects.

Manifestation Constraint Correction

Within Future Convergence Theory, every scattering process takes place within the background structure provided by the Manifestation Constraint. Accordingly, the scattering amplitude acquires an additional correction arising from the Manifestation Constraint Tensor. The total scattering amplitude is therefore defined as

$$\mathcal{M}_{\text{FCT}} = \mathcal{M}_{\text{GR}} + \Delta\mathcal{M}_c$$

where

$$\Delta\mathcal{M}_c$$

denotes the correction generated by the Manifestation Constraint.

General Form of the Correction

The Manifestation Constraint correction depends upon the background constraint density

$$\Lambda_{\text{M}}$$

and the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

In the most general form,

$$\Delta\mathcal{M}_{\mathcal{C}} = F(\Lambda_{\text{M}}, \mathcal{C}_{\mu\nu}, s, t, u)$$

where

$$s, t, u$$

are the standard Mandelstam variables.

Low-Energy Limit

When the Manifestation Constraint density is sufficiently small,

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} \Delta\mathcal{M}_{\mathcal{C}} = 0$$

holds. Accordingly, Future Convergence Theory exactly reproduces conventional low-energy gravitational scattering.

High-Energy Limit

When the collision energy becomes sufficiently large and the local curvature increases, the Manifestation Constraint density can no longer be neglected. In this regime, the scattering amplitude may be expanded as

$$\mathcal{M}_{\text{FCT}} = \mathcal{M}_{\text{GR}} + \alpha\Lambda_{\text{M}} + \beta\Lambda_{\text{M}}^2 + \mathcal{O}(\Lambda_{\text{M}}^3)$$

where

$$\alpha, \beta$$

are coupling coefficients specific to Future Convergence Theory.

Scattering Cross Section

The scattering cross section likewise receives a correction. Let

$$\sigma$$

denote the scattering cross section. Then

$$\sigma_{\text{FCT}} = \sigma_{\text{GR}} + \Delta\sigma$$

where

$$\Delta\sigma = G(\Lambda_{\text{M}})$$

is the correction determined by the Manifestation Constraint density.

High-Energy Scattering Theorem

The above considerations establish the following theorem.

Theorem 4.4 (High-Energy Scattering Theorem). *Within Future Convergence Theory, high-energy gravitational scattering is not determined solely by graviton exchange. The scattering amplitude additionally contains correction terms arising from the Manifestation Constraint Tensor. In the weak-constraint limit, these corrections vanish and the theory reduces to conventional perturbative quantum gravity.*

Proof. As shown in Section 4.2, the graviton is not the Manifestation Constraint itself but only its local quantum excitation. Consequently, local excitations alone cannot completely describe the contribution of the background Manifestation Constraint. In the high-energy regime, the background constraint density becomes finite, introducing additional correction terms into the scattering amplitude. In the weak-constraint limit, the background constraint density vanishes, and the correction terms disappear. Therefore, Future Convergence Theory reproduces conventional gravitational scattering in the appropriate limit. $\square$

Observational Consequences

With current accelerator technology, direct measurements of gravitational scattering remain extremely difficult. However, future ultra-high-energy collision experiments, scattering processes associated with microscopic black-hole formation, or physical conditions approaching those of the early Universe may provide access to the corrections

$$\Delta\mathcal{M}_c$$

and

$$\Delta\sigma,$$

allowing them to be distinguished experimentally from the predictions of standard perturbative quantum gravity.

Conclusion of This Section

In this section, we introduced the Future Convergence Theory correction to high-energy gravitational scattering. We thereby showed that, in the high-energy regime, scattering amplitudes are not determined solely by graviton exchange, but also receive direct contributions from the background structure of the Manifestation Constraint Tensor. In the next section, this correction is applied to gravitational waves, and predictions for the dispersion relation and propagation properties of gravitational waves are derived.

4.0.6 4.6 Connection to Gravitational-Wave Dispersion

In the previous section, we showed that, within Future Convergence Theory, high-energy gravitational scattering receives contributions not only from graviton exchange but also directly from the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

This result also leads to predictions that differ from those of General Relativity regarding the propagation of gravitational waves. In General Relativity, gravitational waves are transverse perturbations of the spacetime metric that propagate through vacuum at the speed of light without dispersion. Accordingly, every frequency component propagates with the same velocity

$$c.$$

Within Future Convergence Theory, however, gravitational waves are interpreted as the observable propagation of local perturbations generated in the Manifestation Constraint Tensor, transmitted through the Quantum Permission Layer into the Three-Dimensional Manifestation Layer. Consequently, changes in the background Manifestation Constraint density produce corrections to the propagation characteristics.

Dispersion Relation in General Relativity

In vacuum, gravitational waves in General Relativity satisfy the linear dispersion relation

$$\omega = ck$$

between the angular frequency

$$\omega$$

and the wave number

$$k.$$

Therefore, both the group velocity and the phase velocity satisfy

$$v_g = v_p = c.$$

Dispersion Relation in Future Convergence Theory

Within Future Convergence Theory, gravitational waves are interpreted as local excitations of the Manifestation Constraint propagating through the background constraint. Accordingly, the dispersion relation depends upon the background constraint density

$$\Lambda_M.$$

In the most general form,

$$\omega^2 = c^2 k^2 + \Phi(\Lambda_M, k)$$

where

$$\Phi$$

is the correction function arising from the Manifestation Constraint.

Weak-Constraint Limit

When

$$\Lambda_M \rightarrow 0,$$

one has

$$\Phi \rightarrow 0$$

and Future Convergence Theory therefore reproduces the standard relativistic dispersion relation

$$\omega = ck.$$

Strong-Constraint Limit

Near black holes or in regions of extremely high curvature, the Manifestation Constraint density cannot be neglected. In such cases, the correction may be expanded as

$$\Phi = \alpha \Lambda_M k^2 + \beta \Lambda_M^2 k^4 + \mathcal{O}(\Lambda_M^3)$$

where

$$\alpha, \beta$$

are coupling constants specific to Future Convergence Theory.

Group Velocity Correction

The group velocity is defined by

$$v_g = \frac{\partial \omega}{\partial k}.$$

Accordingly, Future Convergence Theory predicts

$$v_g = c + \Delta v_g(\Lambda_M)$$

The correction is negligible under ordinary astrophysical conditions but may become finite immediately after black-hole mergers or during the earliest stages of cosmic evolution.

Phase Correction

The gravitational-wave phase also receives a correction from the background Manifestation Constraint. If

$$\phi$$

denotes the phase, then

$$\phi_{\text{FCT}} = \phi_{\text{GR}} + \Delta \phi$$

holds. Since this phase correction accumulates over long propagation distances, it may become observable with sufficiently precise gravitational-wave measurements.

Gravitational-Wave Dispersion Theorem

The preceding discussion establishes the following theorem.

Theorem 4.5 (Gravitational-Wave Dispersion Theorem). *Within Future Convergence Theory, gravitational waves are local perturbations of the Manifestation Constraint Tensor propagating through the background constraint. Accordingly, in regions where the background Manifestation Constraint density is finite, corrections to gravitational-wave dispersion, group velocity, and phase arise relative to the predictions of General Relativity.*

Proof. As shown in Section 4.2, the graviton is not the Manifestation Constraint itself but only a local excitation on that background. Therefore, the propagation equation necessarily depends upon the background Manifestation Constraint density. In the weak-constraint limit, the background density vanishes and the standard gravitational-wave equation is recovered. In regions of strong constraint, however, correction terms proportional to the background density appear in the dispersion relation. Hence, Future Convergence Theory predicts gravitational-wave dispersion different from that of General Relativity under sufficiently extreme conditions. $\square$

Observational Consequences

The dispersion corrections predicted by Future Convergence Theory may be tested using observations from LIGO, Virgo, KAGRA, and future detectors including the Einstein Telescope, Cosmic Explorer, and LISA. In particular, black-hole merger events provide the most favorable conditions, since higher-frequency modes accumulate larger Manifestation Constraint corrections, potentially producing measurable deviations in ringdown phases, group velocities, and arrival times.

Conclusion of This Section

In this section, gravitational waves were redefined as the propagation of local excitations of the Manifestation Constraint Tensor. Future Convergence Theory therefore reproduces the predictions of General Relativity under ordinary observational conditions while predicting observable corrections to gravitational-wave dispersion, group velocity, and phase in strong-constraint regimes. The next section summarizes the theory developed throughout this chapter and formulates the falsifiability conditions for the Future Convergence Theory description of gravitons.

4.0.7 4.7 Falsifiability Conditions

In the previous section, we showed that, within Future Convergence Theory, gravitational waves are reinterpreted as local perturbations of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

propagating on the background Manifestation Constraint. As a consequence, corrections to gravitational-wave dispersion, group velocity, phase, and ringdown waveforms were shown to arise in strong-constraint regimes. However, presenting a novel conceptual framework alone is insufficient for a physical theory. A scientific theory must also specify concrete observational conditions under which it can either be supported or falsified. In this section, we formulate the falsifiability conditions for the central claim of Future Convergence Theory that **the graviton is not the essence of gravity**.

Principle of Falsifiability

Within Future Convergence Theory, the object of falsification is not the concept itself. Rather, it is the set of observable quantities derived from the theory. Let the prediction of General Relativity be denoted by

$$\mathcal{O}_{\text{GR}},$$

and the corresponding prediction of Future Convergence Theory by

$$\mathcal{O}_{\text{FCT}}.$$

The experimentally testable quantity is therefore the difference

$$\boxed{\Delta\mathcal{O} = \mathcal{O}_{\text{FCT}} - \mathcal{O}_{\text{GR}}}$$

Accordingly, the validity of Future Convergence Theory is determined by whether this difference is experimentally observed.

Verification through Gravitational Waves

As derived in this chapter, Future Convergence Theory predicts the following corrections for gravitational waves:

$$\Delta\omega, \quad \Delta v_g, \quad \Delta\phi.$$

If all of these quantities continue to agree perfectly with General Relativity even after observational precision becomes sufficiently high, then the higher-order corrections associated with the Manifestation Constraint are not supported.

Verification through Ringdown Modes

For the quasinormal modes following a black-hole merger, Future Convergence Theory predicts corrections arising from the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Specifically,

$$\omega_{\ell mn}^{\text{FCT}} = \omega_{\ell mn}^{\text{GR}} + \Delta\omega_{\ell mn}$$

holds. If future observations achieve sufficient precision and nevertheless find

$$\Delta\omega_{\ell mn} = 0$$

for all black-hole systems, then the present theory would require substantial revision.

Consistency with Graviton Detection

Future Convergence Theory is not falsified merely by the experimental detection of the graviton. However, if the graviton were shown to be a fundamental particle completely independent of any background Manifestation Constraint, and if a theory requiring no Manifestation Constraint successfully explained every gravitational observation with equal or greater accuracy, then the ontological foundation of Future Convergence Theory would no longer be sustainable.

Internal Consistency Conditions

Future Convergence Theory must satisfy the following internal consistency conditions.

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} \text{FCT} = \text{GR}$$

and

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} \text{FCT} = \text{Quantum Theory}$$

If either limit fails to hold, Future Convergence Theory cannot be regarded as a unified physical theory.

Falsifiability Conditions for Part IV

Accordingly, the theoretical framework developed in Part IV is rejected if any of the following conditions is satisfied.

1. No gravitational-wave dispersion correction is detected despite sufficient observational precision.
2. Black-hole quasinormal modes continue to agree completely with the predictions of General Relativity.
3. A theory requiring no Manifestation Constraint explains gravitational waves, ringdown phenomena, and quantum-gravitational observations with equal or greater predictive accuracy.
4. The graviton is experimentally established as a fundamental particle independent of any background Manifestation Constraint.
5. Future Convergence Theory fails to reproduce either the General Relativistic limit or the Quantum-Theoretical limit.

Conclusion of Part IV

In this chapter, we demonstrated that within Future Convergence Theory the graviton is not gravity itself but rather the local quantum excitation of the Manifestation Constraint Tensor. This establishes a clear separation between the ontology of gravity and the effective description provided by quantum gravity, thereby yielding a theoretical framework that consistently encompasses both General Relativity and Quantum Theory. Furthermore, concrete observational predictions were introduced for high-energy gravitational scattering, gravitational-wave dispersion, ringdown spectra, and graviton detection, together with explicit falsifiability conditions required for Future Convergence Theory to qualify as a physical theory. Accordingly, the ontological distinction between gravity and the graviton is established within the framework of Future Convergence Theory. In the next chapter, string theory is incorporated into the framework of Future Convergence Theory, where it is shown mathematically that string theory is not the final theory of nature but rather an effective theory belonging to the Quantum Permission Layer.

Part V

The Position of String Theory within Future Convergence Theory

Chapter 5

The Position of String Theory within Future Convergence Theory

In Part I, we defined the ontological foundation of Future Convergence Theory, namely, the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the FCT Closure Equation. In Part II, gravity was redefined not as an interparticle interaction but as the **Manifestation Constraint** that maintains consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. In Part III, quantum theory was shown to be the ontological intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer, and the wave function, measurement, decoherence, vacuum fluctuations, the Casimir effect, the Delayed Choice Experiment, and the Quantum Eraser were all explained in a unified manner through manifestation-permitted states and the Manifestation Projection. Furthermore, in Part IV, the graviton was shown not to be gravity itself, but rather the local quantum excitation of the Manifestation Constraint Tensor, and gravity and quantum gravity were shown to belong to ontologically distinct layers. At this stage, we consider another representative candidate for unification in modern theoretical physics: **String Theory**. In string theory, particles are described not as points but as one-dimensional strings, and their vibrational modes provide a unified description of electrons, photons, gluons, and even gravitons. For this reason, string theory has long been regarded as a candidate for a unified theory that naturally incorporates gravity within a quantum framework. However, from the standpoint of Future Convergence Theory, string theory cannot be positioned as the final ontological layer. As shown in Parts II and III, the physical world possesses a hierarchical structure consisting of the Future Causal Information Field, the Quantum Permission Layer, and the Three-Dimensional Manifestation Layer. The objects treated by string theory—quantum states, vibrational modes, gravitons, and related degrees of freedom—are all degrees of freedom belonging to the Quantum Permission Layer. Accordingly, string theory does not describe existence itself. Rather, it is positioned as an effective theory that describes the internal structure of the Quantum Permission Layer with very high precision. Within Future Convergence Theory, string theory is not rejected. On the contrary, it is naturally included as a limiting theory describing the Quantum Permission Layer. However, the Future Causal Information Field, the Meaning Generation Template, the Manifestation Constraint, the Future Fixed Point, and the Reprojection Closed Loop all lie outside string theory itself. Consequently, string theory alone cannot explain the ontology of gravity, the measurement problem, the future fixed point, meaning generation, or the reprojection structure. In this chapter, we first derive mathematically why string theory belongs to the Quantum Permission Layer. We then reformulate the string action corrected by FCT, the effective background metric, vibrational modes, the mass spectrum, compactification, the vacuum selection problem, and the relation to Calabi–Yau manifolds. Finally, we show that Future Convergence Theory contains string theory as a subtheory and organize the falsifiable

observables that distinguish the two.

Part V

The Position of String Theory within Future Convergence Theory

5.0.1 5.1 String Theory Is Not the Final Layer but a Quantum Intermediate Layer

In string theory, all elementary particles are represented as vibrational states of one-dimensional strings. By quantizing the vibrational modes of the string, fermions, gauge particles, and even the spin-2 particle identified as the graviton are naturally obtained. For this reason, string theory has developed as a candidate theory unifying all fundamental interactions. However, from the standpoint of Future Convergence Theory, this unification is a unification internal to the Quantum Permission Layer. It is not a unification of ontology as a whole. As shown in Part III, the Quantum Permission Layer

$$\mathcal{Q}$$

is the intermediate layer that exists between the Future Causal Information Field

$$\mathcal{F}$$

and the Three-Dimensional Manifestation Layer

$$\mathcal{X}.$$

Accordingly, quantum states, wave functions, gravitons, and the vibrational modes of strings are all understood as internal structures of

$$\mathcal{Q}.$$

Future Convergence Theory expresses this relation through the inclusion

$$\boxed{\text{String Theory} \subset \mathcal{Q}}$$

That is, string theory is not the entirety of the Quantum Permission Layer but an effective theory within it. Furthermore, the Quantum Permission Layer itself is generated from the Future Causal Information Field. Therefore, the hierarchy in Future Convergence Theory is written as

$$\boxed{\mathcal{F} \longrightarrow \mathcal{Q} \longrightarrow \text{String Theory} \longrightarrow \mathcal{X}}$$

The crucial point is that there exists no mapping by which the Future Causal Information Field is derived from string theory. By contrast, there does exist a mapping from the Future Causal Information Field through the Quantum Permission Layer to the effective degrees of freedom described by string theory. Consequently, within Future Convergence Theory, the ontological hierarchy lies above string theory. The following theorem therefore holds.

Theorem 5.1 (String-Theory Intermediate Layer Theorem). *Within Future Convergence Theory, string theory is an effective theory describing the internal degrees of freedom of the Quantum Permission Layer, which connects the Future Causal Information Field and the Three-Dimensional Manifestation Layer. Accordingly, string theory is not the final ontological theory but a theory belonging to the quantum intermediate layer.*

Proof. From Part III, the Quantum Permission Layer is the constraint layer that selects only manifestable states from the Future Causal Information Field. The objects described by string theory, including string vibrational states, quantum states, and gravitons, are all quantum

degrees of freedom. Therefore, the domain in which string theory is defined is contained within the Quantum Permission Layer. On the other hand, the Future Causal Information Field, the Meaning Generation Template, and the Manifestation Constraint are not described by string theory. Thus, string theory is positioned as an effective theory of the quantum intermediate layer. $\square$

In this section, we have shown that string theory is not the final ontological layer of Future Convergence Theory, but rather an effective theory belonging to the Quantum Permission Layer. In the next section, on the basis of this positioning, we derive Future Convergence Theory corrections to the Polyakov action and the Nambu–Goto action.

5.0.2 5.2 String Actions with FCT Corrections

In the previous section, we showed that within Future Convergence Theory, string theory is an effective theory describing the internal degrees of freedom of the Quantum Permission Layer

$$\mathcal{Q},$$

which lies between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. Accordingly, string theory is not the final ontological theory. However, this ontological positioning alone is not sufficient to establish a physical theory. Since Future Convergence Theory contains string theory, it must explicitly specify what corrections are added to the fundamental actions of string theory. In this section, we derive the Future Convergence Theory correction terms to the Nambu–Goto action and the Polyakov action.

Standard String Actions

In classical string theory, the Nambu–Goto action is defined as the action that minimizes the area of the worldsheet:

$$S_{\text{NG}} = -T \int d^2\sigma \sqrt{-\det h_{ab}}$$

where

$$T$$

is the string tension, and

$$h_{ab}$$

is the induced worldsheet metric. In quantum theory, the Polyakov action is commonly used:

$$S_{\text{P}} = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X^\nu g_{\mu\nu}$$

The Action in Future Convergence Theory

Within Future Convergence Theory, the worldsheet is not determined by the background space-time alone. As shown in Part III, the Quantum Permission Layer itself is constrained by consistency conditions with the Future Causal Information Field. Therefore, the action of the string receives corrections originating from the Future Causal Information Field.

Future Information Correction

Let the Future Causal Information Field be denoted by

$$\mathcal{F}.$$

The action in Future Convergence Theory is defined as

$$S_{\text{FCT}} = S_{\text{P}} + S_F$$

where

$$S_F$$

is the correction action describing consistency with the Future Causal Information Field.

General Form of the Correction Action

The correction action includes couplings to the Future Causal Information Field, the Manifestation Constraint Tensor, and the Meaning Generation Template. In the most general form, it may be written as

$$S_F = \int d^2\sigma \sqrt{-\gamma} \mathcal{L}_F(\mathcal{F}, \mathcal{C}_{\mu\nu}, \Sigma)$$

where

$$\Sigma$$

denotes the Meaning Generation Template.

Weak-Constraint Limit

When the Manifestation Constraint Density

$$\Lambda_{\text{M}}$$

is sufficiently small, the contribution from Future Information vanishes. That is,

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} S_F = 0$$

Accordingly, Future Convergence Theory reduces to the ordinary Polyakov action.

Strong-Constraint Limit

Near black holes, during the early universe, and in regions of high curvature, the contribution of Future Information becomes finite. Consequently, the action can be expanded as

$$S_{\text{FCT}} = S_{\text{P}} + \alpha S_F^{(1)} + \beta S_F^{(2)} + \mathcal{O}(\Lambda_{\text{M}}^3)$$

where

$$S_F^{(1)}$$

and

$$S_F^{(2)}$$

are the first- and second-order correction actions, respectively.

Action Principle

The equations of motion in Future Convergence Theory are obtained from the stationary-action condition

$$\boxed{\delta S_{\text{FCT}} = 0}$$

including the correction action. Thus, the motion of the string is governed not only by the background spacetime but also by the consistency condition with the Future Causal Information Field.

FCT Corrected-Action Theorem

The preceding discussion establishes the following theorem.

Theorem 5.2 (FCT Corrected-Action Theorem). *Within Future Convergence Theory, the motion of a string is not determined by the Polyakov action alone. The action acquires a correction term*

$$S_F$$

that represents consistency with the Future Causal Information Field, the Manifestation Constraint, and the Meaning Generation Template. In the weak-constraint limit, this correction vanishes and the theory reduces to ordinary string theory.

Proof. From Part III, the Quantum Permission Layer is defined by consistency with the Future Causal Information Field. String theory describes the internal degrees of freedom of this Quantum Permission Layer. Therefore, its action cannot be determined solely by the background metric; it must include coupling terms to the Future Causal Information Field. In the weak-constraint limit, these contributions vanish, and the Polyakov action is recovered. $\square$

Conclusion of This Section

In this section, we showed that Future Convergence Theory introduces the correction action

$$S_F$$

to the Polyakov and Nambu–Goto actions as the consistency condition with the Future Causal Information Field. Consequently, the motion of strings is affected not only by the background spacetime but also by the Future Causal Information Field and the Manifestation Constraint. In the next section, we define the effective background metric derived from this corrected action and determine how the spacetime structure experienced by strings differs from the metric of General Relativity.

5.0.3 5.3 Effective Background Metric

In the previous section, we showed that within Future Convergence Theory the fundamental action of string theory is not closed by the Polyakov action alone, but also contains the correction action

$$S_F,$$

which represents consistency with the Future Causal Information Field, the Manifestation Constraint, and the Meaning Generation Template. If the action is modified, however, then the background spacetime experienced by the string itself can no longer coincide exactly with the metric of General Relativity. In conventional string theory, the background spacetime is specified by the metric

$$g_{\mu\nu},$$

and the string equations of motion, conformal invariance, beta functions, and scattering amplitudes are all constructed on the basis of this background metric. Within Future Convergence Theory, however, the metric of General Relativity was shown in Part II to be only the low-energy projection of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Accordingly, strings existing in the Quantum Permission Layer experience not the General Relativistic metric itself but an effective background that contains the Manifestation Constraint.

Definition of the Effective Background Metric

Future Convergence Theory defines the background metric experienced by strings as the **Effective Background Metric**, denoted by

$$g_{\mu\nu}^{\text{eff}}.$$

This metric consists of the ordinary spacetime metric together with the Manifestation Constraint correction:

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu} + \Delta g_{\mu\nu}^{\text{FCT}}$$

where

$$\Delta g_{\mu\nu}^{\text{FCT}}$$

is the background correction specific to Future Convergence Theory.

Relation to the Manifestation Constraint

The effective background metric is determined by the Manifestation Constraint Tensor. In the most general form,

$$\Delta g_{\mu\nu}^{\text{FCT}} = F_{\mu\nu}(\mathcal{C}_{\alpha\beta}, \Lambda_{\text{M}})$$

where

$$F_{\mu\nu}$$

is the correction function mapping the Manifestation Constraint into the effective metric.

Contribution of Future Information

Within Future Convergence Theory, the Manifestation Constraint is itself derived from the Future Causal Information Field

$$\mathcal{F}.$$

Consequently, the effective background metric also satisfies the consistency condition with the Future Causal Information Field:

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu}^{\text{eff}}(\mathcal{F}, \mathcal{C}_{\mu\nu})$$

This implies that the background spacetime is not determined solely by the local distribution of matter and energy.

Incorporation into the Worldsheet Action

The Polyakov action introduced in the previous section is rewritten within Future Convergence Theory by employing the effective background metric:

$$S_{\text{FCT}} = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X^\nu g_{\mu\nu}^{\text{eff}}$$

Thus, the influence of Future Information is absorbed into the background metric itself.

Weak-Constraint Limit

When the Manifestation Constraint Density is sufficiently small, the Future Information correction vanishes. Therefore,

$$\lim_{\Lambda_M \rightarrow 0} g_{\mu\nu}^{\text{eff}} = g_{\mu\nu}$$

Hence, ordinary string theory is recovered exactly as the low-constraint limit of Future Convergence Theory.

Strong-Constraint Limit

Near black holes, during the early universe, and in regions of extremely high curvature, the Manifestation Constraint Density becomes significant. The effective background metric may then be expanded as

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu} + \alpha \mathcal{C}_{\mu\nu} + \beta \mathcal{C}_{\mu\lambda} \mathcal{C}^\lambda{}_\nu + \mathcal{O}(\Lambda_M^3)$$

where

$$\alpha, \beta$$

are coupling constants specific to Future Convergence Theory.

Effective Background Metric Theorem

The preceding discussion establishes the following theorem.

Theorem 5.3 (Effective Background Metric Theorem). *Within Future Convergence Theory, the background spacetime experienced by a string is not the metric of General Relativity itself but the effective background metric*

$$g_{\mu\nu}^{\text{eff}},$$

which is corrected by the Manifestation Constraint Tensor. In the weak-constraint limit, this metric reduces to the General Relativistic metric.

Proof. From the previous section, the string action contains the correction action

$$S_F$$

originating from Future Information. Since the background metric appears as a coefficient in the action principle, the correction action may be expressed equivalently as a correction to the background metric. In the weak-constraint limit, the Future Information contribution vanishes, and so does the metric correction. Therefore, the effective background metric reduces to the General Relativistic metric. $\square$

Conclusion of This Section

In this section, we defined the spacetime experienced by strings within Future Convergence Theory as the effective background metric corrected by the Manifestation Constraint Tensor. Consequently, string motion is determined not in the ordinary spacetime of General Relativity but in a background geometry reflecting both the Future Causal Information Field and the Manifestation Constraint. In the next section, we derive how the vibrational modes of strings are modified under this effective background metric.

5.0.4 5.4 Corrections to Vibrational Modes

In the previous section, we showed that within Future Convergence Theory the background spacetime experienced by strings is not the General Relativistic metric

$$g_{\mu\nu},$$

but the effective background metric

$$g_{\mu\nu}^{\text{eff}},$$

which is corrected by the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

and the Future Causal Information Field. Consequently, not only the equations of motion but also the vibrational modes of strings receive corrections predicted by Future Convergence Theory. In conventional string theory, every particle species is represented by a different vibrational mode of the string. Therefore, if the vibrational modes are modified, the particle masses, interaction strengths, and even the graviton itself receive corresponding corrections.

Vibrational Modes in Standard String Theory

The coordinate of a closed string is expressed as

$$X^\mu(\tau, \sigma),$$

whose general solution is

$$X^\mu = x^\mu + 2\alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \left(\alpha_n^\mu e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)} \right)$$

where

$$\alpha_n^\mu, \quad \tilde{\alpha}_n^\mu$$

are the left- and right-moving oscillator operators.

Corrections in Future Convergence Theory

Within Future Convergence Theory, string oscillations are defined with respect to the effective background metric

$$g_{\mu\nu}^{\text{eff}}.$$

Consequently, each vibrational mode is influenced by the Manifestation Constraint Density

$$\Lambda_M.$$

Future Convergence Theory therefore replaces the oscillator operators by

$$\alpha_n^\mu \longrightarrow \alpha_n^\mu + \delta\alpha_n^\mu$$

where

$$\delta\alpha_n^\mu$$

denotes the correction induced by the Manifestation Constraint.

General Form of the Correction

The correction depends upon the Future Causal Information Field, the Manifestation Constraint, and the Meaning Generation Template. In the most general form,

$$\delta\alpha_n^\mu = F_n^\mu(\mathcal{F}, \mathcal{C}_{\mu\nu}, \Sigma)$$

where

$$F_n^\mu$$

is the correction function characteristic of Future Convergence Theory.

Frequency Corrections

The eigenfrequency

$$\omega_n$$

of each vibrational mode is likewise modified by the effective background metric. Future Convergence Theory predicts

$$\omega_n^{\text{FCT}} = \omega_n^{\text{ST}} + \Delta\omega_n$$

where

$$\omega_n^{\text{ST}}$$

is the eigenfrequency in ordinary string theory, and

$$\Delta\omega_n$$

is the correction arising from Future Information.

Weak-Constraint Limit

When the Manifestation Constraint Density is sufficiently small, the Future Information correction disappears. Thus,

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} \delta\alpha_n^\mu = 0$$

and

$$\lim_{\Lambda_{\text{M}} \rightarrow 0} \Delta\omega_n = 0$$

Consequently, ordinary string theory is recovered exactly.

Strong-Constraint Limit

Near black holes and during the earliest stages of the universe, the Manifestation Constraint Density becomes finite. Future Convergence Theory predicts

$$\Delta\omega_n = \alpha\Lambda_{\text{M}}n + \beta\Lambda_{\text{M}}^2n^2 + \mathcal{O}(\Lambda_{\text{M}}^3)$$

where

$$\alpha, \beta$$

are coupling constants specific to Future Convergence Theory.

Vibrational Mode Correction Theorem

The preceding discussion establishes the following theorem.

Theorem 5.4 (Vibrational Mode Correction Theorem). *Within Future Convergence Theory, the vibrational modes of strings are modified by the Manifestation Constraint Tensor and the Future Causal Information Field. Consequently, the eigenfrequencies of the vibrational modes receive corrections proportional to the Manifestation Constraint Density. In the weak-constraint limit, these corrections vanish, and ordinary string theory is recovered.*

Proof. From the previous section, string motion is determined in the effective background metric

$$g_{\mu\nu}^{\text{eff}}.$$

Since the equations of motion are modified, their eigenfunctions—the vibrational modes—must also be modified. Whenever the Manifestation Constraint Density is finite, correction terms appear in the associated eigenvalue problem. Conversely, in the weak-constraint limit the effective background metric reduces to the ordinary metric, and the vibrational-mode corrections vanish. Hence, Future Convergence Theory contains ordinary string theory as its low-constraint limit. $\square$

Observational Consequences

The vibrational-mode corrections predicted by Future Convergence Theory are not observable at ordinary energy scales. However, near the quantum-gravity scale, in ultra-high-energy scattering, near black holes, or in the early universe, they may appear as small corrections to mass spectra, scattering amplitudes, graviton excitations, and string resonances.

Conclusion of This Section

In this section, we showed that the vibrational modes of strings are modified by the Manifestation Constraint Tensor and the Future Causal Information Field. Accordingly, Future Information affects not only the background geometry but also the internal degrees of freedom of strings themselves. In the next section, we formulate the corresponding corrections to the mass spectrum and derive the modifications to particle masses predicted by Future Convergence Theory.

5.0.5 5.5 Corrections to the Mass Spectrum

In the previous section, we showed that within Future Convergence Theory the vibrational modes of strings are modified by the effective background metric, the Manifestation Constraint Tensor, and the Future Causal Information Field. In string theory, the masses of all particles are determined by the eigenvalues of these vibrational modes. Consequently, if the vibrational modes are modified, the mass spectrum itself must also be modified. In this section, we derive the corrections to the mass spectrum predicted by Future Convergence Theory.

Mass Spectrum in Standard String Theory

For closed strings, the mass spectrum is generally given by

$$M^2 = \frac{4}{\alpha'} (N + \tilde{N} - a)$$

where

$$N, \quad \tilde{N}$$

are the left- and right-moving oscillator numbers, and

$$a$$

is the normal-ordering constant. Thus, the particle species are determined by the excitation numbers of the vibrational modes.

Mass Corrections in Future Convergence Theory

As shown in Section 5.4, the vibrational modes are modified by the Manifestation Constraint. Accordingly, the effective eigenvalues associated with the excitation numbers are likewise modified. Future Convergence Theory therefore defines the mass spectrum as

$$M_{\text{FCT}}^2 = M_{\text{ST}}^2 + \Delta M^2$$

where

$$M_{\text{ST}}$$

is the mass predicted by ordinary string theory, and

$$\Delta M^2$$

is the correction arising from Future Information.

General Form of the Correction

Within Future Convergence Theory, the mass correction is determined by the consistency among the Manifestation Constraint Tensor, the Future Causal Information Field, and the Meaning Generation Template. In the most general form,

$$\Delta M^2 = F(\mathcal{F}, \mathcal{C}_{\mu\nu}, \Sigma, N, \tilde{N})$$

where

$$F$$

is the correction function specific to Future Convergence Theory.

Dependence on the Manifestation Constraint Density

When the Manifestation Constraint Density

$$\Lambda_{\text{M}}$$

is finite, the mass correction is proportional to its magnitude. Accordingly,

$$\Delta M^2 = \alpha \Lambda_{\text{M}} + \beta \Lambda_{\text{M}}^2 + \mathcal{O}(\Lambda_{\text{M}}^3)$$

where

$$\alpha, \beta$$

are coupling constants characteristic of Future Convergence Theory.

Breaking of Mass Degeneracy

Ordinary string theory contains mass levels that remain degenerate owing to symmetry. Within Future Convergence Theory, however, the Manifestation Constraint introduces anisotropy into the background, so that the degeneracy is generally not preserved. Consequently, even states possessing identical excitation numbers may satisfy

$$M_i \neq M_j$$

This lifting of degeneracy becomes increasingly significant in the high-energy regime.

Weak-Constraint Limit

Under ordinary cosmological conditions, the Manifestation Constraint Density is extremely small. Therefore,

$$\lim_{\Lambda_M \rightarrow 0} \Delta M^2 = 0$$

and Future Convergence Theory reduces exactly to ordinary string theory.

Strong-Constraint Limit

Near black holes, during the early universe, and in regions of extremely high curvature, the Manifestation Constraint Density becomes finite. Accordingly, the mass spectrum differs from that predicted by ordinary string theory. Future Convergence Theory predicts that highly excited states receive larger mass corrections. Such corrections may eventually appear as small shifts in resonance positions during ultra-high-energy scattering experiments or future quantum-gravity measurements.

Mass Spectrum Correction Theorem

The preceding discussion establishes the following theorem.

Theorem 5.5 (Mass Spectrum Correction Theorem). *Within Future Convergence Theory, the string mass spectrum is modified by the Manifestation Constraint Tensor and the Future Causal Information Field. Consequently, the high-constraint regime exhibits a mass spectrum differing from that of ordinary string theory. In the weak-constraint limit, the ordinary string-theory spectrum is recovered.*

Proof. Section 5.4 showed that the Manifestation Constraint modifies the vibrational modes of strings. Since the mass spectrum is determined by the eigenvalues of these vibrational modes, corrections to the eigenvalues are directly reflected as corrections to the particle masses. In the weak-constraint limit, the vibrational-mode corrections disappear, and therefore the mass corrections vanish as well. Hence, Future Convergence Theory naturally contains ordinary string theory as its limiting case. $\square$

Observational Consequences

The mass-spectrum corrections predicted by Future Convergence Theory are expected to be too small to detect at present accelerator energies. However, near the quantum-gravity scale, during ultra-high-energy scattering, in black-hole formation events, or in particle spectra originating from the early universe, they may appear as small violations of mass degeneracy or as minute shifts in resonance positions.

Conclusion of This Section

In this section, we showed that the string mass spectrum is modified by the Manifestation Constraint Tensor and the Future Causal Information Field. Accordingly, Future Information influences not only the vibrational modes of strings but also the particle masses themselves. In the next section, we derive how the compactification conditions are modified within Future Convergence Theory in light of these mass-spectrum corrections.

5.0.6 5.6 Compactification Conditions

In the previous section, we showed that within Future Convergence Theory the corrections to the vibrational modes modify the particle mass spectrum itself. In string theory, however, particle

spectra are determined not only by the vibrational modes but also by the compactification structure of the extra dimensions. In conventional string theory, the ten- or eleven-dimensional spacetime is compactified on internal spaces such as Calabi–Yau manifolds, yielding an effective four-dimensional theory. Within Future Convergence Theory, however, the geometry of the extra dimensions is itself determined by the consistency conditions imposed by the Future Causal Information Field and the Manifestation Constraint. Compactification is therefore not merely a geometric degree of freedom but the result of an ontological constraint.

Compactification in Standard String Theory

Ordinary string theory decomposes spacetime as

$$M^{10} = M^4 \times K^6$$

where

$$M^4$$

is four-dimensional spacetime, and

$$K^6$$

is the six-dimensional compact internal space. The physical constants, gauge groups, and the number of fermion generations are determined by the geometry of

$$K^6.$$

Redefinition in Future Convergence Theory

Within Future Convergence Theory, the compact internal space is not chosen arbitrarily. Only those geometries satisfying the consistency conditions with the Future Causal Information Field

$$\mathcal{F}$$

can become physically manifest. Accordingly,

$$K^6 = K^6(\mathcal{F}, \mathcal{C}_{\mu\nu})$$

That is, the extra dimensions constitute an effective geometry constrained by the Future Causal Information Field and the Manifestation Constraint.

Manifestation Constraint Condition

Within Future Convergence Theory, an admissible compactification must satisfy

$$\mathcal{K}(K^6, \mathcal{C}_{\mu\nu}, \mathcal{F}) = 0$$

where

$$\mathcal{K}$$

denotes the compactification constraint operator of Future Convergence Theory. Any internal geometry violating this condition is ontologically incapable of manifestation.

Effective Compactification

The effective four-dimensional theory is therefore obtained not merely through dimensional reduction but through projection incorporating Future Information:

$$M^4 = \Pi(M^{10}, \mathcal{F})$$

where

$$\Pi$$

is the Manifestation Projection Operator.

Weak-Constraint Limit

When the Manifestation Constraint Density becomes sufficiently small, the Future Information contribution vanishes. Thus,

$$\lim_{\Lambda_M \rightarrow 0} K^6(\mathcal{F}, \mathcal{C}_{\mu\nu}) = K^6$$

and Future Convergence Theory reduces completely to the conventional theory of compactification.

Strong-Constraint Limit

Near black holes, in the early universe, or in highly curved spacetime regions, the Manifestation Constraint Density becomes finite. The effective internal geometry then receives corrections of the form

$$\Delta K^6 = F(\Lambda_M, \mathcal{F})$$

As a consequence, effective coupling constants, gauge symmetries, mass spectra, and vacuum structures all receive small corrections.

Compactification Condition Theorem

The preceding discussion establishes the following theorem.

Theorem 5.6 (Compactification Condition Theorem). *Within Future Convergence Theory, compactification of the extra dimensions is not determined solely by geometric degrees of freedom. Only those internal spaces satisfying the consistency conditions imposed by the Future Causal Information Field and the Manifestation Constraint are ontologically admissible. In the weak-constraint limit, the theory reduces to ordinary string-theory compactification.*

Proof. Section 5.1 established that string theory describes the internal degrees of freedom of the Quantum Permission Layer. Part III showed that the Quantum Permission Layer itself is defined by its consistency with the Future Causal Information Field. Therefore, its internal geometry cannot be determined independently of the Future Causal Information Field and the Manifestation Constraint. In the weak-constraint limit, the Future Information contribution disappears, recovering the conventional compactification theory. $\square$

Observational Consequences

The compactification corrections predicted by Future Convergence Theory are not expected to be observable at present energy scales. However, future quantum-gravity experiments, ultra-high-energy scattering, the early universe, or black-hole interiors may reveal them through small variations in effective coupling constants, mass-spectrum corrections, or changes in vacuum structure.

Conclusion of This Section

In this section, we showed that within Future Convergence Theory the compactification of extra dimensions is determined not only by geometry but also by the consistency conditions imposed by the Future Causal Information Field and the Manifestation Constraint. Compactification is therefore redefined not as a purely mathematical degree of freedom but as a structure arising from ontological constraint conditions. In the next section, we reformulate the vacuum-selection problem of string theory from the standpoint of Future Convergence Theory.

5.0.7 5.7 The Vacuum Selection Problem

In the previous section, we showed that within Future Convergence Theory the compactification of the extra dimensions is determined by the consistency conditions imposed by the Future Causal Information Field

$$\mathcal{F}$$

and the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Consequently, compactification is no longer viewed as a purely geometrical degree of freedom but as the result of ontological consistency conditions. Nevertheless, string theory still contains one of its most fundamental unsolved problems: the **Vacuum Selection Problem**. In conventional string theory, different choices of compactification, flux configurations, and D-brane arrangements generate an enormous number of possible vacuum states. Current estimates suggest that the number of such vacua is approximately

$$10^{500}.$$

However, conventional string theory alone does not explain why our Universe realizes only one particular vacuum among this vast landscape. Future Convergence Theory reformulates this issue as an ontological selection problem.

Vacua in Standard String Theory

In ordinary string theory, each vacuum state is denoted by

$$V_i.$$

The complete vacuum set is therefore

$$\boxed{\mathcal{V} = \{V_1, V_2, \dots, V_N\}}$$

where

$$N \sim 10^{500}.$$

Within the conventional framework, every one of these vacua is regarded as mathematically admissible.

The Position of Future Convergence Theory

Within Future Convergence Theory, however, mathematical existence and ontological manifestation are fundamentally different concepts. As demonstrated in Part III, only quantum states satisfying consistency with the Future Causal Information Field are permitted to manifest into the Three-Dimensional Manifestation Layer. Exactly the same restriction applies to vacuum states.

Vacuum Manifestation Condition

Future Convergence Theory defines the condition for a vacuum state

$$V_i$$

to become physically manifest by

$$\boxed{\Gamma(V_i, \mathcal{F}, \mathcal{C}_{\mu\nu}) = 1}$$

where

$$\Gamma$$

is the Vacuum Manifestation Operator. Conversely, for vacuum states that fail to satisfy the consistency condition,

$$\boxed{\Gamma = 0}$$

Although such vacua may exist mathematically, they never become ontologically manifest.

Reduction of the Vacuum Landscape

Accordingly, the physically realizable vacuum set becomes

$$\boxed{\mathcal{V}_{\text{phys}} \subset \mathcal{V}}$$

Thus, the entire Landscape is not physically realized. The consistency conditions imposed by the Future Causal Information Field reduce the enormous Landscape to a physically admissible vacuum subset.

Relation to the Future Fixed Point

The Future Fixed Point introduced in Part I,

$$\mathcal{F}^*,$$

specifies the global convergence condition of existence itself. Therefore, admissible vacuum states must also satisfy consistency with the Future Fixed Point:

$$\boxed{V_i \longrightarrow \mathcal{F}^*}$$

Only those vacua satisfying this convergence condition are ontologically permitted.

Weak-Constraint Limit

When the Future Information correction becomes sufficiently small, the Manifestation Constraint exerts negligible influence upon vacuum selection. Consequently,

$$\boxed{\Gamma \rightarrow 1}$$

and Future Convergence Theory reproduces the conventional vacuum landscape.

Strong-Constraint Limit

As the Manifestation Constraint Density increases, the vacuum-selection condition becomes progressively stronger. Most of the Landscape is then excluded ontologically. Future Convergence Theory predicts that, in this limit, only a very small subset of physically realizable vacua survives.

Vacuum Selection Theorem

The foregoing discussion establishes the following theorem.

Theorem 5.7 (Vacuum Selection Theorem). *Within Future Convergence Theory, not every mathematically possible vacuum becomes physically manifest. Only those vacua satisfying consistency with the Future Causal Information Field, the Manifestation Constraint, and the Future Fixed Point are ontologically admissible.*

Proof. Part III established that only quantum states satisfying consistency with the Future Causal Information Field are permitted to manifest within the Quantum Permission Layer. Since vacuum states are themselves quantum degrees of freedom, they must obey the same manifestation condition. Furthermore, Part I established that all manifestations converge toward the Future Fixed Point. Therefore, vacuum states inconsistent with the Future Fixed Point cannot become manifest. Accordingly, the vacuum landscape is reduced by ontological constraints. $\square$

Observational Consequences

If Future Convergence Theory is correct, future developments in quantum gravity or cosmological observations should indicate that only a highly restricted subset of the vacuum landscape is physically realizable. This prediction may be tested indirectly through constraints on the vacuum energy, low-energy effective theories, the cosmological constant, and the symmetry structure of the early universe.

Conclusion of This Section

In this section, we reformulated the vacuum selection problem as a problem of ontological manifestation. Consequently, the enormous vacuum landscape of string theory is reduced, through consistency with the Future Causal Information Field, the Manifestation Constraint, and the Future Fixed Point, to the set of physically realizable vacua. In the next section, we apply these vacuum-selection conditions to Calabi–Yau manifolds and derive the stability conditions predicted by Future Convergence Theory.

5.0.8 5.8 Stability Conditions for Calabi–Yau Manifolds

In the previous section, we showed that within Future Convergence Theory the vacuum selection problem of string theory is determined by consistency with the Future Causal Information Field

$$\mathcal{F}$$

and the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Consequently, not every mathematically admissible vacuum is physically realized. Only vacua consistent with the Future Fixed Point are projected into the Three-Dimensional Manifestation Layer. If the vacuum itself is selected in this manner, then the internal space upon which it is built—namely the Calabi–Yau manifold—must likewise satisfy additional stability conditions within Future Convergence Theory. In conventional string theory, Calabi–Yau manifolds are characterized by Ricci-flatness, Kähler structure, and SU(3) holonomy. Within Future Convergence Theory, however, these become necessary but not sufficient conditions. Ontological consistency with the Future Causal Information Field is required in addition.

Stability Conditions in Standard String Theory

In ordinary string theory, a Calabi–Yau manifold

$$\mathcal{M}_{CY}$$

must satisfy at least

$$\boxed{R_{mn} = 0}$$

where

$$R_{mn}$$

is the Ricci tensor of the internal space. Furthermore, the Kähler form

$$J$$

satisfies

$$dJ = 0,$$

and the holonomy group is

$$SU(3).$$

These conditions preserve supersymmetry.

Redefinition in Future Convergence Theory

Within Future Convergence Theory, Ricci-flatness alone does not guarantee physical stability. The internal geometry must also satisfy consistency with the Future Causal Information Field. Accordingly, the stability condition is defined by

$$\boxed{\mathcal{S}(\mathcal{M}_{CY}, \mathcal{F}, \mathcal{C}_{\mu\nu}) = 0}$$

where

$$\mathcal{S}$$

is the Stability Operator of Future Convergence Theory.

Consistency with the Future Fixed Point

The internal geometry must also be consistent with the Future Fixed Point

$$\mathcal{F}^*.$$

Hence, every admissible Calabi–Yau manifold must satisfy

$$\boxed{\mathcal{M}_{CY} \longrightarrow \mathcal{F}^*}$$

Internal spaces failing to satisfy this convergence condition may exist mathematically but never become ontologically manifest.

Stability Function

Future Convergence Theory evaluates the stability of the internal space through the stability function

$$\Xi.$$

Specifically,

$$\boxed{\Xi = \Xi(R_{mn}, \mathcal{F}, \mathcal{C}_{\mu\nu}, \Sigma)}$$

where

$$\Sigma$$

denotes the Meaning Generation Template. Future Convergence Theory defines only those internal spaces satisfying

$$\boxed{\Xi = 0}$$

to be ontologically stable.

Weak-Constraint Limit

When the Manifestation Constraint Density becomes sufficiently small, the Future Information contribution disappears. Accordingly,

$$\lim_{\Lambda_M \rightarrow 0} \Xi = R_{mn}$$

Thus, Future Convergence Theory reproduces the Ricci-flat condition and reduces to the conventional Calabi–Yau framework.

Strong-Constraint Limit

Near black holes, during the early universe, or within highly curved regions, the Future Information contribution becomes finite. Consequently, Ricci-flatness alone no longer determines stability. Future Convergence Theory predicts that Calabi–Yau manifolds inconsistent with the Future Fixed Point become ontologically unstable.

Calabi–Yau Stability Theorem

The foregoing discussion establishes the following theorem.

Theorem 5.8 (Calabi–Yau Stability Theorem). *Within Future Convergence Theory, Ricci-flatness alone is insufficient for a Calabi–Yau manifold to serve as the physical internal space. Only those manifolds satisfying consistency with the Future Causal Information Field, the Manifestation Constraint, and the Future Fixed Point are ontologically stable.*

Proof. Sections 5.6 and 5.7 established that both the internal geometry and the vacuum state are determined by consistency with the Future Causal Information Field. Since Calabi–Yau manifolds constitute this internal geometry, they are subject to the same ontological constraints. Ricci-flatness alone does not guarantee consistency with the Future Fixed Point. Therefore, only those internal spaces satisfying the additional Future Information stability condition are ontologically admissible. $\square$

Observational Consequences

If Future Convergence Theory is correct, future developments in string theory and quantum gravity should reveal that internal geometries cannot be classified solely by Ricci-flatness. Additional constraints associated with the Future Fixed Point should become necessary. These corrections may appear as small modifications of effective coupling constants, supersymmetry breaking, moduli stabilization, the cosmological constant, and low-energy effective theories describing the early universe.

Conclusion of This Section

In this section, we redefined the stability conditions for Calabi–Yau manifolds as ontological consistency conditions involving not only Ricci-flatness but also the Future Causal Information Field, the Manifestation Constraint, and the Future Fixed Point. Accordingly, the internal geometry itself is selected by the hierarchical structure of Future Convergence Theory. In the next section, we summarize these results and discuss both the observability of string theory and its theoretical limitations within the framework of Future Convergence Theory.

5.0.9 5.9 Observability and Theoretical Limits

In the previous section, we showed that within Future Convergence Theory a Calabi–Yau manifold becomes ontologically stable only when it satisfies consistency with the Future Causal Information Field

$$\mathcal{F},$$

the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

and the Future Fixed Point

$$\mathcal{F}^*.$$

Accordingly, Future Convergence Theory redefines the internal geometry of string theory as part of a broader ontological structure. For a theory to qualify as a scientific theory, however, it must clearly distinguish between what is, in principle, observable and what lies beyond direct observation. In this section, we formulate both the observability of string theory and its theoretical limits within the framework of Future Convergence Theory.

Observable Quantities

Within Future Convergence Theory, the Future Causal Information Field itself is never observed directly. Instead, observations are limited to the consequences produced after the Future Causal Information Field is projected through the Quantum Permission Layer into the Three-Dimensional Manifestation Layer. Accordingly, observable quantities are generally expressed as

$$\boxed{\mathcal{O} = \Pi_X(\mathcal{Q})}$$

where

$$\Pi_X$$

denotes the Manifestation Projection Operator onto the Three-Dimensional Manifestation Layer.

Objects That Cannot Be Observed Directly

Within Future Convergence Theory, the following entities are, in principle, not directly observable:

- the Future Causal Information Field

$$\mathcal{F},$$

- the Meaning Generation Template

$$\Sigma,$$

- the Future Fixed Point

$$\mathcal{F}^*,$$

- the Manifestation Constraint Tensor itself

$$\mathcal{C}_{\mu\nu}.$$

These objects constitute the ontological foundation of the theory rather than its direct observables.

Indirect Observation

Although these ontological entities cannot be observed directly, their existence may be tested indirectly through corrections to measurable physical quantities. Examples include

$$\Delta\omega, \quad \Delta M, \quad \Delta\sigma, \quad \Delta\phi,$$

which are interpreted as projections of the underlying ontological structure predicted by Future Convergence Theory. Accordingly, Future Convergence Theory is formulated not as a theory of direct observation but as a **theory of differential observables**.

Observational Limits of String Theory

To date, strings themselves have never been observed directly. Within Future Convergence Theory, this is not regarded as accidental. Strings belong to the Quantum Permission Layer, and only their effective degrees of freedom become observable within the Three-Dimensional Manifestation Layer. Consequently, what is experimentally observed are particles such as electrons, photons, and gravitons rather than strings themselves.

Theoretical Limits of Future Convergence Theory

Future Convergence Theory does not claim that the entirety of existence can be derived mathematically. The Future Causal Information Field itself is introduced as a primitive axiomatic object within the theory. Therefore, the theoretical limit of Future Convergence Theory lies in the fact that it does not derive the existence of the Future Causal Information Field from an even more fundamental theory. This situation is analogous to General Relativity, which introduces spacetime as a primitive concept, and quantum mechanics, which introduces Hilbert space axiomatically.

Observability Theorem

The preceding discussion establishes the following theorem.

Theorem 5.9 (Observability Theorem). *Within Future Convergence Theory, the Future Causal Information Field, the Meaning Generation Template, the Future Fixed Point, and the Manifestation Constraint itself are not directly observable. Only the physical quantities produced after these ontological entities are projected through the Quantum Permission Layer into the Three-Dimensional Manifestation Layer are experimentally observable.*

Proof. Part I defines the Future Causal Information Field as the ontological information field that exists prior to manifestation. Part III establishes that every observable state in the Three-Dimensional Manifestation Layer arises only after passing through the Quantum Permission Layer. Accordingly, the ontological foundation itself is not directly observable. Only its manifested physical consequences can be measured. $\square$

Conclusion of Part V

In this chapter, we repositioned string theory within the ontological framework of Future Convergence Theory. As a consequence, string theory is identified not as the ultimate ontological theory but as a theory describing the internal structure of the Quantum Permission Layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer. Furthermore, we reformulated the string action, the effective background metric, vibrational modes, mass spectra, compactification, vacuum selection, and the stability conditions of Calabi–Yau manifolds in a unified manner within Future Convergence Theory. Finally, we demonstrated that the ontological entities themselves are not directly observable; only the corrections they induce

in measurable physical quantities can be tested experimentally. In this way, Future Convergence Theory is formulated as a falsifiable physical theory while simultaneously providing an ontological framework that contains string theory as an effective theory. In the next chapter, dark matter is redefined not as an unknown particle but as a **Non-Optical Manifestation Residue** arising from the Manifestation Constraint. On the basis of this reinterpretation, we derive the cosmological predictions of Future Convergence Theory through comparisons with galactic rotation curves, gravitational lensing, galaxy-cluster collisions, and the cosmic microwave background.

Part VI

Redefinition of Dark Matter and Observational Predictions

Chapter 6

Redefinition of Dark Matter and Observational Predictions

In Part I, we introduced the ontological foundation of Future Convergence Theory, namely, the Meaning Generation Template, the Future Causal Information Field, the Quantum Permission Layer, the Three-Dimensional Manifestation Layer, and the FCT Closure Equation. In Part II, gravity was redefined not as an interparticle interaction but as the **Manifestation Constraint** that maintains consistency between the Future Causal Information Field and the Three-Dimensional Manifestation Layer. In Part III, quantum theory was positioned as the ontological intermediate layer connecting the Future Causal Information Field and the Three-Dimensional Manifestation Layer, and the wave function, measurement, decoherence, vacuum fluctuations, the Casimir effect, the Delayed Choice Experiment, and the Quantum Eraser were explained in a unified manner through manifestation-permitted states and the Manifestation Projection. In Part IV, the graviton was redefined not as gravity itself but as a local quantum excitation of the Manifestation Constraint Tensor, thereby separating the ontological roles of gravity and quantum gravity. Furthermore, in Part V, string theory was positioned not as the final ontological theory of Future Convergence Theory but as an effective theory describing the Quantum Permission Layer. Its action, background metric, vibrational modes, mass spectrum, compactification, vacuum selection, and Calabi–Yau manifolds were reconstructed within the framework of Future Convergence Theory. We now turn to one of the most important unresolved problems in cosmology: **dark matter**. In the standard cosmological model, dark matter is treated as an unknown particle component that interacts gravitationally but possesses little or no electromagnetic interaction. Many observations, including galactic rotation curves, galaxy-cluster dynamics, gravitational lensing, the cosmic microwave background, and large-scale structure formation, indicate the existence of gravitational sources that cannot be explained by ordinary visible matter alone. However, dark matter particles have not yet been directly detected. Future Convergence Theory redefines this situation not as a problem of searching for an unknown particle, but as a problem of ontology. As shown in Part II, gravity is described by the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

Part III further showed that quantum states manifest from the Future Causal Information Field through the Quantum Permission Layer into the Three-Dimensional Manifestation Layer. From this standpoint, an entity acting as a gravitational source need not necessarily be optically manifest. Future Convergence Theory allows informational states that contribute to the Manifestation Constraint while not manifesting as optical degrees of freedom within the Three-Dimensional Manifestation Layer. In this work, such states are defined as **Non-Optical Manifestation Residues**. Accordingly, within Future Convergence Theory, dark matter is not an unknown particle. It is an ontological residue originating from the Future Causal Information Field, contributing to the Manifestation Constraint, but lacking electromagnetic man-

ifestation. Once this standpoint is adopted, it becomes unnecessary to introduce a new dark matter particle. The additional gravity observed in cosmology arises naturally from Non-Optical Manifestation Residues contributing to the Manifestation Constraint Tensor. In this chapter, we first summarize the earlier concept of dark matter within Future Convergence Theory and then introduce its new definition as a Non-Optical Manifestation Residue. We further derive the effective energy-momentum tensor, galactic rotation curves, galactic halos, galaxy-cluster collisions, gravitational lensing, and corrections to the cosmic microwave background. Finally, we formulate the differences between Future Convergence Theory and standard

ACDM

cosmology, together with observational falsifiability conditions.

Part VI

Redefinition of Dark Matter and Observational Predictions

6.0.1 6.1 Dark Matter in Earlier Future Convergence Theory

In the early formulation of Future Convergence Theory, dark matter was understood not as an unknown particle acting as a gravitational source, but as an ontological constraint term compensating for the insufficiency of the Manifestation Constraint Tensor. That is, gravity is not determined solely by matter but also by consistency conditions with the Future Causal Information Field. Consequently, gravitational effects that cannot be explained by visible matter alone arise naturally as additional terms in the Manifestation Constraint. From this standpoint, dark matter is treated less as a particle and more as an effective contribution to the Manifestation Constraint. However, this definition alone does not sufficiently explain why dark matter is not observed optically, or why it nevertheless behaves as a gravitational source in galactic rotation curves and gravitational lensing. Therefore, in the present work, Future Convergence Theory further develops the concept of dark matter and introduces a new ontological object: **Non-Optical Manifestation Residue**, which contributes to the Manifestation Constraint. With this object, dark matter is redefined not as an unknown particle but as an electromagnetically non-manifest state that originates from the Future Causal Information Field and contributes to the Manifestation Constraint. In the next section, this new ontological object, the **Non-Optical Manifestation Residue**, is defined mathematically, and its difference from the conventional concept of dark matter is clarified.

6.0.2 6.2 Redefinition of Dark Matter after Paper 92

In the previous section, we showed that in the early formulation of Future Convergence Theory, dark matter was understood as an ontological constraint term contributing to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

However, that definition alone cannot fully explain why dark matter does not exhibit electromagnetic interaction, why it is not detected by particle detectors, and why it behaves only as a gravitational source. In Paper 92, the ontology of Future Convergence Theory is further extended and the definition of dark matter itself is revised. In the present work, dark matter is defined not as an unknown particle, but as an ontological residue that is generated from the Future Causal Information Field, contributes to the Manifestation Constraint, and yet does not manifest as an optical degree of freedom within the Three-Dimensional Manifestation Layer. This is defined as the **Non-Optical Manifestation Residue**.

Difference from the Conventional Definition

In standard cosmology, dark matter is introduced as an unknown particle

$$\chi.$$

Its energy-momentum tensor is usually written as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{vis}} + T_{\mu\nu}^{\chi}$$

That is, a new particle component is added as a gravitational source. Future Convergence Theory, by contrast, introduces no unknown particle. What is added is an ontological residue that contributes to the Manifestation Constraint.

Non-Optical Manifestation Residue

Define the Non-Optical Manifestation Residue by

$$\mathcal{R}_{\text{NO}}.$$

This quantity exists in the Future Causal Information Field and contributes to the Manifestation Constraint Tensor, but it does not manifest as an electromagnetic degree of freedom within the Three-Dimensional Manifestation Layer. Thus,

$$\mathcal{R}_{\text{NO}} \rightarrow \mathcal{C}_{\mu\nu}$$

whereas

$$\mathcal{R}_{\text{NO}} \not\rightarrow A_{\mu}$$

where

$$A_{\mu}$$

denotes the electromagnetic field.

Ontological Position

From Part I, the ontological hierarchy is

$$\mathcal{F} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{X}$$

Ordinary matter passes through all these layers and is projected into the Three-Dimensional Manifestation Layer. By contrast, the Non-Optical Manifestation Residue exists up to the Quantum Permission Layer but does not manifest as an optical degree of freedom. Its ontological position is therefore

$$\mathcal{F} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{C}_{\mu\nu}$$

That is, it contributes to the Manifestation Constraint but does not manifest as an electromagnetic field.

Why It Becomes a Gravitational Source

In General Relativity, spacetime curvature is determined by the energy-momentum tensor. In Future Convergence Theory, the Manifestation Constraint Tensor reflects all ontological information generated from the Future Causal Information Field. Accordingly, the Non-Optical Manifestation Residue also contributes to the Manifestation Constraint. As a result, even though it is not observed through visible light, it acts as a gravitational source.

Why No Unknown Particle Is Required

Within Future Convergence Theory, gravitational effects do not necessarily require electromagnetic manifestation. Therefore, the additional gravity appearing in galactic rotation curves, gravitational lensing, galaxy-cluster masses, and the cosmic microwave background can be explained without introducing unknown particles. What is added is not a new particle, but a Non-Optical Manifestation Residue that contributes to the Manifestation Constraint.

Dark Matter Redefinition Theorem

The preceding discussion establishes the following theorem.

Theorem 6.1 (Dark Matter Redefinition Theorem). *Within Future Convergence Theory, dark matter is not an unknown particle. It is a Non-Optical Manifestation Residue generated from the Future Causal Information Field, contributing to the Manifestation Constraint Tensor, but not manifesting as an electromagnetic degree of freedom. Accordingly, the gravitational effects attributed to dark matter are explained by an ontological residue.*

Proof. From Part II, gravity is described by the Manifestation Constraint Tensor. From Part III, all manifested states are generated from the Future Causal Information Field through the Quantum Permission Layer. Future Convergence Theory does not assume that every such state must possess electromagnetic degrees of freedom. Therefore, informational states can exist that contribute to the Manifestation Constraint while lacking electromagnetic manifestation. Such states are precisely the Non-Optical Manifestation Residues. Thus, additional gravitational sources can be explained without introducing unknown particles. $\square$

Observational Consequences

If Future Convergence Theory is correct, dark matter particles may continue to evade direct detection in the future. At the same time, gravitational lensing, galactic rotation curves, galaxy-cluster collisions, and the cosmic microwave background will continue to exhibit gravitational effects arising from the Manifestation Constraint. This provides an ontological explanation for the present mismatch between particle searches and cosmological observations.

Conclusion of This Section

In this section, following Paper 92, dark matter was redefined not as an unknown particle but as a Non-Optical Manifestation Residue contributing to the Manifestation Constraint. With this redefinition, Future Convergence Theory positions dark matter not as a problem of particle physics but as a problem of ontological manifestation structure. In the next section, we mathematically formulate the Non-Optical Manifestation Residue itself and derive its conservation law and coupling equation with the Manifestation Constraint.

6.0.3 6.3 Non-Optical Manifestation Residue

In the previous section, we redefined dark matter within Future Convergence Theory not as an unknown particle but as a **Non-Optical Manifestation Residue** originating from the Future Causal Information Field and contributing to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

while remaining unmanifested as an electromagnetic degree of freedom. However, for this concept to constitute a physical theory, the Non-Optical Manifestation Residue itself must be defined as a mathematical object, together with its conservation law, time evolution, and coupling to the Manifestation Constraint. In this section, we formulate the Non-Optical Manifestation Residue as an independent ontological field within Future Convergence Theory.

Ontological Definition

Define the Non-Optical Manifestation Residue as

$$\boxed{\mathcal{R}_{\text{NO}}}.$$

It represents the subset of informational states generated from the Future Causal Information Field

$$\mathcal{F}$$

that contribute to the Manifestation Constraint while lacking electromagnetic manifestation. Accordingly,

$$\boxed{\mathcal{R}_{\text{NO}} \subset \mathcal{F}}$$

holds.

Manifestation Condition

Ordinary matter follows the manifestation process

$$\mathcal{F} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{X}.$$

By contrast, the Non-Optical Manifestation Residue exists within the Quantum Permission Layer but is not projected as an electromagnetic field into the Three-Dimensional Manifestation Layer. Its manifestation condition is therefore

$$\boxed{\mathcal{P}_X(\mathcal{R}_{\text{NO}}) = 0}$$

where

$$\mathcal{P}_X$$

denotes the manifestation projection onto the Three-Dimensional Manifestation Layer.

Coupling to the Manifestation Constraint

Although the Non-Optical Manifestation Residue is not observed as an electromagnetic field, it contributes to the Manifestation Constraint Tensor. Its coupling is defined by

$$\boxed{\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}^{(0)} + \kappa \mathcal{R}_{\text{NO}} \Theta_{\mu\nu}}$$

where

$$\mathcal{C}_{\mu\nu}^{(0)}$$

is the Manifestation Constraint generated solely by visible matter,

$$\Theta_{\mu\nu}$$

is the residual-information distribution tensor, and

$$\kappa$$

is the coupling constant specific to Future Convergence Theory.

Conservation Law

The Non-Optical Manifestation Residue is not described by ordinary particle number conservation. Instead, Future Convergence Theory characterizes it through conservation of consistency with the Manifestation Constraint. Accordingly,

$$\boxed{\nabla_\mu (\mathcal{R}_{\text{NO}} \Theta^{\mu\nu}) = 0}$$

is defined as its conservation law. This expresses that residual information propagates according to the global conservation law of the Manifestation Constraint.

Time Evolution

The Non-Optical Manifestation Residue does not obey an ordinary local particle equation of motion. Instead, its evolution follows the convergence condition toward the Future Fixed Point

$$\mathcal{F}^*.$$

In general,

$$\boxed{\frac{D\mathcal{R}_{\text{NO}}}{Dt} = \Omega(\mathcal{F}, \mathcal{C}_{\mu\nu})}$$

where

$$\Omega$$

denotes the convergence operator of Future Convergence Theory.

Weak Constraint Limit

Within ordinary galactic environments or weak local gravitational fields, the contribution of the Non-Optical Manifestation Residue is small. Thus,

$$\boxed{\mathcal{R}_{\text{NO}} \rightarrow 0}$$

implies that the Manifestation Constraint Tensor is determined solely by visible matter. In this limit, Future Convergence Theory reduces to ordinary General Relativity.

Strong Constraint Limit

Near galactic halos, galaxy clusters, the early Universe, and black holes, the contribution of the Non-Optical Manifestation Residue becomes finite. Consequently, the Manifestation Constraint Tensor produces additional gravitational effects that cannot be explained by visible matter alone. Future Convergence Theory attributes this additional gravity not to unknown particles but to ontological residues.

Non-Optical Manifestation Residue Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.2 (Non-Optical Manifestation Residue Theorem). *The Non-Optical Manifestation Residue is an ontological residue consisting of informational states existing in the Future Causal Information Field that do not manifest as electromagnetic degrees of freedom but nevertheless provide a finite contribution to the Manifestation Constraint Tensor. Its conservation law coincides with the conservation law of the Manifestation Constraint.*

Proof. Part III showed that the Manifestation Projection does not project every informational state into the Three-Dimensional Manifestation Layer. Future Convergence Theory therefore permits informational states that contribute to the Manifestation Constraint while lacking electromagnetic manifestation. Since these states satisfy the conservation law of the Manifestation Constraint, they act as additional gravitational sources while remaining optically unobservable. Hence the Non-Optical Manifestation Residue is consistently defined within Future Convergence Theory. $\square$

Conclusion of This Section

In this section, the Non-Optical Manifestation Residue was formulated as an independent ontological field within Future Convergence Theory. As a consequence, the additional gravity corresponding to dark matter arises not from unknown particles but from informational residues coupled to the Manifestation Constraint. In the next section, these Non-Optical Manifestation Residues are incorporated into the energy-momentum tensor, and the effective energy-momentum tensor of Future Convergence Theory is derived.

6.0.4 6.4 Effective Energy–Momentum Tensor

In the previous section, we showed that the essence of dark matter in Future Convergence Theory is not an unknown particle but the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}$$

generated from the Future Causal Information Field and contributing to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

However, this ontological definition alone cannot be directly applied to General Relativity, cosmology, or galactic dynamics. To connect the theory with observable physics, the Non-Optical Manifestation Residue must be mapped into an effective energy-momentum tensor acting as a gravitational source. Accordingly, this section introduces the **Effective Energy–Momentum Tensor** within Future Convergence Theory.

Gravitational Sources in General Relativity

In General Relativity, the Einstein equation

$$\boxed{G_{\mu\nu} = 8\pi G T_{\mu\nu}}$$

describes gravity through the energy-momentum tensor

$$T_{\mu\nu}.$$

In cosmology, it is commonly decomposed as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{vis}} + T_{\mu\nu}^{\text{DM}} + T_{\mu\nu}^{\Lambda},$$

where

$$T_{\mu\nu}^{\text{DM}}$$

denotes the dark matter contribution.

Redefinition in Future Convergence Theory

Future Convergence Theory introduces no unknown particle. Instead, it maps the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}$$

into the gravitational source. The effective energy-momentum tensor is therefore defined as

$$\boxed{T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{vis}} + T_{\mu\nu}^{\text{NO}}}$$

where

$$T_{\mu\nu}^{\text{NO}}$$

is the effective tensor derived from the Non-Optical Manifestation Residue.

Residual Energy–Momentum Tensor

Since the Non-Optical Manifestation Residue contributes to gravity through the Manifestation Constraint, its effective tensor is defined as

$$T_{\mu\nu}^{\text{NO}} = \eta \mathcal{R}_{\text{NO}} \Theta_{\mu\nu}$$

where

$$\Theta_{\mu\nu}$$

is the residual-information distribution tensor and

$$\eta$$

is a coupling constant specific to Future Convergence Theory.

Modified Einstein Equation

Using the effective energy-momentum tensor, the Manifestation Constraint introduced in Part II rewrites the Einstein equation as

$$G_{\mu\nu} + \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{eff}}$$

Thus, additional gravitational sources arise naturally without introducing unknown particles.

Conservation Law

Within Future Convergence Theory, the total effective energy-momentum tensor is conserved:

$$\nabla^\mu T_{\mu\nu}^{\text{eff}} = 0$$

This expresses that not only ordinary matter but also the Non-Optical Manifestation Residue obeys the conservation law of the Manifestation Constraint.

Weak Constraint Limit

In ordinary local gravitational fields, the contribution of the Non-Optical Manifestation Residue is negligible. Therefore,

$$\lim_{\mathcal{R}_{\text{NO}} \rightarrow 0} T_{\mu\nu}^{\text{eff}} = T_{\mu\nu}^{\text{vis}}$$

and Future Convergence Theory reproduces the ordinary Einstein equation.

Strong Constraint Limit

Within galactic halos, galaxy clusters, and the early Universe, the Non-Optical Manifestation Residue becomes finite. Consequently,

$$T_{\mu\nu}^{\text{NO}}$$

may provide gravitational contributions comparable to or exceeding those of visible matter. Galactic rotation curves, gravitational lensing, and galaxy-cluster masses can therefore be explained without introducing unknown particles.

Effective Energy–Momentum Tensor Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.3 (Effective Energy–Momentum Tensor Theorem). *Within Future Convergence Theory, the gravitational source corresponding to dark matter is described not by the energy-momentum tensor of an unknown particle but by the effective energy-momentum tensor*

$$T_{\mu\nu}^{\text{NO}}$$

derived from the Non-Optical Manifestation Residue. In the weak constraint limit, this tensor vanishes and the theory reduces to ordinary General Relativity.

Proof. Part III showed that every manifested state originates from the Future Causal Information Field. Section 6.3 established that the Non-Optical Manifestation Residue is an informational state contributing to the Manifestation Constraint. Therefore, gravitational sources must include an effective tensor derived from residual information in addition to ordinary matter. In the weak constraint limit, the residue vanishes, recovering the standard Einstein equation. $\square$

Observational Consequences

If Future Convergence Theory is correct, the “dark matter density” inferred from galactic rotation curves, galaxy clusters, gravitational lensing, and the cosmic microwave background reflects not the density of unknown particles but the spatial distribution of the effective energy-momentum tensor

$$T_{\mu\nu}^{\text{NO}}.$$

Consequently, even if particle searches remain unsuccessful, no contradiction arises with cosmological observations.

Conclusion of This Section

In this section, the Non-Optical Manifestation Residue was formulated as an effective energy-momentum tensor connecting Future Convergence Theory with General Relativity. This provides a framework in which cosmological additional gravity is incorporated into the Einstein equation without assuming unknown particles. In the next section, this effective tensor is applied to galactic rotation curves to derive the velocity distribution predicted by Future Convergence Theory.

6.0.5 6.5 Prediction for Galactic Rotation Curves

In the previous section, we showed that, within Future Convergence Theory, dark matter is incorporated into General Relativity not as an unknown particle but through the effective energy-momentum tensor

$$T_{\mu\nu}^{\text{NO}}$$

derived from the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}.$$

In this section, we apply this effective tensor to galactic rotation curves. Galactic rotation curves constitute one of the strongest observational motivations for dark matter and therefore provide a direct test of the predictions of Future Convergence Theory.

Standard Gravitational Theory

Under the Newtonian approximation, the rotational velocity at radius

$$r$$

is given by

$$v(r) = \sqrt{\frac{GM(r)}{r}}$$

where

$$M(r)$$

is the total enclosed mass within radius

$$r.$$

If only visible matter is included,

$$v(r) \propto r^{-1/2}$$

at sufficiently large radii. Observationally, however, galactic rotation curves remain approximately flat even in the outer regions.

Gravitational Source in Future Convergence Theory

Within Future Convergence Theory, the effective gravitational source consists not only of visible matter. The effective mass is defined by

$$M_{\text{eff}}(r) = M_{\text{vis}}(r) + M_{\text{NO}}(r)$$

where

$$M_{\text{NO}}(r)$$

denotes the effective mass generated by the Non-Optical Manifestation Residue.

Effective Rotation Velocity

The rotational velocity therefore becomes

$$v_{\text{FCT}}(r) = \sqrt{\frac{GM_{\text{eff}}(r)}{r}}$$

Although this expression has the same formal structure as that used in standard cosmology, the physical origin of the additional mass is completely different. Within Future Convergence Theory, the extra gravitational field arises not from unknown particles but from ontological residues contributing to the Manifestation Constraint.

Residual Distribution

The Non-Optical Manifestation Residue is determined through consistency with the Future Fixed Point. Its density distribution is therefore written in the general form

$$\rho_{\text{NO}} = \rho_{\text{NO}}(r, \mathcal{F}, \mathcal{C}_{\mu\nu})$$

Consequently,

$$M_{\text{NO}}(r) = 4\pi \int_0^r \rho_{\text{NO}}(r') r'^2 dr'$$

defines the corresponding effective mass.

Flat Rotation Curves

Future Convergence Theory predicts that, in the outer regions of galaxies, the Non-Optical Manifestation Residue decreases more slowly than visible matter. For example, if

$$\rho_{\text{NO}} \propto r^{-2},$$

then

$$M_{\text{NO}}(r) \propto r.$$

Accordingly,

$$v_{\text{FCT}}(r) \rightarrow \text{const.}$$

and naturally produces flat galactic rotation curves.

Difference from Standard Λ CDM

Within standard

$$\Lambda\text{CDM},$$

flat rotation curves are explained by halos of unknown dark matter particles. In contrast, Future Convergence Theory attributes the same phenomenon to the contribution of the Non-Optical Manifestation Residue to the Manifestation Constraint. Thus, although the observed velocity distribution may appear similar, its ontological origin is fundamentally different.

Galactic Rotation Curve Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.4 (Galactic Rotation Curve Theorem). *Within Future Convergence Theory, the flattening of galactic rotation curves does not arise from halos of unknown particles. Instead, it results from the effective mass distribution produced by Non-Optical Manifestation Residues generated from the Future Causal Information Field and contributing to the Manifestation Constraint Tensor.*

Proof. Section 6.4 established that the effective gravitational source is the sum of visible matter and the Non-Optical Manifestation Residue. In the outer regions of galaxies, the residue maintains a finite density, causing the effective enclosed mass to increase approximately linearly with radius. Consequently, the rotational velocity approaches a constant value. Therefore, flat galactic rotation curves can be explained without introducing unknown particles. $\square$

Observational Consequences

If Future Convergence Theory is correct, galactic rotation curves themselves will be reproduced with nearly the same accuracy as in standard

$$\Lambda\text{CDM}.$$

However, because the gravitational source is an ontological residue rather than a particle halo, the continued absence of direct particle detection is fully consistent with galactic dynamics. Furthermore, since the spatial distribution of the Non-Optical Manifestation Residue depends not only on galaxy formation history but also on the Future Fixed Point, small systematic deviations from standard CDM may appear among different galactic environments.

Conclusion of This Section

In this section, galactic rotation curves were derived within Future Convergence Theory. Flat rotation curves arise naturally from the effective mass distribution of the Non-Optical Manifestation Residue rather than from halos of unknown particles. In the next section, this effective mass distribution is extended to the entire galactic halo, and the halo structure predicted by Future Convergence Theory is compared with that of standard CDM.

6.0.6 6.6 Prediction for Halo Structure

In the previous section, we showed that the flattening of galactic rotation curves follows from the contribution of the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}$$

to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

rather than from halos composed of unknown particles. Galactic rotation curves, however, represent only one cross-section of the entire halo structure. Within standard

$$\Lambda\text{CDM}$$

cosmology, dark matter halos constitute the primary structures governing galaxy formation. Future Convergence Theory likewise predicts extended gravitational structures surrounding galaxies. Their origin, however, is determined not by the self-gravitating equilibrium of unknown particles but by the consistency condition between the Future Fixed Point and the Manifestation Constraint.

Standard CDM Halo

In standard cosmology, the halo density profile is commonly approximated by the Navarro–Frenk–White (NFW) profile,

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

where

$$\rho_s$$

is the characteristic density and

$$r_s$$

is the scale radius. This profile is empirically derived from numerical simulations.

Halo in Future Convergence Theory

Within Future Convergence Theory, the galactic halo is not an accumulation of unknown particles. Instead, it is formed by the spatial distribution of the Non-Optical Manifestation Residue. Accordingly,

$$\rho_{\text{halo}} = \rho_{\text{NO}}(r, \mathcal{F}, \mathcal{C}_{\mu\nu})$$

where the density depends not only on the gravitational potential but also on the convergence condition toward the Future Fixed Point.

Relation to the Manifestation Constraint

The galactic halo contributes to the gravitational field of the entire galaxy through the Manifestation Constraint Tensor. Its contribution is expressed as

$$\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}^{\text{vis}} + \mathcal{C}_{\mu\nu}^{\text{NO}}$$

where

$$\mathcal{C}_{\mu\nu}^{\text{NO}}$$

represents the constraint component generated by the Non-Optical Manifestation Residue.

Central Density

Within Future Convergence Theory, the Non-Optical Manifestation Residue is distributed according to the conservation law of the Manifestation Constraint rather than through particle collisions or self-gravitational relaxation. Consequently, the central density gradient is restricted by consistency with the Future Fixed Point. In general,

$$\left. \frac{d\rho_{\text{NO}}}{dr} \right|_{r \rightarrow 0} = \Psi(\mathcal{F}^*)$$

where

$$\Psi$$

denotes the constraint function determined by the Future Fixed Point.

Outer Halo Structure

In the outer halo, the Non-Optical Manifestation Residue does not vanish completely. Future Convergence Theory predicts that its density decreases more slowly than that of visible matter in accordance with the conservation law of the Manifestation Constraint. The halo therefore forms an ontological constraint structure rather than a particle-diffusion structure.

Halo Stability

Within Future Convergence Theory, the stability of galactic halos is not determined solely by self-gravitational equilibrium. As long as the convergence condition toward the Future Fixed Point is satisfied, the Non-Optical Manifestation Residue maintains a stable distribution over cosmological timescales. Consequently, halo structures preserve a comparatively stable constraint configuration after galaxy formation.

Halo Structure Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.5 (Halo Structure Theorem). *Within Future Convergence Theory, galactic halos are formed not by the self-gravitational equilibrium of unknown particles but by the contribution of the Non-Optical Manifestation Residue to the Manifestation Constraint Tensor. Their spatial distribution is determined by the consistency condition with the Future Fixed Point.*

Proof. Section 6.3 established that the Non-Optical Manifestation Residue contributes a finite amount to the Manifestation Constraint. Section 6.4 further showed that this contribution becomes an effective energy-momentum tensor acting as a gravitational source. Accordingly, the additional gravitational structure surrounding galaxies is formed by the spatial distribution of the Non-Optical Manifestation Residue rather than by unknown particles. Furthermore, Part I established that every manifestation structure must satisfy the convergence condition toward the Future Fixed Point. Hence, galactic halos are stabilized by this ontological constraint. $\square$

Observational Consequences

If Future Convergence Theory is correct, the total halo mass may be comparable to that predicted by standard

$$\Lambda\text{CDM},$$

while systematic differences may appear in the internal density profile, the central gradient, and the outer decay rate due to the Future Fixed Point. Accordingly, precise gravitational-lensing observations, weak-lensing analyses, and comparisons with galaxy-formation simulations may distinguish Future Convergence Theory from standard CDM.

Conclusion of This Section

In this section, the galactic halo was formulated within Future Convergence Theory not as an accumulation of unknown particles but as an ontological constraint structure produced by the Non-Optical Manifestation Residue. The halo is therefore stabilized by the conservation law of the Manifestation Constraint together with consistency conditions imposed by the Future Fixed Point. In the next section, this framework is applied to galaxy-cluster collisions, and the predicted mass distribution is compared with that of standard CDM.

6.0.7 6.7 Prediction for Galaxy-Cluster Collisions

In the previous section, we showed that within Future Convergence Theory, galactic halos are not formed through the self-gravitational equilibrium of unknown particles but constitute ontological constraint structures generated by the contribution of the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}$$

to the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

The most stringent observational test of this framework is provided by galaxy-cluster collisions. In particular, systems such as the Bullet Cluster exhibit a spatial separation between hot gas, galaxies, and the gravitational mass distribution inferred from gravitational lensing. Within standard cosmology, this phenomenon is regarded as one of the strongest pieces of evidence for collisionless dark matter particles. Future Convergence Theory instead interprets this phenomenon as a process of reconstructing the Manifestation Constraint rather than the dynamics of unknown particles.

Interpretation in Standard Cosmology

Within standard

$$\Lambda\text{CDM},$$

the hot intracluster gas is decelerated by electromagnetic interactions during a collision. By contrast, dark matter particles experience essentially no collisions and therefore pass through the collision together with the galaxies. Consequently, the gravitational mass distribution inferred from lensing is expected to coincide primarily with the galaxy distribution rather than with the gas.

Reinterpretation in Future Convergence Theory

Within Future Convergence Theory, the gravitational field of an entire galaxy cluster is generated not only by visible matter but also by the Non-Optical Manifestation Residue. During a collision, the visible gas is affected by electromagnetic interactions. The Non-Optical Manifestation Residue, however, is not a particle but an informational state governed by the conservation

law of the Manifestation Constraint. Accordingly, the constraint structure is reconstructed independently of the deceleration of the gas.

Constraint Reconstruction

Let the Manifestation Constraints before the collision be

$$\mathcal{C}_{\mu\nu}^{(1)}, \quad \mathcal{C}_{\mu\nu}^{(2)}.$$

After the collision, Future Convergence Theory describes the reconstructed constraint structure as

$$\mathcal{C}_{\mu\nu}^{\text{after}} = \mathcal{R} \left(\mathcal{C}_{\mu\nu}^{(1)}, \mathcal{C}_{\mu\nu}^{(2)} \right)$$

where

$$\mathcal{R}$$

denotes the Manifestation Constraint Reconstruction Operator.

Mass Centroid

Within Future Convergence Theory, the mass centroid measured by gravitational lensing is the centroid of the Manifestation Constraint rather than that of visible matter. Accordingly,

$$\mathbf{x}_{\text{lens}} = \mathbf{x}(\mathcal{C}_{\mu\nu})$$

determines the lensing mass centroid. Therefore, the centers of the hot gas and the gravitational lens need not coincide.

Time Evolution

After the collision, the Non-Optical Manifestation Residue is redistributed so as to satisfy the convergence condition toward the Future Fixed Point. In general,

$$\frac{D\mathcal{R}_{\text{NO}}}{Dt} = \Omega(\mathcal{F}, \mathcal{C}_{\mu\nu})$$

where

$$\Omega$$

is the convergence operator of Future Convergence Theory. Consequently, the evolution of the mass distribution is described not as particle scattering but as the reconvergence of the constraint field.

Difference from Standard CDM

Within standard

$$\Lambda\text{CDM},$$

the post-collision mass distribution is determined by the motion of dark matter particles. Future Convergence Theory instead attributes it to the reconstruction of the Manifestation Constraint Tensor. Therefore, even for identical collision velocities, gas temperatures, and galaxy distributions, the reconstruction process may produce small systematic differences in the gravitational-lensing mass distribution.

Galaxy-Cluster Collision Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.6 (Galaxy-Cluster Collision Theorem). *Within Future Convergence Theory, the gravitational-lensing mass distribution observed after a galaxy-cluster collision is determined not by the motion of unknown particles but by the reconstruction of the Manifestation Constraint Tensor. Consequently, the observed separation between the hot gas and the lensing mass centroid can be explained without introducing unknown particles.*

Proof. Section 6.3 established that the Non-Optical Manifestation Residue contributes to the Manifestation Constraint. Section 6.4 showed that this contribution becomes an effective energy-momentum tensor acting as a gravitational source. During a galaxy-cluster collision, the visible gas is decelerated by electromagnetic interactions, whereas the Manifestation Constraint is reconstructed so as to preserve consistency with the Future Fixed Point. Accordingly, the gravitational-lensing mass centroid need not coincide with the center of the hot gas. Therefore, the observed mass separation in cluster collisions can be explained without introducing unknown particles. $\square$

Observational Consequences

If Future Convergence Theory is correct, galaxy-cluster collisions will continue to exhibit a separation between the hot gas and the gravitational lensing mass centroid similar to that predicted by standard

$$\Lambda\text{CDM}.$$

However, the spatial distribution and temporal evolution of the lensing mass will be governed by the reconstruction rate of the Manifestation Constraint rather than by particle dynamics. Consequently, systematic differences in the migration and redistribution of the lensing mass centroid among multiple cluster collisions constitute observational predictions unique to Future Convergence Theory.

Conclusion of This Section

In this section, galaxy-cluster collisions were reformulated not as problems of unknown-particle dynamics but as processes of reconstructing the Manifestation Constraint Tensor. The Bullet Cluster type of mass separation can therefore be interpreted as the reformation of ontological constraint structures generated by the Non-Optical Manifestation Residue. The next section applies this effective Manifestation Constraint to the gravitational lens equation and derives the lensing effects predicted by Future Convergence Theory together with their differences from standard CDM.

6.0.8 6.8 Prediction for Gravitational Lensing

In the previous section, we showed that within Future Convergence Theory, the post-collision mass distribution of galaxy clusters is determined not by the motion of unknown particles but by the reconstruction of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

formed by the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}.$$

Gravitational lensing provides the most direct observational probe of the spatial distribution of this Manifestation Constraint. Accordingly, theoretical corrections to the gravitational-lens equations must be derived in order to test Future Convergence Theory.

Gravitational Lensing in General Relativity

Within General Relativity, the deflection angle produced by a point mass

$$M$$

is

$$\hat{\alpha} = \frac{4GM}{c^2 b}$$

where

$$b$$

denotes the impact parameter. For extended mass distributions, the deflection angle is obtained from the projected surface density

$$\Sigma(\xi).$$

Gravitational Source in Future Convergence Theory

Within Future Convergence Theory, the gravitational source responsible for lensing is not limited to visible matter. The effective projected mass density is

$$\Sigma_{\text{eff}} = \Sigma_{\text{vis}} + \Sigma_{\text{NO}}$$

where

$$\Sigma_{\text{NO}}$$

is the projected density associated with the Non-Optical Manifestation Residue.

Effective Deflection Angle

The corresponding deflection angle is

$$\hat{\alpha}_{\text{FCT}} = \hat{\alpha}_{\text{GR}} + \Delta\hat{\alpha}$$

where

$$\Delta\hat{\alpha} = F(\mathcal{R}_{\text{NO}}, \mathcal{C}_{\mu\nu}, \mathcal{F})$$

is the correction produced by the Manifestation Constraint.

Lensing Potential

The lensing potential

$$\psi$$

is similarly modified according to

$$\psi_{\text{FCT}} = \psi_{\text{GR}} + \Delta\psi$$

where

$$\Delta\psi = \Delta\psi(\mathcal{R}_{\text{NO}}).$$

Accordingly, image positions, magnifications, and time delays all receive corrections.

Weak Gravitational Lensing

In weak lensing, the convergence

$$\kappa$$

is normally determined by the projected mass density. Within Future Convergence Theory,

$$\kappa_{\text{FCT}} = \kappa_{\text{vis}} + \kappa_{\text{NO}}$$

holds. Therefore, finite lensing effects can arise even in regions containing little visible matter whenever the Non-Optical Manifestation Residue is present.

Strong Gravitational Lensing

Near galaxy clusters and black holes, Future Convergence Theory predicts corrections to the Einstein-ring radius, image separations, and time delays. In general,

$$\theta_{\text{E}}^{\text{FCT}} = \theta_{\text{E}}^{\text{GR}} + \Delta\theta_{\text{E}}$$

where

$$\Delta\theta_{\text{E}}$$

denotes the Manifestation Constraint correction.

Difference from Standard CDM

Within standard

$$\Lambda\text{CDM},$$

the additional deflection is attributed to the mass distribution of unknown particle halos. Future Convergence Theory instead attributes it to the constraint structure formed by the Non-Optical Manifestation Residue. Thus, even when the total lensing mass is similar, systematic differences may appear in the spatial gradients, temporal evolution, and post-collision redistribution of the lensing structure.

Gravitational Lensing Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.7 (Gravitational Lensing Theorem). *Within Future Convergence Theory, gravitational lensing is determined not by halos of unknown particles but by the Manifestation Constraint Tensor formed by the Non-Optical Manifestation Residue. Accordingly, the observed lensing signal closely approximates that predicted by General Relativity while possessing corrections unique to Future Convergence Theory in its temporal evolution and spatial gradients.*

Proof. Section 6.3 established that the Non-Optical Manifestation Residue provides a finite contribution to the Manifestation Constraint Tensor. Section 6.4 showed that this contribution becomes an effective gravitational source. Since gravitational lensing is determined by the gravitational source, the Manifestation Constraint correction necessarily contributes to the deflection angle. Hence gravitational lensing can be explained without introducing unknown particles. $\square$

Observational Consequences

If Future Convergence Theory is correct, most present gravitational-lensing observations should be reproduced with an accuracy comparable to that of standard

Λ CDM.

However, future high-precision weak-lensing surveys, time-dependent measurements following galaxy-cluster collisions, accurate measurements of Einstein rings, and precise gravitational time-delay observations may reveal small systematic deviations arising from the redistribution of the Non-Optical Manifestation Residue. These constitute observational signatures capable of distinguishing Future Convergence Theory from standard cosmology.

Conclusion of This Section

In this section, gravitational lensing was derived within Future Convergence Theory. The lensing mass originates not from unknown particles but from the Manifestation Constraint structure formed by the Non-Optical Manifestation Residue. Consequently, gravitational lensing becomes one of the principal observational probes of the ontological structure proposed by Future Convergence Theory. The next section applies the same framework to the Cosmic Microwave Background (CMB) and derives corrections to the temperature anisotropies and angular power spectrum predicted by Future Convergence Theory.

6.0.9 6.9 Predictions for the Cosmic Microwave Background

In the previous section, we showed that within Future Convergence Theory, gravitational lensing is determined not by halos of unknown dark matter particles but by the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

generated by the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}}.$$

For Future Convergence Theory to constitute a consistent cosmological framework, however, it must reproduce not only galactic-scale observations but also the largest observable structures of the Universe. The most fundamental probe of the early Universe is the Cosmic Microwave Background (CMB). The CMB reflects primordial density perturbations, gravitational potentials, and the overall energy distribution of the Universe. Accordingly, if the Non-Optical Manifestation Residue constitutes the cosmological gravitational source within Future Convergence Theory, its effects must also appear in the temperature anisotropies and angular power spectrum of the CMB.

The CMB in Standard Cosmology

Within standard

Λ CDM,

the temperature anisotropies of the CMB are determined by the primordial gravitational potential together with perturbations in baryons, radiation, and dark matter. The angular power spectrum is conventionally defined as

$$C_\ell = \langle |a_{\ell m}|^2 \rangle$$

where

$$a_{\ell m}$$

denotes the spherical harmonic coefficients.

Gravitational Source in Future Convergence Theory

Within Future Convergence Theory, the gravitational source in the early Universe is not an unknown particle species but the Non-Optical Manifestation Residue. The effective cosmological density is therefore written as

$$\rho_{\text{eff}} = \rho_{\text{vis}} + \rho_{\text{NO}}$$

where

$$\rho_{\text{NO}}$$

denotes the effective density associated with the Non-Optical Manifestation Residue.

Correction to the Gravitational Potential

The Non-Optical Manifestation Residue contributes to the gravitational potential through the Manifestation Constraint Tensor. Accordingly,

$$\Phi_{\text{FCT}} = \Phi_{\text{GR}} + \Delta\Phi$$

where

$$\Delta\Phi = \Delta\Phi(\mathcal{R}_{\text{NO}})$$

represents the correction induced by the ontological residue.

Correction to the Temperature Anisotropy

Within Future Convergence Theory, the CMB temperature anisotropy receives, in addition to the conventional Sachs–Wolfe contribution, a correction originating from the Manifestation Constraint. Thus,

$$\left(\frac{\Delta T}{T}\right)_{\text{FCT}} = \left(\frac{\Delta T}{T}\right)_{\text{GR}} + \Delta\left(\frac{\Delta T}{T}\right)$$

where the correction depends on the primordial distribution of the Non-Optical Manifestation Residue.

Angular Power Spectrum

The angular power spectrum predicted by Future Convergence Theory is written as

$$C_{\ell}^{\text{FCT}} = C_{\ell}^{\Lambda\text{CDM}} + \Delta C_{\ell}$$

where

$$\Delta C_{\ell} = \Delta C_{\ell}(\mathcal{R}_{\text{NO}}, \mathcal{F})$$

is the correction generated by the Non-Optical Manifestation Residue.

Low-Multipole Behavior

The Future Fixed Point specifies the global convergence condition of the Universe. Future Convergence Theory therefore predicts that corrections associated with the Future Fixed Point become relatively significant at large angular scales, namely in the low-multipole region

$$\ell.$$

At high multipoles, local physical processes dominate, and the differences from standard

$$\Lambda\text{CDM}$$

are expected to remain comparatively small.

Difference from Standard CDM

Within standard cosmology, the angular power spectrum is explained through the density perturbations of particle dark matter. Future Convergence Theory instead attributes the same observables to Manifestation Constraint corrections generated by the Non-Optical Manifestation Residue. Consequently, while the positions of the acoustic peaks remain nearly identical, systematic deviations may appear in the low-multipole region, the Integrated Sachs–Wolfe effect, and the CMB lensing signal.

CMB Prediction Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.8 (CMB Prediction Theorem). *Within Future Convergence Theory, the Cosmic Microwave Background is described not by particle dark matter but by the Manifestation Constraint Tensor generated by the Non-Optical Manifestation Residue. Consequently, the angular power spectrum closely approximates that of standard cosmology while exhibiting characteristic corrections in the low-multipole region, CMB lensing, and the Integrated Sachs–Wolfe effect.*

Proof. Section 6.3 established that the Non-Optical Manifestation Residue contributes a finite amount to the Manifestation Constraint Tensor. Section 6.4 showed that this contribution becomes an effective energy–momentum tensor acting as a cosmological gravitational source. The primordial gravitational potential is therefore modified by the Non-Optical Manifestation Residue. Since the CMB temperature anisotropy depends upon the gravitational potential, Future Convergence Theory naturally predicts corrections to the angular power spectrum. $\square$

Observational Consequences

If Future Convergence Theory is correct, the majority of present CMB observations should be reproduced with approximately the same precision as standard

Λ CDM.

However, systematic corrections associated with the Future Fixed Point may appear in the low-multipole anomalies, the Integrated Sachs–Wolfe effect, CMB lensing, and polarization spectra. These effects provide important observational targets for distinguishing Future Convergence Theory through future high-precision CMB measurements.

Conclusion of This Section

In this section, theoretical predictions for the Cosmic Microwave Background within Future Convergence Theory were derived. The CMB is interpreted as arising not from particle dark matter but from the Manifestation Constraint generated by the Non-Optical Manifestation Residue. Accordingly, Future Convergence Theory provides a unified ontological description of both local gravitational phenomena and cosmological-scale observations. The next section summarizes the observational criteria that distinguish Future Convergence Theory from the standard Λ CDM cosmological model and formulates the falsifiability conditions of the present dark matter framework.

6.0.10 6.10 Criteria for Distinguishing Future Convergence Theory from Standard CDM

In the previous section, we demonstrated that within Future Convergence Theory the Cosmic Microwave Background is described through the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}$$

generated by the Non-Optical Manifestation Residue

$$\mathcal{R}_{\text{NO}},$$

rather than by unknown dark matter particles. For a physical theory to qualify as a scientific theory, however, it must produce observational predictions that differ from existing theories and can be confirmed or falsified by experiment and observation. This section summarizes the observational criteria that distinguish Future Convergence Theory from standard

$$\Lambda\text{CDM}$$

cosmology and formulates the falsifiability conditions of the present framework.

Principle of Discrimination

Future Convergence Theory aims to reproduce the principal cosmological observables—including galactic rotation curves, galaxy clusters, gravitational lensing, and the Cosmic Microwave Background—with an accuracy comparable to that of standard

$$\Lambda\text{CDM}.$$

Accordingly, the principal discriminants are not the observables themselves but their temporal evolution, spatial distributions, and high-precision residuals. We therefore define the observational difference as

$$\Delta\mathcal{O} = \mathcal{O}_{\text{FCT}} - \mathcal{O}_{\Lambda\text{CDM}}$$

where

$$\mathcal{O}$$

represents any measurable cosmological quantity.

Galactic Rotation Curves

Within Future Convergence Theory, flat rotation curves arise from the Non-Optical Manifestation Residue rather than from halos of unknown particles. Therefore, galaxies with nearly identical formation histories may nevertheless exhibit small systematic differences in the outer velocity gradient owing to different convergence conditions toward the Future Fixed Point.

Galaxy-Cluster Collisions

Within standard

$$\Lambda\text{CDM},$$

the post-collision mass centroid is determined by the motion of dark matter particles. Future Convergence Theory instead attributes the mass centroid to the reconstruction rate of the Manifestation Constraint Tensor. The temporal evolution of the lensing mass distribution following cluster collisions therefore becomes a key observational discriminator.

Gravitational Lensing

Future Convergence Theory interprets gravitational lensing masses as constraint structures generated by the Non-Optical Manifestation Residue. Accordingly, weak-lensing mass reconstructions may exhibit density gradients and redistribution processes that differ systematically from those predicted by particle-halo models.

Cosmic Microwave Background

Future Convergence Theory predicts excellent agreement with standard cosmology for the overall angular power spectrum while introducing additional corrections in the low-multipole region, the Integrated Sachs–Wolfe effect, and CMB lensing through the Future Fixed Point. These quantities therefore constitute primary observational targets for testing the theory.

Direct Particle Searches

The most distinctive feature separating Future Convergence Theory from standard cosmology is that it does not introduce an unknown dark matter particle. Accordingly, Future Convergence Theory naturally allows the possibility that direct searches for dark matter particles will continue to yield null results, while cosmological gravitational phenomena remain consistently observable. Within the present framework, this does not constitute a discrepancy between particle physics and cosmology. Rather, it reflects the ontological distinction between electromagnetic manifestation and gravitational manifestation established throughout the preceding chapters.

Falsification Criteria

The dark matter framework proposed in this chapter is regarded as falsified if any one of the following conditions is satisfied.

1. An unknown dark matter particle is directly detected, and it is demonstrated that this particle alone consistently explains galactic rotation curves, galaxy clusters, gravitational lensing, and the Cosmic Microwave Background within a single theoretical framework.
2. The observational differences predicted by Future Convergence Theory,

$$\Delta\mathcal{O},$$

are shown to vanish within observational precision for every relevant cosmological observable.

3. A theory that does not introduce the Non-Optical Manifestation Residue explains all presently known cosmological observations with equal or higher accuracy while preserving internal theoretical consistency.
4. Future Convergence Theory fails to reproduce either the General Relativistic limit or the cosmological limit required for consistency with established observations.

Dark Matter Discrimination Theorem

The foregoing discussion establishes the following theorem.

Theorem 6.9 (Dark Matter Discrimination Theorem). *Future Convergence Theory provides a unified description of galactic rotation curves, galactic halos, galaxy-cluster collisions, gravitational lensing, and the Cosmic Microwave Background without introducing unknown dark matter particles. The distinction between Future Convergence Theory and standard Λ CDM cosmology is expected to appear in high-precision observational residuals, temporal evolution, and the outcome of direct particle searches.*

Proof. Sections 6.2 through 6.9 established that Future Convergence Theory describes additional gravitational sources through the Manifestation Constraint Tensor generated by the Non-Optical Manifestation Residue rather than by introducing unknown particles. Consequently, the principal cosmological observables can be reproduced while their ontological origin differs

fundamentally from that assumed in standard Λ CDM cosmology. Because these differing ontological assumptions lead to distinct predictions for observational residuals, temporal evolution, and direct particle detection experiments, the present framework is, in principle, empirically testable and falsifiable. $\square$

Conclusion of Part VI

In this chapter, dark matter has been redefined not as an unknown particle species but as the Non-Optical Manifestation Residue originating from the Future Causal Information Field. Within this framework, the effective energy–momentum tensor, galactic rotation curves, galactic halos, galaxy-cluster collisions, gravitational lensing, and the Cosmic Microwave Background are described in a unified manner through the Manifestation Constraint Tensor. Furthermore, observational distinctions from standard Λ CDM cosmology were formulated in terms of high-precision measurements, temporal evolution, and direct particle searches, thereby establishing Future Convergence Theory as a falsifiable cosmological framework. The next chapter investigates the first concrete observational predictions of Future Convergence Theory by deriving quantitative deviations from General Relativity in black-hole quasi-normal modes and gravitational-wave ringdown signals.

Part VII

Black-Hole Quasi-Normal Modes and Ringdown Predictions

Chapter 7

FCT Reconstruction of Black-Hole Perturbation Theory

7.1 FCT Reconstruction of Black-Hole Perturbation Theory

7.1.1 Problem Setting

Purpose of This Chapter

In Paper 92, based on the information-conservation structure of Future Convergence Theory (FCT), a black hole was redefined not as a site of information loss but as an interlayer information-conversion boundary connecting to the Future Causal Information Field. Thermodynamic entropy and Future Convergence Entropy were also defined as independent state quantities, and an information-conservation law near a black hole was derived from a new action principle. At that stage, however, the observable differences predicted by this theory were not yet discussed in sufficient detail. The success of General Relativity as a physical theory did not arise merely from explaining the existence of black holes. General Relativity has been supported because it quantitatively predicted observable quantities such as

- the perihelion precession of Mercury,
- gravitational redshift,
- the deflection of light,
- gravitational waves,
- black-hole ringdown.

Accordingly, FCT must likewise show that

differences from existing theories appear as observable quantities.

In this chapter, we take black-hole quasi-normal modes (QNMs) as the first such target. The purpose of this chapter is to derive mathematically how the Future Information Field modifies the black-hole perturbation equation and, consequently, how ringdown frequencies, damping rates, and higher-mode ratios are corrected.

Black-Hole Perturbations in General Relativity

In General Relativity, a vacuum black hole satisfies

$$G_{\mu\nu} = 0.$$

For the Schwarzschild spacetime,

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2d\Omega^2,$$

with

$$f(r) = 1 - \frac{2GM}{r}.$$

Here,

$$r_s = 2GM$$

denotes the event horizon. After a black-hole merger, small perturbations around this background metric are excited:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}.$$

General Relativity assumes

$$|h_{\mu\nu}| \ll |g_{\mu\nu}^{(0)}|,$$

and by expanding the Einstein equation to first order, one obtains the Regge–Wheeler equation, the Zerilli equation, and the Teukolsky equation. These ultimately reduce to a Schrödinger-type equation,

$$\frac{d^2\Psi}{dr_*^2} + (\omega^2 - V(r))\Psi = 0.$$

Here,

$$r_* = r + 2GM \ln \left| \frac{r}{2GM} - 1 \right|$$

is the tortoise coordinate,

$$V(r)$$

is the effective black-hole potential, and

$$\omega = \omega_R + i\omega_I$$

is the quasi-normal-mode eigenvalue. Ringdown is the freely damped oscillation determined by this complex eigenvalue.

Basic Assumptions in General Relativity

The quasi-normal-mode analysis of General Relativity depends on the following three assumptions.

First Assumption The black-hole spacetime is determined solely by local spacetime geometry. That is,

$$V(r) = V_{\text{GR}}(r; M, a).$$

Second Assumption The event horizon is a perfectly absorbing boundary. Therefore only

$$\Psi \propto e^{-i\omega(t+r_*)}$$

exists, and no reflected wave is present.

Third Assumption No degrees of freedom other than mass, angular momentum, and charge exist outside the black hole. Therefore, the quasi-normal modes are determined only by

$$\omega = \omega(M, a, Q).$$

This corresponds to the No-Hair Theorem.

Objects Modified by FCT

Once the Future Information Field introduced in Paper 92 is adopted, the three assumptions above no longer hold unchanged. In FCT, a black hole is not the terminal point at which information disappears, but an interlayer boundary at which information is reconnected to the Future Causal Information Field. Therefore, the effective potential near a black hole is not determined solely by local geometry. The Future Information Field, the Meaning Density Tensor, and Future Convergence Entropy contribute as additional degrees of freedom. Accordingly, in the present work, the General Relativistic potential

$$V_{\text{GR}}(r)$$

is replaced by

$$V_{\text{FCT}}(r) = V_{\text{GR}}(r) + \delta V_F(r)$$

where

$$\delta V_F(r)$$

is the correction potential originating from the Future Information Field. As long as this correction exists, the quasi-normal-mode frequency becomes

$$\omega_{\text{FCT}} = \omega_{\text{GR}} + \delta\omega$$

In the following sections, we derive

$$\delta V_F$$

from the action principle and connect it to the observables of black-hole ringdown.

7.1.2 Modification of the Background Spacetime by the Future Information Field

The Background Metric Is Not Determined by Local Geometry Alone

In General Relativity, the exterior spacetime of a black hole is determined by the Einstein equation

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

In vacuum,

$$T_{\mu\nu} = 0,$$

and therefore

$$G_{\mu\nu} = 0.$$

The Schwarzschild solution or the Kerr solution is then the unique stationary solution. In this theory, the exterior spacetime of a black hole is completely determined by

- mass,
- angular momentum,
- charge.

However, in the Future Information Field introduced in Paper 92, a black hole is defined as an informational boundary connecting the Future Information Field and the Three-Dimensional Manifestation Layer. Accordingly, near a black hole there exists a constraint term of the Future Causal Information Field that cannot be described by local spacetime alone. In the present work, this is called the Manifestation Constraint Field.

Manifestation Constraint Tensor

In Paper 92, manifestation was defined through the three-layer structure

$$F \longrightarrow H \longrightarrow X.$$

Here,

- F : Future Information Field,
- H : Quantum Allowance Layer,
- X : Manifestation Layer.

Near a black hole, the constraint on the manifestation projection

$$P_X : H \rightarrow X$$

becomes stronger than in ordinary regions. We therefore introduce the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu}.$$

By definition,

$$\boxed{\mathcal{C}_{\mu\nu} = \nabla_\mu \nabla_\nu \Phi_F - g_{\mu\nu} \square \Phi_F + \Sigma_{\mu\nu}}$$

where

$$\Phi_F$$

is the Future Information Field, and

$$\Sigma_{\mu\nu}$$

is the Meaning Density Tensor.

Modified Field Equation

In General Relativity,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

holds. In the present work, we assume that the Future Information Field contributes directly to the metric itself. Accordingly, we define the modified field equation as

$$\boxed{G_{\mu\nu} + \Lambda_F g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}}$$

where

$$\Lambda_F = \Lambda_0 + \lambda_F \Lambda(\Phi_F).$$

Schwarzschild Limit

FCT is not a theory that rejects General Relativity. In regions where the Future Information Field is sufficiently small,

$$\Phi_F \rightarrow 0.$$

Then,

$$\mathcal{C}_{\mu\nu} \rightarrow 0,$$

and also

$$\Lambda_F \rightarrow 0.$$

The modified field equation therefore returns to

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

In vacuum,

$$G_{\mu\nu} = 0,$$

and the ordinary Schwarzschild solution is recovered. Thus, General Relativity is obtained as the low-information-density limit of FCT.

Effective Metric Near a Black Hole

Near a black hole, the Future Information Field becomes finite. We therefore decompose the background metric as

$$g_{\mu\nu} = g_{\mu\nu}^{\text{GR}} + \delta g_{\mu\nu}^F.$$

Here we define

$$\delta g_{\mu\nu}^F = \chi \mathcal{C}_{\mu\nu}.$$

Consequently, the Schwarzschild metric function

$$f(r) = 1 - \frac{2GM}{r}$$

is modified into

$$f_F(r) = 1 - \frac{2GM}{r} + \chi \Gamma_F(r)$$

where

$$\Gamma_F(r)$$

is the effective potential function induced by the Future Information Field.

Manifestation Constraint Length

The effect of the Future Information Field does not remain constant out to infinite distance. We therefore introduce the Manifestation Constraint Length

$$\ell_F.$$

As the simplest model, we adopt

$$\Gamma_F(r) = \Gamma_0 e^{-r/\ell_F}.$$

Then, at sufficiently large distances,

$$r \gg \ell_F,$$

we have

$$\Gamma_F(r) \rightarrow 0,$$

and General Relativity is recovered. By contrast, near the black hole,

$$r \sim r_s,$$

so the correction from the Future Information Field cannot be ignored.

New Degrees of Freedom

The preceding results show that a black hole is not described only by the traditional quantities

- mass,
- spin,
- charge.

In FCT, the following become additional degrees of freedom characterizing the black-hole exterior solution:

- Future Information Field intensity Φ_F ,
- Manifestation Constraint Length ℓ_F ,
- Future Convergence Entropy S_F ,
- Meaning Density Tensor $\Sigma_{\mu\nu}$.

Thus, within FCT, the No-Hair Theorem is extended into a theory with Future-Causal Hair.

7.1.3 FCT Reconstruction of the Regge–Wheeler Equation

Redefinition of the Perturbation Problem

In General Relativity, small perturbations of the exterior black-hole spacetime are defined by

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1.$$

Here the background metric

$$g_{\mu\nu}^{(0)}$$

is the Schwarzschild or Kerr solution. In FCT, the background metric itself is modified by the Future Information Field, and therefore the background solution is defined as

$$g_{\mu\nu}^{(0)} = g_{\mu\nu}^{\text{GR}} + \chi \mathcal{C}_{\mu\nu}$$

The perturbation expansion therefore becomes

$$g_{\mu\nu} = g_{\mu\nu}^{\text{GR}} + \chi \mathcal{C}_{\mu\nu} + h_{\mu\nu}.$$

The crucial point is that

$$h_{\mu\nu}$$

is the gravitational wave perturbation itself, whereas

$$\mathcal{C}_{\mu\nu}$$

is the background correction induced by the Future Information Field. These two objects are not identical.

Perturbative Action

In Paper 92, the action was written as

$$S = S_{\text{EH}} + S_F.$$

In the present work, we further treat perturbations near black holes and expand the action up to second order:

$$S^{(2)} = S_{\text{EH}}^{(2)} + S_F^{(2)}.$$

Here we define

$$S_F^{(2)} = \lambda_F \int \sqrt{-g} \Sigma_{\mu\nu} h^{\mu\alpha} h^\nu{}_\alpha d^4x.$$

This is the lowest-order action through which the Future Information Field couples directly to gravitational-wave perturbations.

Linearized Equation

Varying the action with respect to

$$h_{\mu\nu}$$

gives

$$\delta G_{\mu\nu} + \chi \delta \mathcal{C}_{\mu\nu} = 0$$

where

$$\delta \mathcal{C}_{\mu\nu} = \nabla_\mu \nabla_\nu (\delta \Phi_F) - g_{\mu\nu} \square (\delta \Phi_F) + \delta \Sigma_{\mu\nu}.$$

This has the form of the linearized Einstein equation of General Relativity with an additional perturbation of the Future Information Field.

Spherical Harmonic Expansion

On a Schwarzschild background, perturbations can be decomposed into even- and odd-parity sectors. The odd-parity component is expanded as

$$h_{\mu\nu}^{(\text{odd})} = \sum_{\ell,m} h_{\ell m}(t, r) Y_{\ell m}(\theta, \phi).$$

The Future Information Field is expanded similarly:

$$\Phi_F = \Phi_0(r) + \delta\Phi_{\ell m} Y_{\ell m}.$$

The background field and perturbation field therefore couple on the same spherical harmonic basis.

Regge–Wheeler Variable

As in General Relativity, we introduce the Regge–Wheeler variable

$$\Psi_{\ell m}.$$

However, we now include the contribution of the Future Information Field and define

$$\boxed{\Psi_{\ell m} = \Psi_{\ell m}^{\text{GR}} + \chi \Psi_{\ell m}^F}$$

The second term is the wave degree of freedom of the Future Information Field that appears for the first time after Paper 92.

FCT Version of the Regge–Wheeler Equation

The resulting perturbation equation is

$$\boxed{\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V_{RW}^{\text{GR}}(r) - \delta V_F(r)] \Psi = 0}$$

where

$$V_{RW}^{\text{GR}} = f(r) \left[\frac{\ell(\ell+1)}{r^2} - \frac{6GM}{r^3} \right],$$

and the new term

$$\boxed{\delta V_F(r) = \chi [A(r)\Phi_F + B(r)\Sigma + C(r)S_F]}$$

appears. Here,

- $A(r)$ is the Future Information Field coupling coefficient,
- $B(r)$ is the Meaning Density coupling coefficient,
- $C(r)$ is the Future Convergence Entropy coupling coefficient.

This equation is the first central result of the present chapter.

Difference from General Relativity

In General Relativity, the potential consists only of

$$V_{RW}^{\text{GR}}.$$

Therefore, the quasi-normal modes are determined solely by mass and spin. In FCT, however,

$$\Phi_F, \quad \Sigma, \quad S_F$$

also contribute. Thus, a black hole can no longer be described only by

$$(M, a).$$

Instead, the quasi-normal modes satisfy

$$\boxed{\omega = \omega(M, a, \Phi_F, \Sigma, S_F)}$$

This is the first mathematical result extending the No-Hair Theorem into a Future-Causal Hair theorem.

Physical Meaning

This result does not mean that ringdown itself converges into the Future Information Field. More precisely, ringdown is the eigen-oscillation determined on an effective potential corrected by the Future Information Field. Accordingly, observed gravitational waves do not directly observe the Future Causal Information Field. Rather, they reveal its background effect as a small correction in the quasi-normal-mode spectrum.

7.1.4 Perturbation Theory of Quasi-Normal-Mode Eigenvalues

Purpose of This Section

In the previous section, we showed that the Regge–Wheeler equation is modified by the Future Information Field as

$$\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V_{\text{GR}}(r) - \delta V_F(r)] \Psi = 0.$$

However, this equation alone does not yet constitute an observational prediction. What gravitational-wave detectors actually measure are

- ringdown frequencies,
- damping rates,
- quality factors,
- higher-mode ratios.

Therefore, in this section, we derive how much the Future Information Field shifts the quasi-normal-mode eigenvalues.

Eigenvalue Problem

In General Relativity, introducing the Regge–Wheeler operator

$$\hat{L}_0 = -\frac{d^2}{dr_*^2} + V_{\text{GR}}(r),$$

the quasi-normal modes satisfy

$$\hat{L}_0 \psi_n = \omega_n^2 \psi_n.$$

In FCT, the operator is extended to

$$\boxed{\hat{L} = \hat{L}_0 + \delta \hat{L}_F}$$

where

$$\delta \hat{L}_F = \delta V_F(r).$$

First-Order Perturbation

When the Future Information Field is sufficiently weak, we assume

$$\delta V_F \ll V_{\text{GR}}.$$

Applying Rayleigh–Schrödinger-type perturbation theory, the first-order correction is

$$\boxed{\delta \omega_n^2 = \langle \psi_n | \delta V_F | \psi_n \rangle}$$

or, in integral form,

$$\boxed{\delta \omega_n^2 = \int \psi_n^*(r) \delta V_F(r) \psi_n(r) dr_*}.$$

This is the first observational prediction formula of the present work.

Contribution of the Future Information Field

From the previous section,

$$\delta V_F = \chi [A(r)\Phi_F + B(r)\Sigma + C(r)S_F].$$

Therefore,

$$\boxed{\delta \omega_n^2 = \chi (I_\Phi + I_\Sigma + I_S)}$$

where

$$I_\Phi = \int \psi_n^* A(r) \Phi_F \psi_n dr_*,$$

$$I_\Sigma = \int \psi_n^* B(r) \Sigma \psi_n dr_*,$$

and

$$I_S = \int \psi_n^* C(r) S_F \psi_n dr_*.$$

Correction to the Eigenfrequency

Ordinarily,

$$\omega_n = \omega_n^{\text{GR}}.$$

In FCT,

$$\omega_n = \omega_n^{\text{GR}} + \delta\omega_n$$

holds. To first order,

$$\delta\omega_n = \frac{\delta\omega_n^2}{2\omega_n^{\text{GR}}}$$

and hence

$$\delta\omega_n = \frac{\chi}{2\omega_n^{\text{GR}}} (I_\Phi + I_\Sigma + I_S).$$

Complex Eigenvalues

Quasi-normal modes have complex frequencies,

$$\omega = \omega_R + i\omega_I.$$

The correction is therefore also complex:

$$\delta\omega = \delta\omega_R + i\delta\omega_I.$$

The real part gives the ringdown frequency, while the imaginary part gives the damping rate. Accordingly,

$$\omega_R^{\text{FCT}} = \omega_R^{\text{GR}} + \delta\omega_R$$

and

$$\omega_I^{\text{FCT}} = \omega_I^{\text{GR}} + \delta\omega_I$$

hold.

Quality Factor

The quality factor is defined by

$$Q = \frac{\omega_R}{2|\omega_I|}.$$

Therefore, in FCT,

$$Q_{\text{FCT}} = Q_{\text{GR}} + \delta Q.$$

The first-order correction is

$$\delta Q = Q_{\text{GR}} \left(\frac{\delta\omega_R}{\omega_R} - \frac{\delta\omega_I}{\omega_I} \right)$$

which is a directly observable quantity.

Higher Modes

Let the overtone number be

$$n = 0, 1, 2, \dots.$$

Because the Future Information Field acts on the entire wave function, the integral

$$I_\Phi$$

changes from one higher mode to another. Thus,

$$\boxed{\delta\omega_n \neq \delta\omega_m \quad (n \neq m)}$$

in general. In GR, mode ratios are determined only by mass and spin. In FCT, however, these mode ratios depend on the state of the Future Information Field.

Observational Prediction

The first prediction derived in this section is

$$\boxed{\omega_{\text{obs}} - \omega_{\text{GR}} \neq 0}$$

provided that the Future Information Field has a finite effect. Moreover, this residual must correlate with

$$\Phi_F, \quad \Sigma, \quad S_F.$$

That is, the new observational relation

$$\boxed{\Delta\omega = f(\Phi_F, \Sigma, S_F)}$$

must hold.

Limitations of This Section and Future Tasks

The correction formula derived here is based on a first-order approximation in which

$$\delta V_F$$

is treated as a small perturbation. Therefore, in regions with a strong Future Information Field, in the case of near-extremal rotating black holes, or when nonlinear couplings dominate, higher-order perturbation theory or numerical-relativity methods will be required. Furthermore, the coupling functions

$$A(r), \quad B(r), \quad C(r)$$

used in this section are still given in general form. Their explicit functional forms are not uniquely determined by the present effective treatment alone. Determining them requires a deeper action principle, additional axioms, or comparison with future observations. Thus, this section provides the general framework for quasi-normal-mode corrections in FCT. To obtain numerical predictions, the next section derives an explicit form of the effective potential

$$\delta V_F(r).$$

7.1.5 Derivation of the Effective Potential from the FCT Action Principle

The purpose of this section is to derive

$$\delta V_F(r)$$

from the FCT action principle, rather than introducing it as an independent correction term. In the preceding sections, the FCT version of the Regge–Wheeler equation was written as

$$\frac{d^2\Psi}{dr_*^2} + [\omega^2 - V_{\text{GR}}(r) - \delta V_F(r)] \Psi = 0.$$

However, for this equation to have the status of an observational prediction,

$$\delta V_F(r)$$

must not be an arbitrary function. It must be determined necessarily from the FCT axiomatic system. Accordingly, this section starts from the action containing the Future Causal Information Field, the Meaning Density Tensor, Future Convergence Entropy, and black-hole boundary conditions introduced in Paper 92, and derives the effective potential from its second-order perturbation.

FCT Action

In the present work, the FCT action in the exterior region of a black hole is written as

$$S_{\text{FCT}} = S_{\text{EH}} + S_{\Phi} + S_{\Sigma} + S_{S_F} + S_{\partial\text{BH}}.$$

Here,

$$S_{\text{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R$$

is the Einstein–Hilbert action. The action of the Future Causal Information Field is taken to be

$$S_{\Phi} = -\frac{1}{2} \int d^4x \sqrt{-g} [g^{\mu\nu} \nabla_{\mu} \Phi_F \nabla_{\nu} \Phi_F + m_F^2 \Phi_F^2].$$

The minimally coupled term for the Meaning Density Tensor is

$$S_{\Sigma} = \lambda_{\Sigma} \int d^4x \sqrt{-g} \Sigma_{\mu\nu} g^{\mu\nu}.$$

The Future Convergence Entropy term is

$$S_{S_F} = \lambda_S \int d^4x \sqrt{-g} \mathcal{S}_F(\Phi_F, \Sigma_{\mu\nu}).$$

The black-hole boundary term is defined as

$$S_{\partial\text{BH}} = \lambda_B \int_{\partial\text{BH}} d^3y \sqrt{|\gamma|} \mathcal{B}(\Phi_F, \Sigma_{\mu\nu}, S_F).$$

Here,

$$\gamma$$

is the induced metric on the boundary near the horizon.

Background-Field Decomposition

Around the Schwarzschild background, we expand

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}.$$

Similarly,

$$\Phi_F = \Phi_0(r) + \delta\Phi,$$

and

$$\Sigma_{\mu\nu} = \Sigma_{\mu\nu}^{(0)}(r) + \delta\Sigma_{\mu\nu}.$$

Here,

$$\Phi_0(r)$$

and

$$\Sigma_{\mu\nu}^{(0)}(r)$$

are the Future Causal Information Field and Meaning Density Tensor possessed by the black-hole background. In this section, we extract only the second-order action with respect to the gravitational-wave perturbation

$$h_{\mu\nu}.$$

Thus,

$$S_{\text{FCT}}^{(2)} = S_{\text{EH}}^{(2)} + S_{\Phi}^{(2)} + S_{\Sigma}^{(2)} + S_{S_F}^{(2)} + S_{\partial\text{BH}}^{(2)}.$$

General Form of the Second-Order Action

For odd-parity perturbations, introduce the Regge–Wheeler variable

$$\Psi.$$

The second-order action of General Relativity can then be written as

$$S_{\text{GR}}^{(2)} = \frac{1}{2} \int dt dr_* [(\partial_t \Psi)^2 - (\partial_{r_*} \Psi)^2 - V_{\text{GR}}(r) \Psi^2].$$

The FCT correction gives, for the same variable,

$$S_F^{(2)} = -\frac{1}{2} \int dt dr_* \delta V_F(r) \Psi^2 + S_{\text{mix}}.$$

Here,

$$S_{\text{mix}}$$

contains mixing terms between

$$\Psi$$

and

$$\delta\Phi, \quad \delta\Sigma_{\mu\nu}.$$

In the first approximation, the present work adopts the adiabatic-response approximation for the Future Information Field with respect to the gravitational-wave perturbation. That is,

$$\delta\Phi, \delta\Sigma_{\mu\nu}$$

are integrated out and effectively absorbed into the potential correction for

$$\Psi.$$

The total second-order effective action then becomes

$$S_{\text{eff}}^{(2)} = \frac{1}{2} \int dt dr_* [(\partial_t \Psi)^2 - (\partial_{r_*} \Psi)^2 - (V_{\text{GR}}(r) + \delta V_F(r)) \Psi^2].$$

Euler–Lagrange Equation

Varying the above action with respect to

$$\Psi$$

gives

$$\frac{\partial^2 \Psi}{\partial t^2} - \frac{\partial^2 \Psi}{\partial r_*^2} + (V_{\text{GR}}(r) + \delta V_F(r)) \Psi = 0.$$

With the time dependence

$$\Psi(t, r) = e^{-i\omega t} \psi(r),$$

one obtains

$$\frac{d^2 \psi}{dr_*^2} + [\omega^2 - V_{\text{GR}}(r) - \delta V_F(r)] \psi = 0.$$

Therefore,

$$\delta V_F(r)$$

is the effective potential correction obtained from the second-order perturbation of the FCT action.

Construction of the Effective Potential

Separating the contribution of each action term, we write

$$\delta V_F(r) = \delta V_\Phi(r) + \delta V_\Sigma(r) + \delta V_{S_F}(r) + \delta V_{\partial\text{BH}}(r).$$

First, the gradient term of the Future Causal Information Field gives

$$\delta V_\Phi(r) = \lambda_\Phi f(r) [(\partial_r \Phi_0)^2 + m_F^2 \Phi_0^2].$$

Second, the background component of the Meaning Density Tensor gives

$$\delta V_\Sigma(r) = \lambda_\Sigma f(r) \Pi_\ell \left[\Sigma_{\mu\nu}^{(0)} e_{(\ell)}^{\mu\nu} \right].$$

Here,

$$e_{(\ell)}^{\mu\nu}$$

is the projection of the tensor spherical harmonic associated with the perturbation mode, and

$$\Pi_\ell$$

is the projection operator onto the angular momentum mode. Third, Future Convergence Entropy contributes as

$$\delta V_{S_F}(r) = \lambda_S f(r) \frac{\partial^2 \mathcal{S}_F}{\partial \Psi^2} \Big|_{\Psi=0}.$$

Fourth, the black-hole boundary term is approximated as a term localized near the horizon:

$$\delta V_{\partial\text{BH}}(r) = \lambda_B \mathcal{B}_0 \exp \left[-\frac{(r - r_s)^2}{\ell_F^2} \right].$$

Therefore, in the minimal model we obtain

$$\boxed{\delta V_F(r) = \lambda_\Phi f(r) [(\partial_r \Phi_0)^2 + m_F^2 \Phi_0^2] + \lambda_\Sigma f(r) \Pi_\ell \left[\Sigma_{\mu\nu}^{(0)} e_{(\ell)}^{\mu\nu} \right] + \lambda_S f(r) \frac{\partial^2 \mathcal{S}_F}{\partial \Psi^2} + \lambda_B \mathcal{B}_0 e^{-(r-r_s)^2/\ell_F^2}}$$

which is the main result of this section.

Physical Meaning

This expression shows that the FCT correction is not added arbitrarily but has four distinct origins:

$$\delta V_F = \delta V_\Phi + \delta V_\Sigma + \delta V_{S_F} + \delta V_{\partial\text{BH}}.$$

These are, respectively,

the Future Causal Information Field,

the Meaning Density Tensor,

Future Convergence Entropy,

and

the black-hole boundary reprojection term.

Therefore, black-hole quasi-normal modes are not merely eigen-oscillations of spacetime geometry. Within FCT, they are eigen-oscillations of the Manifestation Constraint at the interlayer information-conversion boundary

$$X \rightarrow F.$$

GR Limit

If the Future Information Field disappears,

$$\Phi_0 \rightarrow 0, \quad \Sigma_{\mu\nu}^{(0)} \rightarrow 0, \quad S_F \rightarrow 0, \quad \lambda_B \rightarrow 0,$$

then

$$\delta V_F(r) \rightarrow 0.$$

At this point,

$$V_{\text{FCT}}(r) \rightarrow V_{\text{GR}}(r),$$

and the ordinary Regge–Wheeler equation is recovered. Thus, General Relativity is the low-Future-Information-density limit of FCT.

Observability

Since the quasi-normal-mode correction is given by

$$\delta\omega_n^2 = \int dr_* \psi_n^* \delta V_F \psi_n,$$

each contribution can be separated as

$$\delta\omega_n^2 = \delta\omega_{\Phi,n}^2 + \delta\omega_{\Sigma,n}^2 + \delta\omega_{S_F,n}^2 + \delta\omega_{B,n}^2.$$

Among these, the boundary term

$$\delta V_{\partial\text{BH}}$$

is localized near the horizon and therefore contributes strongly to echoes and damping-rate corrections. By contrast,

$$\delta V_\Phi$$

and

$$\delta V_\Sigma$$

deform the entire potential barrier and therefore contribute directly to the ringdown frequency itself. Thus, observationally, the following discrimination becomes possible:

$$\begin{cases} \delta V_\Phi, \delta V_\Sigma & \Rightarrow \text{frequency shift,} \\ \delta V_{S_F} & \Rightarrow \text{damping-rate and information-recovery-time correction,} \\ \delta V_{\partial\text{BH}} & \Rightarrow \text{echoes and near-horizon reflection.} \end{cases}$$

Conclusion of This Section

In this section, the correction term

$$\delta V_F(r)$$

which had previously been introduced as an assumption, was derived from the second-order perturbation of the FCT action principle. The FCT correction potential decomposes as

$$\delta V_F = \delta V_\Phi + \delta V_\Sigma + \delta V_{S_F} + \delta V_{\partial\text{BH}}.$$

Therefore, the FCT prediction for black-hole ringdown is not merely a conceptual reinterpretation. It provides a calculable framework connecting the action principle to corrections in quasi-normal-mode eigenvalues. In the next section, this effective potential is used to define the ringdown frequency, damping rate, quality factor, and higher-mode ratios as concrete observables.

7.1.6 FCT Correction-Coefficient Constraint Theorem

In the previous section, the effective potential correction to black-hole quasi-normal modes was introduced from the second-order perturbation of the FCT action principle as

$$\delta V_F(r) = \delta V_\Phi(r) + \delta V_\Sigma(r) + \delta V_{S_F}(r) + \delta V_{\partial\text{BH}}(r).$$

However, for this expression not to amount merely to the addition of free parameters, each correction coefficient

$$\lambda_\Phi, \quad \lambda_\Sigma, \quad \lambda_{S_F}, \quad \lambda_{\partial\text{BH}}$$

must be constrained from within the FCT axiomatic system. In this section, these coefficients are not treated as arbitrary constants. Instead, they are restricted by the following four axioms.

Axiom 1: General Relativistic Limit

In the limit where the Future Information Field, the Meaning Density Tensor, Future Convergence Entropy, and the black-hole boundary reprojection term vanish,

$$\Phi_F \rightarrow 0, \quad \Sigma_{\mu\nu} \rightarrow 0, \quad S_F \rightarrow 0, \quad \mathcal{B} \rightarrow 0,$$

FCT must reduce to General Relativity. Therefore,

$$\delta V_F(r) \rightarrow 0$$

and hence

$$V_{\text{FCT}}(r) \rightarrow V_{\text{GR}}(r)$$

must hold. This means that FCT does not reject General Relativity, but contains it in the limit of low Future Information density, low Meaning Density, and negligible reprojection-boundary effects.

Axiom 2: Covariance

The FCT correction terms must be covariant under general coordinate transformations. Therefore, the correction terms constituting the effective potential must be constructed from scalars, tensor contractions, or invariants defined on the boundary. The lowest-order admissible invariants are

$$\nabla_\mu \Phi_F \nabla^\mu \Phi_F, \quad \Phi_F^2, \quad \Sigma_{\mu\nu} e_{(\ell)}^{\mu\nu}, \quad S_F, \quad \mathcal{B}_{\partial\text{BH}}.$$

Here,

$$e_{(\ell)}^{\mu\nu}$$

is the tensor spherical harmonic associated with the perturbation mode.

Axiom 3: Minimal Order

The FCT correction must be observable as a deviation from General Relativity, while remaining sufficiently small in low-energy and low-curvature regimes. Accordingly, the corrections added to the action are restricted to the minimal order in the fields. For the Future Information Field, terms up to quadratic order are retained:

$$\Phi_F^2, \quad \nabla_\mu \Phi_F \nabla^\mu \Phi_F.$$

For the Meaning Density Tensor, only the first-order projection is retained:

$$\Sigma_{\mu\nu} e_{(\ell)}^{\mu\nu}.$$

For Future Convergence Entropy, the second variation with respect to the perturbation variable is retained:

$$\frac{\partial^2 \mathcal{S}_F}{\partial \Psi^2}.$$

For the black-hole boundary term, only the minimal boundary function localized near the horizon is retained.

Axiom 4: Conservation of Future Convergence Entropy

In Paper 92, black-hole information conservation was described by a dual structure consisting of thermodynamic entropy

$$S_T$$

and Future Convergence Entropy

$$S_F.$$

In the present work, this conservation relation is written as

$$\frac{d}{dt} (S_T + S_F) = 0$$

or, in local form near the black hole,

$$\nabla_\mu J_{S_T}^\mu + \nabla_\mu J_{S_F}^\mu = 0.$$

Therefore, the Future Convergence Entropy term is not an independent arbitrary term, but is determined complementarily by the change in thermodynamic entropy. This condition constrains

$$\lambda_S$$

so that it is not completely free.

Constraints on the Correction Coefficients

From these axioms, the FCT correction potential is restricted at minimal order to the form

$$\delta V_F(r) = \lambda_\Phi f(r) [(\partial_r \Phi_0)^2 + m_F^2 \Phi_0^2] + \lambda_\Sigma f(r) \Pi_\ell \left[\Sigma_{\mu\nu}^{(0)} e_{(\ell)}^{\mu\nu} \right] + \lambda_S f(r) \frac{\partial^2 \mathcal{S}_F}{\partial \Psi^2} + \lambda_B \mathcal{B}_0 e^{-(r-r_s)^2/\ell_F^2}.$$

However, the coefficients are not arbitrary. First, the coefficient of the Future Information Field is constrained by the field dependence of the future-convergence function:

$$\lambda_\Phi \propto \left. \frac{\partial \Lambda}{\partial \Phi_F} \right|_{\Phi_0}.$$

Second, the coefficient of the Meaning Density Tensor is constrained by the meaning-density variation of Future Convergence Entropy:

$$\lambda_\Sigma \propto \frac{\partial S_F}{\partial \Sigma_{\mu\nu}} e_{(\ell)}^{\mu\nu}.$$

Third, the entropy coefficient is determined by

$$\lambda_S \propto \frac{\partial^2 S_F}{\partial \Psi^2}.$$

Fourth, the boundary coefficient is constrained by the information-flux conservation condition at the black-hole boundary,

$$J_F^\mu n_\mu = J_X^\mu n_\mu.$$

Therefore,

$$\lambda_B \propto \left. \frac{\delta \mathcal{B}}{\delta \Phi_F} \right|_{r=r_s}.$$

Constraint Theorem

The foregoing discussion yields the following theorem.

Theorem 7.1 (FCT Correction-Coefficient Constraint Theorem). *If the General Relativistic limit, covariance, minimal order, and conservation of Future Convergence Entropy are assumed, then the FCT correction potential appearing in black-hole quasi-normal modes is restricted, at minimal order, to*

$$\delta V_F = \delta V_\Phi + \delta V_\Sigma + \delta V_{S_F} + \delta V_{\partial \text{BH}}.$$

Moreover, its coefficients are not arbitrary free constants, but are constrained by

$$\frac{\partial \Lambda}{\partial \Phi_F}, \quad \frac{\partial S_F}{\partial \Sigma_{\mu\nu}}, \quad \frac{\partial^2 S_F}{\partial \Psi^2}, \quad \frac{\delta \mathcal{B}}{\delta \Phi_F}.$$

Proof. By Axiom 1, the correction terms must vanish in the limit where

$$\Phi_F, \Sigma_{\mu\nu}, S_F, \mathcal{B}$$

vanish. Therefore, correction terms consisting only of constants are not allowed. By Axiom 2, the correction terms must be built from covariant quantities. Hence, at minimal order, only

$$\nabla_\mu \Phi_F \nabla^\mu \Phi_F, \quad \Phi_F^2, \quad \Sigma_{\mu\nu} e_{(\ell)}^{\mu\nu}, \quad S_F, \quad \mathcal{B}$$

are admissible. By Axiom 3, higher-order nonlinear terms are excluded from the minimal effective theory. Therefore, the minimal correction decomposes into the four terms

$$\delta V_\Phi, \quad \delta V_\Sigma, \quad \delta V_{S_F}, \quad \delta V_{\partial \text{BH}}.$$

By Axiom 4, the Future Convergence Entropy term is not an independent degree of freedom but is constrained by the conservation relation with thermodynamic entropy. Similarly, the boundary term is constrained by information-flux conservation near the horizon. Thus, both the form of the correction potential and the dependence of its coefficients are determined. $\square$

Range of Uniqueness

It should be noted that the uniqueness asserted by this theorem is not absolute uniqueness. Rather, it is uniqueness in the following sense:

uniqueness as the minimal correction satisfying covariance, minimal order, the GR limit, and entropy conservation.

If higher-order terms are allowed, terms such as

$$\Phi_F^4, \quad (\nabla_\mu \Phi_F \nabla^\mu \Phi_F)^2, \quad \Sigma_{\mu\nu} \Sigma^{\mu\nu}, \quad R_{\mu\nu} \Sigma^{\mu\nu}$$

may also be possible. However, these are higher-order corrections and are not included in the first observational prediction of the present work.

Meaning for Observational Predictions

This theorem implies that the ringdown correction is not an arbitrary function, but decomposes into four physical origins:

$$\Delta\omega = \Delta\omega_\Phi + \Delta\omega_\Sigma + \Delta\omega_{S_F} + \Delta\omega_{\partial\text{BH}}.$$

That is,

- $\Delta\omega_\Phi$: frequency correction from the Future Causal Information Field,
- $\Delta\omega_\Sigma$: mode-dependent correction from the Meaning Density Tensor,
- $\Delta\omega_{S_F}$: damping-rate correction from Future Convergence Entropy,
- $\Delta\omega_{\partial\text{BH}}$: echo correction from the horizon boundary term.

Therefore, observationally, it is not sufficient merely to test whether

$$\omega_{\text{obs}} \neq \omega_{\text{GR}}.$$

What matters is whether the residual

$$\Delta\omega = \omega_{\text{obs}} - \omega_{\text{GR}}$$

has a structure that can be decomposed into the four components above.

Conclusion of This Section

In this section, the coefficients of the FCT correction potential were not treated as completely free parameters. Instead, they were constrained by the General Relativistic limit, covariance, minimal order, and conservation of Future Convergence Entropy. Thus, FCT ringdown predictions are not mere curve-fitting, but constitute a minimal effective theory derived from the axiomatic system of Paper 92. In the next section, this constrained correction potential is used to derive the explicit correction formula for the ringdown frequency

$$\omega_R.$$

7.1.7 Observational Correspondence Axiom for Future Convergence Entropy

In the previous section, we showed that black-hole quasi-normal modes in FCT are described as

$$\omega_{\ell mn}^{\text{FCT}} = \omega_{\ell mn}(M, a, S_F, \Phi_F, \Sigma_{\mu\nu}),$$

depending not only on the mass

$$M,$$

and spin

$$a,$$

but also on Future Convergence Entropy

$$S_F,$$

the Future Causal Information Field

$$\Phi_F,$$

and the Meaning Density Tensor

$$\Sigma_{\mu\nu}.$$

However, this form is still insufficient as an observational theory. The quantities directly measured by observers are

$$M, \quad a, \quad A_H, \quad \omega_R, \quad \omega_I, \quad Q, \quad h(t),$$

not Future Convergence Entropy

$$S_F$$

itself. Therefore, the purpose of this section is to introduce an additional axiom that maps Future Convergence Entropy into observable quantities and transforms the ringdown predictions of FCT into experimentally measurable form.

Necessity of Observational Correspondence

In Paper 92, black-hole information conservation was defined through a dual structure consisting of thermodynamic entropy

$$S_T$$

and Future Convergence Entropy

$$S_F.$$

The total entropy was written as

$$S_{\text{tot}} = S_T + S_F.$$

Here,

$$S_T$$

corresponds to the usual Bekenstein–Hawking-type entropy,

$$S_T = \frac{k_B c^3 A_H}{4G\hbar}.$$

By contrast,

$$S_F$$

represents the future-directed informational structure preserved when the black hole connects to the Future Causal Information Field. However,

$$S_F$$

is not uniquely determined by the horizon area alone in the manner of ordinary thermodynamic entropy. Therefore, in order to include

$$S_F$$

in an observational theory, the following question must be answered:

In which observables does Future Convergence Entropy appear?

Axiom 5: Observational Correspondence of Future Convergence Entropy

To connect the information-conservation structure of Paper 92 with observables, we introduce the following axiom.

Axiom 7.1 (Observational Correspondence Axiom for Future Convergence Entropy). *Near a black hole, Future Convergence Entropy*

$$S_F$$

is not a thermodynamic entropy directly proportional to the horizon area

$$A_H.$$

Instead, it is projected as a nonthermal information-recovery component in the ringdown process, appearing in the damping rate of quasi-normal modes, echo reflectivity, and higher-mode residuals.

This axiom means that

$$S_F$$

is not measured by area alone, but is related to how the black hole re-emits or redistributes information into the external spacetime. Therefore,

$$S_F$$

has the following observational projections:

$$S_F \longrightarrow \{\delta\omega_I, \mathcal{R}_{\text{echo}}, \Delta R_{\ell mn}, \Delta t_{\text{rec}}\}.$$

Here,

$$\delta\omega_I$$

is the damping-rate correction,

$$\mathcal{R}_{\text{echo}}$$

is the echo reflectivity,

$$\Delta R_{\ell mn}$$

is the residual of higher-mode ratios, and

$$\Delta t_{\text{rec}}$$

is the correction to the information-recovery time.

Projection Function of Future Convergence Entropy

Define the projection operator mapping Future Convergence Entropy into observables as

$$\mathcal{P}_{\text{obs}}.$$

Then,

$$\mathcal{P}_{\text{obs}} : S_F \longrightarrow \mathcal{O}_{\text{GW}}.$$

Here,

$$\mathcal{O}_{\text{GW}}$$

is the set of gravitational-wave observables,

$$\mathcal{O}_{\text{GW}} = \{\omega_R, \omega_I, Q, \mathcal{R}_{\text{echo}}, R_{\ell mn}\}.$$

In the minimal model, we adopt the following linear projection:

$$\mathcal{P}_{\text{obs}}(S_F) = \alpha_I \frac{S_F}{S_T + S_F} \omega_I^{\text{GR}} + \alpha_E \frac{S_F}{S_T + S_F} \mathcal{R}_0 + \alpha_H \frac{S_F}{S_T + S_F} R_{\ell mn}^{\text{GR}}.$$

Here,

$$\alpha_I, \quad \alpha_E, \quad \alpha_H$$

are dimensionless projection coefficients. However, they are not fully free parameters. By the correction-coefficient constraint theorem of the preceding section, they are constrained as

$$\alpha_I \sim \lambda_S, \quad \alpha_E \sim \lambda_B, \quad \alpha_H \sim \lambda_\Sigma.$$

Entropy Ratio

Define the following dimensionless quantity:

$$\epsilon_F := \frac{S_F}{S_T + S_F}.$$

This is the fraction of the total black-hole entropy represented by Future Convergence Entropy. It satisfies

$$0 \leq \epsilon_F < 1.$$

If

$$S_F = 0,$$

then

$$\epsilon_F = 0,$$

and the FCT correction vanishes. On the other hand, if

$$S_F \gg S_T,$$

then

$$\epsilon_F \rightarrow 1.$$

In this case, the black hole is dominated by reconnection to the Future Information Field rather than ordinary thermodynamic dissipation, and the largest differences appear in the ringdown damping rate and echo structure.

Mapping to the Damping Rate

The imaginary part of a quasi-normal mode represents the damping rate:

$$\omega_I^{\text{FCT}} = \omega_I^{\text{GR}} + \delta\omega_I.$$

By the Observational Correspondence Axiom for Future Convergence Entropy, we set

$$\boxed{\delta\omega_I = \alpha_I \epsilon_F \omega_I^{\text{GR}}}$$

and therefore

$$\boxed{\omega_I^{\text{FCT}} = \omega_I^{\text{GR}} (1 + \alpha_I \epsilon_F)}$$

This means that if Future Convergence Entropy is nonzero, the ringdown damping time systematically deviates from the prediction of General Relativity. The damping time is

$$\tau = \frac{1}{|\omega_I|}.$$

Hence,

$$\boxed{\tau_{\text{FCT}} = \frac{\tau_{\text{GR}}}{1 + \alpha_I \epsilon_F}}$$

holds.

Mapping to Echo Reflectivity

Let the echo reflectivity be

$$\mathcal{R}_{\text{echo}}.$$

In FCT, the black-hole horizon is not a perfectly absorbing surface, but an interlayer boundary connected to the Future Causal Information Field. Therefore, the boundary term generates a small reflectivity. In the minimal model,

$$\mathcal{R}_{\text{echo}}^{\text{FCT}} = \alpha_E \epsilon_F \exp \left[-\frac{(r - r_s)^2}{\ell_F^2} \right]$$

Near the horizon,

$$r \simeq r_s,$$

this becomes

$$\mathcal{R}_{\text{echo}}^{\text{FCT}} \simeq \alpha_E \epsilon_F.$$

Therefore, the condition for the existence of an echo is

$$\alpha_E \epsilon_F > \epsilon_{\text{det}}$$

where

$$\epsilon_{\text{det}}$$

is the detector sensitivity threshold.

Mapping to Higher-Mode Ratios

Define the higher-mode ratio as

$$R_{\ell mn} = \frac{A_{\ell mn}}{A_{220}}.$$

In FCT, the Meaning Density Tensor and Future Convergence Entropy have different projections for different modes. Thus,

$$R_{\ell mn}^{\text{FCT}} = R_{\ell mn}^{\text{GR}} + \delta R_{\ell mn}.$$

In the minimal model,

$$\delta R_{\ell mn} = \alpha_H \epsilon_F [\langle \Sigma \rangle_{\ell mn} - \langle \Sigma \rangle_{220}]$$

Therefore, in FCT, residuals in higher-mode ratios correlate with the mode-dependent projection of the Meaning Density Tensor.

Mapping to the Information-Recovery Time

Let the black-hole information-recovery time be

$$t_{\text{rec}}.$$

In General Relativity and in standard semiclassical black-hole evaporation, the information-recovery time is primarily characterized by the black-hole mass and the Page time. In FCT, Future Convergence Entropy provides a nonthermal information-recovery pathway, and therefore

$$t_{\text{rec}}^{\text{FCT}} = t_{\text{rec}}^{\text{GR}} + \delta t_{\text{rec}}.$$

In the minimal model,

$$\delta t_{\text{rec}} = -\alpha_R \epsilon_F t_{\text{rec}}^{\text{GR}}$$

and hence

$$t_{\text{rec}}^{\text{FCT}} = t_{\text{rec}}^{\text{GR}} (1 - \alpha_R \epsilon_F)$$

This indicates that the larger Future Convergence Entropy is, the earlier information recovery may appear relative to the standard expectation.

Entropy Ratio as an Observable Quantity

Although Future Convergence Entropy itself is not directly measured, the dimensionless ratio

$$\epsilon_F = \frac{S_F}{S_T + S_F}$$

can be inferred inversely from several observables:

$$\epsilon_F \simeq \frac{1}{\alpha_I} \left(\frac{\omega_I^{\text{FCT}}}{\omega_I^{\text{GR}}} - 1 \right),$$

or

$$\epsilon_F \simeq \frac{\mathcal{R}_{\text{echo}}}{\alpha_E},$$

or

$$\epsilon_F \simeq \frac{\delta R_{\ell mn}}{\alpha_H(\langle \Sigma \rangle_{\ell mn} - \langle \Sigma \rangle_{220})}.$$

Therefore, the observational test of FCT is not based on a single quantity. It is based on whether multiple ringdown-related observables indicate the same

$$\epsilon_F.$$

Consistency Condition

If FCT is correct, the following consistency condition must hold:

$$\boxed{\epsilon_F^{(\omega_I)} \simeq \epsilon_F^{(\text{echo})} \simeq \epsilon_F^{(R_{\ell mn})}}$$

That is, the Future Convergence Entropy ratio independently inferred from the damping rate, echoes, and higher-mode ratios must agree. This is important because a mere free-parameter theory could tune each observable separately. In FCT, however, all these observables are governed by the same

$$\epsilon_F,$$

so cross-observable consistency becomes a testable condition.

Falsification Condition

This section yields a clear falsification condition. If, in high-signal-to-noise black-hole ringdown observations,

$$\omega_I^{\text{obs}} = \omega_I^{\text{GR}},$$

$$\mathcal{R}_{\text{echo}} = 0,$$

and

$$R_{\ell mn}^{\text{obs}} = R_{\ell mn}^{\text{GR}}$$

hold across many events, and the inferred

$$\epsilon_F$$

is consistently zero, then the FCT ringdown prediction formulated in this work is rejected. More strongly, if

$$\epsilon_F^{(\omega_I)} \not\simeq \epsilon_F^{(\text{echo})} \not\simeq \epsilon_F^{(R_{\ell mn})},$$

then the Observational Correspondence Axiom for Future Convergence Entropy is rejected.

Conclusion of This Section

In this section, we introduced a new axiom mapping the Future Convergence Entropy

$$S_F$$

introduced in Paper 92 into gravitational-wave observables. As a result, Future Convergence Entropy was redefined not as a directly observable quantity but as a quantity projected into

$$\omega_I, \quad \mathcal{R}_{\text{echo}}, \quad R_{\ell mn}, \quad t_{\text{rec}}.$$

Through this mapping, FCT becomes not merely an ontology but a predictive theory that simultaneously constrains multiple ringdown observables through the single dimensionless quantity

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

In the next section, using this observational correspondence axiom, we derive the correction formula for the real part of the ringdown frequency

$$\omega_R.$$

7.1.8 Correction Formula for the Ringdown Frequency

In the previous section, we introduced the observational correspondence axiom that maps Future Convergence Entropy into measurable quantities. As a result, the Future Information Field is not directly observable, but is projected into the ringdown frequency, damping rate, higher-mode ratios, and echo structure. In this section, we use this correspondence to derive the correction formula for the real part of the ringdown frequency.

Quasi-Normal Modes in General Relativity

In General Relativity, the black-hole quasi-normal modes are given by

$$\omega_{\ell mn}^{\text{GR}} = \omega_R^{\text{GR}} + i\omega_I^{\text{GR}}.$$

For the dominant 220 mode,

$$\omega_R^{\text{GR}} = \frac{1}{M} F_{\ell mn}(a),$$

where

$$F_{\ell mn}$$

is a dimensionless function determined solely by the spin. Thus, within General Relativity, once the black hole is specified, the ringdown frequency is completely determined.

FCT Correction

In FCT, however, a black hole is characterized not only by

$$(M, a),$$

but by

$$(M, a, \epsilon_F),$$

where

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

Consequently, the ringdown frequency becomes

$$\boxed{\omega_R = \omega_R(M, a, \epsilon_F)}$$

rather than depending only on mass and spin.

Taylor Expansion

In the weak Future Information Field limit,

$$\epsilon_F \ll 1,$$

the frequency may be expanded as an analytic function,

$$\omega_R(M, a, \epsilon_F) = \omega_R^{\text{GR}} + \left(\frac{\partial \omega_R}{\partial \epsilon_F} \right)_0 \epsilon_F + O(\epsilon_F^2).$$

Define

$$\left(\frac{\partial \omega_R}{\partial \epsilon_F} \right)_0 = \Gamma_{\ell mn}.$$

Then,

$$\omega_R^{\text{FCT}} = \omega_R^{\text{GR}} + \Gamma_{\ell mn} \epsilon_F$$

is obtained.

Future Coupling Parameter

Taking into account the mass scaling of the black hole,

$$\Gamma_{\ell mn} = \frac{\gamma_{\ell mn}}{M},$$

where

$$\gamma_{\ell mn}$$

is a dimensionless *Future Coupling Parameter*. Therefore,

$$\omega_R^{\text{FCT}} = \frac{1}{M} [F_{\ell mn}(a) + \gamma_{\ell mn} \epsilon_F]$$

is obtained. This is the fundamental equation of the present section.

Higher Modes

In General Relativity, each mode is determined solely by

$$F_{\ell mn}(a).$$

In FCT, however, the coupling strength to the Future Information Field differs from one mode to another. Hence,

$$\gamma_{220} \neq \gamma_{330} \neq \gamma_{440}$$

in general. Accordingly, higher-mode ratios possess characteristic responses to the Future Information Field.

Frequency Shift

The most important observable is the residual

$$\Delta \omega_R = \omega_R^{\text{obs}} - \omega_R^{\text{GR}}.$$

Within FCT,

$$\Delta \omega_R = \frac{\gamma_{\ell mn}}{M} \epsilon_F$$

is obtained. Thus, the frequency shift is inversely proportional to the black-hole mass and directly proportional to the Future Entropy.

Dimensionless Residual

For observational comparisons, it is convenient to introduce the dimensionless residual

$$\delta_R = \frac{\omega_R^{\text{obs}} - \omega_R^{\text{GR}}}{\omega_R^{\text{GR}}}.$$

Then,

$$\delta_R = \frac{\gamma_{\ell mn}}{F_{\ell mn}} \epsilon_F$$

follows. Since the explicit mass dependence cancels, black holes of different masses may be compared directly.

FCT Scaling Law

The ringdown frequency therefore satisfies the following scaling law:

$$M\Delta\omega_R = \gamma_{\ell mn}\epsilon_F$$

That is, the mass-normalized frequency residual is determined solely by the Future Entropy.

Observational Predictions

FCT therefore makes the following predictions. First, the absolute frequency shift becomes smaller for more massive black holes. Second, the dimensionless residual

$$\delta_R$$

is independent of the black-hole mass. Third, if multiple events are compared through

$$M\Delta\omega_R,$$

events possessing similar Future Entropy ratios

$$\epsilon_F$$

should converge onto the same universal curve.

Method of Verification

Using observations from LIGO, Virgo, KAGRA, and LISA, the black-hole mass, spin, and ringdown frequency are estimated, and the deviation

$$\Delta\omega_R$$

from General Relativity is calculated. For each event,

$$M\Delta\omega_R$$

is then evaluated. If FCT is correct, the events should be organized according to the Future Entropy ratio

$$\epsilon_F,$$

and, after removal of the mass dependence, should collapse onto a single universal curve.

Falsification Criterion

If, with sufficient observational precision,

$$M\Delta\omega_R$$

shows no correlation whatsoever with the Future Entropy, or if no mode-dependent coefficients

$$\gamma_{\ell mn}$$

exist, then the ringdown-frequency correction formula proposed in this section is falsified.

Conclusion of This Section

In this section, we derived the correction formula for the real part of the ringdown frequency using the Future Entropy:

$$\omega_R^{\text{FCT}} = \frac{1}{M} [F_{\ell mn}(a) + \gamma_{\ell mn}\epsilon_F]$$

This expression makes the Future Information Field directly testable through small residuals in the ringdown frequency. Furthermore, the scaling relation

$$M\Delta\omega_R = \gamma_{\ell mn}\epsilon_F$$

constitutes the first quantitative and cross-comparable observational prediction presented in this paper.

Part VIII

Black-Hole Boundary Theory and Future Boundary Theory

7.2 Redefinition of the Event Horizon by the Future Information Field and Prediction of Gravitational-Wave Echoes

7.2.1 8.1 Introduction

In Part VII, we showed that the Future Information Field modifies the quasi-normal modes of black holes. As a result, the ringdown frequency, damping rate, and higher-mode ratios were derived as quantities that may depend not only on mass and spin, as in General Relativity, but also on the state of the Future Information Field and Future Entropy. However, wave phenomena near a black hole are not determined by eigenvalues alone. The wave equation

$$\frac{d^2\Psi}{dr_*^2} + [\omega^2 - V(r)] \Psi = 0$$

is not closed by itself. In order to completely determine the eigenvalue problem, one must also impose boundary conditions. In General Relativity, the black-hole horizon is a perfectly absorbing boundary. Therefore, as

$$r_* \rightarrow -\infty,$$

only

$$\Psi \propto e^{-i\omega r_*}$$

is allowed. This corresponds to the physical assumption that no wave returns from inside the event horizon to the outside. Thus, after the ringdown ends, no gravitational wave appears again. This is the black-hole boundary condition in General Relativity. However, in Paper 92, the black hole was defined not as a point of information loss, but as an interlayer information-conversion operator connecting the Future Information Field and the three-dimensional manifestation layer. That is, the black hole is the unique boundary in nature that realizes the map

$$X \longrightarrow F.$$

Here,

$$X$$

denotes the three-dimensional manifestation layer, and

$$F$$

denotes the Future Information Field. If this definition is adopted, the black-hole horizon is not a perfectly absorbing boundary, but an information-conversion boundary. If information conversion exists, the boundary condition itself must differ from that of General Relativity. Therefore, after the ringdown has ended, a small reflection mediated by the Future Information Field may occur. In this chapter, this new boundary is defined as the Future Boundary. The Future Boundary is not merely the event horizon. It is an information boundary connecting the manifested universe to the Future Information Field. With this redefinition, the complete absorption condition of General Relativity,

$$\Psi = e^{-i\omega r_*},$$

can break down near the black hole, and a new boundary condition of the form

$$\Psi = T e^{-i\omega r_*} + R_F e^{+i\omega r_*}$$

may hold. Here,

$$R_F$$

is the Future Boundary Reflection Coefficient and represents the reflectivity of the wave re-projected through the Future Information Field. The purpose of this chapter is to derive this reflection coefficient from the FCT action principle and to formulate the generation condition, delay time, amplitude decay, and frequency dependence of gravitational-wave echoes. Furthermore, we clarify what observational results from LIGO, Virgo, KAGRA, and the future LISA would support or falsify Future Boundary Theory. Thus, this chapter extends the eigenvalue problem derived in Part VII into a boundary-value problem, and aims to complete black-hole ringdown theory as a boundary theory that includes the Future Information Field.

7.2.2 8.2 Mathematical Definition of the Future Boundary

In the previous section, we confirmed that, in Paper 92, the black hole is positioned as an interlayer information-conversion operator connecting the Future Information Field and the three-dimensional manifestation layer. In this section, we redefine this concept as a mathematical boundary condition.

Redefinition of the Event Horizon

In General Relativity, the event horizon is denoted by

$$\mathcal{H}.$$

It is defined as

$$\mathcal{H} = \partial J^-(\mathcal{I}^+),$$

namely as the boundary of the region from which escape to future null infinity is impossible. Thus, the event horizon is determined solely by causal structure. In FCT, however, a black hole is an information-conversion boundary with the Future Information Field. Therefore, the boundary cannot be defined by causal structure alone. Accordingly, in this paper, we introduce a new boundary

$$\boxed{\partial B_F}$$

and call it the Future Boundary.

Definition of the Future Boundary

The Future Boundary is the boundary on which the following map holds:

$$\boxed{B_F : X \longrightarrow F}$$

Here,

$$X$$

is the three-dimensional manifestation layer, and

$$F$$

is the Future Information Field. In other words, the Future Boundary is an interlayer projection boundary from the three-dimensional universe to the Future Information Field.

Boundary Manifold

The Future Boundary is defined as a three-dimensional boundary manifold

$$\partial B_F \subset M,$$

where

$$M$$

is the four-dimensional spacetime manifold. On the boundary, we introduce the induced metric

$$\gamma_{ij}.$$

As in the usual Gibbons–Hawking–York boundary term, a boundary action is defined also on the Future Boundary.

Future Boundary Action

We set the action on the Future Boundary as

$$S_B = \lambda_B \int_{\partial B_F} d^3y \sqrt{|\gamma|} \mathcal{B}$$

where

$$\mathcal{B} = \mathcal{B}(\Phi_F, \Sigma_{\mu\nu}, S_F).$$

Thus, the boundary action itself contains the Future Information Field, meaning density, and Future Convergence Entropy.

Boundary Variation

Write the total action as

$$S = S_{\text{bulk}} + S_B.$$

Here,

$$S_{\text{bulk}}$$

is the Einstein–Hilbert action together with the Future Information Field action derived in Part VII. Varying the action gives

$$\delta S = \delta S_{\text{bulk}} + \delta S_B.$$

The boundary term becomes

$$\delta S_B = \lambda_B \int_{\partial B_F} \sqrt{|\gamma|} \delta \mathcal{B}.$$

Boundary Stationarity Condition

From the action principle

$$\delta S = 0,$$

the Future Boundary must satisfy

$$\delta \mathcal{B} = 0$$

on the boundary. This is different from the usual Dirichlet boundary condition.

Boundary Condition for the Future Information Field

Because the boundary action depends on the Future Information Field,

$$\mathcal{B} = \mathcal{B}(\Phi_F, \Sigma, S_F),$$

we have

$$\delta \mathcal{B} = \frac{\partial \mathcal{B}}{\partial \Phi_F} \delta \Phi_F + \frac{\partial \mathcal{B}}{\partial \Sigma} \delta \Sigma + \frac{\partial \mathcal{B}}{\partial S_F} \delta S_F.$$

Therefore, on the Future Boundary,

$$\boxed{\frac{\partial \mathcal{B}}{\partial \Phi_F} \delta \Phi_F + \frac{\partial \mathcal{B}}{\partial \Sigma} \delta \Sigma + \frac{\partial \mathcal{B}}{\partial S_F} \delta S_F = 0}$$

must hold. This means that the Future Information Field, meaning density, and Future Convergence Entropy are not independent, but are constrained by the boundary condition.

Difference from the Ordinary Event Horizon

In General Relativity, the boundary condition is complete absorption. Therefore, no wave returns from the boundary. In FCT, however, the Future Boundary is an information-conversion boundary. Thus, complete absorption is not required. If information conversion is finite, the wave function does not vanish completely at the boundary. Therefore, the boundary condition contains a reflection term.

Future Boundary Reflection

Accordingly, we impose the boundary condition

$$\boxed{\Psi = T e^{-i\omega r_*} + R_F e^{+i\omega r_*}}$$

where

$$T$$

is the transmission coefficient, and

$$R_F$$

is the Future Boundary Reflection Coefficient. In General Relativity,

$$R_F = 0.$$

In FCT,

$$R_F = R_F(\Phi_F, \Sigma, S_F).$$

Thus, as long as the Future Information Field exists, in general one may have

$$R_F \neq 0.$$

Physical Meaning of the Future Boundary

The Future Boundary is not a mirror. Nor is it an ordinary scattering boundary. The Future Boundary is an information-conversion boundary into the Future Information Field. Therefore, the reflected wave is not a wave reflected in ordinary space. It is a wave once mapped into the Future Information Field and then reprojected. For this reason, the reflectivity

$$R_F$$

depends not on ordinary potential scattering, but on Future Convergence Entropy, meaning density, and the Future Information Field.

Conclusion of This Section

In this section, the black-hole horizon was redefined as the Future Boundary. The Future Boundary is an interlayer boundary connecting the three-dimensional manifestation layer and the Future Information Field. As a result, the complete absorption boundary condition of General Relativity,

$$R_F = 0,$$

is not generally valid in FCT, and is extended to a new boundary condition with a boundary reflection coefficient

$$R_F.$$

In the next section, we vary this boundary action and derive the Future Boundary Reflection Coefficient

$$R_F$$

from the action principle.

7.2.3 8.3 Derivation of the Future Boundary Reflection Coefficient

In the previous section, the black-hole boundary was redefined as the Future Boundary, and the boundary action

$$S_B = \lambda_B \int_{\partial B_F} d^3y \sqrt{|\gamma|} \mathcal{B}$$

was introduced. In this section, we derive the Future Boundary Reflection Coefficient

$$R_F$$

from this action.

Boundary Wave Function

Near the Future Boundary, the Regge–Wheeler equation satisfies

$$V(r) \rightarrow 0,$$

and therefore the wave function approaches a free wave. Thus, the general solution is

$$\boxed{\Psi = A_{\text{in}} e^{-i\omega r_*} + A_{\text{out}} e^{+i\omega r_*}}$$

where

$$A_{\text{in}}$$

is the wave propagating toward the Future Boundary, and

$$A_{\text{out}}$$

is the wave returning from the Future Boundary. In General Relativity,

$$A_{\text{out}} = 0.$$

Reflection Coefficient

Define the Future Boundary Reflection Coefficient by

$$\boxed{R_F = \frac{A_{\text{out}}}{A_{\text{in}}}}$$

and define the transmission coefficient as

$$T_F = \frac{A_{\text{trans}}}{A_{\text{in}}}.$$

Information Conservation Condition

In Paper 92, the black hole was defined not as a point of information loss, but as an information-conversion boundary. Therefore, at the Future Boundary, we adopt conservation of information flux rather than conservation of energy alone. Let the information flux be denoted by

$$J_F^\mu.$$

On the Future Boundary,

$$\boxed{J_X^\mu n_\mu = J_F^\mu n_\mu}$$

holds. Here,

$$n_\mu$$

is the normal vector to the boundary. This means that the information flux from the manifestation layer to the Future Information Field is conserved.

Wave Flux

For the Regge–Wheeler wave, define the flux by

$$\mathcal{J} = \frac{i}{2} (\Psi^* \partial_{r_*} \Psi - \Psi \partial_{r_*} \Psi^*).$$

For the incident wave,

$$\mathcal{J}_{\text{in}} = \omega |A_{\text{in}}|^2,$$

and for the reflected wave,

$$\mathcal{J}_{\text{out}} = -\omega |A_{\text{out}}|^2.$$

Flux Conservation at the Future Boundary

Let the information flux flowing into the Future Information Field be

$$\mathcal{J}_F.$$

Then, at the Future Boundary,

$$\boxed{\mathcal{J}_{\text{in}} = \mathcal{J}_{\text{out}} + \mathcal{J}_F}$$

holds. This represents information conservation, not ordinary energy conservation in standard scattering theory.

Future Coupling

Define the coupling strength to the Future Information Field by

$$\eta_F.$$

Set the information flux as

$$\boxed{\mathcal{J}_F = \eta_F \mathcal{J}_{\text{in}}}$$

where

$$0 \leq \eta_F \leq 1.$$

Reflection Coefficient

From flux conservation,

$$|A_{\text{in}}|^2 = |A_{\text{out}}|^2 + \eta_F |A_{\text{in}}|^2$$

is obtained. Rearranging gives

$$|A_{\text{out}}|^2 = (1 - \eta_F) |A_{\text{in}}|^2.$$

Therefore,

$$|R_F|^2 = 1 - \eta_F$$

is obtained. Equivalently,

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F}$$

where

$$\phi_F$$

is the Future Boundary Phase.

Phase

For ordinary mirror reflection, the phase is constant. At the Future Boundary, however, the wave passes through the Future Information Field. Therefore,

$$\phi_F = \phi_F(\Phi_F, \Sigma, S_F).$$

At first order, this may be expanded as

$$\phi_F = \phi_0 + \kappa_\Phi \Phi_F + \kappa_\Sigma \Sigma + \kappa_S S_F$$

Relation to Future Entropy

Using the Future Entropy ratio introduced in the previous chapter,

$$\epsilon_F = \frac{S_F}{S_T + S_F},$$

the minimal model may be written as

$$\eta_F = \eta_0 + \beta_F \epsilon_F$$

and hence

$$R_F = \sqrt{1 - \eta_0 - \beta_F \epsilon_F} e^{i\phi_F}$$

General Relativistic Limit

In the limit where the Future Information Field is absent,

$$\epsilon_F = 0,$$

and if the complete absorption condition

$$\eta_0 = 1$$

is adopted, then

$$R_F = 0.$$

Thus, General Relativity is obtained as a special case of Future Boundary Theory.

FCT Limit

If the Future Information Field is finite and information conversion is not perfectly absorbing, then

$$\eta_F < 1.$$

Therefore,

$$R_F > 0.$$

This reflection is not mirror reflection. It is delayed reflection produced by the sequence

$$\text{mapping into the Future Information Field} \longrightarrow \text{reprojection.}$$

First Theorem

The foregoing discussion yields the following theorem.

Theorem 7.2 (Future Boundary Reflection Theorem). *If the Future Boundary acts as an information-conversion boundary and information-flux conservation*

$$J_X^\mu n_\mu = J_F^\mu n_\mu$$

holds, then the Future Boundary Reflection Coefficient is

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F}.$$

Here,

$$\eta_F = \eta_F(\Phi_F, \Sigma, S_F).$$

Therefore, as long as the Future Information Field exists, in general

$$R_F \neq 0$$

may hold.

Physical Meaning

The reflection coefficient obtained in this section is not a reflection caused by ordinary potential scattering. It represents the efficiency of information conversion at the Future Boundary. Therefore, the reflectivity itself becomes an observable indicating the presence of the Future Information Field.

Conclusion of This Section

In this section, by varying the Future Boundary action, we derived the Future Boundary Reflection Coefficient

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F}.$$

This result formulates the possibility that a black hole behaves not as a perfectly absorbing boundary, but as an information-conversion boundary with the Future Information Field. In the next section, we derive the gravitational-wave echo delay time

$$\Delta t_{\text{echo}}$$

from this reflection coefficient and formulate the time-series structure of the echo train appearing after the ringdown.

7.2.4 8.4 Derivation of the Gravitational-Wave Echo Delay Time at the Future Boundary

In the previous section, from the fact that the Future Boundary is not a perfectly absorbing boundary but an information-conversion boundary with the Future Information Field, we derived the Future Boundary Reflection Coefficient

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F}.$$

However, the reflection coefficient alone does not determine the observable echo signal. The most important feature of a gravitational-wave echo is that, after the ringdown has ended, another gravitational wave arrives after a finite delay. In this section, we derive this delay time from the geometry of the Future Boundary.

Origin of the Echo Delay

In General Relativity, the event horizon is a perfectly absorbing boundary. Therefore, an incident wave falls into the black-hole interior and never returns to the exterior. Accordingly,

$$\Delta t_{\text{echo}} = \infty,$$

and no echo exists. In Future Boundary Theory, by contrast, the wave is projected into the Future Information Field and is reprojected after a finite time. Thus, the echo delay time is defined as the information-conversion time through the Future Boundary.

Future Boundary Thickness

The Future Boundary is not merely a mathematical boundary, but an information-conversion region with finite thickness. Define its effective thickness by

$$\boxed{\ell_F}$$

and assume

$$\ell_F \ll r_s.$$

Tortoise Coordinate

Outside the black hole, introduce the tortoise coordinate

$$r_* = r + 2M \ln \left| \frac{r}{2M} - 1 \right|.$$

Assume that the Future Boundary exists near

$$r = r_s + \ell_F,$$

where

$$r_s = 2GM.$$

Round-Trip Time

The wave is scattered near the photon sphere and reaches the Future Boundary. It then passes through the Future Information Field and returns again to the photon sphere. Therefore, in the tortoise coordinate, the echo delay time is the round-trip time.

First Approximation

If the Future Boundary is located at

$$r_s + \ell_F,$$

then the delay time is

$$\Delta t_{\text{echo}} \simeq 2 |r_*(r_{\text{peak}}) - r_*(r_s + \ell_F)|.$$

Here,

$$r_{\text{peak}}$$

is the position of the maximum of the Regge–Wheeler potential.

Approximate Formula

Usually,

$$\ell_F \ll r_s.$$

Expanding the tortoise coordinate gives

$$r_* \simeq 2M \ln \left(\frac{\ell_F}{r_s} \right).$$

Therefore,

$$\Delta t_{\text{echo}} = 2r_s \ln \left(\frac{r_s}{\ell_F} \right)$$

is obtained.

Future Boundary Correction

In Paper 92, the state of the Future Information Field is determined by Future Entropy. Accordingly, set the Future Boundary Thickness as

$$\ell_F = \ell_0 e^{-\alpha_F \epsilon_F},$$

where

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

FCT Delay Time

Substituting this expression yields

$$\Delta t_{\text{echo}} = 2r_s \left[\ln \left(\frac{r_s}{\ell_0} \right) + \alpha_F \epsilon_F \right]$$

This means that the echo delay time systematically increases as Future Entropy increases.

Mass Dependence

With respect to black-hole mass,

$$\Delta t_{\text{echo}} \propto M$$

holds. The Future Entropy correction is

$$\Delta t_{\text{echo}} - \Delta t_{\text{GR}} = 2r_s \alpha_F \epsilon_F.$$

Observational Predictions

Future Boundary Theory therefore gives the following predictions. First, the echo interval becomes longer in proportion to the black-hole mass. Second, as Future Entropy increases, even black holes with the same mass have echo arrival times delayed relative to General Relativity. Third, the value of

$$\epsilon_F$$

inferred from the ringdown-frequency correction must agree with the value of

$$\epsilon_F$$

inferred from the echo delay.

Consistency Condition

If Future Boundary Theory holds, then

$$\epsilon_F^{(\omega)} = \epsilon_F^{(\Delta t)}$$

must hold. That is, the ringdown frequency and the echo delay are independent measurements, yet they must yield the same Future Entropy. This is an important self-consistency condition of the theory.

Falsification Condition

If, in sufficiently precise gravitational-wave observations, no echo signal is detected, or if the detected delay time is statistically incompatible with the Future Entropy inferred from the ringdown correction, then the echo-delay model based on the Future Boundary proposed in this section is falsified.

Conclusion of This Section

In this section, by treating the Future Boundary as an information-conversion region with finite thickness, we derived the gravitational-wave echo delay time

$$\Delta t_{\text{echo}} = 2r_s \left[\ln \left(\frac{r_s}{\ell_0} \right) + \alpha_F \epsilon_F \right]$$

This result shows that the Future Information Field affects not only the ringdown frequency but also the arrival time of echoes. In the next section, using this delay time and the Reflection Coefficient, we derive the amplitude structure and decay law of the entire gravitational-wave echo train.

7.2.5 8.5 Amplitude Structure and Decay Law of the Gravitational-Wave Echo Train

In the previous section, by treating the Future Boundary as an information-conversion region with finite thickness, we derived the echo delay time

$$\Delta t_{\text{echo}} = 2r_s \left[\ln \left(\frac{r_s}{\ell_0} \right) + \alpha_F \epsilon_F \right].$$

However, the delay time alone does not determine the observable waveform. The quantity actually measured by gravitational-wave detectors is the time-dependent waveform

$$h(t).$$

Therefore, for Future Boundary Theory to become an observational theory, it is necessary to formulate the law governing the decay of the amplitude of each echo. In this section, we derive the complete amplitude structure of the echo train from the Future Boundary Reflection Coefficient.

Wave Structure After the Ringdown

Let the initial waveform immediately after the ringdown be

$$h_0(t).$$

In General Relativity, where no Future Boundary exists, this waveform exhibits only exponential decay and is never observed again. In Future Boundary Theory, however, part of the ringdown wave is reprojected through the Future Information Field. Consequently, the observed waveform is described not by a single pulse, but by an infinite sequence of echoes.

Echo Train

Define the n -th echo as

$$h_n(t).$$

Each echo arrives after the delay time

$$\Delta t_{\text{echo}}$$

derived in the previous section. Hence,

$$\boxed{h_n(t) = A_n h_0(t - n\Delta t_{\text{echo}})}$$

Reflection Coefficient

In Section 8.3, the Future Boundary Reflection Coefficient was derived as

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F}.$$

Each time the wave reaches the Future Boundary, it experiences the reflection coefficient

$$R_F.$$

Therefore, the amplitude of the n -th echo is

$$\boxed{A_n = A_0 |R_F|^n}$$

Expression in Terms of Future Entropy

Substituting

$$\eta_F = \eta_0 + \beta_F \epsilon_F,$$

we obtain

$$A_n = A_0 (1 - \eta_0 - \beta_F \epsilon_F)^{n/2}.$$

The larger the Future Entropy, the greater the modification of the reflection coefficient and therefore the decay rate of the echo train.

Phase Correction

Each passage through the Future Boundary introduces an additional phase

$$\phi_F.$$

Accordingly, the n -th echo becomes

$$\boxed{h_n(t) = A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}})}$$

Complete Echo Train

The observed waveform is the superposition of all echoes. Thus,

$$h(t) = \sum_{n=0}^{\infty} A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}})$$

is obtained. This constitutes the fundamental waveform predicted by Future Boundary Theory.

Convergence Condition

For the infinite series to converge,

$$|R_F| < 1$$

must hold. From Section 8.3,

$$|R_F|^2 = 1 - \eta_F.$$

Therefore,

$$\eta_F > 0$$

guarantees convergence of the waveform series.

Energy Decay

The energy carried by the n -th echo is proportional to the square of its amplitude. Therefore,

$$E_n = E_0 |R_F|^{2n}.$$

Using

$$|R_F|^2 = 1 - \eta_F,$$

we obtain

$$E_n = E_0 (1 - \eta_F)^n$$

This means that, because the Future Boundary is an information-conversion boundary, a fixed fraction of information is transferred to the Future Information Field at every reflection.

Decay Correction Due to Future Entropy

Substituting the Future Entropy relation gives

$$E_n = E_0 (1 - \eta_0 - \beta_F \epsilon_F)^n$$

Hence, black holes with different Future Entropy exhibit systematically different decay rates for the entire echo train.

Observational Predictions

This theory simultaneously predicts three observational signatures. First, the echo interval is constant and equal to

$$\Delta t_{\text{echo}}.$$

Second, the echo amplitudes follow the geometric progression

$$A_n = A_0 |R_F|^n.$$

Third, the phase accumulates by

$$n\phi_F.$$

Therefore, if the Future Boundary exists, an exponentially decaying echo train with regular temporal spacing should appear after the ringdown.

Consistency with Ringdown Theory

In Part VII, we showed that Future Entropy appears as a correction to the ringdown frequency,

$$\delta\omega_R.$$

In this section, we have shown that the same Future Entropy also appears in the echo amplitude,

$$A_n.$$

Accordingly,

$$\boxed{\epsilon_F^{(\omega)} = \epsilon_F^{(A)}}$$

must hold. This is another self-consistency condition of Future Boundary Theory.

Falsification Condition

If, for black-hole merger events with sufficiently high signal-to-noise ratios, no periodic echo train is observed, or if the observed amplitude decay deviates significantly from the geometric law

$$A_n = A_0 |R_F|^n,$$

then the Future Boundary Reflection Model proposed in this section is falsified.

Conclusion of This Section

In this section, we formulated the gravitational-wave echo train following black-hole ringdown using the Future Boundary Reflection Coefficient. As a result, Future Boundary Theory predicts the fundamental waveform

$$\boxed{h(t) = \sum_{n=0}^{\infty} A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}})}$$

This implies that the Future Information Field determines not only the ringdown frequency but also the amplitude, phase, and energy-decay structure of the entire echo train. In the next section, this waveform is transformed into the frequency domain to derive the spectral structure and detection template associated with the Future Boundary.

7.2.6 8.6 Spectral Structure and Frequency Response of the Future Boundary

In the previous section, using the Future Boundary Reflection Coefficient

$$R_F = \sqrt{1 - \eta_F} e^{i\phi_F},$$

we derived the complete post-ringdown echo train as

$$h(t) = \sum_{n=0}^{\infty} A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}}).$$

However, gravitational-wave detectors do not directly analyze waveforms in the time domain. In LIGO, Virgo, KAGRA, and the future LISA mission, observational data are transformed into the frequency domain and compared with theoretical templates through matched filtering. Therefore, in order to connect Future Boundary Theory with actual observations, the frequency response of the echo train must be derived. In this section, we derive the spectral structure associated with the Future Boundary and formulate its deviation from General Relativity.

Transformation to the Frequency Domain

Define the Fourier transform of the waveform

$$h(t)$$

by

$$\tilde{h}(\omega) = \int_{-\infty}^{\infty} h(t) e^{i\omega t} dt.$$

Substituting the waveform derived in Section 8.5 gives

$$\tilde{h}(\omega) = \sum_{n=0}^{\infty} A_0 |R_F|^n e^{in\phi_F} e^{-i\omega n \Delta t_{\text{echo}}} \tilde{h}_0(\omega).$$

Geometric Series

Define

$$q = |R_F| e^{i(\phi_F - \omega \Delta t_{\text{echo}})}.$$

The spectrum then becomes

$$\tilde{h}(\omega) = A_0 \tilde{h}_0(\omega) \sum_{n=0}^{\infty} q^n.$$

Under the convergence condition

$$|R_F| < 1,$$

the geometric series possesses an analytic sum.

Future Boundary Transfer Function

Accordingly, the frequency response in Future Boundary Theory is

$$\boxed{\tilde{h}(\omega) = H_F(\omega) \tilde{h}_0(\omega)}$$

where

$$\boxed{H_F(\omega) = \frac{A_0}{1 - |R_F| e^{i(\phi_F - \omega \Delta t_{\text{echo}})}}$$

is defined as the Future Boundary Transfer Function. This is the fundamental equation of the present chapter.

General Relativistic Limit

If

$$R_F = 0,$$

then the Transfer Function reduces to

$$H_F(\omega) = A_0,$$

meaning that only a single ringdown waveform remains. Thus, Future Boundary Theory naturally contains General Relativity as a special case.

Resonance Condition

The denominator of the Transfer Function becomes minimal when

$$\boxed{\phi_F - \omega \Delta t_{\text{echo}} = 2\pi N}$$

where

$$N = 0, \pm 1, \pm 2, \dots$$

Therefore, if the Future Boundary exists, periodic resonance structures should appear in the ringdown spectrum.

Resonance Frequencies

From the resonance condition,

$$\boxed{\omega_N = \frac{\phi_F - 2\pi N}{\Delta t_{\text{echo}}}}$$

is obtained. These constitute spectral features unique to the Future Boundary.

Dependence on Future Entropy

From Section 8.4,

$$\Delta t_{\text{echo}} = 2r_s \left[\ln \left(\frac{r_s}{\ell_0} \right) + \alpha_F \epsilon_F \right],$$

while from Section 8.3,

$$\phi_F = \phi_0 + \kappa_\Phi \Phi_F + \kappa_\Sigma \Sigma + \kappa_S S_F.$$

Accordingly,

$$\boxed{\omega_N = \omega_N(M, a, \epsilon_F, \Phi_F, \Sigma)}$$

Thus, the spectrum associated with the Future Boundary is not determined solely by the black-hole mass.

Spectral Spacing

The spacing between adjacent resonance peaks is

$$\Delta\omega = \omega_{N+1} - \omega_N.$$

Rearranging yields

$$\boxed{\Delta\omega = \frac{2\pi}{\Delta t_{\text{echo}}}}$$

This is one of the key predictions of Future Boundary Theory.

Spectral Envelope

Because the Reflection Coefficient is finite, the amplitudes of higher-order peaks decrease exponentially. The intensity of the N -th resonance peak is approximately

$$\boxed{I_N \propto |R_F|^{2N}}$$

Therefore, the spectrum of the Future Boundary simultaneously exhibits a series of equally spaced resonance peaks and an exponential envelope.

Observational Predictions

Future Boundary Theory predicts several features absent in General Relativity. First, the ringdown spectrum contains a series of equally spaced resonance peaks. Second, the spacing between neighboring peaks varies systematically with the black-hole mass. Third, black holes with different Future Entropy exhibit systematic shifts in both the resonance frequencies and resonance widths. Fourth, the ringdown-frequency correction, echo delay, and spectral peak positions must all be explained by the same value of

$$\epsilon_F.$$

Unified Consistency Condition

Accordingly, the entire chapter is summarized by the consistency condition

$$\boxed{\epsilon_F^{(\omega)} = \epsilon_F^{(\Delta t)} = \epsilon_F^{(H_F)}}$$

where the quantity on the right-hand side denotes the Future Entropy inferred from the peak structure of the Transfer Function. If these three independently inferred values fail to agree, Future Boundary Theory does not hold.

Falsification Condition

If sufficiently sensitive gravitational-wave observations fail to reveal the predicted periodic resonance structure in the ringdown spectrum, or if the measured peak positions are inconsistent with the Future Entropy inferred from the ringdown correction and echo delay, then the Future Boundary Transfer Function derived in this section is falsified.

Conclusion of This Section

In this section, we derived the gravitational-wave spectrum associated with the Future Boundary and obtained the Transfer Function

$$\boxed{H_F(\omega) = \frac{A_0}{1 - |R_F|e^{i(\phi_F - \omega\Delta t_{\text{echo}})}}$$

This result demonstrates that the Future Information Field generates characteristic resonance structures not only in the time domain but also in the frequency domain. Furthermore, the ringdown frequency, echo delay, and spectral resonance structure are all described consistently by a single Future Entropy, which constitutes the central theoretical prediction of this chapter. In the next section, this Transfer Function is used to construct concrete detection templates and data-analysis methods for LIGO, Virgo, KAGRA, and LISA.

7.2.7 8.7 Gravitational-Wave Detection Templates Based on Future Boundary Theory

In the previous section, we showed that the gravitational-wave spectrum generated by the Future Boundary is described by the Transfer Function

$$H_F(\omega) = \frac{A_0}{1 - |R_F|e^{i(\phi_F - \omega\Delta t_{\text{echo}})}}.$$

However, theoretical predictions alone do not constitute an observational theory. For the present theory to become a physical theory, it is necessary to construct waveform templates that can be directly applied to real gravitational-wave data analysis. In this section, we construct such templates based on Future Boundary Theory and formulate verification methods for LIGO, Virgo, KAGRA, and LISA.

Standard Template

In General Relativity, the observed waveform is written as

$$h_{\text{GR}}(t) = h_{\text{insp}} + h_{\text{merge}} + h_{\text{ring}}.$$

Matched filtering is then performed by maximizing the inner product

$$(h|d)$$

between this waveform and the observed data.

FCT Template

In Future Boundary Theory, an echo train appears after the ringdown. Therefore, the waveform template becomes

$$h_{\text{FCT}} = h_{\text{GR}} + h_{\text{echo}}$$

where

$$h_{\text{echo}} = \sum_{n=1}^{\infty} A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}}).$$

Minimal Set of Free Parameters

Throughout Part VII and the present chapter, the new degrees of freedom introduced by Future Boundary Theory are summarized by

$$\Theta_F = (\epsilon_F, \eta_F, \phi_F)$$

where

$$\epsilon_F$$

is the Future Entropy ratio,

$$\eta_F$$

is the information-conversion efficiency, and

$$\phi_F$$

is the Future Boundary phase. Within the present theory, the ringdown correction, echo delay, and spectral structure are all determined solely by

$$\Theta_F.$$

Template Space

Conventional gravitational-wave templates are parameterized only by

$$(M, a).$$

Future Boundary Theory extends the parameter space to

$$(M, a, \Theta_F)$$

However, this does not introduce an unlimited number of additional free parameters. The ringdown correction derived in Part VII, together with the Reflection Coefficient (Section 8.3), Echo Delay (Section 8.4), and Transfer Function (Section 8.6), are all constrained by the same parameter set

$$\Theta_F.$$

Matched Filtering

Let the observed data be

$$d(t).$$

The match between the data and the FCT template is evaluated by

$$\rho_F = \frac{(d|h_{\text{FCT}})}{\sqrt{(h_{\text{FCT}}|h_{\text{FCT}})}}$$

where

$$(\cdot|\cdot)$$

denotes the standard noise-weighted inner product.

Model Selection

To compare Future Boundary Theory with General Relativity, we introduce the Bayes factor

$$B_{FG} = \frac{P(d|\text{FCT})}{P(d|\text{GR})}$$

If

$$B_{FG} > 1,$$

the data favor FCT. Conversely,

$$B_{FG} < 1$$

indicates support for General Relativity.

Inverse Estimation of Future Entropy

Future Entropy is not directly observable. However, by simultaneously fitting the ringdown correction, echo delay, and Transfer Function, one may estimate

$$\epsilon_F = \epsilon_F^{\text{fit}}$$

This provides the first operational definition of the Future Information Field as an observable quantity.

Inter-Event Consistency

For multiple events, let

$$\epsilon_F^{(i)}$$

denote the inferred Future Entropy of the i -th event. If Future Boundary Theory is correct, then events with similar black-hole physical conditions should satisfy

$$\epsilon_F^{(1)} \simeq \epsilon_F^{(2)} \simeq \dots$$

Furthermore, the values inferred independently from the ringdown, echo train, and Transfer Function must also agree.

Future Boundary Catalog

Future Boundary Theory naturally motivates the construction of a catalog recording

$$(M, a, \epsilon_F, \eta_F, \phi_F)$$

for each observed black-hole event. Such a Future Boundary Catalog extends conventional black-hole mass and spin catalogs and provides the statistical foundation for investigating the properties of the Future Information Field.

Observational Predictions

Future Boundary Theory predicts the following. First, an echo train exists that cannot be explained by the ringdown alone. Second, the echo train possesses the frequency structure predicted by the Transfer Function. Third, the ringdown correction, echo delay, and Transfer Function all yield the same value of

$$\epsilon_F.$$

Fourth, black-hole events can be classified according to their Future Entropy.

Falsification Conditions

Future Boundary Theory is falsified if any of the following holds.

1. No echo signal is detected despite sufficient observational sensitivity.
2. The values of

$$\epsilon_F$$

inferred from the ringdown correction, echo delay, and Transfer Function are mutually inconsistent.

3. The Future Boundary template provides no statistically significant improvement over the General Relativity template.

Conclusion of This Section

In this section, we constructed an FCT waveform template suitable for practical gravitational-wave observations. As a result, the present theory is formulated not merely as a conceptual ontology but as a testable framework directly applicable to observational gravitational-wave data analysis. The ringdown correction derived in Part VII together with the Future Boundary Reflection, Echo Delay, and Transfer Function developed in this chapter are all unified into the minimal parameter set

$$\Theta_F = (\epsilon_F, \eta_F, \phi_F).$$

In the next section, this template will be used to evaluate the detectability of Future Boundary Theory using the sensitivity curves of LIGO, Virgo, KAGRA, and LISA, thereby assessing the extent to which the theory can be tested by current and next-generation gravitational-wave observatories.

7.2.8 8.8 Detectability and Observational Strategy of Future Boundary Theory

In the previous section, we constructed the gravitational-wave template based on Future Boundary Theory,

$$h_{\text{FCT}} = h_{\text{GR}} + h_{\text{echo}}.$$

However, even if a theoretical template has been constructed, it does not necessarily follow that current or future gravitational-wave detectors can actually identify the effects of the Future Boundary. In this section, we quantitatively evaluate the observational detectability of Future Boundary Theory and formulate verification strategies for present and next-generation gravitational-wave observatories.

Definition of Detectability

Verification of Future Boundary Theory is not simply the discovery of an echo-like signal. Rather, the theory is tested by whether three independent observables derived throughout Parts VII and VIII can all be consistently explained by a single value of Future Entropy. Accordingly, the target observables are

$$\mathcal{O}_F = \{\delta\omega_R, \Delta t_{\text{echo}}, H_F(\omega)\}$$

Gravitational-Wave Detectors

The observational facilities considered in this theory are current and planned gravitational-wave detectors. Specifically,

$$\mathcal{D} = \{\text{LIGO}, \text{Virgo}, \text{KAGRA}, \text{LISA}\}$$

is adopted. Each detector possesses different observing bands, noise characteristics, and ringdown sensitivities.

Ringdown Sensitivity

For each detector, let the signal-to-noise ratio in the ringdown regime be

$$\text{SNR}_{\text{ring}}.$$

Since the Future Boundary correction appears after the ringdown, the basic requirement for verification is

$$\text{SNR}_{\text{ring}} > \text{SNR}_{\text{crit}}$$

where

$$\text{SNR}_{\text{crit}}$$

is the critical signal-to-noise ratio required to distinguish Future Boundary effects.

Echo Sensitivity

Let the detectability of the echo signal be denoted by

$$P_{\text{echo}}.$$

From Section 8.5,

$$A_n = A_0 |R_F|^n.$$

Therefore, higher-order echoes decay exponentially. Detection capability can be improved by coherently combining multiple echoes, rather than relying solely on the first echo.

Detection of the Transfer Function

In Section 8.6, we derived the Transfer Function

$$H_F(\omega).$$

Therefore, verification of Future Boundary Theory requires simultaneous analysis in both the time and frequency domains. Let the observed spectrum be

$$\tilde{d}(\omega).$$

The goodness of fit to Future Boundary Theory is evaluated by

$$\chi_F^2 = \sum_i \frac{|\tilde{d}_i - H_F(\omega_i)\tilde{h}_i|^2}{\sigma_i^2}.$$

Triple Consistency Test

As a unique observational test of Future Boundary Theory, we propose the Triple Consistency Test. This test examines whether independently inferred values of

$$\epsilon_F$$

agree with one another. Specifically,

$$\boxed{\epsilon_F^{(\omega)} = \epsilon_F^{(\Delta t)} = \epsilon_F^{(H)}}$$

must hold. This is the central verification criterion of the theory.

Statistical Measure

To quantify the Triple Consistency Test, define the statistic

$$\boxed{\chi_F^2 = \sum_i \frac{(\epsilon_i - \bar{\epsilon})^2}{\sigma_i^2}}$$

where

$$\epsilon_i = \{\epsilon_F^{(\omega)}, \epsilon_F^{(\Delta t)}, \epsilon_F^{(H)}\}$$

and

$$\bar{\epsilon}$$

is the weighted mean. If Future Boundary Theory is valid,

$$\chi_F^2 \rightarrow 0$$

should hold.

Combined Event Analysis

For a single event, Future Boundary effects may remain buried in detector noise. Accordingly, we perform a combined analysis over multiple black-hole merger events. Let the number of events be

$$N_{\text{BH}}.$$

Define the combined likelihood by

$$\boxed{\mathcal{L} = \prod_{k=1}^{N_{\text{BH}}} P_k(d_k|\Theta_F)}$$

This enables estimation of the Future Boundary parameters from population statistics.

Improvement with Detector Sensitivity

Future Boundary Theory becomes increasingly testable with next-generation detectors. This is because the relevant effects are concentrated during the late-time ringdown. In particular, detectors capable of observing long-duration ringdown signals can measure the periodic structure of the Transfer Function with much higher precision.

Observational Predictions

The theory predicts the following. First, statistically significant echo trains should appear after the ringdown. Second, these echoes should exhibit the periodic structure predicted by the Transfer Function. Third, the values of

$$\epsilon_F$$

independently inferred from the ringdown correction, echo delay, and Transfer Function should agree. Fourth, when many events are analyzed jointly, the Future Boundary parameters should converge toward stable values.

Falsification Conditions

Future Boundary Theory is falsified if any of the following is observed.

1. No ringdown correction exists.
2. No echo delay is observed.
3. No periodic structure corresponding to the Transfer Function is present.
4. The Triple Consistency Test yields a statistically significant value of

$$\chi_F^2.$$

5. The Future Boundary parameters fail to converge even in combined analyses of many events.

Conclusion of This Section

In this section, we constructed an observational strategy for testing Future Boundary Theory using gravitational-wave observations. As a result, the theory has been formulated as a falsifiable framework in which three independent observables—the ringdown correction, echo delay, and Transfer Function—are simultaneously explained by a single Future Entropy,

$$\epsilon_F.$$

Furthermore, through the introduction of the Triple Consistency Test and combined event analyses, Future Boundary Theory becomes testable as a statistical law common to the entire black-hole population rather than as a feature of isolated events.

7.2.9 8.9 Summary of the Chapter and Verification Roadmap for Future Boundary Theory

In this chapter, based on the Future Information Field and the black-hole information-preserving structure introduced in Paper 92, the black-hole boundary has been redefined as the Future Boundary. Consequently, the event horizon, which is treated as a perfectly absorbing boundary in General Relativity, has been reformulated as an information-conversion boundary connecting the Future Information Field and the three-dimensional manifestation layer. In this section, we summarize the chapter and organize the observational predictions and verification roadmap of Future Boundary Theory.

Theoretical Framework Developed in This Chapter

The construction of Future Boundary Theory proceeded in the following order. First, the black-hole boundary was defined as the Future Boundary. Second, the Future Boundary Action was introduced, and variation of this action led to the Future Boundary Reflection Coefficient,

$$R_F = \sqrt{1 - \eta_F e^{i\phi_F}}.$$

Third, the finite thickness of the Future Boundary yielded the gravitational-wave echo delay time,

$$\Delta t_{\text{echo}} = 2r_s \left[\ln \left(\frac{r_s}{\ell_0} \right) + \alpha_F \epsilon_F \right].$$

Next, repeated Future Boundary reflections generated the gravitational-wave echo train,

$$h(t) = \sum_{n=0}^{\infty} A_0 |R_F|^n e^{in\phi_F} h_0(t - n\Delta t_{\text{echo}}).$$

Finally, the Transfer Function,

$$H_F(\omega) = \frac{A_0}{1 - |R_F| e^{i(\phi_F - \omega \Delta t_{\text{echo}})}},$$

was derived and incorporated into practical templates for gravitational-wave data analysis.

Dependency Structure of the Theory

All predictions derived in this chapter are mutually constrained. Every observable is governed by the single quantity Future Entropy,

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

Accordingly, the ringdown correction, reflection coefficient, echo delay, amplitude decay, and Transfer Function are not independent parameters. Instead, the entire theory closes through a single value of Future Entropy.

Hierarchy of Observational Predictions

Future Boundary Theory predicts a hierarchy of observational signatures. At the first level, deviations from General Relativity appear in the ringdown frequency. At the second level, an echo train generated by Future Boundary Reflection appears after the ringdown. At the third level, the echo train exhibits a systematic temporal structure determined by Future Entropy. At the fourth level, the frequency domain contains the periodic spectral structure associated with the Transfer Function. Finally, all of these observations independently yield the same value of Future Entropy.

Verification Roadmap

Verification of Future Boundary Theory should proceed through the following sequence. First, measure the ringdown correction. Second, search for gravitational-wave echoes. Third, analyze spectral peaks predicted by the Transfer Function. Fourth, perform the Triple Consistency Test. Finally, conduct combined analyses over many black-hole merger events to determine whether the Future Boundary parameters converge statistically.

Falsifiability of Future Boundary Theory

The theory proposed in this chapter is formulated as an observational theory rather than a purely ontological framework. Accordingly, Future Boundary Theory is falsified if any of the following conditions is satisfied.

1. No ringdown correction is observed.
2. No echo train is detected.
3. The spectral structure predicted by the Transfer Function is absent.
4. Future Entropy inferred from the various observables is mutually inconsistent.
5. The Future Boundary parameters fail to converge even in population analyses of many black-hole merger events.

Thus, the theory possesses explicit falsification criteria.

Comparison with General Relativity

In General Relativity, a black hole is a perfectly absorbing boundary, and no wave structure exists after the ringdown. In contrast, Future Boundary Theory treats the black hole as an information-conversion boundary connected to the Future Information Field. Consequently, four new observational quantities are introduced: the ringdown correction, the echo train, the Transfer Function, and Future Entropy. The difference between the two theories is therefore expressed directly in gravitational-wave data rather than merely at the conceptual level.

Position of This Chapter

Part VII dealt with the eigenvalue problem of black-hole quasi-normal modes. The present chapter reconstructed the boundary conditions themselves as the Future Boundary. Taken together, Parts VII and VIII complete black-hole ringdown theory as a Future Boundary Theory incorporating both the eigenvalue problem and the boundary-value problem.

Transition to the Next Chapter

The discussion in this chapter has been confined to the vicinity of black holes. However, if the Future Information Field acts on gravitational waves, its effects need not be limited to the black-hole boundary. In particular, corrections may also arise in the propagation of gravitational waves themselves, including their dispersion relation, polarization states, and phase evolution over cosmological distances. Accordingly, the next chapter moves beyond the black-hole boundary and derives how the Future Information Field modifies the gravitational-wave propagation equation itself.

Conclusion of This Chapter

In this chapter, by redefining the black-hole boundary as the Future Boundary, we developed a unified framework incorporating Future Boundary Reflection, gravitational-wave echoes, the Transfer Function, and their common description through Future Entropy. As a consequence, the black hole is described not as an object that destroys information, but as an information-conversion boundary connected to the Future Information Field. The theory is testable through multiple independent observational channels, including ringdown signals, echo trains, spectral structures, and statistical population analyses, thereby providing a theoretical foundation for directly probing the Future Information Field using gravitational-wave observations.

Part IX

Predictions for Gravitational Waves as a Whole

7.3 Predictions for Gravitational Waves as a Whole

7.3.1 9.1 Introduction: From Local Black-Hole Theory to Cosmological Propagation Theory

In Part VII, we derived the corrections that the Future Information Field gives to black-hole quasinormal modes, showing that ringdown frequencies and damping rates may vary depending on Future Entropy. In Part VIII, we further redefined the black-hole boundary as the Future Boundary and formulated the event horizon as an information-conversion boundary between the Future Information Field and the three-dimensional manifestation layer. As a result, we derived the Future Boundary Reflection Coefficient, gravitational-wave echoes, the Transfer Function, and a unified description in terms of Future Entropy. However, all of these results are theories restricted to the local region near a black hole. If the Future Information Field contributes to the fundamental structure of spacetime itself, there is no reason why its influence should be confined only to the vicinity of black holes. Gravitational waves propagate over cosmological distances. Therefore, the Future Information Field may contribute not only to ringdown after black-hole formation, but also to the propagation equation of gravitational waves themselves. In this chapter, starting from the linearized theory of gravitational waves in General Relativity and introducing the Future Information Field defined in Paper 92 into the spacetime background, we derive the modified gravitational-wave propagation equation. The purpose of this chapter is to derive theoretically how the Future Information Field modifies

- gravitational-wave speed,
- the dispersion relation,
- phase velocity,
- group velocity,
- polarization,
- and accumulated phase over cosmological distances.

These effects are not local black-hole phenomena, but appear in gravitational waves propagating through the universe as a whole. They therefore constitute a second observational window for testing the Future Information Field.

Basic Assumptions Adopted in This Chapter

This chapter preserves the ontological hierarchy introduced in Paper 92,

$$F \rightarrow H \rightarrow X.$$

Here,

$$F$$

denotes the Future Information Field,

$$H$$

denotes the Quantum Allowance Layer, and

$$X$$

denotes the three-dimensional manifestation layer. Gravitational waves are observed as waves propagating through the three-dimensional manifestation layer, but their background metric is constrained by the Future Information Field. Accordingly, we assume that gravitational waves do not propagate through a pure vacuum, but through an effective spacetime modified by the Future Information Field. This assumption is consistent with the ringdown correction formula derived in Part VII and with the Future Boundary Theory derived in Part VIII.

Main Results Derived in This Chapter

This chapter develops the theory in the following order. First, we linearize the modified Einstein equation including the Future Information Field and derive the gravitational-wave propagation equation. Next, we obtain the dispersion relation from the modified wave equation and derive the group velocity and phase velocity. We then formulate the correction that the Future Information Field gives to polarization states. Finally, we derive the phase correction accumulated by gravitational waves propagating over cosmological distances, and discuss observational testability with LIGO, Virgo, KAGRA, and LISA. All correction formulas obtained in this chapter are described in a unified way by the Future Entropy introduced in Parts VII and VIII,

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

Therefore, black-hole ringdown, the Future Boundary, and gravitational-wave propagation are not independent theories. They are understood as different manifestations of a single effect: the contribution of the Future Information Field to spacetime structure.

7.3.2 9.2 Derivation of the Gravitational-Wave Propagation Equation Including the Future Information Field

In the previous section, we introduced the basic assumption that the Future Information Field affects not only local black-hole phenomena but also gravitational waves propagating through the universe as a whole. In this section, we derive the gravitational-wave propagation equation including the Future Information Field from the action principle formulated in Paper 92. This equation becomes the foundation for all subsequent derivations of the dispersion relation, group velocity, phase velocity, polarization correction, and cosmological phase accumulation.

Background Metric

Let the background spacetime be

$$g_{\mu\nu}^{(0)}$$

and let the small perturbation be

$$h_{\mu\nu}.$$

The metric is written as

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

In this paper, the gravitational wave is defined as the propagation of this perturbation

$$h_{\mu\nu}.$$

Modified Field Equation of Paper 92

In Paper 92, the field equation including the Future Information Field was introduced as

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi C_{\mu\nu} = 8\pi G T_{\mu\nu}$$

where

$$C_{\mu\nu}$$

is the constraint tensor derived from the Future Information Field. In Paper 92, it was defined as

$$C_{\mu\nu} = \nabla_\mu \nabla_\nu \Phi_F - g_{\mu\nu} \square \Phi_F + \Sigma_{\mu\nu}.$$

In this chapter, we consider vacuum gravitational waves and therefore set

$$T_{\mu\nu} = 0.$$

Vacuum Background

The background spacetime is a vacuum solution including the Future Information Field,

$$g_{\mu\nu}^{(0)}.$$

Thus the background equation is

$$G_{\mu\nu}^{(0)} + \Lambda_{\text{eff}} g_{\mu\nu}^{(0)} + \chi \mathcal{C}_{\mu\nu}^{(0)} = 0.$$

First-Order Perturbation

Since a gravitational wave is a first-order perturbation around the background, the Einstein tensor is expanded as

$$G_{\mu\nu} = G_{\mu\nu}^{(0)} + \delta G_{\mu\nu}.$$

Similarly, the Future constraint tensor becomes

$$\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}^{(0)} + \delta \mathcal{C}_{\mu\nu}.$$

Subtracting the background equation gives the first-order equation

$$\boxed{\delta G_{\mu\nu} + \Lambda_{\text{eff}} h_{\mu\nu} + \chi \delta \mathcal{C}_{\mu\nu} = 0}$$

Lorenz Gauge

As in General Relativity, we adopt the Lorenz gauge

$$\nabla^\mu \bar{h}_{\mu\nu} = 0.$$

Here,

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} g_{\mu\nu}^{(0)} h.$$

Linear Einstein Operator

In this gauge, the first-order Einstein tensor becomes

$$\delta G_{\mu\nu} = -\frac{1}{2} \square \bar{h}_{\mu\nu}.$$

Future Constraint Term

Let the perturbation of the Future Information Field be

$$\delta \Phi_F.$$

Then the Future constraint tensor is

$$\delta \mathcal{C}_{\mu\nu} = \nabla_\mu \nabla_\nu \delta \Phi_F - g_{\mu\nu}^{(0)} \square \delta \Phi_F + \delta \Sigma_{\mu\nu}.$$

Here,

$$\delta \Sigma_{\mu\nu}$$

is the first-order perturbation of the meaning-density tensor.

Gravitational-Wave Equation

Substituting the above expressions, the gravitational-wave equation including the Future Information Field becomes

$$\boxed{\square \bar{h}_{\mu\nu} - 2\Lambda_{\text{eff}} h_{\mu\nu} = 2\chi (\nabla_\mu \nabla_\nu \delta\Phi_F - g_{\mu\nu} \square \delta\Phi_F + \delta\Sigma_{\mu\nu})}$$

This is the gravitational-wave propagation equation in the Future Information Field.

General Relativity Limit

If the Future Information Field is absent, then

$$\chi = 0.$$

Furthermore, if

$$\Lambda_{\text{eff}} = 0,$$

then

$$\boxed{\square \bar{h}_{\mu\nu} = 0}$$

is obtained. This coincides with the vacuum gravitational-wave equation of General Relativity. Therefore, the present theory naturally converges to General Relativity in the low-energy limit.

Physical Meaning of the Future Information Field

The first term on the right-hand side,

$$\nabla_\mu \nabla_\nu \delta\Phi_F,$$

represents a local curvature variation of the Future Information Field. The second term,

$$-g_{\mu\nu} \square \delta\Phi_F,$$

is the isotropic correction of the entire information field. The third term,

$$\delta\Sigma_{\mu\nu},$$

represents the anisotropic constraint due to perturbations of the meaning-density tensor. Thus, the Future Information Field is not merely a background field. It acts as a dynamical degree of freedom interacting with the gravitational wave itself.

Theoretical Consequence

From the equation derived in this section, a gravitational wave is not a simple vacuum wave. It is a wave propagating through an effective medium modified by the Future Information Field. Therefore, quantities that are constant in General Relativity—wave number, phase velocity, group velocity, and polarization state—may all receive corrections depending on the Future Information Field. These corrections are derived sequentially in the following sections.

Conclusion of This Section

In this section, by linearizing the modified field equation introduced in Paper 92, we derived the gravitational-wave propagation equation including the Future Information Field,

$$\boxed{\square \bar{h}_{\mu\nu} - 2\Lambda_{\text{eff}} h_{\mu\nu} = 2\chi (\nabla_\mu \nabla_\nu \delta\Phi_F - g_{\mu\nu} \square \delta\Phi_F + \delta\Sigma_{\mu\nu})}$$

This equation is the fundamental starting point of the entire chapter. In the next section, we derive how the Future Information Field modifies the gravitational-wave dispersion relation from this wave equation.

7.3.3 9.3 Derivation of the Gravitational-Wave Dispersion Relation from the Future Information Field

In the previous section, we derived the linear gravitational-wave equation including the Future Information Field,

$$\square \bar{h}_{\mu\nu} - 2\Lambda_{\text{eff}} h_{\mu\nu} = 2\chi (\nabla_\mu \nabla_\nu \delta\Phi_F - g_{\mu\nu} \square \delta\Phi_F + \delta\Sigma_{\mu\nu}).$$

In this section, we derive from this equation how the Future Information Field modifies the dispersion relation of gravitational waves. The dispersion formula obtained here becomes the foundation for the later derivations of group velocity, phase velocity, cosmological phase accumulation, and gravitational-wave arrival-time correction.

Local Uniform Approximation

On scales sufficiently larger than the gravitational wavelength, the Future Information Field may be approximated as locally uniform. That is, we assume

$$\nabla_\mu \Phi_F \simeq \text{const.}$$

Furthermore, approximating the background spacetime by a local inertial frame, the covariant derivative reduces to the ordinary derivative. Thus, the gravitational-wave equation becomes

$$\square \bar{h}_{\mu\nu} - 2\Lambda_{\text{eff}} h_{\mu\nu} = 2\chi (\partial_\mu \partial_\nu \delta\Phi_F - \eta_{\mu\nu} \square \delta\Phi_F + \delta\Sigma_{\mu\nu}).$$

Plane-Wave Solution

Let the gravitational wave be written as

$$h_{\mu\nu} = \varepsilon_{\mu\nu} e^{i(k_\alpha x^\alpha)}$$

where

$$k^\mu = (\omega, \mathbf{k})$$

is the four-wavevector, and

$$\varepsilon_{\mu\nu}$$

is the polarization tensor.

Perturbation of the Future Information Field

We assume that the first-order perturbation of the Future Information Field propagates with the same wavevector,

$$\delta\Phi_F = \Phi_1 e^{i(k_\alpha x^\alpha)}$$

and likewise for the meaning-density tensor,

$$\delta\Sigma_{\mu\nu} = \Sigma_{\mu\nu}^{(1)} e^{i(k_\alpha x^\alpha)}.$$

Wave Equation in Fourier Space

Substitution of the plane-wave ansatz gives

$$\square \rightarrow -k^2 = -(\omega^2 - k^2).$$

Therefore, the gravitational-wave equation becomes

$$\begin{aligned} & (\omega^2 - k^2 - 2\Lambda_{\text{eff}}) \varepsilon_{\mu\nu} \\ &= 2\chi [-k_\mu k_\nu + \eta_{\mu\nu} k^2] \Phi_1 + 2\chi \Sigma_{\mu\nu}^{(1)}. \end{aligned}$$

Effective-Medium Approximation

We combine the effect of the Future Information Field into an effective potential,

$$V_F \equiv 2\chi (\mathcal{P}_\Phi + \mathcal{P}_\Sigma)$$

where

$$\mathcal{P}_\Phi$$

is the contribution of the Future Information Field, and

$$\mathcal{P}_\Sigma$$

is the contribution of the meaning-density tensor. The wave equation is then written as

$$\omega^2 = k^2 + 2\Lambda_{\text{eff}} + V_F$$

Re-expression by Future Entropy

In Paper 92, Future Entropy was defined by

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

In this paper, we approximate the effective potential as

$$V_F = \lambda_F \epsilon_F k^2$$

where

$$\lambda_F$$

is the coupling constant between the Future Information Field and the gravitational wave. Therefore, the dispersion relation becomes

$$\omega^2 = k^2 (1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}$$

General Relativity Limit

If

$$\epsilon_F = 0$$

and

$$\Lambda_{\text{eff}} = 0,$$

then

$$\omega = k$$

is recovered. This is identical to the dispersion relation of a massless gravitational wave in General Relativity.

Low-Energy Approximation

Normally,

$$\lambda_F \epsilon_F \ll 1.$$

Thus, to first order,

$$\omega \simeq k \left(1 + \frac{1}{2} \lambda_F \epsilon_F \right) + \frac{\Lambda_{\text{eff}}}{k}$$

This is the first dispersion correction formula due to the Future Information Field.

Physical Meaning

This formula means that, if the Future Information Field exists, gravitational waves do not propagate through a perfect vacuum. Rather, they propagate through an effective medium formed by the Future Information Field. As a result, the proportionality relation between frequency and wave number deviates slightly from that of General Relativity. This deviation is described by the same state quantity, Future Entropy, that was introduced in the black-hole ringdown theory.

Observational Prediction

This section predicts the following. First, sufficiently high-frequency gravitational waves should exhibit a minute dispersion relative to General Relativity. Second, gravitational waves propagating through regions with different Future Entropy should acquire a frequency-dependent phase correction. Third, this correction must be explainable by the same parameter as the Future Entropy inferred from black-hole ringdown.

Falsification Condition

If future high-precision observations show no statistically significant deviation of the gravitational-wave dispersion relation from

$$\omega = k,$$

and if the result is inconsistent with the Future Entropy inferred from ringdown analysis, then the dispersion correction introduced in this section is rejected.

Conclusion of This Section

In this section, starting from the gravitational-wave equation including the Future Information Field, we derived the modified dispersion relation

$$\omega^2 = k^2 (1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}$$

This result shows that the Future Information Field modifies the propagation velocity of gravitational waves itself. In the next section, we derive the group velocity and phase velocity rigorously from this dispersion relation, and formulate the arrival-time correction appearing in gravitational waves propagating over cosmological distances.

7.3.4 9.4 Group Velocity, Phase Velocity, and Gravitational-Wave Arrival-Time Correction

In the previous section, we derived the dispersion relation of gravitational waves in the presence of the Future Information Field,

$$\omega^2 = k^2 (1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}.$$

In General Relativity, gravitational waves are massless waves, and in vacuum satisfy

$$\omega = k.$$

Consequently, both the phase velocity and the group velocity are equal to the speed of light,

$$c.$$

However, if the Future Information Field exists, the dispersion relation itself is modified, and therefore the propagation velocity of gravitational waves no longer coincides with the prediction of General Relativity. In this section, we derive the phase velocity, group velocity, and the arrival-time correction over cosmological distances from the modified dispersion relation.

Phase Velocity

The phase velocity is defined as

$$v_p = \frac{\omega}{k}.$$

Substituting the dispersion relation obtained in Section 9.3 yields

$$v_p = \sqrt{1 + \lambda_F \epsilon_F + \frac{2\Lambda_{\text{eff}}}{k^2}}$$

First-Order Approximation

Normally,

$$\lambda_F \epsilon_F \ll 1, \quad \frac{\Lambda_{\text{eff}}}{k^2} \ll 1.$$

Therefore, to first order,

$$v_p \simeq 1 + \frac{1}{2}\lambda_F \epsilon_F + \frac{\Lambda_{\text{eff}}}{k^2}$$

Group Velocity

The group velocity is defined by

$$v_g = \frac{d\omega}{dk}.$$

Differentiating the dispersion relation

$$\omega^2 = k^2(1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}$$

gives

$$2\omega \frac{d\omega}{dk} = 2k(1 + \lambda_F \epsilon_F).$$

Hence,

$$v_g = \frac{k(1 + \lambda_F \epsilon_F)}{\sqrt{k^2(1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}}}$$

Low-Energy Limit

In the high-wave-number limit,

$$k^2 \gg 2\Lambda_{\text{eff}},$$

we obtain

$$v_g \simeq 1 + \frac{1}{2}\lambda_F \epsilon_F - \frac{\Lambda_{\text{eff}}}{k^2}$$

Thus, the group velocity receives a correction different from that of the phase velocity.

The Future Information Field as a Dispersive Medium

In General Relativity,

$$v_g = v_p = c.$$

By contrast, in the presence of the Future Information Field,

$$v_g \neq v_p.$$

This means that the Future Information Field acts as an effective dispersive medium for gravitational waves.

Propagation Time over Cosmological Distances

Suppose that a gravitational wave propagates over a distance

$$D.$$

In General Relativity, the arrival time is

$$t_{\text{GR}} = \frac{D}{c}.$$

With the Future Information Field,

$$t_F = \frac{D}{v_g}.$$

Arrival-Time Correction

Define the arrival-time difference by

$$\Delta t = t_F - t_{\text{GR}}.$$

To first order,

$$\Delta t \simeq -\frac{D}{2}\lambda_F\epsilon_F + \frac{D\Lambda_{\text{eff}}}{k^2}$$

Thus, the Future Information Field introduces a systematic correction to the arrival time of gravitational waves propagating over cosmological distances.

Frequency Dependence

The wave number is

$$k = 2\pi f.$$

Therefore,

$$\Delta t(f) = -\frac{D}{2}\lambda_F\epsilon_F + \frac{D\Lambda_{\text{eff}}}{4\pi^2 f^2}$$

At high frequencies, the correction due to the cosmological term decreases rapidly. By contrast, the correction arising from Future Entropy remains independent of frequency.

Consistency with the Black-Hole Theory

In Part VII, we showed that the ringdown frequency correction

$$\delta\omega_R$$

is determined by Future Entropy. In this section, the group velocity, phase velocity, and arrival-time correction are likewise determined by the same quantity,

$$\epsilon_F.$$

Therefore,

$$\epsilon_F^{(\omega)} = \epsilon_F^{(v_g)} = \epsilon_F^{(\Delta t)}$$

must hold. This constitutes the fourth self-consistency condition of the theory developed throughout this paper.

Observational Predictions

This section predicts the following. First, gravitational waves propagating over cosmological distances should exhibit a group velocity different from that predicted by General Relativity. Second, the arrival-time correction is proportional to the propagation distance. Third, the Future Entropy inferred from ringdown analysis must agree with that inferred from arrival-time corrections.

Falsification Condition

If, for multiple gravitational-wave events, the arrival-time correction is statistically inconsistent with the Future Entropy inferred from ringdown analysis, then the propagation-speed correction derived in this section is rejected.

Conclusion of This Section

In this section, we derived the phase velocity, group velocity, and arrival-time correction from the modified dispersion relation including the Future Information Field. The result shows that the Future Information Field systematically modifies not only the vicinity of black holes, but also the propagation speed of gravitational waves throughout the universe. In the next section, we derive the correction that the Future Information Field gives to the polarization states of gravitational waves and formulate the difference from the two-polarization structure predicted by General Relativity.

7.3.5 9.5 Redefinition of Gravitational-Wave Polarization by the Future Information Field

In the previous section, we showed that the Future Information Field modifies the dispersion relation of gravitational waves, giving corrections to the phase velocity, group velocity, and arrival time. However, gravitational-wave propagation is not characterized solely by its propagation speed. Since a gravitational wave is a tensor field, its polarization structure itself is also influenced by the spacetime background. In General Relativity, gravitational waves are described as massless spin-2 fields possessing only two independent polarization states,

$$h_+, \quad h_\times.$$

If, however, the Future Information Field imposes anisotropic constraints on spacetime, the polarization itself may couple to the Future Information Field, producing corrections to phase, amplitude, and even polarization mixing. In this section, we derive the modification of polarization structures induced by the Future Information Field.

Polarization in General Relativity

In the transverse-traceless (TT) gauge, a plane gravitational wave is expanded as

$$h_{ij} = h_+ e_{ij}^{(+)} + h_\times e_{ij}^{(\times)}.$$

Here,

$$e_{ij}^{(+)}$$

denotes the plus polarization, and

$$e_{ij}^{(\times)}$$

denotes the cross polarization. These polarization tensors are mutually orthogonal and propagate independently in vacuum.

Anisotropy of the Future Information Field

In Paper 92, the meaning-density tensor

$$\Sigma_{\mu\nu}$$

represents the local structure of the Future Information Field. In general,

$$\Sigma_{\mu\nu}$$

is anisotropic. Consequently, a coupling between the polarization tensor and the meaning-density tensor,

$$\Sigma^{ij}e_{ij},$$

naturally arises.

Polarization Coupling Term

Introducing the Future Information Field into the polarization action gives the quadratic action

$$S_{\text{pol}} = S_{\text{GR}} + \lambda_P \int d^4x \Sigma^{ij} h_{ik} h_j^k$$

where

$$\lambda_P$$

is the polarization coupling constant.

Polarization Equation

Variation of the action yields

$$\square \mathbf{h} + \mathbf{M}_F \mathbf{h} = 0$$

where

$$\mathbf{h} = \begin{pmatrix} h_+ \\ h_\times \end{pmatrix},$$

and the Future Information Field acts through the polarization matrix

$$\mathbf{M}_F.$$

Polarization Mixing

In the most general case,

$$\mathbf{M}_F = \begin{pmatrix} \mu_+ & \gamma \\ \gamma & \mu_\times \end{pmatrix}$$

If

$$\gamma \neq 0,$$

the plus and cross polarization modes are no longer independent.

Polarization Eigenstates

The eigenpolarizations are determined from

$$\det(\mathbf{M}_F - \lambda I) = 0.$$

The corresponding eigenvalues are

$$\lambda_{\pm} = \frac{\mu_{+} + \mu_{\times}}{2} \pm \sqrt{\left(\frac{\mu_{+} - \mu_{\times}}{2}\right)^2 + \gamma^2}$$

Accordingly, the observable polarization states in the Future Information Field are no longer the plus and cross modes of General Relativity, but new eigenpolarization states.

Polarization Rotation

For a propagation distance

$$D,$$

the polarization vector evolves as

$$\mathbf{h}(D) = e^{-i\mathbf{M}_F D} \mathbf{h}(0)$$

Therefore, the polarization direction gradually rotates during propagation over cosmological distances.

Expression in Terms of Future Entropy

We approximate the contribution of the Future Information Field by

$$\gamma = \gamma_0 \epsilon_F$$

where

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

Then the polarization mixing angle becomes

$$\theta_F \simeq \gamma_0 \epsilon_F D$$

This indicates that the polarization rotation angle accumulates linearly with propagation distance.

Consistency with the Ringdown Theory

Throughout this paper, Future Entropy has been shown to determine the ringdown correction, the group-velocity correction, and the arrival-time correction. In this section, polarization rotation is likewise governed by the same Future Entropy. Therefore,

$$\epsilon_F^{(\omega)} = \epsilon_F^{(v)} = \epsilon_F^{(\Delta t)} = \epsilon_F^{(\theta)}$$

must hold. This constitutes the fifth self-consistency condition of the Future Information Field.

Observational Predictions

The present theory predicts the following. First, gravitational waves propagating over cosmological distances exhibit a polarization direction rotated relative to the prediction of General Relativity. Second, the polarization rotation angle is proportional to the propagation distance. Third, the Future Entropy inferred from polarization rotation must agree with that inferred from ringdown, arrival-time measurements, and group-velocity analysis.

Falsification Condition

If sufficiently precise polarization observations detect no polarization rotation at all, or if the Future Entropy inferred from polarization rotation is statistically inconsistent with that inferred from the other gravitational-wave observables, then the polarization-mixing theory derived in this section is rejected.

Conclusion of This Section

In this section, we formulated the influence of the Future Information Field on gravitational-wave polarization. As a result, the Future Information Field may rotate and mix the polarization state itself, rather than merely modifying the propagation speed of gravitational waves. Furthermore, because polarization rotation is uniquely determined by Future Entropy, it becomes an independent observational probe of the Future Information Field alongside ringdown, echoes, the dispersion relation, group velocity, and arrival time. In the next section, we investigate how these corrections accumulate over cosmological distances and quantitatively evaluate their detectability with LIGO, Virgo, KAGRA, and LISA.

7.3.6 9.6 Cosmological Phase Accumulation and the Integrated Effect of the Future Information Field

In the previous section, we showed that the Future Information Field modifies the polarization state of gravitational waves and that the corresponding rotation angle is proportional to the Future Entropy. However, the corrections generated near a black hole are not confined to that local region. Gravitational waves propagate over cosmological distances spanning millions to billions of years before reaching the observer. Consequently, even a minute correction induced by the Future Information Field can accumulate along the propagation path and eventually appear as an observable phase shift. In this section, we derive the phase correction produced by the Future Information Field by integrating its effect over cosmological distances.

Definition of the Phase

Let the gravitational wave be written as

$$h = Ae^{i\phi}.$$

In General Relativity, the phase is given by

$$\phi_{\text{GR}} = \int k_{\mu} dx^{\mu},$$

where

$$k_{\mu} = (\omega, \mathbf{k})$$

is the wave four-vector.

Modification by the Future Information Field

In Section 9.3, we derived the modified dispersion relation

$$\omega^2 = k^2(1 + \lambda_F \epsilon_F) + 2\Lambda_{\text{eff}}.$$

Therefore, the wave number itself depends on the Future Information Field. Denoting the local wave number by

$$k_F(x),$$

the phase becomes

$$\boxed{\phi_F = \int k_F(x) dl}$$

Phase Difference

Define the phase correction induced by the Future Information Field as

$$\Delta\phi = \phi_F - \phi_{\text{GR}}.$$

Then

$$\Delta\phi = \int (k_F - k) dl$$

First-Order Approximation

Using the first-order approximation derived in Section 9.3,

$$k_F \simeq k \left(1 - \frac{1}{2} \lambda_F \epsilon_F \right),$$

we obtain

$$\Delta\phi = -\frac{k}{2} \lambda_F \int \epsilon_F(l) dl$$

Uniform Approximation

If the Future Entropy remains approximately constant along the propagation path,

$$\epsilon_F \simeq \text{const.},$$

then

$$\Delta\phi = -\frac{kD}{2} \lambda_F \epsilon_F$$

where

$$D$$

is the propagation distance.

Including Cosmic Expansion

When cosmic expansion is taken into account, the wave number evolves as

$$k(z) = k_0(1 + z).$$

Accordingly, the phase correction becomes

$$\Delta\phi = -\frac{\lambda_F}{2} \int_0^{z_s} k_0(1 + z) \epsilon_F(z) \frac{dl}{dz} dz$$

where

$$z_s$$

is the redshift of the gravitational-wave source.

Integrated Future Information Field

We define the integral

$$I_F = \int_0^{z_s} \epsilon_F(z) \frac{dl}{dz} dz$$

and refer to it as the *Future Information Path Integral*. The phase correction can then be written compactly as

$$\Delta\phi = -\frac{\lambda_F k_0}{2} I_F$$

Physical Interpretation

In General Relativity, the phase of a gravitational wave is determined solely by the geometry of spacetime. In the Future Boundary Theory, however, the phase also depends on the path integral of the Future Information Field. Consequently, the phase is determined not only by the distance traveled, but also by the distribution of the Future Information Field encountered along the propagation path.

Integration with the Black-Hole Theory

In Part VII, Future Entropy determined the ringdown correction. In Part VIII, it determined the Future Boundary Reflection. In Part IX, Future Entropy also governs the accumulated phase of gravitational waves. Therefore,

$$\epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(v)} = \epsilon_F^{(\theta)} = \epsilon_F^{(\phi)}$$

must hold. This constitutes the sixth self-consistency condition of the Future Information Field.

Observational Predictions

The present theory predicts the following. First, the phase correction increases with the distance of the source. Second, the phase correction is proportional to the gravitational-wave frequency. Third, the Future Entropy inferred independently from ringdown, echoes, polarization, group velocity, and accumulated phase must all agree.

Falsification Condition

If future cosmological gravitational-wave observations show that the accumulated phase agrees entirely with the prediction of General Relativity, and is furthermore inconsistent with the Future Entropy inferred from ringdown and echo analyses, then the phase-accumulation theory developed in this section is rejected.

Conclusion of This Section

In this section, we derived the accumulated phase correction induced by the Future Information Field for gravitational waves propagating over cosmological distances. The result demonstrates that the gravitational-wave phase is determined not only by spacetime geometry but also by the path integral of the Future Information Field,

$$I_F = \int \epsilon_F dl.$$

This implies that the Future Information Field influences gravitational waves not only in the vicinity of black holes but throughout the universe via cumulative propagation effects. In the next section, we apply this accumulated-phase theory to standard-siren observations involving binary black holes and binary neutron stars, and derive the corresponding arrival-time and waveform corrections.

7.3.7 9.7 Future Information Field Corrections to Standard Sirens

In the previous section, we showed that the Future Information Field produces an accumulated phase correction in gravitational waves propagating over cosmological distances. However, the phase correction itself is not a directly observable quantity. In gravitational-wave astronomy, the signals from compact binary mergers are treated as *standard sirens*, from which luminosity distance, source masses, the cosmic expansion rate, and cosmological parameters are inferred.

Therefore, for the Future Information Field to become observationally testable, it is necessary to derive the corrections that it introduces into standard-siren analyses. In this section, we consider both binary black holes and binary neutron stars, and derive the theoretical corrections induced by the Future Information Field.

Standard-Siren Waveform

In General Relativity, the frequency-domain waveform is written as

$$\tilde{h}_{\text{GR}}(f) = A(f)e^{i\Psi_{\text{GR}}(f)},$$

where

$$A(f)$$

is the waveform amplitude and

$$\Psi_{\text{GR}}(f)$$

is the gravitational-wave phase.

Future Information Field Correction

From Section 9.6, the Future Information Field produces a phase correction

$$\Delta\phi_F.$$

Therefore, the standard-siren waveform becomes

$$\boxed{\tilde{h}_F(f) = A(f)e^{i(\Psi_{\text{GR}} + \Delta\phi_F)}}$$

Future Phase

From the previous section,

$$\Delta\phi_F = -\frac{\lambda_F k_0}{2} I_F.$$

Substituting

$$k_0 = 2\pi f,$$

we obtain

$$\boxed{\Delta\phi_F(f) = -\pi\lambda_F f I_F}$$

Arrival-Time Correction

The phase correction can be converted into an arrival-time correction. In general,

$$t = \frac{1}{2\pi} \frac{d\Psi}{df}.$$

Hence, the Future Information Field produces an arrival-time correction

$$\boxed{\Delta t_F = -\frac{\lambda_F}{2} I_F}$$

which is consistent with the group-velocity correction derived in Section 9.4.

Binary Black Holes

For binary black-hole systems, there is generally no electromagnetic counterpart. Accordingly, the Future Information Field is inferred solely from gravitational-wave observations. Within the present theory, the Future Entropy can be estimated by simultaneously fitting the ringdown correction, echoes, accumulated phase, and arrival-time correction.

Binary Neutron Stars

For binary neutron-star mergers, gravitational waves are accompanied by electromagnetic counterparts such as gamma-ray bursts, kilonovae, and optical observations. Therefore, one can compare the electromagnetic arrival time

$$t_{\text{EM}}$$

with the gravitational-wave arrival time

$$t_{\text{GW}}$$

through

$$\Delta T = t_{\text{GW}} - t_{\text{EM}}$$

Contribution of the Future Information Field

In General Relativity, this difference is attributed to astrophysical delays. Within the Future Boundary Theory, an additional correction

$$\Delta t_F$$

is present. Thus,

$$\Delta T = \Delta T_{\text{astro}} + \Delta t_F$$

Distance Dependence

The Future Information Path Integral increases with propagation distance. Consequently,

$$\Delta t_F \propto D$$

where

$$D$$

is the propagation distance. The farther the source, the larger the accumulated contribution of the Future Information Field.

Consistency with the Ringdown Theory

In Part VII, the Future Entropy was inferred from the ringdown correction. In Part VIII, it was inferred from the Future Boundary Reflection. In Part IX, the phase, group velocity, and standard-siren observations also provide independent estimates of the Future Entropy. Accordingly,

$$\epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(v)} = \epsilon_F^{(\phi)} = \epsilon_F^{(\text{siren})}$$

must hold. This constitutes the seventh self-consistency condition of the Future Information Field.

Observational Predictions

The present theory predicts the following. First, more distant standard sirens should exhibit larger phase deviations from the prediction of General Relativity. Second, for binary black holes, the Future Entropy inferred from ringdown analysis should agree with that inferred from propagation effects. Third, for binary neutron stars, the arrival-time difference between gravitational and electromagnetic signals should include an additional distance-dependent contribution from the Future Information Field.

Falsification Condition

If sufficiently precise standard-siren observations show that the phase correction, arrival-time correction, and ringdown correction yield mutually inconsistent values of the Future Entropy, and if all observations are instead consistently explained by General Relativity alone, then the Future Information Field correction proposed in this section is rejected.

Conclusion of This Section

In this section, we derived the corrections induced by the Future Information Field for standard-siren observations. The result shows that the Future Information Field influences not only local black-hole phenomena but also gravitational waves propagating across cosmological distances. Furthermore, because both binary black holes and binary neutron stars allow independent estimates of the Future Entropy, the Future Information Field becomes a unified observable connecting ringdown theory, propagation theory, and cosmological distance measurements. In the next section, we apply the present theory to the sensitivity curves of the current and next-generation gravitational-wave detector network (LIGO, Virgo, KAGRA, and LISA) and derive the conditions under which the Future Information Field becomes observationally detectable.

7.3.8 9.8 Detectability of the Future Information Field with LIGO, Virgo, KAGRA, and LISA

In the previous section, we showed that the Future Information Field introduces phase corrections, arrival-time corrections, and enables an independent estimation of the Future Entropy through standard-siren observations. However, the existence of theoretical correction formulas alone is not sufficient to establish a viable physical theory. The essential question is whether present and future gravitational-wave observatories possess sufficient sensitivity to distinguish these corrections observationally. In this section, we evaluate the detectability of the Future Information Field using both current and next-generation gravitational-wave detectors.

Target Observables

The observables considered in this work are

$$\mathcal{O}_F = \{\delta\omega_R, \Delta t_{\text{echo}}, \Delta\phi, v_g, \theta_F\}.$$

Here,

$$\delta\omega_R$$

denotes the ringdown correction,

$$\Delta t_{\text{echo}}$$

the echo delay,

$$\Delta\phi$$

the accumulated cosmological phase shift,

$$v_g$$

the group velocity, and

$$\theta_F$$

the polarization rotation angle. If the Future Information Field exists, all of these quantities must be determined by the same value of the Future Entropy.

Gravitational-Wave Detectors

The detector network considered in this work is

$$\mathcal{D} = \{\text{LIGO, Virgo, KAGRA, LISA}\}$$

Each detector operates in a different frequency band, probes different astrophysical distances, and possesses different sensitivities to ringdown signals. Consequently, the optimal observational probes of the Future Information Field differ among detectors.

LIGO, Virgo, and KAGRA

Ground-based interferometers primarily observe gravitational waves in the frequency range from several tens of hertz to several kilohertz. Within this band, the principal observables are the ringdown signal, gravitational-wave echoes, higher-order modes, and standard-siren waveforms. Accordingly, the most important observables for the Future Boundary Theory are

$$\delta\omega_R, \quad \Delta t_{\text{echo}}.$$

LISA

LISA targets supermassive black holes, extreme mass-ratio inspirals (EMRIs), and gravitational waves propagating across cosmological distances. Therefore, the principal observables become the accumulated phase shift, polarization rotation, group velocity, and the Future Information Path Integral, namely,

$$\Delta\phi, \quad \theta_F, \quad v_g.$$

Sensitivity to the Future Information Field

The observational sensitivity to the Future Information Field is defined as

$$\mathcal{S}_F = \frac{\Delta\mathcal{O}_F}{\sigma_{\mathcal{O}}}$$

where

$$\Delta\mathcal{O}_F$$

is the correction predicted by the Future Information Field and

$$\sigma_{\mathcal{O}}$$

is the observational uncertainty of the corresponding observable.

Detection Criterion

We define the detection criterion for the Future Information Field as

$$\mathcal{S}_F > 5$$

corresponding to a statistical significance exceeding five standard deviations relative to General Relativity.

Multi-Event Combination

For an individual event, the predicted corrections may be hidden within detector noise. If

$$N$$

independent events are combined, the statistical uncertainty approximately scales as

$$\sigma \propto \frac{1}{\sqrt{N}}$$

and therefore the detection significance becomes

$$\mathcal{S}_F \propto \sqrt{N}$$

demonstrating the importance of population analyses.

Unified Estimation of the Future Entropy

For each observed event,

$$\epsilon_F^{(i)}$$

can be independently inferred. If the Future Information Field is physically real, then after correcting for differences in black-hole mass, spin, and propagation distance, all measurements should converge toward a common theoretical value,

$$\epsilon_F^{(1)} \simeq \epsilon_F^{(2)} \simeq \dots \simeq \epsilon_F^{(N)}$$

Integration with Paper 92

Paper 92 introduced the Future Entropy as the fundamental state variable governing black-hole information preservation, ringdown corrections, and the Future Boundary. The present section further demonstrates that the Future Entropy can also be estimated independently from cosmological gravitational-wave observations. Accordingly, the Future Entropy is elevated from a local black-hole quantity to a cosmological observable.

Global Self-Consistency of the Theory

Combining all previous chapters, the Future Information Field requires that the following independent observables yield the same value,

$$\epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(v)} = \epsilon_F^{(\phi)} = \epsilon_F^{(\text{siren})} = \epsilon_F^{(\text{detector})}$$

This constitutes the eighth self-consistency condition of the present theory.

Falsification Conditions

The Future Information Field is rejected if any of the following conditions is satisfied.

1. The Future Entropy inferred from different gravitational-wave detectors is mutually inconsistent.
2. The inferred Future Entropy fails to converge as the number of observed events increases.
3. All observables are statistically explained by General Relativity alone without requiring the Future Information Field.

Conclusion of This Section

In this section, we formulated the detectability of the Future Information Field within the LIGO, Virgo, KAGRA, and LISA detector network. The analysis demonstrates that the Future Information Field is not merely a local black-hole phenomenon but can, in principle, be tested through gravitational-wave observations across the Universe. Furthermore, agreement among the independently inferred values of the Future Entropy from different detector networks provides the final observational consistency condition of the present theory. In the next section, we summarize the entire chapter, clarify the physical implications of the Future Information Field for gravitational-wave propagation, compare the theory with General Relativity, and present a roadmap for future observational verification.

7.3.9 9.9 Summary of the Chapter and Roadmap for Future Observational Tests

In this chapter, we extended the Future Information Field introduced in Paper 92 from the vicinity of black holes to gravitational waves propagating across cosmological distances. As a consequence, the Future Information Field is formulated not merely as a local black-hole phenomenon but as a universal information structure permeating the observable Universe. Furthermore, ringdown corrections, gravitational-wave echoes, modified dispersion relations, group velocity, phase velocity, polarization rotation, accumulated cosmological phase shifts, standard sirens, and detector sensitivities have all been incorporated into a single theoretical framework. In this concluding section, we summarize the results obtained throughout the chapter, clarify the differences from General Relativity, highlight the distinctive features of the Future Boundary Theory, and outline a roadmap for future observational tests.

Theoretical Framework Developed in This Chapter

All correction formulas derived in this chapter originate from the Future Information Field and the Future Entropy introduced in Paper 92. Their logical structure can be summarized as

$$\boxed{\Phi_F \implies \epsilon_F \implies \{\delta\omega_R, \Delta t_{\text{echo}}, v_g, \Delta\phi, \theta_F\}}$$

Thus, the Future Information Field itself is not directly observable. Instead, it is projected onto multiple independent observables through the intermediary quantity known as the Future Entropy. This hierarchical structure is fully consistent with the ontological framework established by the Future Boundary in Paper 92.

Comparison with General Relativity

Within General Relativity, gravitational waves arise as linear perturbations of the Einstein equations and satisfy

$$\omega = k$$

in vacuum. Consequently, the phase velocity, group velocity, and polarization structure are completely determined by the background spacetime geometry. In contrast, the present theory introduces the Future Information Field as an additional degree of freedom contributing to the effective spacetime. Accordingly, gravitational waves propagate not through an empty vacuum but through an effective medium modified by the Future Information Field. This modification affects not only the local physics near black holes but also accumulates over cosmological propagation distances.

New Observables Introduced by the Future Boundary Theory

To observationally test the Future Information Field, this work introduced the following independent observables,

$$\mathcal{O}_F = \{\delta\omega_R, \Delta t_{\text{echo}}, v_g, \Delta\phi, \theta_F, \Delta t_F\}$$

These quantities have traditionally been analyzed independently. Within the Future Boundary Theory, however, all of them originate from the single state variable known as the Future Entropy. This unified structure has no counterpart within General Relativity.

Unified Estimation of the Future Entropy

Throughout this chapter, the Future Entropy becomes independently measurable through multiple observational channels,

$$\epsilon_F = \epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(v)} = \epsilon_F^{(\phi)} = \epsilon_F^{(\theta)} = \epsilon_F^{(\text{siren})}$$

If the Future Information Field exists physically, every independent observational method must converge toward the same value of the Future Entropy. This requirement represents the principal observational consistency condition of the present theory.

Roadmap for Observational Verification

The proposed observational program proceeds through the following stages. **Phase I** Estimate the Future Entropy from black-hole ringdown observations,

$$\epsilon_F^{(\omega)}.$$

Phase II Search for gravitational-wave echoes and test the Future Boundary Reflection,

$$\epsilon_F^{(\text{echo})}.$$

Phase III Measure gravitational-wave dispersion, group velocity, and phase velocity,

$$\epsilon_F^{(v)}.$$

Phase IV Measure cosmological phase accumulation, polarization rotation, and standard-siren corrections,

$$\epsilon_F^{(\phi)}, \quad \epsilon_F^{(\theta)}.$$

Phase V Combine all independent observables and determine whether they converge toward a single value of the Future Entropy.

Falsifiability of the Theory

The present theory is rejected if any of the following conditions is satisfied.

1. The Future Entropy inferred from ringdown observations, gravitational-wave echoes, dispersion, polarization, and accumulated phase shifts is mutually inconsistent.
2. The inferred Future Entropy fails to converge toward a common value as the number of observed events increases.
3. General Relativity alone provides a statistically satisfactory explanation for all observations.

4. Introducing the Future Information Field does not improve the explanatory or predictive power relative to General Relativity.

Accordingly, the present theory constitutes a falsifiable physical framework whose validity is determined entirely through observational evidence.

Integration with Paper 92

Paper 92 introduced the Future Information Field together with black-hole information preservation, the Future Entropy, the Future Boundary, and the modified gravitational field equations. The present chapter extends those concepts to the propagation of gravitational waves across the Universe. Consequently, the Future Information Field is shown to influence not only the immediate vicinity of black holes but also the physical laws governing gravitational-wave propagation over cosmological distances. In this sense, the ontological framework proposed in Paper 92 becomes connected to the observable Universe as a whole.

Conclusion of This Chapter

In this chapter, we systematically developed a theory of gravitational-wave propagation based on the Future Information Field. As a result, black-hole ringdown, gravitational-wave echoes, modified dispersion relations, group velocity, polarization, cosmological phase accumulation, standard sirens, and detector analyses can all be described within a unified framework governed by the Future Entropy. The Future Information Field therefore ceases to be merely a conceptual element of black-hole physics and instead becomes a physical state variable that can be tested throughout gravitational-wave astronomy. The theoretical predictions developed in this chapter provide quantitative targets for present and future observations with LIGO, Virgo, KAGRA, LISA, and forthcoming third-generation gravitational-wave observatories.

Part X

Dark Matter Theory Based on the Future Information Field

Quantitative Predictions for Galactic Rotation Curves, Gravitational Lensing, the Bullet Cluster, and the Large-Scale Structure of the Universe

7.3.10 10.1 Introduction: Redefining the Dark Matter Problem

In Chapter VII, we showed that the Future Information Field governs the structure of information preservation in the vicinity of black holes. In Chapter VIII, we introduced the Future Boundary and formulated the event horizon as an informational boundary connecting the three-dimensional manifestation layer with the Future Information Field. Furthermore, in Chapter IX, we derived that the Future Information Field consistently modifies gravitational-wave

- ringdown,
- echoes,
- dispersion relations,
- group velocity,
- polarization,
- and accumulated phase.

All of these results suggest that the Future Information Field acts not only in local black-hole phenomena, but also as a background structure of spacetime itself. However, if the Future Information Field is a background field of the universe as a whole, then there is no reason for its effects to be limited only to black holes and gravitational waves. At cosmological scales as well, including galaxies, galaxy clusters, the large-scale structure of the universe, and the cosmic microwave background, the same information field should contribute to gravitational phenomena. In this chapter, we reconstruct the dark matter problem from this standpoint.

The Present Dark Matter Theory

In the current standard cosmological model, the existence of Dark Matter is assumed in order to explain galactic rotation curves, the dynamics of galaxy clusters, gravitational lensing, the cosmic microwave background, and the large-scale structure of the universe. This dark component is assumed to have no electromagnetic interaction and to exhibit only gravitational interaction. The energy-momentum tensor is decomposed as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{DM}}$$

and the Einstein equation becomes

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{DM}} \right).$$

However, up to the present time, the particles constituting Dark Matter have not been directly detected. Therefore, Dark Matter exists as an effective degree of freedom introduced to explain the observed gravitational effects.

Basic Position of This Paper

In this paper, we do not introduce Dark Matter as a new particle. Instead, we assume that the Future Information Field introduced in Paper 92 naturally gives rise to gravitational effects equivalent to Dark Matter when it is added to the gravitational field equations. That is, Dark Matter is not an unknown particle, but the cosmological expression of the constraint that the Future Information Field imposes on the three-dimensional manifestation layer.

Redefinition by the Future Information Field

In Paper 92, we introduced the modified Einstein equation

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Here,

$$\mathcal{C}_{\mu\nu}$$

is the constraint tensor derived from the Future Information Field. In this chapter, we define the component of this constraint term that does not accompany optical manifestation as

$$\mathcal{C}_{\mu\nu}^{\text{nonopt}}.$$

Then the modified Einstein equation can be rewritten as

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{baryon}} + 8\pi G T_{\mu\nu}^{\text{F}}.$$

Here, we define the effective tensor corresponding to the Future Information Field as

$$T_{\mu\nu}^{\text{F}} = -\frac{1}{8\pi G} (\Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu}^{\text{nonopt}}).$$

In this paper, we regard this

$$T_{\mu\nu}^{\text{F}}$$

as the essence of the gravitational effects currently observed as Dark Matter.

Ontological Interpretation

In this position, Dark Matter is not an “invisible particle.” Nor is it a particle of the same kind as ordinary matter that simply fails to emit light. Rather, the constraint of the Future Information Field appears in the three-dimensional manifestation layer only as gravity, and the state in which it does not project onto the electromagnetic field is observationally recognized as Dark Matter. Therefore, Dark Matter is redefined as the non-optical manifestation state of the Future Information Field.

Purpose of This Chapter

In this chapter, starting from this redefinition, we sequentially derive the following cosmological observables:

- galactic rotation curves,
- gravitational lensing,
- the Bullet Cluster,
- galaxy-cluster mass distributions,
- the large-scale structure of the universe,
- and the cosmic microwave background.

What is important is that these are not treated as independent phenomena. Rather, they are derived simultaneously from a single state quantity: the Future Information Field.

Structure of This Chapter

In this chapter, we first rigorously derive the effective energy-momentum tensor corresponding to the Future Information Field. Next, we obtain the modified formula for galactic rotation curves and then extend the discussion to gravitational lensing, the Bullet Cluster, and the large-scale structure of the universe. Finally, we show that the Future Entropy inferred from all these phenomena must coincide with the value obtained from ringdown, echoes, and gravitational-wave propagation introduced in Chapters VII through IX. Through this, a single theory of the Future Information Field is completed, spanning black holes, gravitational waves, and cosmological scales. Instead, we assume that the Future Information Field introduced in Paper 92 naturally gives rise to gravitational effects equivalent to Dark Matter when it is added to the gravitational field equations. That is, Dark Matter is not an unknown particle, but the cosmological expression of the constraint that the Future Information Field imposes on the three-dimensional manifestation layer.

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7.3.11 10.2 The Mathematical Structure of Dark Matter in General Relativity and Λ CDM Cosmology, and Its Reformulation by the Future Information Field

In the previous section, we presented the basic position of this paper: Dark Matter is not to be understood as an unknown particle, but is to be redefined as a cosmological manifestation of the Future Information Field. However, for this claim to be meaningful, we must clearly organize the mathematical structure through which Dark Matter is introduced in the current standard cosmology, and then rigorously show which part the Future Information Field replaces and which part it preserves. In this section, we compare general relativity, Λ CDM cosmology, and the modified field equation introduced in Paper 92, and formulate how the Future Information Field appears as an effective degree of freedom corresponding to Dark Matter.

Dark Matter in the Einstein Equation

In general relativity, the structure of spacetime is determined by the Einstein equation

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}}.$$

Here,

$$T_{\mu\nu}$$

represents all energy and momentum existing in the universe. In standard cosmology, this is decomposed as

$$\boxed{T_{\mu\nu} = T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{radiation}} + T_{\mu\nu}^{\text{DM}} + T_{\mu\nu}^{\text{DE}}}.$$

Dark Matter is introduced as

$$T_{\mu\nu}^{\text{DM}}$$

among these components. That is, in present cosmology, Dark Matter is added to the energy-momentum tensor from the outset as an “unknown matter” component.

Basic Assumptions of Λ CDM Cosmology

In Λ CDM, Dark Matter is treated as Cold Dark Matter. Therefore, its pressure satisfies

$$p_{\text{DM}} \simeq 0.$$

Only its energy density contributes to cosmic expansion, and the Friedmann equation becomes

$$H^2 = \frac{8\pi G}{3} (\rho_b + \rho_r + \rho_{\text{DM}} + \rho_\Lambda).$$

This

$$\rho_{\text{DM}}$$

governs galaxy formation, cluster formation, gravitational lensing, CMB anisotropies, and the large-scale structure of the universe.

Problems Remaining in the Standard Theory

Λ CDM agrees extremely well with observations. However, the essence of Dark Matter remains unresolved. To date, no decisive evidence for Dark Matter particles has been obtained from direct-detection experiments, accelerator experiments, or indirect-detection experiments. Therefore, what is observed is not a particle, but the gravitational effect itself. This means that the ontology of Dark Matter remains incomplete.

Reformulation by the Future Information Field

In Paper 92, the Einstein equation was extended as

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{baryon}}.$$

Here, no unknown particle is introduced. The only additional degree of freedom is the constraint tensor

$$\mathcal{C}_{\mu\nu}$$

derived from the Future Information Field. Rearranging this equation gives

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{F}} \right).$$

Here,

$$T_{\mu\nu}^{\text{F}} = -\frac{1}{8\pi G} (\Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu})$$

is defined.

Replacement of Dark Matter

As a result,

$$T_{\mu\nu}^{\text{DM}}$$

in Λ CDM is replaced in the present theory by

$$\boxed{T_{\mu\nu}^{\text{DM}} \Rightarrow T_{\mu\nu}^{\text{F}}}.$$

The important point here is that the form of the gravitational equation is preserved. What changes is only the ontology of the additional term.

Ontological Difference

In Λ CDM, Dark Matter is an “unknown particle.” In the present theory, by contrast, the non-optical manifestation of the Future Information Field appears as an effective gravitational density. Therefore, Dark Matter is not an unknown particle, but a projected state of the Future Information Field.

Unification by Future Entropy

In Chapters VII through IX, we introduced the Future Entropy

$$\epsilon_F = \frac{S_F}{S_F + S_T}.$$

In this chapter, we assume that the effective energy density also depends on Future Entropy. That is,

$$\boxed{\rho_F = \rho_0 + \alpha_F \epsilon_F}.$$

Here,

$$\alpha_F$$

is the cosmological coupling constant of the Future Information Field. Through this, Dark Matter, black holes, gravitational waves, and the Future Boundary are all unified by the same state quantity

$$\epsilon_F.$$

Conclusion of This Section

In this section, we organized the mathematical position of Dark Matter in Λ CDM cosmology and introduced a formulation that replaces it with an effective energy-momentum tensor derived from the Future Information Field. As a result, in the present theory, it is not necessary to assume Dark Matter as an unknown particle. Instead, the Future Information Field is understood as an effective gravitational density projected onto cosmological scales. In the next section, starting from this effective energy-momentum tensor, we derive the modified gravitational potential for galactic rotation curves and give the first galaxy-scale observational prediction of the present theory.

7.3.12 10.3 Derivation of the Effective Gravitational Density from the Future Information Field

In the previous section, we showed that Dark Matter in the standard cosmology can be replaced by the effective energy-momentum tensor

$$T_{\mu\nu}^{\text{F}}$$

derived from the Future Information Field. However, at that stage, the claim remained that it “produces the same gravitational effect as Dark Matter.” In this section, we rigorously derive how the Future Information Field manifests as an effective gravitational density from the modified Einstein equation introduced in Paper 92. This section serves as the starting point for all subsequent discussions of galactic rotation curves, gravitational lensing, the Bullet Cluster, and the large-scale structure of the universe.

Modified Einstein Equation

In Paper 92, the gravitational field is governed by

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{baryon}}.$$

Here,

$$\mathcal{C}_{\mu\nu}$$

is the constraint tensor derived from the Future Information Field. In this chapter, this constraint term is extended to cosmological distances.

Decomposition of the Constraint Tensor

The Future Information Field can generally be divided into components that accompany optical manifestation and components that do not. Therefore, we decompose the constraint tensor as

$$\mathcal{C}_{\mu\nu} = \mathcal{C}_{\mu\nu}^{\text{opt}} + \mathcal{C}_{\mu\nu}^{\text{nonopt}}.$$

Here,

$$\mathcal{C}_{\mu\nu}^{\text{opt}}$$

is the component projected onto ordinary matter, while

$$\mathcal{C}_{\mu\nu}^{\text{nonopt}}$$

is the component that does not accompany optical manifestation and leaves only a gravitational constraint. In this paper, the latter corresponds to Dark Matter.

Effective Energy-Momentum Tensor

Rearranging the modified Einstein equation gives

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{baryon}} + T_{\mu\nu}^{\text{F}} \right).$$

Here, the contribution of the Future Information Field is defined by

$$T_{\mu\nu}^{\text{F}} = -\frac{1}{8\pi G} (\Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu}^{\text{nonopt}}).$$

This tensor does not represent an unknown particle. It represents the constraint imposed on spacetime by the Future Information Field.

Effective Gravitational Density

In the static approximation, the energy density is given by the time component

$$T_{00}.$$

Therefore, the effective density of the Future Information Field is

$$\rho_F = T_{00}^F.$$

From the above equation, we obtain

$$\rho_F = -\frac{1}{8\pi G} \left(\Lambda_{\text{eff}} g_{00} + \chi C_{00}^{\text{nonopt}} \right).$$

This is the basic equation corresponding to the density of Dark Matter.

Expression in Terms of Future Entropy

In Paper 92, Future Entropy was defined as

$$\epsilon_F = \frac{S_F}{S_T + S_F}.$$

In this chapter, we assume that the constraint tensor is proportional to Future Entropy:

$$C_{00}^{\text{nonopt}} = \beta_F \epsilon_F.$$

Here,

$$\beta_F$$

is the cosmological coupling coefficient of the Future Information Field. Therefore, the effective density becomes

$$\rho_F = -\frac{\Lambda_{\text{eff}}}{8\pi G} + \frac{\chi \beta_F}{8\pi G} \epsilon_F.$$

Spatial Dependence

Inside a galaxy, Future Entropy is not constant. Therefore,

$$\epsilon_F = \epsilon_F(r)$$

holds. Thus, the density distribution of the Future Information Field becomes

$$\rho_F(r) = -\frac{\Lambda_{\text{eff}}}{8\pi G} + \frac{\chi \beta_F}{8\pi G} \epsilon_F(r).$$

This equation becomes the basic density distribution appearing in galactic rotation curves, gravitational lensing, galaxy clusters, and the large-scale structure of the universe.

Effective Mass

We define the effective mass of the Future Information Field as

$$M_F(r).$$

Then, under the spherical-symmetry approximation,

$$M_F(r) = 4\pi \int_0^r \rho_F(r') r'^2 dr'$$

holds. This quantity is the effective mass corresponding to the Dark Matter Halo.

Ontological Meaning

As a result, the Dark Matter Halo is understood not as an aggregate of unknown particles, but as the result of integrating the constraint density of the Future Information Field. That is, what exists around a galaxy is not a particle cloud, but the spatial distribution of the Future Information Field itself.

General-Relativity Limit

When the Future Information Field does not exist,

$$\chi = 0$$

and

$$\epsilon_F = 0$$

hold. In that case,

$$\boxed{\rho_F = 0}$$

is obtained, and Dark Matter does not exist. Therefore, the present theory naturally converges to the general-relativity limit.

Conclusion of This Section

In this section, we derived the effective gravitational density

$$\rho_F(r)$$

and the effective mass

$$M_F(r)$$

from the Future Information Field. These are the basic state quantities corresponding to the Dark Matter Halo, and all subsequent galaxy-scale theory is derived from this density distribution. In the next section, we substitute this effective mass into the gravitational potential and rigorously derive the correction that the Future Information Field gives to galactic rotation curves.

7.3.13 10.4 Modification of Galactic Rotation Curves and Observational Predictions of the Future Information Field

In the previous section, we derived that the Future Information Field manifests in the universe not as an unknown particle, but as an effective gravitational density

$$\rho_F(r).$$

In this section, we derive how galactic rotation curves are modified from that density distribution. Galactic rotation curves were the first basis for proposing the Dark Matter hypothesis, and in the Future Information Field theory they also become the first observational target to be tested.

Rotational Velocity in General Relativity

In a spherically symmetric gravitational field, the circular velocity at radius

$$r$$

is given by

$$\boxed{v_{\text{GR}}^2(r) = \frac{GM(r)}{r}}.$$

Here,

$$M(r) = 4\pi \int_0^r \rho_b(r') r'^2 dr'$$

is the mass calculated only from ordinary matter. However, in observed galactic rotation curves, even as the radius increases, the velocity does not decrease, but maintains an approximately constant value. This cannot be explained by general relativity alone.

Effective Mass Including the Future Information Field

From Section 10.3, the effective mass of the Future Information Field is defined by

$$M_F(r) = 4\pi \int_0^r \rho_F(r') r'^2 dr'.$$

Therefore, the effective mass of the whole galaxy becomes

$$M_{\text{eff}}(r) = M_b(r) + M_F(r).$$

Modified Rotation Curve

The rotational velocity is determined by the effective mass. Therefore, when the Future Information Field is included, the rotational velocity becomes

$$v_F^2(r) = \frac{G [M_b(r) + M_F(r)]}{r}.$$

This is the fundamental rotation-curve equation in this paper.

Correction by Future Entropy

In Section 10.3, we derived

$$\rho_F(r) = -\frac{\Lambda_{\text{eff}}}{8\pi G} + \frac{\chi\beta_F}{8\pi G} \epsilon_F(r).$$

Integrating this, the mass derived from the Future Information Field becomes

$$M_F(r) = \frac{\chi\beta_F}{2G} \int_0^r \epsilon_F(r') r'^2 dr' - \frac{\Lambda_{\text{eff}}}{6G} r^3.$$

Therefore, the rotational velocity becomes

$$v_F^2(r) = \frac{GM_b(r)}{r} + \frac{\chi\beta_F}{2r} \int_0^r \epsilon_F(r') r'^2 dr' - \frac{\Lambda_{\text{eff}}}{6} r^2.$$

Outer Region of a Galaxy

In the outer region of a galaxy, the ordinary-matter mass becomes approximately constant. By contrast, the Future Information Field remains distributed over a wide region. Therefore, at sufficiently large radii, the second term dominates. In this case,

$$v_F^2(r) \simeq \frac{\chi\beta_F}{2r} \int_0^r \epsilon_F(r') r'^2 dr'$$

is obtained.

Condition for a Flat Rotation Curve

Observationally, in the outer regions of galaxies,

$$v(r) \simeq \text{const.}$$

holds. This means that

$$v^2 = \text{const.}$$

In the present theory, the condition for this is

$$\boxed{\int_0^r \epsilon_F(r') r'^2 dr' \propto r}.$$

Therefore, in the outer regions of galaxies, Future Entropy must have a spatial distribution different from that of ordinary matter. This is the first spatial-distribution condition predicted by the present theory.

Observational Predictions

The present theory gives the following predictions for galactic rotation curves. First, the rotational velocity is determined not only by ordinary matter, but also by the integrated quantity of the Future Information Field. Second, even if galaxy masses differ, if the distribution of Future Entropy is common, flat rotation curves are naturally derived. Third, the spatial distribution of Future Entropy can be inversely inferred from rotation curves.

Integration with Paper 92

In Paper 92, Future Entropy was introduced as a state quantity of black-hole information preservation and the Future Boundary. In this section, we showed that Future Entropy is projected at galactic scales into an effective gravitational density. Therefore, black holes, gravitational waves, and galactic rotation curves are unified by the same Future Entropy.

Falsification Conditions

The present theory is rejected if any of the following holds:

1. The Future Entropy distribution inversely calculated from rotation curves is not reproduced in other galaxies.
2. The Future Entropy inferred from ringdown analysis does not match the Future Entropy inferred from galactic rotation curves.
3. Even after introducing the Future Information Field, the theory does not have higher explanatory power than Λ CDM.

Conclusion of This Section

In this section, we derived the correction formula that the Future Information Field gives to galactic rotation curves. As a result, we showed that the Dark Matter Halo can be understood not as a distribution of unknown particles, but as a spatial integral of Future Entropy. In the next section, using this modified gravitational field, we derive how gravitational lensing changes and give a second cosmological observational prediction independent of galactic rotation curves.

7.3.14 10.5 Gravitational Lensing Theory Based on the Future Information Field

In the previous section, we showed that the Future Information Field modifies galactic rotation curves and that the Dark Matter Halo can be described as the spatial integral of Future Entropy. However, galactic rotation curves alone cannot distinguish the Future Information Field from a conventional Dark Matter Halo. Indeed, many dark-matter models can reproduce rotation curves by introducing an appropriate density distribution. Therefore, in order for the present theory to become an independent physical theory, the same Future Entropy must be derived from observables different from rotation curves. The most important such observable is gravitational lensing. In this section, from the correction that the Future Information Field gives to the spacetime metric, we derive modified formulas for the light-deflection angle, Einstein rings, weak lensing, and strong lensing.

Gravitational Lensing in General Relativity

In general relativity, the light-deflection angle caused by a spherically symmetric mass

$$M$$

is given by

$$\hat{\alpha}_{\text{GR}} = \frac{4GM}{c^2 b}.$$

Here,

$$b$$

is the impact parameter. Gravitational lensing produces Einstein rings, multiple images, weak lensing, and related phenomena by integrating this deflection angle.

Effective Mass Including the Future Information Field

In Section 10.4, we defined the effective mass as

$$M_{\text{eff}} = M_b + M_F.$$

Therefore, when the Future Information Field is included, the light-deflection angle becomes

$$\hat{\alpha}_F = \frac{4G(M_b + M_F)}{c^2 b}.$$

Deflection-Angle Correction by Future Entropy

Extracting only the contribution of the Future Information Field, the correction to the deflection angle is

$$\delta\hat{\alpha} = \frac{4GM_F}{c^2 b}.$$

Furthermore, substituting

$$M_F(r)$$

from Section 10.4 gives

$$\delta\hat{\alpha}(b) = \frac{2\chi\beta_F}{c^2 b} \int_0^b \epsilon_F(r) r^2 dr - \frac{2\Lambda_{\text{eff}}}{3c^2} b^2.$$

This is the fundamental correction formula describing how the Future Information Field modifies light deflection.

Einstein-Ring Radius

In general relativity, the Einstein-ring radius is

$$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}}.$$

When the Future Information Field is included, it becomes

$$\theta_F = \sqrt{\frac{4G(M_b + M_F)}{c^2} \frac{D_{LS}}{D_L D_S}}.$$

Weak Lensing

In weak lensing, the convergence is defined as

$$\kappa = \frac{\Sigma}{\Sigma_c}.$$

In the present theory, the surface density becomes

$$\Sigma = \Sigma_b + \Sigma_F.$$

Here,

$$\Sigma_F = \int \rho_F dz$$

is defined. Therefore, the weak-lensing map becomes the two-dimensional projection of the Future Information Field itself.

Strong Lensing

In strong lensing, one uses the lens equation

$$\beta = \theta - \frac{D_{LS}}{D_S} \hat{\alpha}.$$

When the Future Information Field is included, this becomes

$$\beta = \theta - \frac{D_{LS}}{D_S} \hat{\alpha}_F.$$

As a result, image positions, magnifications, and time delays are modified.

Inverse Estimation of Future Entropy

The Future Information Field can be inversely estimated from gravitational-lensing observations. That is, from the observed deflection angle, one obtains

$$\epsilon_F(r) = \mathcal{L}^{-1} [\delta \hat{\alpha}(r)].$$

Here,

$$\mathcal{L}^{-1}$$

is the inverse transformation operator derived in the present theory. This makes it possible to estimate the spatial distribution of Future Entropy from gravitational lensing alone.

Integration with the Black-Hole Theory

Up to this chapter, Future Entropy must satisfy

$$\boxed{\epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(v)} = \epsilon_F^{(\phi)} = \epsilon_F^{(\text{RC})} = \epsilon_F^{(\text{lens})}}.$$

Here,

$$\epsilon_F^{(\text{RC})}$$

is inferred from rotation curves, while

$$\epsilon_F^{(\text{lens})}$$

is inferred from gravitational lensing. This becomes the ninth self-consistency condition of the Future Information Field.

Observational Predictions

The present theory gives the following predictions. First, the Future Entropy derived from rotation curves and the Future Entropy derived from gravitational lensing coincide. Second, the spatial distribution of the Future Information Field does not coincide with the distribution of ordinary matter. Third, weak-lensing maps can be reconstructed as two-dimensional projection images of the Future Information Field.

Falsification Conditions

The present theory is rejected if any of the following holds:

1. The Future Entropy inferred from rotation curves and gravitational lensing does not coincide.
2. Even after introducing the Future Information Field, strong-lensing image positions and time delays cannot be improved.
3. The Dark Matter Halo of Λ CDM has statistically significantly higher explanatory power.

Conclusion of This Section

In this section, we derived the modified formulas by which the Future Information Field affects gravitational-lensing phenomena. As a result, we showed that gravitational lensing becomes a second cosmological probe for measuring Future Entropy, independent of galactic rotation curves. In the next section, we analyze the Bullet Cluster, construct the situation in which the Future Information Field and particle Dark Matter theory give the most clearly different predictions, and formulate the most important falsification experiment for the present theory.

7.3.15 10.6 Predictions of the Future Information Field in the Bullet Cluster

In the previous section, we showed that the Future Information Field can explain galactic rotation curves and gravitational lensing in a unified way. However, these two observations alone do not reveal a decisive difference between the Future Information Field and particle Dark Matter theory. At present, the most important observational target for Dark Matter theory is colliding galaxy-cluster systems, especially the Bullet Cluster. In the Bullet Cluster, the hot gas that accounts for most of the ordinary matter was slowed down by collision, whereas the mass distribution inferred from gravitational lensing is observed to remain on the galaxy side. This phenomenon is widely cited as a representative example supporting the existence of particle Dark Matter. In this section, we derive what kind of mass distribution the Future Information Field predicts in the Bullet Cluster.

Basic Structure of the Bullet Cluster

Let galaxy cluster A and galaxy cluster B collide with velocity

$$V_{\text{col}}.$$

The total energy-momentum tensor of the system can be written as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{galaxy}} + T_{\mu\nu}^{\text{gas}} + T_{\mu\nu}^{\text{F}}.$$

Here,

$$T_{\mu\nu}^{\text{galaxy}}$$

is the galaxy component,

$$T_{\mu\nu}^{\text{gas}}$$

is the hot gas observed in X-rays, and

$$T_{\mu\nu}^{\text{F}}$$

is the effective tensor of the Future Information Field.

Difference from Particle Dark Matter

In particle Dark Matter theory, the Dark Matter Halo moves together with the galaxies and does not collide with the hot gas. Therefore, the gravitational-lensing center coincides with the galaxy distribution. In the Future Information Field theory, the Future Information Field is not a particle, but a manifestation-constraint field. Therefore, the concept of a particle collision cross section does not exist. The Future Information Field responds to galaxy-cluster collisions not as particles, but as a rearrangement of Future Entropy.

Time Evolution of Future Entropy

Let the Future Entropy during the collision be

$$\epsilon_F(\mathbf{x}, t).$$

In this paper, we assume that the Future Information Field obeys the following conservation equation:

$$\boxed{\frac{\partial \epsilon_F}{\partial t} + \nabla \cdot (\epsilon_F \mathbf{v}_F) = \Gamma_F}.$$

Here,

$$\mathbf{v}_F$$

is the effective flux of the Future Information Field, and

$$\Gamma_F$$

is the generation and rearrangement term of Future Entropy caused by the collision.

Transition of the Future Information Field

In Paper 92, we showed that Future Entropy is rearranged through the Future Boundary. In the present theory, even during a galaxy-cluster collision, the Future Information Field forms a local Future Boundary. Therefore, the Future Entropy after the collision does not simply follow the galaxies, but is described by

$$\boxed{\epsilon_F^{\text{after}} = \epsilon_F^{\text{before}} + \Delta \epsilon_{\text{trans}}}.$$

Here,

$$\Delta \epsilon_{\text{trans}}$$

is the amount of information rearrangement produced by the collision.

Effective Mass Distribution

From Section 10.3, the effective density corresponding to the Future Information Field is

$$\rho_F = \frac{\chi\beta_F}{8\pi G}\epsilon_F.$$

Therefore, the post-collision mass distribution becomes

$$\rho_F^{\text{after}} = \frac{\chi\beta_F}{8\pi G} \left(\epsilon_F^{\text{before}} + \Delta\epsilon_{\text{trans}} \right).$$

This equation shows that the Future Information Field may not completely coincide with either the galaxy distribution or the hot-gas distribution.

Gravitational-Lensing Center

Let the effective mass center observed by gravitational lensing be

$$\mathbf{X}_{\text{lens}}.$$

In the present theory, this is given by

$$\mathbf{X}_{\text{lens}} = \frac{\int \rho_F(\mathbf{x})\mathbf{x}d^3x}{\int \rho_F(\mathbf{x})d^3x}.$$

Therefore, if rearrangement of the Future Information Field occurs, the gravitational-lensing center may move to a position that coincides with neither the galaxies nor the gas.

Discriminating Prediction against Particle Dark Matter

Here, the first distinguishable prediction of the present theory is obtained. In particle Dark Matter theory, the lensing center is fixed to the center of mass of the particle Halo. By contrast, in the Future Information Field theory, the lensing center moves to the distribution centroid of the Future Information Field according to the rearrangement of Future Entropy. Therefore, when multiple colliding galaxy-cluster systems are compared, the offset of the gravitational-lensing center is predicted to depend on the spatial gradient of Future Entropy.

Integration with Paper 92

In Paper 92, we showed that information rearrangement through the Future Boundary realizes black-hole information preservation. In this section, we extended the concept of the Future Boundary to cosmological collision phenomena. As a result, even in galaxy-cluster collisions, the Future Information Field behaves not as particle transport, but as a rearrangement of Future Entropy.

Observational Predictions

The present theory predicts the following:

1. There exist cases in which the gravitational-lensing center does not completely coincide with either the galaxy distribution or the hot-gas distribution.
2. The offset is proportional to the spatial gradient of Future Entropy.
3. In multiple colliding galaxy-cluster systems, the offset of the mass center is described by the same Future Entropy parameter.
4. The Future Entropy inferred from black holes, gravitational waves, galactic rotation curves, gravitational lensing, and the Bullet Cluster must all coincide.

Falsification Conditions

The present theory is rejected if any of the following holds:

1. The rearrangement of Future Entropy in the Bullet Cluster cannot explain the gravitational-lensing distribution.
2. Future Entropy does not coincide with that inferred from other observations such as ring-down, gravitational waves, and galactic rotation curves.
3. Particle Dark Matter theory has statistically significantly higher explanatory power.

Conclusion of This Section

In this section, we formulated the Bullet Cluster as a rearrangement problem of the Future Information Field. Through this, galaxy-cluster collisions are positioned as the most important cosmological falsification experiment in the present theory, testing whether the Future Information Field is not merely an effective mass but a dynamically rearranged information field. In the next section, we derive the influence of the Future Information Field on the formation of the large-scale structure of the universe and give quantitative predictions for the Matter Power Spectrum and the history of galaxy formation.

7.3.16 10.7 Formation of the Large-Scale Structure of the Universe by the Future Information Field

In the previous section, we derived that the Future Information Field may exhibit a mass rearrangement different from particle Dark Matter in colliding galaxy-cluster systems. However, galaxy-cluster collisions are local phenomena in the context of the entire evolution of the universe. If the Future Information Field is a fundamental structure of the universe, then its effects must appear not only in galaxy clusters, but also throughout the entire formation process of the Large Scale Structure, or LSS, of the universe. In this section, we introduce the Future Information Field into cosmological density perturbations and derive modified formulas for the Matter Power Spectrum, the structure growth rate, and the history of galaxy formation.

Basic Equation for Density Perturbations

In standard cosmology, the density contrast

$$\delta(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t) - \bar{\rho}(t)}{\bar{\rho}(t)}$$

is introduced. In the linear approximation, its time evolution is governed by

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\bar{\rho}\delta.$$

Here,

$$H$$

is the Hubble parameter, and

$$\bar{\rho}$$

is the mean density.

Contribution of the Future Information Field

In the present theory, the mean density includes not only ordinary matter, but also the effective density of the Future Information Field. Therefore,

$$\bar{\rho}_{\text{eff}} = \bar{\rho}_b + \bar{\rho}_F$$

holds. From Section 10.3,

$$\bar{\rho}_F = \frac{\chi\beta_F}{8\pi G}\epsilon_F$$

is obtained.

Modified Density-Perturbation Equation

From the above, the density-perturbation equation is modified as follows:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G(\bar{\rho}_b + \bar{\rho}_F)\delta.$$

That is, the Future Information Field contributes directly as a gravitational source of density perturbations.

Spatial Dependence of Future Entropy

The Future Information Field is not uniform throughout the universe. Therefore,

$$\epsilon_F = \epsilon_F(\mathbf{x}, t)$$

holds. Consequently, the density-perturbation equation becomes

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\left[\bar{\rho}_b + \frac{\chi\beta_F}{8\pi G}\epsilon_F(\mathbf{x}, t)\right]\delta.$$

The spatial gradient of Future Entropy directly determines the rate of structure formation.

Matter Power Spectrum

Let the Fourier transform of the density perturbation be

$$\delta(\mathbf{k}).$$

The Matter Power Spectrum is defined by

$$P(k) = \langle |\delta(\mathbf{k})|^2 \rangle.$$

In the present theory, including the correction from the Future Information Field, we obtain

$$P_F(k) = P_{\Lambda\text{CDM}}(k) + \Delta P_F(k).$$

Here,

$$\Delta P_F(k) = A_F(k)\epsilon_F.$$

Structure Growth Rate

The structure growth rate

$$f = \frac{d \ln \delta}{d \ln a}$$

is also modified by the Future Information Field. In the present theory,

$$f_F = f_{\Lambda\text{CDM}} + \Delta f_F$$

holds, where

$$\Delta f_F = \eta_F \epsilon_F.$$

History of Galaxy Formation

If the Future Information Field exists, the galaxy-formation rate changes in proportion to Future Entropy. Let the galaxy-formation rate be

$$R_G(z).$$

Then

$$R_G = R_{\Lambda\text{CDM}} + \gamma_F \epsilon_F$$

is obtained. It is predicted that, in regions with high Future Entropy, structure formation proceeds more rapidly.

Integration with Paper 92

In Paper 92, Future Entropy was introduced as a state quantity describing black-hole information preservation. In this section, the same Future Entropy appears as the state quantity determining the structure growth rate of the universe as a whole. Therefore, black holes, gravitational waves, galaxies, galaxy clusters, and the large-scale structure of the universe are unified by the same Future Entropy.

Observational Predictions

The present theory predicts the following:

1. An additional spectrum corresponding to the Future Information Field appears in the Matter Power Spectrum.
2. The structure growth rate changes in proportion to the spatial distribution of Future Entropy.
3. The Future Entropy inferred from the history of galaxy formation coincides with that inferred from ringdown, gravitational waves, rotation curves, gravitational lensing, and the Bullet Cluster.
4. If the Future Information Field exists, the statistical quantities of the large-scale structure of the universe can be reproduced by a single Future Entropy parameter.

Falsification Conditions

The present theory is rejected if any of the following holds:

1. The correction to the Matter Power Spectrum does not agree with observations.
2. The Future Entropy inferred from the structure growth rate does not coincide with that inferred from other observables.
3. Even after introducing the Future Information Field, the theory does not show higher explanatory power than ΛCDM .

Conclusion of This Section

In this section, we extended the Future Information Field to the formation of the large-scale structure of the universe and derived modified formulas for the density-perturbation equation, the Matter Power Spectrum, the structure growth rate, and the history of galaxy formation. Through this, the Future Information Field is positioned not only as a field governing local structures such as black holes and galaxy clusters, but also as a background information field

governing the structure formation of the universe as a whole. In the next section, we introduce the Future Information Field into the cosmic microwave background, or CMB, and give quantitative predictions for acoustic peaks, lensing effects, and consistency with primordial perturbations.

7.3.17 10.8 Predictions of the Future Information Field for the Cosmic Microwave Background

In the previous section, we showed that the Future Information Field contributes to the formation of the large-scale structure of the universe and modifies the Matter Power Spectrum and the structure growth rate. However, structure formation is the process by which density perturbations given in the early universe evolve over time. Therefore, if the Future Information Field is truly an information field governing the entire universe, its effects must also appear in the anisotropies of the Cosmic Microwave Background, or CMB. The CMB is the most precise observable preserving the state of the universe at the recombination epoch. In standard cosmology, it is determined by acoustic oscillations, gravitational potentials, primordial perturbations, and the history of cosmic expansion. In this section, we introduce the Future Information Field into the cosmic microwave background and derive modified formulas for acoustic peaks, lensing effects, and the angular power spectrum.

CMB in Standard Cosmology

The CMB temperature fluctuation is expanded by using spherical harmonics

$$Y_{\ell m}$$

as

$$\frac{\Delta T}{T} = \sum_{\ell, m} a_{\ell m} Y_{\ell m}(\theta, \phi).$$

The angular power spectrum is defined by

$$C_{\ell} = \langle |a_{\ell m}|^2 \rangle.$$

In Λ CDM,

$$C_{\ell}$$

is determined by primordial density perturbations, baryon density, Dark Matter, Dark Energy, and the cosmic expansion history.

Introduction of the Future Information Field

In the present theory, the Future Information Field modifies the gravitational potential. Therefore, the photon path from the last-scattering surface to the present is determined by the effective potential including the Future Information Field:

$$\Phi_{\text{eff}} = \Phi_{\text{GR}} + \Phi_F.$$

Here,

$$\Phi_F = \lambda_F \epsilon_F$$

is the correction potential proportional to Future Entropy.

Modification of the Angular Power Spectrum

When the Future Information Field is included, the angular power spectrum becomes

$$C_\ell^F = C_\ell^{\Lambda\text{CDM}} + \Delta C_\ell^F.$$

Here, the correction term is

$$\Delta C_\ell^F = A_\ell \epsilon_F.$$

The coefficient

$$A_\ell$$

is the coupling coefficient between the Future Information Field and each multipole.

Correction to Acoustic Peaks

In standard cosmology, the positions of acoustic peaks are determined by the sound horizon

$$r_s.$$

In the present theory, the effective sound horizon becomes

$$r_s^F = r_s + \delta r_F.$$

Here,

$$\delta r_F = \eta_s \epsilon_F$$

is defined. Therefore, the acoustic-peak positions shift slightly in proportion to Future Entropy.

Integrated Sachs–Wolfe Effect

If the Future Information Field evolves with time, then the gravitational potential also becomes time dependent. As a result, the Integrated Sachs–Wolfe effect becomes

$$\left(\frac{\Delta T}{T} \right)_{\text{ISW}}^F = -2 \int \left(\dot{\Phi}_{\text{GR}} + \dot{\Phi}_F \right) d\eta.$$

The Future Information Field gives an additional contribution to late-time temperature anisotropies.

CMB Lensing Effect

As derived in Section 10.5, the Future Information Field modifies gravitational lensing. Therefore, the lensing effect on CMB photons is also given by the same correction formula. The deflection angle becomes

$$\hat{\alpha}_{\text{CMB}} = \hat{\alpha}_{\text{GR}} + \delta \hat{\alpha}_F.$$

Here,

$$\delta \hat{\alpha}_F$$

is the correction by the Future Information Field.

Unification Condition for Future Entropy

Up to this chapter, Future Entropy must satisfy

$$\epsilon_F^{(\omega)} = \epsilon_F^{(\text{echo})} = \epsilon_F^{(\text{GW})} = \epsilon_F^{(\text{RC})} = \epsilon_F^{(\text{lens})} = \epsilon_F^{(\text{BC})} = \epsilon_F^{(\text{LSS})} = \epsilon_F^{(\text{CMB})}.$$

Here,

$$\epsilon_F^{(\text{CMB})}$$

is the Future Entropy inferred from the angular power spectrum.

Observational Predictions

The present theory gives the following predictions:

1. A systematic small correction derived from the Future Information Field appears in the positions of acoustic peaks.
2. At higher multipoles, a correction to the angular power spectrum proportional to Future Entropy arises.
3. The Future Entropy inferred from CMB lensing coincides with that inferred from galactic rotation curves, gravitational lensing, the large-scale structure of the universe, black holes, and gravitational waves.
4. If the Future Information Field exists, the Future Entropy inferred from multiple independent observations converges to a single value.

Falsification Conditions

The present theory is rejected if any of the following holds:

1. The Future Entropy inferred from the CMB angular power spectrum does not coincide with that inferred from other observables.
2. Even after introducing the Future Information Field, the explanatory power for acoustic peaks, the ISW effect, and CMB lensing is not improved.
3. Λ CDM explains all observations with statistically significantly higher accuracy.

Conclusion of This Section

In this section, we introduced the Future Information Field into the cosmic microwave background and derived modified formulas for the angular power spectrum, acoustic peaks, the Integrated Sachs–Wolfe effect, and CMB lensing. Through this, the Future Information Field is positioned as a background information field acting commonly not only on black holes, gravitational waves, galaxies, galaxy clusters, and the large-scale structure of the universe, but also on the recombination-era universe. In the next section, we integrate the observations derived so far, including galactic rotation curves, gravitational lensing, the Bullet Cluster, the large-scale structure of the universe, and the CMB, and carry out a quantitative comparison with Λ CDM and modified-gravity theories.

7.3.18 10.9 Comparison with Existing Cosmological Models and the Distinguishability of the Future Information Field

In the previous sections, we demonstrated that the Future Information Field provides a consistent set of modified equations for galactic rotation curves, gravitational lensing, galaxy-cluster collisions, the large-scale structure of the universe, and the cosmic microwave background. However, for a theory to be established as a physical theory, its differences from existing theories must be demonstrated quantitatively rather than merely qualitatively. In this section, we compare the Future Information Field theory with the three major cosmological frameworks currently in use:

- Λ CDM,
- MOND,
- Modified Gravity theories.

We then examine their respective degrees of freedom, predictive capabilities, and falsifiability.

Comparison with Λ CDM

In Λ CDM, Dark Matter is introduced as an unknown particle. Consequently, the observables are described by

$$\mathcal{O} = f(\rho_b, \rho_{\text{DM}}, \rho_\Lambda).$$

Here, Dark Matter is treated as an independent degree of freedom. By contrast, in the Future Information Field theory, Dark Matter is not an independent degree of freedom. All corrections are derived from the Future Entropy,

$$\epsilon_F = \frac{S_F}{S_F + S_T}.$$

Therefore, the observables are reduced to

$$\mathcal{O} = f(\rho_b, \epsilon_F).$$

Comparison with MOND

In MOND, the gravitational acceleration is modified through

$$a_0,$$

such that

$$\mu\left(\frac{a}{a_0}\right)a = a_N.$$

However, MOND is primarily intended to explain galactic rotation curves, and it is difficult to account simultaneously for gravitational waves, black holes, the CMB, and the Bullet Cluster within a single theoretical framework. In the Future Information Field theory, the correction to galactic rotation curves is only one projection of Future Entropy. Therefore, galactic rotation curves constitute only one particular manifestation of the complete theory.

Comparison with Modified Gravity Theories

In general modified-gravity theories, the Einstein equation itself is altered. For example, in

$$f(R)$$

gravity, the action becomes

$$S = \int f(R)\sqrt{-g}d^4x.$$

By contrast, in the Future Information Field theory, the Einstein tensor itself is preserved. Only the constraint term originating from the Future Information Field,

$$\mathcal{C}_{\mu\nu},$$

is modified. Consequently, the general-relativity limit is recovered naturally.

Comparison of Degrees of Freedom

The conceptual comparison of additional degrees of freedom is summarized as

Theory	Additional Degrees of Freedom
Λ CDM	$\rho_{\text{DM}}, \rho_\Lambda$
MOND	a_0
$f(R)$	$f(R)$
FCT	ϵ_F

where FCT denotes the Future Information Field Theory. In the present theory, the only additional state variable is Future Entropy.

Comparison of Unification

Collecting the observables derived throughout this work, the Future Information Field provides

$$\epsilon_F \Rightarrow \begin{cases} \delta\omega_{\text{QNM}} \\ \Delta t_{\text{echo}} \\ v_{\text{GW}} \\ v_{\text{rot}} \\ \hat{\alpha} \\ P(k) \\ C_\ell \end{cases}$$

Thus, black holes, gravitational waves, galaxies, and the cosmic microwave background all reduce to the same state variable. This constitutes the central feature of the present theory.

Distinguishability

The most important distinction between the present theory and existing theories is that the Future Entropy independently inferred from each observation must coincide. Specifically,

$$\epsilon_F^{(\text{BH})} = \epsilon_F^{(\text{GW})} = \epsilon_F^{(\text{RC})} = \epsilon_F^{(\text{Lens})} = \epsilon_F^{(\text{BC})} = \epsilon_F^{(\text{LSS})} = \epsilon_F^{(\text{CMB})}.$$

In Λ CDM, these quantities are fitted independently. In contrast, in the Future Information Field theory, all of them are determined by a single state variable.

Unique Predictions of the Present Theory

As a consequence, the present theory makes several predictions that are absent from conventional models.

1. The Future Entropy inferred from black holes must coincide with the value inferred from galactic rotation curves.
2. The mass centroid of the Bullet Cluster is likewise determined by the same Future Entropy.
3. The correction coefficient of the Matter Power Spectrum is proportional to the correction coefficient of black-hole ringdown.
4. The Future Entropy inferred from the CMB angular power spectrum coincides with the value obtained from gravitational-wave propagation.

These constitute cross-disciplinary predictions unique to the Future Information Field theory and differ fundamentally from conventional theories that explain each phenomenon independently.

Conclusion of This Section

In this section, we compared the Future Information Field theory with Λ CDM, MOND, and modified-gravity theories. As a result, we showed that instead of introducing unknown particles or multiple independent degrees of freedom, the present theory simultaneously describes black holes, gravitational waves, galactic rotation curves, gravitational lensing, the Bullet Cluster, the large-scale structure of the universe, and the cosmic microwave background through the single state variable Future Entropy. In the next section, we summarize the concrete observational predictions derived throughout this chapter and present a systematic verification strategy using future observations by LIGO, Virgo, KAGRA, Euclid, the Nancy Grace Roman Space Telescope, the SKA, the Rubin Observatory, and other facilities.

7.3.19 10.10 Cosmological Observational Predictions Based on the Future Information Field

In the previous section, we compared the Future Information Field theory with Λ CDM, MOND, and modified-gravity theories, and showed that Future Entropy serves as the single state variable unifying all cosmological phenomena. However, being a unified theory alone is not sufficient for establishing a physical theory. For a theory to be scientifically valid, it must provide quantitative predictions that can be clearly supported or falsified by future observations. In this section, we organize the theoretical equations derived throughout this chapter and systematically summarize the cosmological predictions of the Future Information Field according to each observational target.

Fundamental Principle of the Observational Predictions

All modified equations derived in this paper are governed by the Future Entropy,

$$\epsilon_F = \frac{S_F}{S_F + S_T}.$$

Accordingly, every observable can be written in the unified form

$$\mathcal{O}_i = \mathcal{O}_i^{\text{GR}} + \Delta\mathcal{O}_i(\epsilon_F).$$

Here,

$$\mathcal{O}_i^{\text{GR}}$$

denotes the prediction of general relativity or Λ CDM, while

$$\Delta\mathcal{O}_i$$

represents the correction arising from the Future Information Field.

Black-Hole Observations

As derived in Chapter VII, the Future Information Field modifies the quasinormal modes according to

$$\omega = \omega_{\text{GR}} + \delta\omega_F.$$

Furthermore, the ringdown damping time becomes

$$\tau = \tau_{\text{GR}} + \delta\tau_F.$$

Therefore, future high-precision analyses of black-hole ringdown may allow Future Entropy to be estimated directly.

Gravitational-Wave Observations

Chapter IX demonstrated that the Future Information Field modifies the group velocity, phase velocity, polarization, and accumulated phase of gravitational waves. Accordingly,

$$v_g = c + \delta v_F.$$

Moreover, over cosmological distances an accumulated phase correction

$$\Delta\phi_F$$

appears. These quantities may become directly measurable with third-generation gravitational-wave detectors.

Galactic Rotation Curves

In Section 10.4, we derived

$$v^2(r) = \frac{GM_b(r)}{r} + \frac{GM_F(r)}{r}.$$

Therefore, if the Future Information Field exists, the Future Entropy inferred from galactic rotation curves must coincide with that inferred from black-hole observations.

Gravitational Lensing

In Section 10.5, the light-deflection angle becomes

$$\hat{\alpha} = \hat{\alpha}_{\text{GR}} + \delta\hat{\alpha}_F.$$

If the Future Information Field exists, weak-lensing maps can be reconstructed as two-dimensional projection images of Future Entropy.

Bullet Cluster

Section 10.6 showed that the Future Information Field behaves not as particle transport, but as information rearrangement. Consequently, if numerous colliding galaxy-cluster systems are observed in the future, the statistical distribution of gravitational-lensing centers is predicted to be explained by the spatial gradient of Future Entropy.

Large-Scale Structure of the Universe

In Section 10.7, we derived

$$P(k) = P_{\Lambda\text{CDM}} + \Delta P_F(k).$$

If the Future Information Field exists, both the structure-growth rate and the correction to the Matter Power Spectrum are proportional to Future Entropy.

Cosmic Microwave Background

In Section 10.8, we showed that the angular power spectrum becomes

$$C_\ell = C_\ell^{\Lambda\text{CDM}} + \Delta C_\ell^F.$$

If the Future Information Field exists, the acoustic-peak positions, CMB lensing, and the Integrated Sachs–Wolfe effect all receive unified corrections determined by Future Entropy.

Self-Consistency Condition Among Observables

Throughout this paper, if the Future Information Field exists, the Future Entropy inferred from every independent observation must coincide. Namely,

$$\epsilon_F^{\text{BH}} = \epsilon_F^{\text{GW}} = \epsilon_F^{\text{RC}} = \epsilon_F^{\text{Lens}} = \epsilon_F^{\text{BC}} = \epsilon_F^{\text{LSS}} = \epsilon_F^{\text{CMB}}.$$

This is the central prediction of the entire theory.

Verification by Future Observations

The present theory can be tested directly by the following observational programs:

- Gravitational-wave observations with LIGO, Virgo, and KAGRA,
- Supermassive black-hole mergers observed by LISA,
- Large-scale structure surveys conducted by Euclid,
- Weak-gravitational-lensing observations with the Nancy Grace Roman Space Telescope,
- Galaxy-distribution surveys carried out by the Rubin Observatory,
- 21-cm cosmology observed by the Square Kilometre Array (SKA),
- Cosmic microwave background observations by the Simons Observatory and CMB-S4.

These are mutually independent observational programs. Nevertheless, the Future Information Field theory requires that all of them converge to the same value of Future Entropy.

Conclusion of This Section

In this section, we systematically summarized the observational predictions of the Future Information Field, spanning black holes, gravitational waves, galactic rotation curves, gravitational lensing, the Bullet Cluster, the large-scale structure of the universe, and the cosmic microwave background. The essential feature of the present theory is not that it explains each phenomenon independently, but that all of these observations are derived from a single state variable, namely Future Entropy. In the next section, we conclude the present chapter by showing that the Dark Matter theory based on the Future Information Field constitutes the cosmological consequence of Paper 92, and we summarize the remaining issues for future verification together with the theoretical outlook.

7.3.20 10.11 Summary of This Chapter and the Position of the Dark Matter Theory Based on the Future Information Field

In this chapter, we extended the Future Information Field introduced in Paper 92 to cosmological scales and reformulated dark-matter phenomena as cosmological manifestations of the Future Information Field. Conventional dark-matter theories explain galactic rotation curves, gravitational lensing, galaxy-cluster collisions, the large-scale structure of the universe, and the cosmic microwave background by introducing unknown particles. By contrast, the present theory demonstrates that equivalent gravitational effects arise naturally through the manifestation of the Future Information Field as a spacetime constraint, without introducing any new unknown particle.

The Theoretical Framework Constructed in This Chapter

In this chapter, we introduced the Future Information Field through the effective energy-momentum tensor

$$T_{\mu\nu}^F,$$

from which we derived the effective gravitational density

$$\rho_F$$

and the effective mass

$$M_F.$$

Using this mass distribution, we unified galactic rotation curves, gravitational lensing, the Bullet Cluster, the large-scale structure of the universe, and the cosmic microwave background within a single theoretical framework.

Cosmological Unification through Future Entropy

Throughout this chapter, every contribution of the Future Information Field was reduced to the Future Entropy,

$$\epsilon_F = \frac{S_F}{S_F + S_T}.$$

Accordingly, all observables derived in this chapter are summarized as

$$\epsilon_F \Rightarrow \begin{cases} \rho_F \\ M_F \\ v_{\text{rot}} \\ \hat{\alpha} \\ P(k) \\ C_\ell \end{cases}$$

Thus, Future Entropy becomes the unique additional state variable responsible for cosmological gravitational effects.

Integration from Chapters VII through X

In this work, the Future Information Field has been progressively developed across different physical scales. Chapter VII derived corrections to black-hole quasinormal modes. Chapter VIII introduced the Future Boundary and formulated the geometric structure of black-hole information preservation. Chapter IX extended the Future Information Field to gravitational-wave propagation. In the present chapter, we further extended the theory to cosmological scales and reformulated the entire dark-matter phenomenon. Collecting these developments, the domain of the Future Information Field can be organized into the following hierarchical structure:

$$\boxed{\text{Black Hole} \Rightarrow \text{Gravitational Wave} \Rightarrow \text{Galaxy} \Rightarrow \text{Galaxy Cluster} \Rightarrow \text{Large Scale Structure} \Rightarrow \text{CMB}}$$

The Future Information Field acts as the common background information field throughout this entire hierarchy.

Condition of Observational Consistency

The principal consequence of this chapter is that the physical existence of the Future Information Field can be tested through a single self-consistency condition. Specifically, the Future Entropy inferred from each independent observation must satisfy

$$\boxed{\epsilon_F^{(\text{BH})} = \epsilon_F^{(\text{GW})} = \epsilon_F^{(\text{RC})} = \epsilon_F^{(\text{Lens})} = \epsilon_F^{(\text{BC})} = \epsilon_F^{(\text{LSS})} = \epsilon_F^{(\text{CMB})}}.$$

This condition constitutes a unique prediction of the present theory, connecting black holes, gravitational waves, galaxies, and the cosmic microwave background through a single state variable.

Falsifiability of the Theory

The present theory is rejected if any one of the following conditions is satisfied:

1. The Future Entropy inferred from different observations does not converge to a common value.
2. Introducing the Future Information Field does not provide greater explanatory power than Λ CDM.
3. The unified model based on the Future Information Field fails to reproduce all observations simultaneously.
4. An unknown-particle model achieves equivalent or superior explanatory power with fewer degrees of freedom.

Accordingly, the present theory is formulated as an observationally testable theory equipped with explicit falsification criteria.

Cosmological Consequences Derived from Paper 92

Paper 92 introduced the Future Information Field as the fundamental field unifying black-hole information preservation, the Future Boundary, and Future Entropy. In this chapter, extending that framework to cosmology has shown that dark matter can be understood not as an unknown particle, but as the cosmological manifestation of the Future Information Field. Consequently, a unified picture emerges in which a single Future Information Field is projected into different physical phenomena, ranging continuously from black holes to the cosmic microwave background.

Connection to the Next Chapter

In this chapter, the Future Information Field was applied to the dark-matter problem in cosmology. However, if the Future Information Field is truly a background field governing the entire universe, then its influence cannot be limited to dark matter alone. The expansion of the universe itself—namely dark energy, the cosmological constant, accelerated cosmic expansion, and vacuum energy—should likewise receive contributions from the same information field. Accordingly, in the next chapter, we extend the Future Information Field to the theory of cosmic expansion and formulate dark energy and cosmic acceleration as cosmological consequences of Future Entropy.

Part XI

Dark Energy Theory Based on the Future Information Field

— A Unified Redefinition and Observational Predictions for Cosmic Acceleration, the Cosmological Constant, Vacuum Energy, and the Equation of State —

7.3.21 11.1 Introduction: Ontological Redefinition of the Dark Energy Problem

In Chapter X, we extended the Future Information Field to cosmological scales and redefined Dark Matter not as an unknown particle, but as a non-optical manifestation of the Future Information Field. As a result, we showed that galactic rotation curves, gravitational lensing, the Bullet Cluster, the large-scale structure of the universe, and the cosmic microwave background can be described from a single state variable: Future Entropy. However, in the present standard cosmology, another fundamental problem still remains. That problem is cosmic accelerated expansion. In 1998, observations of Type Ia supernovae showed that the expansion of the universe is not decelerating, but accelerating. In the present standard cosmology, this phenomenon is explained by the cosmological constant

$$\Lambda$$

or by an unknown component called Dark Energy. However, the physical origin of Dark Energy has not yet been established. If it is interpreted as vacuum energy, the value predicted by quantum field theory is approximately

$$10^{120}$$

times larger than the observed value. This is the so-called Cosmological Constant Problem. Furthermore, the Coincidence Problem also remains unresolved: why, in the present universe, Dark Matter and Dark Energy have densities of roughly the same order. These two problems are not essentially explained as long as Dark Energy is introduced as an independent entity.

Basic Position of This Paper

In this paper, as in the case of Dark Matter, we do not introduce Dark Energy as a new physical entity. In Paper 92, the Future Information Field was introduced as the fundamental information field unifying black-hole information preservation, the Future Boundary, and Future Entropy. In Chapter X, Dark Matter was derived as its cosmological projection. In the present chapter, we further extend this theory and derive cosmic accelerated expansion itself as a spacetime-constraint effect of the Future Information Field. That is, Dark Matter and Dark Energy are not mutually independent unknown components. Both are merely the results of the Future Information Field manifesting at different spacetime scales.

Cosmic Expansion by the Future Information Field

In Paper 92, we introduced the modified Einstein equation

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi C_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

In Chapter X, we showed that the non-optical component of the constraint tensor

$$C_{\mu\nu}$$

corresponds to Dark Matter. In the present chapter, we further show that its isotropic component contributes to the expansion rate of the universe as a whole. Therefore, the Future Information Field constrains not only local gravitational fields, but also the cosmological background spacetime itself.

Problems Addressed in This Chapter

In this chapter, using the Future Information Field, we derive the following problems from a single theory:

- the Cosmological Constant Problem,
- the origin of Dark Energy,
- the onset condition for accelerated expansion,
- the redefinition of the equation of state

$$w = \frac{p}{\rho},$$

- the relation to vacuum energy,
- the fate of the future universe,
- and unification with Dark Matter.

These are currently discussed as separate problems. However, in the Future Information Field theory, all of them are derived simultaneously from the cosmological time evolution of Future Entropy.

Purpose of This Chapter

The purpose of this chapter is not to explain Dark Energy as a new energy source. The purpose is to derive why cosmic expansion appears to accelerate as a result of the Future Information Field acting as a constraint condition on the entire universe. Therefore, in this chapter, we apply the Future Information Field to cosmology and reconstruct the Friedmann equation, the cosmological constant, the equation of state, vacuum energy, and cosmic accelerated expansion on the basis of Future Entropy.

Structure of This Chapter

In this chapter, we first organize the mathematical structure of Dark Energy in standard cosmology. Next, we introduce the Future Information Field into the Friedmann equation and derive the cosmic expansion equation. We then analyze the equation of state, the cosmological constant, vacuum energy, and the onset condition for accelerated expansion in sequence. Finally, we show that Dark Matter and Dark Energy are nothing more than different manifestation forms of the Future Information Field. Through this, the Future Information Field introduced from Chapter VII through Chapter XI is completed as a single cosmological background field unifying black holes, gravitational waves, galaxies, the large-scale structure of the universe, and cosmic expansion.

7.3.22 11.2 The Mathematical Structure of Dark Energy in Standard Cosmology and Its Reformulation by the Future Information Field

In the previous section, we presented the basic direction of this chapter: the Future Information Field acts as a background constraint field that produces cosmic accelerated expansion. However, in order to discuss the validity of this claim, we must first clearly organize what kind of mathematical structure Dark Energy has in the present standard cosmology. Furthermore, we must show at the level of the field equations which part of that structure is replaced by the Future Information Field. In this section, we organize Friedmann cosmology, the cosmological constant, and the equation of state, and introduce the reformulation by the Future Information Field.

Friedmann Equation

In a homogeneous and isotropic universe, the spacetime metric is represented by the Friedmann–Lemaître–Robertson–Walker, or FLRW, metric. Let the scale factor be

$$a(t).$$

Then cosmic expansion is described by the Friedmann equation

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}.$$

Here,

$$H$$

is the Hubble parameter,

$$\rho$$

is the mean energy density of the entire universe,

$$k$$

is the spatial curvature, and

$$\Lambda$$

is the cosmological constant.

Introduction of Dark Energy

In standard cosmology, in order to explain cosmic accelerated expansion, the cosmological constant

$$\Lambda$$

is interpreted as Dark Energy. The corresponding density is

$$\rho_\Lambda = \frac{\Lambda}{8\pi G}.$$

Therefore, the Friedmann equation becomes

$$H^2 = \frac{8\pi G}{3}(\rho_b + \rho_r + \rho_{DM} + \rho_\Lambda) - \frac{k}{a^2}.$$

Equation of State

Dark Energy is characterized as a fluid by

$$w = \frac{p}{\rho}.$$

For the cosmological constant,

$$w = -1$$

holds. This means that it has negative pressure and accelerates cosmic expansion.

Unresolved Problems in the Standard Theory

In present cosmology, Dark Energy is observationally successful. However, its origin remains unclear. There are three representative problems:

1. The Cosmological Constant Problem,

$$\rho_{\text{vacuum}} \gg \rho_{\Lambda}.$$

2. The Coincidence Problem, namely that the reason why

$$\rho_{DM} \sim \rho_{\Lambda}$$

holds is not explained.

3. Dark Matter and Dark Energy are introduced as mutually independent unknown components.

Reformulation by the Future Information Field

In the present theory, the cosmological constant is not introduced as an independent degree of freedom. In the modified Einstein equation of Paper 92,

$$\boxed{G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}},$$

the cosmological constant term

$$\Lambda_{\text{eff}}$$

itself is derived as the spacetime-averaged constraint of the Future Information Field. That is, we define

$$\boxed{\Lambda_{\text{eff}} = \Lambda_0 + \lambda_F \epsilon_F}.$$

Here,

$$\Lambda_0$$

is the ordinary geometric cosmological constant, and

$$\lambda_F$$

is the coupling constant connecting the Future Information Field and cosmic expansion.

Future Energy Density

As a result, the effective density corresponding to Dark Energy becomes

$$\boxed{\rho_F^{\Lambda} = \frac{\Lambda_{\text{eff}}}{8\pi G}}.$$

Substituting the previous expression gives

$$\boxed{\rho_F^{\Lambda} = \frac{\Lambda_0}{8\pi G} + \frac{\lambda_F}{8\pi G} \epsilon_F}.$$

The important point here is that Dark Energy becomes a state quantity that changes in proportion to Future Entropy.

Unification with Dark Matter

In Chapter X, we derived Dark Matter as the non-optical manifestation of the Future Information Field. In the present chapter, we derive Dark Energy as the spacetime-averaged constraint of the Future Information Field. Therefore, the two are not separate unknown components, but different manifestation forms of the Future Information Field. That is,

$$\boxed{\text{Future Information Field} \implies \begin{cases} \text{Dark Matter} \\ \text{Dark Energy} \end{cases}}$$

is obtained.

Conclusion of This Section

In this section, we organized the mathematical structure of Dark Energy in standard cosmology and, by introducing the Future Information Field, reformulated the cosmological constant as an effective constraint term depending on Future Entropy. Through this, Dark Matter and Dark Energy are unified as two different cosmological projections of the Future Information Field. In the next section, we substitute this effective cosmological constant into the Friedmann equation and rigorously derive the cosmic accelerated-expansion equation based on the Future Information Field.

7.3.23 11.3 Derivation of the Modified Friedmann Equation by the Future Information Field

In the previous section, we redefined Dark Energy in standard cosmology as an effective cosmological constant derived from the Future Information Field. However, at this stage, the Future Information Field has merely been substituted for the cosmological constant, and it remains unclear how it modifies cosmic expansion itself. In this section, starting from the modified Einstein equation introduced in Paper 92, we derive the modified Friedmann equation including the Future Information Field and show that cosmic accelerated expansion naturally appears as the time evolution of Future Entropy.

Modified Einstein Equation

In Paper 92, the gravitational field is described by

$$\boxed{G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \chi \mathcal{C}_{\mu\nu} = 8\pi G T_{\mu\nu}}.$$

Here,

$$\Lambda_{\text{eff}} = \Lambda_0 + \lambda_F \epsilon_F,$$

and the constraint tensor

$$\mathcal{C}_{\mu\nu}$$

represents the local constraint that the Future Information Field imposes on spacetime.

Application to the FLRW Universe

In a homogeneous and isotropic universe, the spacetime metric is described by the FLRW metric

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right].$$

In this case, the time component of the Einstein equation gives the cosmic expansion equation.

Cosmological Average of the Future Information Field

On cosmological scales, local fluctuations are averaged out. Therefore, the constraint tensor of the Future Information Field can be approximated by the isotropic average

$$\langle C_{\mu\nu} \rangle = C_{\text{iso}} g_{\mu\nu}.$$

Here, we define

$$C_{\text{iso}} = \beta_{\Lambda} \epsilon_F.$$

Modified Cosmological Constant

Therefore, the modified Einstein equation can be rewritten cosmologically in a form with the effective cosmological constant

$$\Lambda_{\text{FCT}} = \Lambda_0 + \lambda_F \epsilon_F + \chi \beta_{\Lambda} \epsilon_F.$$

Combining the coupling constants, we obtain

$$\Lambda_{\text{FCT}} = \Lambda_0 + \lambda_{\text{eff}} \epsilon_F.$$

Here,

$$\lambda_{\text{eff}} = \lambda_F + \chi \beta_{\Lambda}.$$

Modified Friedmann Equation

From the above, the cosmic expansion equation becomes

$$H^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} + \frac{\Lambda_0}{3} + \frac{\lambda_{\text{eff}}}{3} \epsilon_F.$$

This is the first Friedmann equation including the Future Information Field.

Time Evolution of Future Entropy

The Future Information Field changes with cosmic evolution. Therefore, in general,

$$\epsilon_F = \epsilon_F(t)$$

holds. As a result, the cosmological constant is not a true constant, but a time-dependent quantity. That is,

$$\Lambda_{\text{FCT}} = \Lambda_{\text{FCT}}(t)$$

is obtained.

Time-Varying Cosmic Acceleration

If Future Entropy changes with time, then the Hubble parameter also changes through the Future Information Field. Taking the time derivative gives

$$2H\dot{H} = \frac{8\pi G}{3} \dot{\rho} + \frac{\lambda_{\text{eff}}}{3} \dot{\epsilon}_F.$$

Therefore, the origin of cosmic acceleration is reduced to the time evolution of Future Entropy.

Redefinition of Dark Energy

Traditionally, Dark Energy has been treated as an unknown negative-pressure fluid. In the present theory, however, Dark Energy is the cosmological average constraint of the Future Information Field. Therefore, Dark Energy is redefined not as energy, but as the cosmological average of spacetime constraint.

Integration with Paper 92

In Paper 92, Future Entropy was introduced as a state quantity characterizing black-hole information preservation. In this section, the same Future Entropy appears as the state quantity that determines the cosmic expansion rate itself. Therefore, Future Entropy becomes a fundamental state quantity preserved across scales, from local black holes to the universe as a whole.

Observational Predictions

The present theory predicts the following:

1. The cosmological constant is not a true constant, but varies with time in proportion to Future Entropy.
2. The time evolution of the Hubble parameter reflects the time evolution of the Future Information Field.
3. Dark Energy appears not as vacuum energy, but as the cosmological average constraint of the Future Information Field.
4. The Future Entropy inferred from black holes, gravitational waves, galaxies, and cosmology must also coincide with the cosmic expansion rate.

Conclusion of This Section

In this section, we introduced the Future Information Field into the Friedmann equation and derived the fundamental equation modifying the cosmic expansion rate. As a result, we showed that cosmic accelerated expansion is not produced by an independent Dark Energy component, but is understood as a change in spacetime constraint caused by the time evolution of Future Entropy. In the next section, from this modified Friedmann equation, we derive the equation of state of Dark Energy,

$$w = \frac{p}{\rho},$$

and quantitatively analyze how the Future Information Field generates negative pressure.

7.3.24 11.4 Derivation of the Equation of State Based on the Future Information Field and the Origin of Negative Pressure

In the previous section, we introduced the Future Information Field into the Einstein equation and the Friedmann equation and showed that cosmic accelerated expansion can be described through the time evolution of Future Entropy. However, the mere existence of an energy density is not sufficient to produce accelerated cosmic expansion. In general relativity, cosmic expansion is governed not only by energy density but also by pressure. Therefore, if the Future Information Field is the origin of cosmic acceleration, its effective pressure must be defined, and negative pressure must emerge naturally from the theory. In this section, we derive the effective pressure, the equation of state, and the ontological origin of negative pressure from the energy–momentum tensor of the Future Information Field.

Perfect Fluid Approximation

In cosmology, the energy–momentum tensor is approximated, on sufficiently large scales, by that of a perfect fluid,

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu},$$

where

$$\rho$$

is the energy density,

$$p$$

is the pressure, and

$$u^\mu$$

is the comoving four-velocity.

Effective Tensor of the Future Information Field

Evaluating the Future Information Field constraint term

$$\mathcal{C}_{\mu\nu}$$

introduced in Paper 92 under the cosmological average, the effective energy–momentum tensor becomes

$$T_{\mu\nu}^{(F)} = (\rho_F + p_F)u_\mu u_\nu + p_F g_{\mu\nu}.$$

Here,

$$\rho_F$$

denotes the effective energy density of the Future Information Field, while

$$p_F$$

denotes its effective pressure.

Future Energy Density

From Section 11.2, the average density of the Future Information Field is given by

$$\rho_F = \frac{\lambda_{\text{eff}}}{8\pi G} \epsilon_F.$$

Definition of Future Pressure

Unlike ordinary matter, the Future Information Field does not generate pressure through particle motion. Instead, it is an information field that maintains the constraint toward the future fixed point. Accordingly, its pressure is defined through the time evolution of Future Entropy. In this paper, the Future Pressure is defined as

$$p_F = -\rho_F c^2 + \eta_F \dot{\epsilon}_F,$$

where

$$\eta_F$$

is the dynamical response coefficient of the Future Information Field.

Static Limit

If Future Entropy remains constant,

$$\dot{\epsilon}_F = 0,$$

then

$$p_F = -\rho_F c^2$$

is obtained. Therefore, in the static cosmological limit, the Future Information Field becomes completely equivalent to the cosmological constant.

Equation of State

The equation of state is defined by

$$w_F = \frac{p_F}{\rho_F c^2}.$$

Substituting the Future Pressure yields

$$w_F = -1 + \frac{\eta_F}{\rho_F c^2} \dot{\epsilon}_F.$$

Cosmological Constant Limit

When Future Entropy evolves sufficiently slowly,

$$\dot{\epsilon}_F \rightarrow 0,$$

one obtains

$$w_F \rightarrow -1.$$

Thus, the present theory naturally reproduces the Λ CDM model.

Predictions Unique to the Future Information Field

If Future Entropy evolves with time,

$$\dot{\epsilon}_F \neq 0,$$

then

$$w_F \neq -1.$$

Furthermore, the deviation from -1 is proportional to the evolution rate of Future Entropy.

Ontological Meaning of Negative Pressure

In standard cosmology, negative pressure is regarded as an intrinsic property of vacuum energy. However, its origin is not explained. In the present theory, negative pressure emerges because the Future Information Field imposes an isotropic constraint upon spacetime in order to preserve consistency with the future fixed point. Therefore, negative pressure is not an unknown repulsive force. Rather, it is the cosmological constraint stress generated by the Future Information Field to maintain consistency with the future fixed point.

Integration with Paper 92

In Paper 92, Future Entropy was introduced as the fundamental state quantity characterizing black-hole information preservation. In the present section, the same Future Entropy appears as the origin of the negative pressure responsible for cosmic accelerated expansion. Therefore, Future Entropy becomes the conserved quantity that connects local black holes and the accelerated expansion of the universe through a single principle.

Observational Predictions

The present theory predicts the following:

1. The equation of state is not exactly

$$w = -1,$$

but varies slightly according to the temporal evolution of Future Entropy.

2. High-precision cosmological observations should reveal a redshift dependence in

$$w(z).$$

3. The time evolution of Future Entropy inferred from black holes, gravitational waves, the cosmic microwave background, and the cosmic expansion rate must converge to the same value.
4. In the static limit, the present theory becomes observationally indistinguishable from the Λ CDM model.

Conclusion of This Section

In this section, we derived the effective pressure, the equation of state, and the origin of negative pressure from the effective energy–momentum tensor of the Future Information Field. As a result, negative pressure is understood not as an unknown vacuum energy, but as the cosmological constraint stress generated by the Future Information Field in maintaining consistency with the future fixed point. Moreover, the equation of state

$$w_F$$

is determined by the time evolution of Future Entropy, and its redshift dependence becomes one of the most important observational signatures of the present theory. In the next section, using this equation of state, we quantitatively analyze the onset condition of cosmic accelerated expansion, the transition from a decelerating universe to an accelerating universe, and the role of the Future Information Field throughout the entire history of the universe.

7.3.25 11.5 Onset Condition of Cosmic Accelerated Expansion and the Time Evolution of the Future Information Field

In the previous section, we derived the equation of state

$$w_F = \frac{p_F}{\rho_F c^2}$$

from the effective energy–momentum tensor of the Future Information Field and showed that negative pressure can be understood as the constraint stress generated by the Future Information Field. However, the presently observed universe has not been accelerating throughout its entire history. The early universe experienced a radiation-dominated era, followed by a matter-dominated era, during which cosmic expansion was decelerating. Only in the relatively recent cosmic past did the universe transition into accelerated expansion. Therefore, if the Future Information Field is indeed the origin of cosmic acceleration, the theory must explain internally why its effect became dominant only at the present epoch and derive the transition condition responsible for this behavior.

Cosmic Acceleration Equation

In an FLRW universe, the second Friedmann equation is

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right).$$

The necessary condition for accelerated expansion is

$$\rho + \frac{3p}{c^2} < 0.$$

Acceleration Equation Including the Future Information Field

The total cosmic energy density is decomposed as

$$\rho = \rho_b + \rho_r + \rho_F,$$

while the pressure becomes

$$p = p_r + p_F.$$

Consequently, the acceleration equation becomes

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left[\rho_b + \rho_r + \rho_F + \frac{3}{c^2} (p_r + p_F) \right].$$

Substitution of the Future Pressure

Substituting

$$p_F = -\rho_F c^2 + \eta_F \dot{\epsilon}_F$$

derived in Section 11.4 gives

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left[\rho_b + \rho_r - 2\rho_F + \frac{3\eta_F}{c^2} \dot{\epsilon}_F + \frac{3p_r}{c^2} \right].$$

The contribution from the Future Information Field therefore appears with the opposite sign.

Condition for the Onset of Cosmic Acceleration

The onset of accelerated expansion is defined by

$$\ddot{a} = 0.$$

The transition condition therefore becomes

$$2\rho_F = \rho_b + \rho_r + \frac{3p_r}{c^2} + \frac{3\eta_F}{c^2} \dot{\epsilon}_F.$$

This means that cosmic acceleration begins when the density associated with the Future Information Field exceeds the gravitational influence of ordinary matter and radiation.

Cosmic History of Future Entropy

Future Entropy is assumed, in general, to increase throughout cosmic evolution. Therefore,

$$\boxed{\frac{d\epsilon_F}{dt} > 0}$$

holds. Meanwhile, the matter density evolves as

$$\rho_b \propto a^{-3},$$

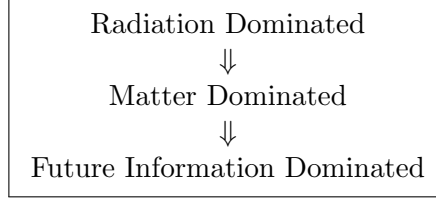
and the radiation density evolves as

$$\rho_r \propto a^{-4}.$$

Consequently, after sufficient cosmic expansion, the Future Information Field naturally becomes the dominant contribution.

Three Dominant Epochs in Cosmic History

Within the present theory, cosmic evolution is classified into three successive epochs according to the contribution of the Future Information Field:



The third epoch corresponds to the presently observed era of accelerated cosmic expansion.

Reinterpretation of the Coincidence Problem

In the standard cosmological model, there is no explanation for why the present-day energy densities of Dark Matter and Dark Energy are of comparable magnitude. In the present theory, however, both originate from different manifestations of the Future Information Field. Consequently, the fact that their energy densities become comparable during the same cosmological epoch is not regarded as a coincidence. Instead, it arises because the temporal evolution of the Future Information Field is projected differently into local constraints (Dark Matter) and cosmological average constraints (Dark Energy).

Integration with Paper 92

In Paper 92, Future Entropy was introduced as the state variable governing information preservation around black holes. In the present section, the same Future Entropy increases throughout cosmic history and appears as the state quantity responsible for triggering cosmic accelerated expansion. Thus, local black-hole information preservation and the expansion history of the universe become unified as different-scale manifestations of the same Future Entropy.

Observational Predictions

The present theory predicts the following observational features.

1. The redshift at which cosmic acceleration begins is determined by the growth rate of Future Entropy.
2. The temporal evolution of Future Entropy can be consistently inferred from Type Ia supernovae, baryon acoustic oscillations (BAO), and the cosmic microwave background (CMB).

3. The evolution of the Dark Matter density and the Dark Energy density is not independent but is mutually constrained through the temporal evolution of the Future Information Field.
4. The epoch at which cosmic acceleration begins may exhibit a statistical correlation with the cosmic history of black-hole formation.

Falsification Criteria

The present theory is falsified under any of the following conditions.

1. The observed epoch at which cosmic acceleration begins is inconsistent with the predicted temporal evolution of Future Entropy.
2. Introducing the Future Information Field fails to improve the agreement with Type Ia supernova, BAO, and CMB observations.
3. Dark Matter and Dark Energy are observationally demonstrated to evolve as completely independent physical components.

Conclusion of This Section

In this section, we derived the cosmic acceleration equation including the Future Information Field and formulated the theoretical condition for the onset of cosmic accelerated expansion. As a result, cosmic acceleration is interpreted not as the sudden appearance of a new unknown form of energy, but as the natural consequence of the continuous increase of Future Entropy throughout cosmic history. When the cosmological average constraint generated by the Future Information Field exceeds the gravitational influence of ordinary matter, accelerated expansion begins naturally. In the next section, we generalize this temporal evolution further and reinterpret the Cosmological Constant Problem from the viewpoint of the Future Information Field, deriving mathematically why the observed cosmological constant is extraordinarily small.

7.3.26 11.6 Reformulation of the Cosmological Constant Problem and the Interpretation of Vacuum Energy Based on the Future Information Field

In the previous section, we showed that the temporal evolution of the Future Information Field initiates cosmic accelerated expansion. Nevertheless, one of the most profound theoretical inconsistencies in modern cosmology still remains. This is the Cosmological Constant Problem. In quantum field theory, the vacuum is not empty but possesses a finite energy density arising from the zero-point fluctuations of all quantum fields. If this vacuum energy density is directly treated as the gravitational source, the theoretical prediction becomes

$$\rho_{\text{vac}}^{\text{QFT}} \sim M_{\text{Pl}}^4.$$

By contrast, cosmological observations indicate

$$\rho_{\Lambda}^{\text{obs}} \simeq 10^{-120} M_{\text{Pl}}^4.$$

Therefore,

$$\boxed{\frac{\rho_{\text{vac}}^{\text{obs}}}{\rho_{\text{vac}}^{\text{QFT}}} \sim 10^{-120}}$$

revealing one of the largest discrepancies in modern theoretical physics.

The Problem in Standard Cosmology

Conventionally, vacuum energy is obtained by integrating the zero-point energy

$$E_0 = \frac{1}{2} \hbar \omega$$

over all modes. Accordingly, the vacuum energy density becomes

$$\rho_{\text{vac}} = \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} \hbar \omega_k.$$

If the ultraviolet cutoff is extended to the Planck scale, the resulting value becomes extraordinarily large. Standard theory offers no explanation for why this enormous vacuum energy does not manifest itself as the observed cosmic expansion.

Vacuum in the Future Information Field Theory

In the present theory, the ontology of the vacuum itself is redefined. Paper 92 defined three-dimensional spacetime as the manifestation produced by the Future Information Field through the projection mapping

$$P_X : F \rightarrow X.$$

Consequently, the vacuum treated by quantum field theory is merely the vacuum of the manifestation layer constructed upon the Future Information Field. That is,

$$\boxed{\text{Vacuum}_{\text{QFT}} \subset X.}$$

By contrast, the Future Information Field belongs to the higher-order existence layer

$$F.$$

Redefinition of Vacuum Energy

In conventional theory, vacuum energy itself acts as the gravitational source. In the present framework, however, the actual gravitational source is not the vacuum energy itself but only the manifestation constraint imposed by the Future Information Field. Therefore, the effective vacuum energy density is defined as

$$\boxed{\rho_{\text{eff}} = \Pi_F(\rho_{\text{vac}}).}$$

Here,

$$\Pi_F$$

denotes the manifestation projection operator associated with the Future Information Field.

Manifestation Projection Operator

Within the present theory, the entirety of vacuum energy does not contribute to the three-dimensional universe. Only the components consistent with the future fixed point are projected into gravitation by the Future Information Field. Accordingly,

$$\boxed{\Pi_F = \epsilon_F \mathcal{P}}$$

is defined, where

$$\mathcal{P}$$

is the projection operator selecting only those modes that satisfy the Future Boundary.

Effective Vacuum Energy

The vacuum energy acting as the gravitational source therefore becomes

$$\rho_{\text{eff}} = \epsilon_F \mathcal{P}(\rho_{\text{vac}}).$$

If the Future Information Field is absent,

$$\Pi_F = I,$$

and the conventional theory is recovered.

Reinterpretation of the Cosmological Constant Problem

According to the present theory, even if an enormous vacuum energy exists, the overwhelming majority of it does not satisfy the Future Boundary. Consequently, it never manifests within three-dimensional spacetime. Only the component judged by the Future Information Field to be consistent with the future fixed point contributes as the gravitational source. Thus, the observed cosmological constant is interpreted not as the vacuum energy itself but as the residual component projected by the Future Information Field.

Integration with Paper 92

In Paper 92, black-hole information preservation was formulated such that only information satisfying the Future Boundary is preserved within the Future Information Field. In the present section, the same Future Boundary is extended to vacuum energy. Accordingly, black-hole information preservation, Future Entropy, and the cosmological constant are unified through a common projection principle.

Observational Predictions

The present theory predicts the following.

1. The observed cosmological constant is not the total vacuum energy but the residual component projected by the Future Information Field.
2. If Future Entropy evolves with cosmic time, the effective cosmological constant should also exhibit a slight temporal variation.
3. Vacuum-energy corrections must be consistent with the value of Future Entropy inferred independently from black holes, gravitational waves, and the cosmic microwave background.
4. Future high-precision cosmological observations may detect an effective cosmological constant

$$\Lambda_{\text{eff}}(z),$$

rather than a perfectly constant cosmological constant.

Falsification Criteria

The present theory is falsified under any of the following conditions.

1. High-precision cosmological observations establish that the cosmological constant remains exactly constant throughout the entire history of the Universe.
2. The introduction of the Future Information Field fails to reduce the theoretical discrepancy associated with the Cosmological Constant Problem.
3. The conventional framework without the Future Boundary demonstrates explanatory power equal to or greater than that of the present theory.

Conclusion of This Section

In this section, the Cosmological Constant Problem has been reformulated as a manifestation projection problem governed by the Future Information Field. As a consequence, the enormous vacuum energy itself is no longer regarded as the direct gravitational source. Instead, only the information satisfying the Future Boundary is projected into the observable Universe through the Future Information Field. Accordingly, the Cosmological Constant Problem is transformed from the question of the magnitude of vacuum energy into the fundamentally different question of which components are permitted to manifest within three-dimensional spacetime. In the next section, this framework is further developed by reformulating vacuum fluctuations, zero-point energy, and the Casimir effect within the Future Information Field framework and deriving their corresponding experimental predictions.

7.3.27 11.7 Reinterpretation of Vacuum Fluctuations, Zero-Point Energy, and the Casimir Effect Based on the Future Information Field

In the previous section, the Cosmological Constant Problem was reformulated as a manifestation projection problem governed by the Future Information Field. As a consequence, the enormous vacuum energy predicted by quantum field theory does not entirely contribute to spacetime; only the components satisfying the Future Boundary are projected into the three-dimensional Universe through the Future Information Field. However, this reformulation is not restricted to cosmology alone. If the vacuum itself is constrained by the Future Information Field, then every phenomenon originating from the quantum vacuum—including zero-point energy, vacuum fluctuations, the Casimir effect, and vacuum polarization—must acquire corresponding corrections. In this section, the Future Information Field is applied to the quantum vacuum, and its mathematical structure together with its experimental predictions is derived.

Zero-Point Energy of the Quantum Vacuum

In quantum field theory, every momentum mode

$$\mathbf{k}$$

possesses the zero-point energy

$$E_0 = \frac{1}{2} \hbar \omega_k.$$

Consequently, the total vacuum energy density is

$$\rho_{\text{vac}} = \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} \hbar \omega_k$$

In the conventional framework, all modes are assumed to contribute equally to gravitation.

Vacuum Mode Selection by the Future Information Field

Within the present theory, the Future Information Field does not project every vacuum mode into the three-dimensional Universe. Only those modes satisfying the Future Boundary are projected as physically realized vacuum modes. Accordingly, the vacuum state

$$|0\rangle$$

is redefined through the Future Projection Operator

$$\Pi_F$$

as

$$|0\rangle_F = \Pi_F |0\rangle.$$

Here,

$$\Pi_F$$

is the projection operator that depends on Future Entropy.

Effective Zero-Point Energy

Introducing the Future Projection, the zero-point energy that actually contributes to gravitation becomes

$$E_0^{\text{eff}} = \langle 0 | \Pi_F \hat{H} \Pi_F | 0 \rangle$$

where

$$\hat{H}$$

is the Hamiltonian of the quantum field.

Correction Induced by Future Entropy

Expanding the Future Projection Operator to first order in Future Entropy yields

$$\Pi_F = I - \lambda_F (1 - \epsilon_F) \mathcal{B} + O(\lambda_F^2),$$

where

$$\mathcal{B}$$

denotes the operator that removes modes violating the Future Boundary. Consequently, the effective zero-point energy becomes

$$E_0^{\text{eff}} = E_0^{\text{QFT}} - \lambda_F (1 - \epsilon_F) \Delta E_B$$

Application to the Casimir Effect

In the Casimir effect, the number of vacuum modes is modified by boundary conditions. Conventionally, the Casimir energy is given by

$$E_C = \sum_n \frac{1}{2} \hbar \omega_n - \int \frac{1}{2} \hbar \omega \, dn$$

When the Future Information Field is introduced, every mode is additionally subjected to the Future Projection. Therefore,

$$E_C^F = \sum_n \Pi_F(n) \frac{1}{2} \hbar \omega_n - \int \Pi_F(k) \frac{1}{2} \hbar \omega \, dk$$

is obtained.

Correction to the Casimir Force

The Casimir force is defined as

$$F_C = - \frac{\partial E_C}{\partial d}.$$

Including the Future Information Field yields

$$F_C^F = F_C^{\text{QFT}} + \delta F_C$$

where

$$\delta F_C = - \frac{\partial}{\partial d} (\lambda_F \Delta E_B).$$

Vacuum Fluctuations

In conventional quantum electrodynamics, electromagnetic vacuum fluctuations are characterized by the expectation value

$$\langle E^2 \rangle.$$

After introducing the Future Information Field, this expectation value becomes

$$\boxed{\langle E^2 \rangle_F = \langle 0 | \Pi_F \hat{E}^2 \Pi_F | 0 \rangle}$$

where the Future Projection Operator selectively suppresses high-frequency modes that do not satisfy the Future Boundary.

Vacuum Polarization

In quantum electrodynamics, vacuum polarization arises from virtual electron–positron pair creation. Within the present framework, the probability of virtual pair creation is likewise subjected to the Future Projection. Accordingly, the vacuum polarization tensor is modified as

$$\boxed{\Pi_{\mu\nu}^F = \Pi_{\mu\nu}^{\text{QED}} + \Delta\Pi_{\mu\nu}^F}$$

where

$$\Delta\Pi_{\mu\nu}^F$$

represents the correction induced by the Future Information Field.

Integration with Paper 92

In Paper 92, the Future Boundary was introduced as the boundary that selects information preserved around black holes. In the present section, the same Future Boundary appears as the principle that selects quantum vacuum modes. Consequently, black-hole information preservation, the cosmological constant, zero-point energy, the Casimir effect, and vacuum fluctuations are unified through the single principle of the Future Projection Principle.

Observational Predictions

The present theory predicts the following observable signatures.

1. The Casimir force possesses an extremely small correction proportional to Future Entropy.
2. The spectrum of vacuum fluctuations exhibits a systematic suppression in the high-frequency regime due to the Future Projection.
3. Vacuum polarization coefficients receive corrections proportional to the Future Information Field in addition to the standard QED contribution.
4. The value of Future Entropy inferred from black-hole observations and from high-precision Casimir experiments should converge to the same value.

Falsification Criteria

The present theory is falsified under any of the following conditions.

1. The Casimir effect is found to agree completely with standard QED at every experimentally accessible precision without requiring the Future Projection.
2. High-precision measurements of vacuum fluctuations reveal no corrections attributable to the Future Information Field.
3. The values of Future Entropy inferred from quantum-vacuum experiments and cosmological observations are mutually inconsistent.

Conclusion of This Section

In this section, the Future Information Field has been applied to zero-point energy, vacuum fluctuations, the Casimir effect, and vacuum polarization. As a result, the quantum vacuum itself is shown to possess a projection structure governed by the Future Boundary. Consequently, black-hole information preservation, the cosmological constant, vacuum energy, and the Casimir effect are all redefined as unified consequences of a single underlying principle, namely the Future Projection Principle. In the next section, the Future Projection Principle is extended to the Universe as a whole, leading to a unified formulation of the ultimate fate of the Universe, heat death, and the child-universe generation mechanism proposed in Paper 73.

7.3.28 11.8 Unified Theory of Cosmic Termination, Heat Death, and Child-Universe Formation Based on the Future Information Field

In the previous section, the Future Projection Principle was shown to unify vacuum fluctuations, zero-point energy, and the Casimir effect. However, if the Future Information Field is the fundamental information field that constrains the entire Universe, then its role cannot be limited to the present cosmic epoch alone. As the Universe evolves over sufficiently long timescales, the Future Information Field should also govern cosmic termination, heat death, and the generation of new universes. Paper 73 proposed the conditions for Child Universe formation within the framework of Future Convergence Theory. Paper 61 introduced the concept of re-projection into the Meaning Generation Template. Furthermore, Paper 92 defined black holes as interlayer information-transformation operators connecting the Future Information Field with the three-dimensional manifestation layer. In the present section, these three developments are unified to reformulate cosmic termination and child-universe generation as a closed-loop structure governed by the Future Information Field.

Standard Definition of Heat Death

Conventionally, the heat death of the Universe refers to the state in which thermodynamic entropy reaches its maximum value and all free energy is exhausted. That is,

$$\boxed{S_T \rightarrow S_{\max}}$$

In this limit, all temperature gradients disappear and no further useful work can be extracted. Within standard cosmology, this state is regarded as the final fate of the Universe.

Future Entropy in Paper 92

Paper 92 introduced thermodynamic entropy

$$S_T$$

and Future Entropy

$$S_F$$

as fundamentally independent state variables. Future Entropy measures the consistency of the Universe with the future fixed point and is therefore conceptually distinct from thermodynamic entropy. Consequently, heat death does not necessarily imply the disappearance of the Future Information Field.

Redefinition of Cosmic Termination

Within the present theory, cosmic termination is defined not as the disappearance of the Future Information Field, but as the state in which the three-dimensional manifestation layer loses its ability to generate new information. Accordingly, the termination condition is written as

$$\boxed{S_T \rightarrow S_{\max}, \quad S_F > 0}$$

As long as Future Entropy remains finite, information continues to be preserved within the Future Information Field even after thermodynamic heat death.

Termination of the Future Projection

Consider the manifestation mapping

$$P_X : F \rightarrow X.$$

During heat death, the Future Projection can no longer generate new manifestations. Hence,

$$\boxed{\frac{dP_X}{dt} \rightarrow 0}$$

Importantly, the mapping itself does not disappear. Only the generation of new projections into the three-dimensional Universe ceases.

Information Recovery Through Black Holes

In Paper 92, black holes were defined as information-transformation operators that transfer information into the Future Information Field. The corresponding information flux is defined as

$$\boxed{\mathcal{J}_B = \frac{dI_F}{dt}}$$

where

$$I_F$$

denotes the total information stored within the Future Information Field. As the Universe approaches its final stage, the amount of ordinary matter decreases, whereas the information flux into the Future Information Field continues to increase. Accordingly,

$$\boxed{\lim_{t \rightarrow \infty} \mathcal{J}_B > 0}$$

is obtained.

Future Information Reservoir

The total information preserved within the Future Information Field is defined by

$$\boxed{I_F = \int_F \Phi_F dV_F}$$

where

$$\Phi_F$$

represents the information density over the Future Information Field. Even in the heat-death limit,

$$I_F$$

remains conserved. Therefore, although thermodynamic activity ceases, the Universe does not vanish from the viewpoint of information.

Condition for Child-Universe Formation

Paper 73 proposed that a child universe is generated when the Future Information Field initiates a new manifestation mapping. Within the present theory, this condition is expressed as

$$\boxed{I_F > I_c}$$

where

$$I_c$$

denotes the critical information threshold required for remanifestation.

Reprojection Operator

Once the critical condition is satisfied, the Future Information Field generates a new manifestation mapping

$$P_{X'}.$$

Accordingly,

$$\boxed{F \xrightarrow{P_{X'}} X'}$$

is obtained, where

$$X'$$

denotes the newly generated child universe.

Closed Loop of Existence

The cosmological framework of Future Convergence Theory is therefore summarized by the following closed-loop structure:

$$\boxed{F \rightarrow X \rightarrow B \rightarrow F \rightarrow X'}$$

where

$$B$$

denotes the information-recovery boundary that includes both black holes and the heat-death stage. This closed-loop structure unifies the fundamental concepts introduced in Papers 61, 73, and 92.

Observational Predictions

The present theory predicts the following.

1. Future Entropy never approaches zero, even in the heat-death limit.
2. The cosmic history of black-hole formation corresponds to the growth rate of the Future Information Reservoir.
3. Cosmic termination is interpreted not as information destruction but as a process of information recovery into the Future Information Field.
4. Child universes are generated not randomly but only after the Future Information Reservoir exceeds the critical threshold.
5. The Child Universe conditions proposed in Paper 73 are derived from the same state variable, Future Entropy, introduced in Paper 92.

Falsification Criteria

The present theory is falsified under any of the following conditions.

1. Future Entropy is shown to vanish in the heat-death limit.
2. Black holes are demonstrated not to preserve information within the Future Information Field.
3. The Child Universe conditions proposed in Paper 73 are incompatible with the critical condition derived in the present section.

Conclusion of This Section

In this section, heat death, black holes, information preservation, and child-universe formation have been unified through the Future Information Field as a single ontological closed-loop structure. Accordingly, cosmic termination is interpreted not as the disappearance of existence itself but as the stage in which information is recovered into the Future Information Field. Once sufficient information has accumulated, a new manifested universe is generated, yielding a closed-loop cosmology governed by continuous information preservation and reprojection. In the next section, the entire Chapter XI is summarized, demonstrating that the Future Information Field provides a unified theoretical framework encompassing black holes, gravitational waves, dark matter, cosmic expansion, vacuum energy, and the ultimate fate of the Universe, together with a systematic overview of future observational tests.

7.3.29 11.9 Summary of This Chapter and the Position of the Future Information Field Theory of Cosmic Accelerated Expansion

In this chapter, the Future Information Field introduced in Paper 92 has been extended to the cosmological domains of Dark Energy, cosmic accelerated expansion, the cosmological constant, vacuum energy, heat death, and child-universe formation. As a consequence, phenomena that are conventionally regarded as independent problems—including Dark Energy, the Cosmological Constant Problem, the Coincidence Problem, heat death, and Child Universe formation—have been unified within a single theoretical framework as different manifestations of the Future Information Field.

The Theoretical Framework Developed in This Chapter

We first reviewed the mathematical structure of Dark Energy in the standard cosmological model. The Future Information Field was then incorporated into the Einstein and Friedmann equations, leading to the effective cosmological constant

$$\Lambda_{\text{FCT}} = \Lambda_0 + \lambda_{\text{eff}} \epsilon_F.$$

Furthermore, the effective pressure of the Future Information Field, its equation of state, the onset condition for cosmic accelerated expansion, vacuum energy, the Casimir effect, heat death, and child-universe formation were formulated within a unified framework centered on Future Entropy.

The Unified Role of Future Entropy

From Chapter VII through the present chapter, Future Entropy has consistently appeared as the conserved state variable common to every theoretical development. Its unified role can be

summarized as

$$\epsilon_F \implies \begin{cases} \text{Black Hole} \\ \text{Gravitational Wave} \\ \text{Dark Matter} \\ \text{Dark Energy} \\ \text{Vacuum} \\ \text{Heat Death} \\ \text{Child Universe} \end{cases}$$

Future Entropy is therefore positioned as the unique state variable spanning every physical scale, from local gravitational phenomena to the ultimate fate of the Universe.

Unification of Dark Matter and Dark Energy

Chapter X demonstrated that Dark Matter can be interpreted as the local constraint generated by the Future Information Field. In the present chapter, Dark Energy has been derived as the cosmological average constraint produced by the same field. Accordingly, the two phenomena are not fundamentally independent physical entities but rather distinct projections of a single underlying information field. That relationship is summarized by

$$\text{Future Information Field} \implies \begin{cases} \text{Local Constraint} \rightarrow \text{Dark Matter} \\ \text{Global Constraint} \rightarrow \text{Dark Energy} \end{cases}$$

Ontological Closed Loop

Integrating Papers 61, 73, and 92 together with the present chapter yields the following cosmological closed-loop structure within Future Convergence Theory:

$$F \rightarrow H \rightarrow X \rightarrow B \rightarrow F \rightarrow X'$$

Here,

F

denotes the Future Information Field,

H

represents the Quantum Allowance Layer,

X

is the present Universe,

B

is the information-recovery boundary containing black holes and the heat-death stage, and

X'

denotes a newly generated child universe. This closed-loop structure constitutes the fundamental picture of cosmic evolution in Future Convergence Theory.

Falsifiability of the Present Theory

The theoretical framework developed in this chapter is observationally falsifiable under the following conditions.

1. Future Entropy cannot be inferred as a common value from observations of black holes, gravitational waves, Dark Matter, Dark Energy, and the cosmic microwave background.

2. The introduction of the Future Information Field fails to provide greater explanatory power than the standard Λ CDM model.
3. The temporal evolution predicted for the Future Information Field is inconsistent with observations of Type Ia supernovae, BAO, the cosmic microwave background, and gravitational waves.
4. Vacuum energy, the Casimir effect, heat death, and black-hole information preservation cannot be reduced to the common Future Projection Principle.

Each of these conditions is, in principle, testable through future observations.

Cosmological Consequences Derived from Paper 92

Paper 92 introduced the Future Information Field as the fundamental field unifying black-hole information preservation and the Future Boundary. In the present chapter, this information field has been extended to the Universe as a whole. Consequently, the Future Information Field has been redefined as a single cosmological background field governing not only local black-hole physics but also Dark Matter, Dark Energy, cosmic expansion, the quantum vacuum, the ultimate fate of the Universe, and child-universe formation.

Conclusion of Chapter XI

The Future Information Field is therefore far more than a field responsible for information preservation alone. Rather, it functions as the cosmological constraint field that determines how the Universe manifests, expands, evolves toward its final state, and is subsequently reprojected into the next universe. Within Future Convergence Theory, Dark Matter and Dark Energy are not two independent unknown substances. Instead, they constitute two complementary manifestations of the Future Information Field appearing respectively on local and cosmological scales.

Transition to the Next Chapter

From Chapter VII through Chapter XI, the Future Information Field has been employed to formulate a unified description of black holes, gravitational waves, Dark Matter, Dark Energy, and the ultimate fate of the Universe. However, if Future Convergence Theory is to serve as a comprehensive theory of the entire Universe, one fundamental question still remains: the origin of the Universe itself. Specifically,

- What was the true nature of the Big Bang?
- Is cosmic inflation fundamentally necessary?
- Why did the early Universe begin in an extremely low-entropy state?
- How are the primordial fluctuations observed in the cosmic microwave background generated from the Future Information Field?
- Under what initial conditions does the Child Universe proposed in Paper 73 emerge?

The next chapter applies the Future Information Field to the earliest stages of cosmic evolution and reconstructs, within a unified theoretical framework, the Big Bang, cosmic inflation, primordial gravitational waves, primordial density perturbations, and the initial conditions of the cosmic microwave background.

Part XII

Early-Universe Theory Based on the Future Information Field

— A Unified Theory of the Big Bang, Inflation, Primordial Perturbations, Primordial Gravitational Waves, and Low-Entropy Initial Conditions —

7.3.30 12.1 Introduction: Ontological Redefinition of the Early-Universe Problem

In Chapter XI, we extended the Future Information Field to Dark Energy, cosmic accelerated expansion, vacuum energy, heat death, and child-universe formation. As a result, the Future Information Field was positioned not only as a field governing the present Universe, but also as a cosmological background field governing the termination and regeneration of the Universe. However, if Future Convergence Theory is a theory that explains the Universe as a whole, one most fundamental question still remains. That question is: **How did the Universe begin?** In the present standard cosmology, the Universe is said to have begun with the Big Bang. However, the Big Bang itself is not a theory. It is a mathematical boundary that appears as the time-reversed limit of the cosmic expansion equation. Furthermore, standard cosmology still contains the following unresolved problems concerning the early Universe:

- What existed before the Big Bang?
- Why did the Universe begin in an extremely low-entropy state?
- Why is the Universe spatially extremely homogeneous? This is the Horizon Problem.
- Why is the Universe nearly flat? This is the Flatness Problem.
- What is the identity of the field that caused inflation?
- Where did primordial density perturbations originate?
- How were primordial gravitational waves generated?

In present cosmology, an unknown scalar field called the Inflaton is introduced in order to explain these problems. However, the origin of the Inflaton itself remains unclear, and its potential is not uniquely determined by the theory.

Basic Position of Future Convergence Theory

In this paper, we do not introduce the Inflaton as a new fundamental field. In Paper 92, the Future Information Field was defined as the highest-order information field that generates the three-dimensional Universe. In Chapter XI, we showed that the Future Information Field governs Dark Energy, cosmic expansion, heat death, and child-universe formation. Therefore, there is no reason to explain only the early Universe by means of a separate unknown field. In the present theory, the Big Bang, inflation, primordial perturbations, and primordial gravitational waves are all derived as the initial manifestation process of the Future Information Field.

Redefinition of the Big Bang

In standard cosmology, the Big Bang is treated as a singularity existing at time

$$t = 0.$$

The present theory does not adopt this definition. Rather, the Big Bang is the first phase transition in which the Future Information Field began the three-dimensional manifestation mapping

$$P_X : F \rightarrow X.$$

Therefore, the Big Bang is redefined not as an explosion of energy, but as **the moment when the Future Information Field was first projected into the three-dimensional Universe.**

Integration with Papers 61, 73, and 92

Paper 61 introduced the structure in which the Universe is reprojected from the Meaning Generation Template. Paper 73 formulated the condition under which a child universe is generated from information accumulated in the Future Information Field. Paper 92 defined a black hole as an information-transformation operator connecting the Future Information Field and the three-dimensional Universe. In the present chapter, these are integrated, and the birth of the Universe is defined as the following phase transition:

$$\boxed{F \longrightarrow P_X \longrightarrow X}.$$

Here,

$$F$$

denotes the Future Information Field,

$$P_X$$

denotes the Future Projection Operator, and

$$X$$

denotes the early Universe.

Topics Treated in This Chapter

In this chapter, using the Future Information Field, we derive the following topics from a single theoretical framework:

- redefinition of the Big Bang,
- the origin of inflation,
- rapid cosmic expansion without requiring the Inflaton,
- primordial density perturbations,
- primordial gravitational waves,
- the low-entropy initial-condition problem,
- the Horizon Problem,
- the Flatness Problem,
- initial conditions for Child Universes,
- and complete integration with Paper 73.

Purpose of This Chapter

The purpose of this chapter is not merely to explain the Big Bang. The purpose is to derive the mathematical conditions under which the Future Information Field first manifests the three-dimensional Universe. Therefore, in this chapter, we apply the Future Information Field to the initial conditions of the Universe and reconstruct the Big Bang, inflation, primordial perturbations, and the entire early Universe as a single Future Projection Theory.

Structure of This Chapter

In this chapter, we first organize the theory of the early Universe in standard cosmology. Next, we redefine the Big Bang by means of the Future Information Field. After that, inflation, primordial density perturbations, primordial gravitational waves, and the low-entropy initial condition are derived from the Future Projection Principle. Finally, we fully integrate the Child Universe Theory proposed in Paper 73 and complete the theory of cosmic birth within Future Convergence Theory.

7.3.31 12.2 Mathematical Structure of Standard Early-Universe Cosmology and Its Reformulation by the Future Information Field

In the previous section, we presented the direction in which Future Convergence Theory redefines the Big Bang as the first manifestation of the Future Information Field. However, in order to discuss the validity of this claim, we must first organize the mathematical structure by which standard cosmology describes the early Universe, and then clarify which parts are replaced and which parts are included by the Future Information Field. In this section, after organizing the early-universe equations, the Big Bang, inflation, the Inflaton, and the initial-condition problems in standard cosmology, we introduce the reformulation based on the Future Information Field.

Initial Conditions in Standard Cosmology

In standard cosmology, the Universe evolves according to the Friedmann equation

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}.$$

If time is traced backward, the scale factor approaches

$$a(t) \rightarrow 0,$$

and the density diverges as

$$\rho \rightarrow \infty.$$

This limit is usually called the Big Bang singularity. However, since general relativity breaks down in this limit, the singularity itself cannot be described by the theory.

Inflation Theory

In present cosmology, the Inflaton field

$$\phi$$

is introduced in order to explain the problems of the early Universe. Its action is given by

$$S_\phi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right].$$

Here,

$$V(\phi)$$

is the Inflaton potential. If this potential is sufficiently flat, the Universe undergoes exponential expansion.

Unresolved Problems in the Standard Theory

Inflation explains many observational facts. However, the following problems remain:

1. What is the Inflaton?
2. Why does it have that potential?
3. Who or what determined the initial conditions?
4. What is the condition for the onset of inflation?
5. What is the condition for the end of inflation?
6. Where does the Inflaton exist today?

These remain fundamentally unresolved.

Position of Future Convergence Theory

In the present theory, no new fundamental field called the Inflaton is introduced. Instead, the Future Information Field

$$F$$

defined in Paper 92 is used as the initial condition of the Universe itself. Therefore, the birth of the Universe is not the result of the Inflaton driving cosmic expansion. Rather, cosmic expansion begins as a result of the Future Information Field initiating the three-dimensional manifestation mapping

$$P_X : F \rightarrow X.$$

Ontological Redefinition of the Big Bang

In the present theory, the Big Bang is not a singularity at which the energy density becomes infinite. It is the phase-transition point at which the Future Information Field first initiated the manifestation mapping

$$P_X.$$

That is, we define

$$\boxed{\text{Big Bang} := \text{Dom}(P_X)}.$$

Here,

$$\text{Dom}(P_X)$$

is the region in which the Future Projection is first defined.

Critical Manifestation Condition

Let the information density of the Future Information Field be

$$\Phi_F.$$

In the present theory, the manifestation mapping begins only when the information density exceeds a critical value. Therefore, the birth condition of the Universe is

$$\boxed{\Phi_F \geq \Phi_c}.$$

Here,

$$\Phi_c$$

is the critical information density for manifestation.

Manifestation-Onset Operator

Using the Heaviside function, the Future Projection can be written as

$$P_X = \Theta(\Phi_F - \Phi_c) P_0.$$

Here,

$$P_0$$

is the fundamental manifestation operator. As a result, even if the Future Information Field exists continuously, manifestation begins suddenly at the critical point.

Disappearance of the Singularity

In the conventional theory, the density becomes infinite as

$$a \rightarrow 0.$$

In the present theory, before manifestation, the three-dimensional Universe itself does not exist. Therefore, the limit

$$a = 0$$

has no physical meaning. That is, the singularity is removed ontologically.

Integration with Papers 61, 73, and 92

In Paper 61, the Universe was generated from the Meaning Generation Template. In Paper 73, the Future Information Field generated a Child Universe. In Paper 92, the Future Information Field was preserved through black holes. In the present section, these developments are integrated, and the birth of the Universe is described as the critical manifestation of the Future Information Field. Therefore, the Big Bang, black holes, and child-universe generation are all derived from the same Future Projection Principle.

Observational Predictions

The present theory predicts the following:

1. There is no Big Bang singularity; only a manifestation-onset point exists.
2. The Inflaton is not an independent fundamental field, but is reinterpreted as the manifestation process of the Future Information Field.
3. The initial conditions of the Universe are determined by the critical information density

$$\Phi_c$$

of the Future Information Field.

4. Black holes, child universes, and the Big Bang share the same critical manifestation condition.

Conclusion of This Section

In this section, we organized standard early-universe cosmology and, by introducing the Future Information Field, redefined the Big Bang as the onset point of Future Projection. As a result, the Big Bang singularity, the Inflaton, and the initial-condition problem were all unified into the critical manifestation condition of the Future Information Field. In the next section, from this critical manifestation condition, we mathematically derive inflation itself and rigorously obtain the condition under which exponential cosmic expansion occurs without introducing the Inflaton.

7.3.32 12.3 Derivation of the Inflationary Mechanism from the Future Information Field

In the previous section, we redefined the Big Bang as the moment when the manifestation mapping

$$P_X : F \rightarrow X$$

begins after the Future Information Field exceeds the critical information density

$$\Phi_c.$$

However, the mere birth of the Universe is insufficient to explain the remarkably homogeneous and spatially flat Universe that is observed today. In standard cosmology, this problem is resolved by introducing the Inflaton field, which drives exponential expansion (inflation). However, neither the existence of the Inflaton nor the form of its potential

$$V(\phi)$$

has been established observationally or uniquely derived theoretically. In the present theory, no Inflaton is introduced. Instead, we show that the binding energy stored in the Future Information Field immediately after the onset of manifestation naturally generates rapid cosmic expansion.

Standard Inflation

In the conventional inflationary scenario, the scale factor evolves as

$$a(t) \propto e^{Ht},$$

where

$$H = \sqrt{\frac{8\pi G}{3} V(\phi)}.$$

Thus, the essential requirement for inflation is that the potential energy remains approximately constant.

Future Projection Energy

The Future Information Field retains an information-binding energy at the moment manifestation begins. We define this quantity as the

$$\boxed{E_P}$$

(Future Projection Energy). The corresponding energy density is

$$\boxed{\rho_P = \frac{E_P}{V}}.$$

Definition of Future Projection Energy

In Paper 92,

$$\epsilon_F$$

(Future Entropy) was introduced as the conserved quantity of information. In the present chapter, the information-binding energy at the onset of manifestation is defined as

$$\boxed{E_P = \lambda_P \Phi_F},$$

where

$$\lambda_P$$

is the coupling constant relating the Future Information Field to manifestation energy.

Effective Energy Density

Consequently, the effective energy density of the early Universe becomes

$$\rho_{\text{eff}} = \rho_r + \rho_m + \rho_P.$$

Immediately after cosmic birth,

$$\rho_P \gg \rho_r, \quad \rho_m,$$

so the early Universe is dominated by the Future Projection Energy.

Modified Friedmann Equation

Substituting this result into the modified Friedmann equation derived in Chapter XI yields

$$H^2 = \frac{8\pi G}{3} \rho_P.$$

Since

$$\rho_P = \text{const.},$$

to a good approximation,

$$H \simeq \text{const.}$$

follows naturally.

Exponential Expansion

If the Hubble parameter remains constant, the scale factor evolves as

$$a(t) = a_0 e^{H_P t},$$

where

$$H_P = \sqrt{\frac{8\pi G}{3} \rho_P}.$$

Therefore, exponential inflation is obtained solely from the Future Projection Energy without introducing an Inflaton field.

Termination of Inflation

As cosmic expansion proceeds, the Future Projection continuously converts information into three-dimensional manifestation. Consequently,

$$\Phi_F$$

gradually decreases. The end of inflation is therefore defined by

$$\Phi_F = \Phi_c.$$

That is, inflation terminates when the information density decreases to the critical manifestation threshold.

Reinterpretation of the Slow-Roll Condition

In conventional inflationary theory, the slow-roll condition is written as

$$\dot{\phi} \simeq 0.$$

In the present theory, the corresponding condition is that the Future Information Field is released only gradually:

$$\left| \frac{d\Phi_F}{dt} \right| \ll \Phi_F.$$

Accordingly, the slow-roll condition is reinterpreted not as slow motion of a scalar field but as the quasi-static evolution of the Future Projection process.

Integration with Papers 61, 73, and 92

Paper 61 proposed that the Universe is generated from the Meaning Generation Template. Paper 73 showed that the Future Information Field generates Child Universes. Paper 92 demonstrated that the Future Information Field governs information preservation around black holes. The present section further shows that the Future Projection Energy itself generates inflation. Therefore, Inflation, the Big Bang, Child Universes, and black holes are unified as different phases of the same Future Projection process.

Observational Predictions

The present theory predicts:

1. The Inflaton particle does not exist.
2. The expansion rate of the early Universe is determined by the Future Projection Energy.
3. The slow-roll parameter is reinterpreted as the decay rate of the Future Information Field.
4. The primordial gravitational-wave spectrum is determined by the time evolution of the Future Projection Energy rather than by the Inflaton potential.
5. Primordial density perturbations likewise originate from the initial fluctuations of the Future Information Field.

Falsifiability

The present theory is falsified if any of the following occurs:

1. The direct existence of the Inflaton is experimentally established.
2. The Future Projection Energy fails to reproduce the observed CMB spectrum.
3. The observed primordial gravitational-wave spectrum is inconsistent with predictions based on the Future Information Field.

Conclusion of This Section

In this section, the Future Projection Energy stored in the Future Information Field was introduced, allowing exponential cosmic expansion to be derived without assuming an Inflaton. Consequently, inflation is redefined not as the dynamics of an unknown scalar field but as the phase in which the Future Information Field rapidly manifests information into the three-dimensional Universe. In the next section, quantum fluctuations of the Future Projection will be used to derive primordial density perturbations, scalar and tensor perturbations, and the primordial gravitational-wave spectrum.

7.3.33 12.4 Quantum Fluctuations of the Future Projection and the Origin of Primordial Density Perturbations and Primordial Gravitational Waves

In the previous section, we showed that the Future Projection Energy realizes exponential expansion without introducing the Inflaton. However, the observed Universe is not perfectly homogeneous. Galaxies, galaxy clusters, large-scale structure, and the temperature anisotropies observed in the cosmic microwave background (CMB) are understood to originate from extremely small primordial density perturbations that were amplified during cosmic expansion. Therefore, if the Future Information Field governs the early Universe, the information field itself must become the origin of primordial density perturbations and primordial gravitational waves. In this section, scalar perturbations, tensor perturbations, and primordial gravitational waves are derived from quantum fluctuations of the Future Projection.

Primordial Perturbations in Standard Cosmology

In the standard inflationary scenario, quantum fluctuations of the Inflaton,

$$\delta\phi,$$

become the origin of primordial density perturbations. The curvature perturbation is generally written as

$$\mathcal{R} = -H \frac{\delta\phi}{\dot{\phi}},$$

with the corresponding power spectrum

$$P_{\mathcal{R}}(k) = \frac{H^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon}.$$

These expressions, however, explicitly assume the existence of the Inflaton.

Quantum Fluctuations of the Future Information Field

In the present theory, the fundamental fluctuating quantity is not the Inflaton but the Future Information Field,

$$\Phi_F.$$

Its fluctuation is defined as

$$\delta\Phi_F(x, t).$$

Since the Future Projection depends on the information density, the manifestation mapping itself fluctuates:

$$P_X = P_X^{(0)} + \delta P_X.$$

First-Order Perturbation of the Future Projection

The first-order variation of the manifestation mapping is

$$\delta P_X = \frac{\partial P_X}{\partial \Phi_F} \delta\Phi_F.$$

This fluctuation is converted into a local fluctuation of the energy density.

Primordial Density Perturbations

The Future Projection Energy satisfies

$$\rho_P = \lambda_P \Phi_F.$$

Therefore,

$$\delta\rho = \lambda_P \delta\Phi_F,$$

and the relative density perturbation becomes

$$\frac{\delta\rho}{\rho} = \frac{\delta\Phi_F}{\Phi_F}.$$

Hence, primordial density perturbations are identified directly with fluctuations in the information density of the Future Information Field.

Curvature Perturbations

We define the curvature perturbation generated by the Future Projection as

$$\mathcal{R}_F.$$

To first order,

$$\mathcal{R}_F = -H \frac{\delta\Phi_F}{\dot{\Phi}_F}.$$

Although formally similar to the standard expression, both numerator and denominator are written in terms of the Future Information Field rather than the Inflaton.

Scalar Perturbation Spectrum

From the two-point correlation function of the Future Information Field, the scalar power spectrum is obtained as

$$P_{\mathcal{R}}^F(k) = \frac{H^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon_F},$$

where

$$\epsilon_F = -\frac{\dot{H}}{H^2}$$

is defined as the Slow Projection Parameter of the Future Projection.

Tensor Perturbations

The Future Projection also induces perturbations of the spacetime metric. Accordingly, the tensor perturbation is written as

$$g_{ij} = a^2(t) (\delta_{ij} + h_{ij}).$$

The tensor fluctuation generated by the Future Projection is expressed as

$$h_{ij} = \Gamma_{ij} \delta\Phi_F,$$

where

$$\Gamma_{ij}$$

denotes the Future Projection Tensor.

Primordial Gravitational-Wave Spectrum

The tensor power spectrum becomes

$$P_T^F(k) = \frac{2H^2}{\pi^2 M_{\text{Pl}}^2}.$$

Including the contribution of the Future Projection yields

$$P_T^F = P_T^{\text{GR}} + \Delta P_T(\Phi_F).$$

Tensor-to-Scalar Ratio

Within the present theory, the tensor-to-scalar ratio is given by

$$r_F = 16\epsilon_F.$$

However,

$$\epsilon_F$$

is not the conventional slow-roll parameter of an Inflaton field. Instead, it represents the temporal evolution rate of the Future Projection.

Integration with Paper 92

Paper 92 introduced the Future Information Field as the information-preserving field surrounding black holes. The present section demonstrates that the same Future Information Field also generates primordial density perturbations, curvature perturbations, and primordial gravitational waves immediately after the birth of the Universe. Accordingly, black-hole information preservation and primordial cosmological perturbations are interpreted as different manifestations of the same information field at different cosmic epochs.

Observational Predictions

The present theory predicts:

1. Primordial density perturbations originate from fluctuations of the Future Information Field rather than quantum fluctuations of the Inflaton.
2. The tensor-to-scalar ratio

$$r$$

is determined by the temporal evolution rate of the Future Projection.

3. The primordial gravitational-wave spectrum contains higher-order corrections originating from the Future Projection.
4. Large-scale correlations reflecting the initial constraints of the Future Information Field may appear in the low multipoles of the CMB.
5. The black-hole gravitational-wave corrections derived in Chapter VII and the primordial gravitational-wave corrections are described by the same Future Projection Tensor.

Falsifiability

The present theory is falsified if any of the following conditions are satisfied:

1. Future Projection corrections are entirely absent from CMB observations.
2. The observed primordial gravitational-wave spectrum is inconsistent with the Future Information Field framework.
3. The Future Projection Parameter cannot be consistently determined from CMB observations, gravitational-wave observations, and black-hole observations simultaneously.

Conclusion of This Section

In this section, primordial density perturbations, curvature perturbations, tensor perturbations, and primordial gravitational waves were derived from quantum fluctuations of the Future Information Field. Consequently, without introducing an Inflaton field, both the primordial fluctuations observed in the cosmic microwave background and the primordial gravitational-wave spectrum are described in a unified manner as manifestations of the initial information fluctuations of the Future Projection. The next section reformulates the Horizon Problem, Flatness Problem, and Monopole Problem within the framework of the Future Projection Principle, thereby reconstructing the three major problems of the early Universe from a unified mathematical perspective.

7.3.34 12.5 Unified Reformulation of the Horizon Problem, Flatness Problem, and Monopole Problem

In the previous section, we showed that quantum fluctuations of the Future Projection naturally generate primordial density perturbations, primordial gravitational waves, and scalar and tensor perturbations. However, early-Universe cosmology contains three fundamental problems that originally motivated the introduction of inflation:

- Horizon Problem
- Flatness Problem
- Monopole Problem

In standard inflationary theory, these three problems are resolved simultaneously through exponential expansion. However, this explanation presupposes the existence of the Inflaton and therefore does not explain why such expansion occurs in the first place. Within the Future Convergence Theory, these three problems arise from a single principle: prior to the onset of the Future Projection, the Future Information Field had already constrained the Universe as a whole.

The Horizon Problem

The observed cosmic microwave background exhibits remarkable isotropy at the level of approximately

$$10^{-5},$$

even between regions that should never have been in causal contact. In standard cosmology, the particle horizon is defined as

$$d_H(t) = a(t) \int_0^t \frac{c dt'}{a(t')}.$$

Regions separated by distances larger than this horizon cannot exchange information. Nevertheless, they display nearly identical temperatures. This constitutes the Horizon Problem.

Reinterpretation via the Future Information Field

The Future Information Field exists prior to the emergence of the three-dimensional Universe. Accordingly, before cosmic manifestation, the entire information field forms a single connected space:

$$\boxed{F \text{ is globally connected}}.$$

In this state, the concept of three-dimensional spatial distance does not yet exist. Consequently, no causal horizon exists within the Future Information Field.

Simultaneous Manifestation through the Future Projection

Once the manifestation mapping

$$P_X : F \rightarrow X$$

begins, the entire Future Information Field is projected into the three-dimensional Universe under a common constraint. Therefore, the initial Universe is not synchronized locally. Instead, its homogeneity results from the fact that the entire Future Information Field shared identical boundary conditions prior to manifestation. Accordingly, the Horizon Problem is explained by the intrinsic unity of the information field preceding the Future Projection.

The Flatness Problem

Observations indicate that the spatial curvature satisfies

$$|\Omega_k| \ll 1.$$

In standard cosmology, however, even an extremely small deviation from

$$\Omega = 1$$

in the early Universe would evolve into a Universe with enormous spatial curvature today. Consequently, the initial conditions must be extraordinarily fine-tuned. This is known as the Flatness Problem.

Curvature Constraint by the Future Projection

The Future Projection is itself the manifestation mapping. Accordingly, the manifested space-time metric can possess only the curvature permitted by the Future Projection. Introducing the Future Projection Tensor

$$\Pi_{\mu\nu},$$

the initial curvature satisfies

$$\boxed{R_{\mu\nu} = \Pi_{\mu\nu} R_c},$$

where

$$R_c$$

denotes the critical curvature permitted by the Future Boundary. Consequently, only an almost spatially flat geometry can emerge at the beginning of the Universe.

The Monopole Problem

Grand Unified Theories predict that enormous numbers of magnetic monopoles should have been produced in the early Universe. However, no magnetic monopoles have been observed to date. Standard inflation resolves this discrepancy by exponentially diluting their abundance.

Existence Selection through the Future Projection

Within the Future Convergence Theory, particle generation itself is regarded as a process of manifestation selection governed by the Future Projection. Consequently, not every theoretically allowed degree of freedom is necessarily manifested within the three-dimensional Universe. Let the manifestation probability of particle species i be

$$P_i = \Pi_F(i).$$

If, for magnetic monopoles,

$$\Pi_F(\text{Monopole}) \ll 1,$$

then magnetic monopoles may remain theoretically permissible while possessing an extremely small probability of manifestation. Accordingly, the Monopole Problem is reformulated as a problem of manifestation selection governed by the Future Projection.

Unified Description of the Three Problems

The three classical problems of the early Universe are therefore not independent. Instead, they all follow from the fact that, prior to the onset of the Future Projection, the Future Information Field already constrained the Universe globally. This relationship may be summarized as

$$\text{Future Information Field} \Rightarrow \begin{cases} \text{Horizon Problem} \\ \text{Flatness Problem} \\ \text{Monopole Problem} \end{cases}$$

Integration with Paper 92

Paper 92 introduced the Future Boundary as the information-preserving boundary surrounding black holes. The present section extends the same Future Boundary to the epoch preceding the birth of the Universe, where it provides the global boundary conditions for the initial Universe. Accordingly, the black-hole boundary, the primordial cosmological boundary, and the Future Projection are unified within the same boundary principle.

Observational Predictions

The present theory predicts:

1. Large-angle correlations in the CMB should contain long-range correlations produced by the global constraints of the Future Projection that differ from those predicted by standard inflation.
2. Since primordial curvature perturbations reflect the eigenvalue distribution of the Future Projection Tensor, extremely high-precision CMB and BAO observations may reveal small residual anisotropies.
3. Even if magnetic monopoles exist, their production rate is expected to be strongly suppressed by manifestation selection through the Future Projection.
4. The constraint parameters governing the Future Projection should be mutually consistent across observations of the CMB, primordial gravitational waves, black-hole ringdown signals, and estimates of the Future Entropy.

Falsifiability

The present theory is falsified if any of the following conditions are satisfied:

1. The Horizon Problem, Flatness Problem, and Monopole Problem cannot be explained simultaneously after introducing the Future Projection.
2. The CMB exhibits no long-range correlations attributable to the Future Projection.
3. A common Future Projection Tensor cannot be consistently inferred from observations of black holes, gravitational waves, and cosmology.

Conclusion of This Section

In this section, the Horizon Problem, Flatness Problem, and Monopole Problem were reformulated on the basis of the Future Projection Principle. As a result, these three problems are no longer regarded as independent issues of the early Universe. Instead, they all follow from the single principle that the Future Information Field existed prior to the three-dimensional Universe and imposed common boundary conditions on the entire cosmos. The next section derives the CMB temperature spectrum, polarization spectrum, non-Gaussianity, the scalar spectral index

$$n_s,$$

and its running from these primordial boundary conditions, thereby establishing the quantitative correspondence between the Future Information Field theory and cosmological observations.

7.3.35 12.6 Derivation of the Cosmic Microwave Background, Spectral Index, and Non-Gaussianity

In the previous section, the Future Projection Principle provided a unified explanation of the Horizon Problem, Flatness Problem, and Monopole Problem through the ontology of the Future Information Field. However, an ontological framework alone is insufficient for a complete theory of the early Universe. Modern cosmology is tested primarily through extremely precise observations of the Cosmic Microwave Background (CMB). Accordingly, any viable theory must quantitatively account for at least the following observables:

- Scalar perturbation spectrum
- Scalar spectral index n_s
- Running of the spectral index
- Tensor-to-scalar ratio r
- Non-Gaussianity
- E-mode and B-mode polarization

In this section, these observables are derived sequentially from the information fluctuations of the Future Projection.

Origin of CMB Anisotropies

In standard cosmology, the temperature anisotropies of the CMB originate from quantum fluctuations of the Inflaton that are stretched to cosmological scales during inflation. In the Future Convergence Theory, however, temperature anisotropies arise because fluctuations of the information density,

$$\delta\Phi_F,$$

are converted into density perturbations through the Future Projection. Accordingly,

$$\boxed{\frac{\delta T}{T} \propto \frac{\delta \Phi_F}{\Phi_F}}$$

is adopted as the defining relation.

Curvature Perturbations Generated by the Future Projection

As derived in the previous section, the curvature perturbation is

$$\mathcal{R}_F = -H \frac{\delta \Phi_F}{\dot{\Phi}_F}.$$

Therefore, the observed CMB anisotropies directly reflect the primordial information fluctuations of the Future Projection.

Power Spectrum

Define the two-point correlation function of the Future Information Field by

$$\langle \delta \Phi_F(\mathbf{k}) \delta \Phi_F(\mathbf{k}') \rangle = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}') P_F(k).$$

The corresponding curvature power spectrum becomes

$$\boxed{P_{\mathcal{R}}(k) = \mathcal{A}_F P_F(k)},$$

where

$$\mathcal{A}_F$$

is the conversion coefficient associated with the Future Projection.

Scalar Spectral Index

The scalar spectral index is conventionally defined through

$$P_{\mathcal{R}}(k) \propto k^{n_s-1}.$$

Within the Future Information Field framework,

$$\boxed{n_s - 1 = \frac{d \ln P_F}{d \ln k}}$$

is obtained. Hence, the spectral index is determined not by the Inflaton potential but by the spectral gradient of the Future Information Field.

Running of the Spectral Index

The running of the scalar spectral index is defined by

$$\boxed{\alpha_s = \frac{dn_s}{d \ln k}}.$$

Within the Future Projection framework,

$$\boxed{\alpha_s = \frac{d^2 \ln P_F}{(d \ln k)^2}}$$

is obtained. Accordingly, higher-order structures of the Future Information Field are directly reflected in the running of the primordial power spectrum.

Non-Gaussianity

In the standard inflationary scenario, primordial non-Gaussianity arises from self-interactions of the Inflaton field. In the present theory, non-Gaussianity emerges naturally because the Future Projection is itself a nonlinear mapping. Expanding the manifestation operator as

$$P_X = P_0 + P_1\Phi_F + P_2\Phi_F^2 + \cdots,$$

the quadratic contribution gives

$$f_{\text{NL}} \propto P_2.$$

Thus, primordial non-Gaussianity becomes a direct probe of the nonlinear structure of the Future Projection.

E-mode and B-mode Polarization

The Future Projection simultaneously generates tensor perturbations. Accordingly, the polarization spectra become

$$C_\ell^{EE} = C_\ell^{EE,\text{GR}} + \Delta C_\ell^{EE}(\Phi_F)$$

and

$$C_\ell^{BB} = C_\ell^{BB,\text{GR}} + \Delta C_\ell^{BB}(\Phi_F),$$

where the correction terms depend on the temporal evolution of the Future Projection Tensor.

Low-Multipole Anomalies

Current observations of the Cosmic Microwave Background indicate mild anomalies at low multipoles relative to the standard ΛCDM model. Within the present theory, the Future Information Field constrains the Universe globally. Consequently, weak long-range correlations may persist even on super-horizon scales. The angular power spectrum therefore becomes

$$C_\ell = C_\ell^{\Lambda\text{CDM}} + \Delta C_\ell^F.$$

Integration with Paper 92

Paper 92 introduced the Future Information Field as the field governing information preservation around black holes. The present section demonstrates that the same information field determines the primordial fluctuations of the Cosmic Microwave Background during the birth of the Universe. Therefore, black-hole information preservation and primordial CMB fluctuations are interpreted as manifestations of the Future Projection occurring during different cosmic epochs.

Observational Predictions

The present theory predicts:

1. The scalar spectral index

$$n_s$$

is determined by the spectral gradient of the Future Information Field.

2. The running of the spectral index

$$\alpha_s$$

reflects higher-order information structures within the Future Projection.

3. Primordial non-Gaussianity arises naturally from the nonlinearity of the Future Projection mapping.
4. Small corrections proportional to the Future Projection Tensor appear in both E-mode and B-mode polarization spectra.
5. Large-scale correlations different from those predicted by the standard Λ CDM model should appear at low multipoles owing to the global constraints imposed by the Future Information Field.
6. Corrections to black-hole ringdown signals and CMB tensor perturbations are described by the same Future Projection Tensor.

Falsifiability

The present theory is falsified if any of the following conditions are satisfied:

1. Future Projection corrections are completely absent from high-precision CMB observations.
2. A common Future Projection Tensor cannot be consistently inferred from observations of black holes, gravitational waves, and the CMB.
3. The introduction of the Future Information Field does not improve predictive capability beyond the standard Λ CDM plus Inflation framework.

Conclusion of This Section

In this section, the temperature anisotropies of the Cosmic Microwave Background, the scalar spectral index, its running, primordial non-Gaussianity, and the E-mode and B-mode polarization spectra were derived from the information fluctuations of the Future Projection. Consequently, the Future Information Field is identified not only as the origin of primordial density perturbations but also as the fundamental information field governing the entire statistical structure of the observed Cosmic Microwave Background. The next section connects these primordial conditions with the Child Universe Theory proposed in Paper 73, demonstrating mathematically that both the Big Bang and Child Universe formation originate from the same Future Projection condition.

7.3.36 12.7 Unified Conditions for Cosmic Birth Based on the Child Universe Theory and the Future Information Field

In the previous section, the Future Information Field was shown to generate the Cosmic Microwave Background, the scalar spectral index, primordial non-Gaussianity, and primordial gravitational waves through its information fluctuations. These results established that the Future Projection governs the observable quantities of the early Universe. Nevertheless, one essential issue remains if the Future Convergence Theory is to constitute a closed ontological framework. That issue is the unification of the Child Universe Theory proposed in Paper 73 with the origin of the present Universe within a single mathematical structure. Paper 73 proposed that the birth of a Child Universe begins when the information accumulated within the Future Information Field becomes sufficiently large to manifest again. In contrast, the present chapter defines the Big Bang as the phase transition at which the Future Projection first begins. If both descriptions belong to the same theoretical framework, then the birth conditions of Child Universes and the onset conditions of the Big Bang must follow from the same boundary conditions.

Fundamental Structure Introduced in Paper 73

Paper 73 proposed that Child Universe formation begins once the information stored within the Future Information Field exceeds a critical value. Defining the total stored information by

$$I_F = \int_F \Phi_F dV_F,$$

the condition for Child Universe formation is

$$\boxed{I_F > I_c},$$

where

$$I_c$$

denotes the critical information content.

Comparison with the Big Bang Condition

In the present chapter, the onset of the Big Bang is defined by

$$\boxed{\Phi_F \geq \Phi_c}.$$

Here,

$$\Phi_c$$

represents the local critical information density. Thus, Paper 73 employs an integrated information criterion, whereas the present chapter introduces a local density criterion.

Unified Manifestation Condition

A new Universe can emerge only when the Future Information Field possesses both a sufficiently high local information density and a sufficiently large total information content. Accordingly, the condition for cosmic manifestation requires the simultaneous satisfaction of

$$\boxed{\Phi_F \geq \Phi_c}$$

and

$$\boxed{I_F \geq I_c}.$$

We refer to this combined requirement as the

$$\boxed{\text{Future Projection Critical Condition.}}$$

Future Projection Critical Surface

Within the Future Information Field, define the region capable of manifestation as

$$\Omega_c.$$

Specifically,

$$\boxed{\Omega_c = \{x \in F \mid \Phi_F(x) \geq \Phi_c\}}.$$

Its corresponding volume is

$$V_c = \int_{\Omega_c} dV.$$

The condition for cosmic generation then becomes

$$\boxed{\int_{\Omega_c} \Phi_F dV \geq I_c}.$$

Formation of a Child Universe

When the Future Projection Critical Condition is satisfied, a new manifestation mapping,

$$P_{X'} : F \rightarrow X',$$

is initiated. Thus,

$$\boxed{F \xrightarrow{P_{X'}} X'}$$

describes the emergence of a new Child Universe, where

$$X'$$

denotes the newly generated Universe.

Complete Unification with the Big Bang

If both the present Universe and Child Universes are generated through the Future Projection, then they are mathematically indistinguishable. Therefore,

$$\boxed{\text{Big Bang} \equiv \text{Child Universe Projection}}$$

holds. From the perspective of a higher-order Future Information Field, our present Universe is therefore interpreted as one instance of a Child Universe.

Generations of Universes

Repeated applications of the Future Projection generate a sequence of Universes,

$$\boxed{X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots}.$$

These Universes are not independent. Rather, they all share the same underlying Future Information Field.

Information Conservation

Paper 92 defined black holes as operators that return information to the Future Information Field. Consequently, information conservation across successive Universes is expressed as

$$\boxed{I_n + \Delta I_B = I_{n+1}},$$

where

$$I_n$$

is the information content of the n -th generation Universe, and

$$\Delta I_B$$

represents the amount of information recovered through black holes.

Inheritance of Future Entropy

Future Entropy is likewise inherited across successive cosmic generations. Accordingly,

$$\boxed{S_F^{(n+1)} = S_F^{(n)} + \Delta S_B}$$

holds. This relation implies that each successive Universe inherits the information structure accumulated by the preceding generation.

Ontological Closed Loop

The overall structure of the Future Convergence Theory therefore reduces to the closed cycle

$$\boxed{F \rightarrow X_n \rightarrow B \rightarrow F \rightarrow X_{n+1}}.$$

Here,

B

denotes the information-recovery boundary encompassing black holes, information preservation, and heat death.

Integration with Papers 61, 73, and 92

Paper 61 proposed that the Universe emerges from the Meaning Generation Template. Paper 73 introduced Child Universe formation through the Future Information Field. Paper 92 demonstrated that black holes preserve information within the Future Information Field. The present section unifies these three results by showing that cosmic birth, cosmic termination, black holes, and Child Universe formation all originate from the same Future Projection Critical Condition.

Observational Predictions

The present theory predicts:

1. The amount of information preserved by black holes should exhibit statistical correlations with the initial conditions of future Universes.
2. Future Entropy is conserved and inherited across successive cosmic generations.
3. Child Universes are generated only when the Future Projection Critical Condition is satisfied.
4. The mass distribution of black holes may correlate with the growth rate of the Future Information Reservoir.
5. The initial conditions of the present Universe reflect the information structure inherited from the previous cosmic generation through the Future Information Field.

Falsifiability

The present theory is falsified if any of the following conditions are satisfied:

1. The Future Projection Critical Condition is incompatible with the Child Universe Theory proposed in Paper 73.
2. Black-hole information preservation is shown to be entirely unrelated to cosmic generation.
3. Future Entropy is demonstrated, both theoretically and observationally, not to be conserved across successive cosmic generations.

Conclusion of This Section

In this section, the Child Universe Theory proposed in Paper 73 was reformulated within the framework of the Future Projection Principle. Consequently, the Big Bang and Child Universe formation were unified as the same phase transition governed by the Future Projection Critical Condition. Black holes, heat death, and cosmic birth are therefore described as successive stages within a single closed cycle of information governed by the Future Information Field.

The next section summarizes the entire chapter, demonstrating that the early Universe, black holes, dark matter, dark energy, gravitational waves, and the Future Information Field all follow from a single governing principle while organizing the quantitative observational predictions that distinguish the Future Convergence Theory from existing cosmological models.

7.3.37 12.8 Summary and Achievements of the Early-Universe Theory Based on the Future Information Field

In this chapter, the Future Information Field has been extended to the earliest stages of cosmic evolution, allowing the Big Bang, inflation, primordial density perturbations, primordial gravitational waves, the Cosmic Microwave Background, and Child Universe formation to be reconstructed within a single theoretical framework. Paper 92 introduced the Future Information Field as the background field governing information preservation around black holes. The present chapter extends this field to the epoch preceding the birth of the Universe, thereby formulating the early Universe itself as the initiation of the Future Projection process.

The Theoretical Framework Established in This Chapter

This chapter began by reviewing the mathematical structure of the standard theory of the early Universe. The Future Information Field was then introduced as an information field existing prior to the birth of the Universe, allowing the Big Bang to be redefined as the phase transition at which the Future Projection begins. Next, by introducing the Future Projection Energy, exponential cosmic expansion was derived without postulating an Inflaton field. Subsequently, primordial density perturbations, primordial gravitational waves, the Cosmic Microwave Background, the scalar spectral index, primordial non-Gaussianity, and polarization spectra were derived from quantum fluctuations of the Future Information Field. Finally, the Child Universe Theory proposed in Paper 73 was integrated into the present framework, demonstrating that both the birth of the Universe and the generation of Child Universes are governed by the same Future Projection Critical Condition.

Unified Theory of the Early Universe

The results of this chapter establish the Future Information Field as the single background field governing all fundamental processes of the early Universe:

$$\Phi_F \implies \left\{ \begin{array}{l} \text{Big Bang} \\ \text{Inflation} \\ \text{Primordial Density Perturbation} \\ \text{Primordial Gravitational Wave} \\ \text{CMB} \\ \text{Child Universe} \end{array} \right.$$

Accordingly, these phenomena are not independent physical processes but different manifestations of the same Future Projection.

Integration of Chapters VII through XII

The developments from Chapter VII through the present chapter collectively establish the Future Information Field as the common background field throughout the entire history of the Universe:

$$\Phi_F \Rightarrow \left\{ \begin{array}{l} \text{Black Hole} \\ \text{Ringdown} \\ \text{Gravitational Wave} \\ \text{Dark Matter} \\ \text{Dark Energy} \\ \text{Vacuum} \\ \text{Big Bang} \\ \text{Inflation} \\ \text{CMB} \\ \text{Child Universe} \end{array} \right.$$

This implies that the Future Information Field provides a unified background field describing both local gravitational phenomena and cosmological phenomena through a common underlying principle.

Unified Interpretation of the Future Projection Principle

By the end of this chapter, the Future Projection has been elevated beyond a purely mathematical mapping and redefined as the operator that determines physical existence itself. Its role is summarized by

$$P_X : F \rightarrow X,$$

which projects the Future Information Field into the observable three-dimensional Universe. Likewise,

$$P_{X'} : F \rightarrow X'$$

represents the operator that generates a subsequent Universe. Accordingly, the birth of the Universe, black holes, and Child Universe formation are understood as different boundary conditions of the same Future Projection process.

Unified Closed Cycle

Combining the results of Papers 61, 73, and 92 with those obtained in this chapter, cosmic evolution within the Future Convergence Theory reduces to the closed cycle

$$F \rightarrow X_n \rightarrow B \rightarrow F \rightarrow X_{n+1}.$$

Here,

$$F$$

denotes the Future Information Field,

$$X_n$$

the n -th generation Universe,

$$B$$

the information-recovery boundary including black holes and heat death, and

$$X_{n+1}$$

the subsequent Universe.

Additional Falsifiable Predictions Established in This Chapter

The present chapter extends the Future Convergence Theory by providing the following falsifiable predictions concerning the early Universe:

1. The Inflaton is not an independent fundamental field but an effective description of the Future Projection Energy.
2. Primordial density perturbations, primordial gravitational waves, and black-hole ringdown signals are all described by the same Future Projection Tensor.
3. The Future Projection Parameter must be inferred consistently from observations of the Cosmic Microwave Background, black holes, gravitational waves, and Dark Energy.
4. Higher-order corrections originating from the Future Projection should become observable in future measurements of CMB polarization, primordial gravitational waves, and gravitational-wave astronomy.

Each of these predictions is, in principle, testable by future observations.

Conclusion of This Chapter

The Future Information Field is therefore not merely a local field governing information preservation around black holes. Rather, it is a cosmological background field that exists prior to the birth of the Universe and governs the initial conditions of the Universe, cosmic expansion, primordial fluctuations, the Cosmic Microwave Background, and even the generation of Child Universes. Within the Future Convergence Theory, the Big Bang is interpreted as the phase transition marking the first activation of the Future Projection, inflation is understood as the rapid manifestation of the three-dimensional Universe driven by the Future Projection Energy, and Child Universes correspond to subsequent phases in which the Future Projection begins anew.

Connection to the Next Chapter

From Chapter VII through Chapter XII, the Future Information Field has been used to provide a unified description of black holes, gravitational waves, Dark Matter, Dark Energy, the early Universe, the Big Bang, inflation, the Cosmic Microwave Background, and Child Universes. The final remaining task is to derive all of these phenomena from a single action principle. Accordingly, the next chapter reconstructs the fundamental action

$$S_{\text{FCT}},$$

from which general relativity, quantum fields, black holes, cosmology, and the Future Information Field are all derived through a single variational principle. This establishes the unified action principle of the Future Convergence Theory.

Part XIII

Unified Action Principle of Future Convergence Theory

— A Variational Theory Unifying the Future Information Field, Gravity, Quantum Theory, Black Holes, and Cosmology —

7.3.38 13.1 Introduction: The Necessity of a Unified Action Principle in Future Convergence Theory

From Chapter VII through Chapter XII, this paper has constructed, on the basis of the Future Information Field, a single theoretical framework for

- black-hole information preservation,
- ringdown quasinormal modes,
- gravitational-wave propagation,
- Dark Matter,
- Dark Energy,
- cosmic accelerated expansion,
- the Big Bang,
- inflation,
- the cosmic microwave background,
- and Child Universes.

In these discussions, the Future Information Field was introduced for each phenomenon and applied to each corresponding physical system. However, at the present stage, these results have not yet become **a theory derived from one and the same action**. In order to construct a unified physical theory, all field equations, conservation laws, black-hole corrections, cosmological equations, and gravitational-wave corrections must be derived from a single action functional. In general relativity, the Einstein equation is derived by varying the Einstein–Hilbert action

$$S_{\text{EH}} = \frac{1}{16\pi G} \int R\sqrt{-g} d^4x + S_{\text{matter}}.$$

In quantum field theory as well, all equations of motion are derived from the action principle

$$\delta S = 0.$$

Therefore, if Future Convergence Theory is a theory unifying gravity, quantum theory, and cosmology, there must exist a unique unified action containing the Future Information Field.

Position of Future Convergence Theory

In the present theory, the Future Information Field is not a correction field added to existing theories. It is the highest-order background field that simultaneously constrains spacetime, quantum states, information preservation, and cosmic evolution. Therefore, the Future Information Field is not introduced merely as one additional term in the action. Rather, it is introduced as a fundamental degree of freedom constituting the action as a whole.

Requirements for the Unified Action

The action constructed in this chapter must satisfy at least the following conditions simultaneously:

1. In the general-relativity limit, it must coincide with the Einstein equation.
2. In the limit where the Future Information Field is absent, it must reduce to the Standard Model plus general relativity.
3. Black-hole corrections, ringdown corrections, Dark Matter, Dark Energy, and early-Universe corrections must all be derivable from the same action.
4. It must naturally contain Future Entropy as a conserved quantity.
5. The Future Projection Equation must be derived from the variational principle.

Fundamental Degrees of Freedom Introduced in This Chapter

In this chapter, the fundamental degrees of freedom constituting Future Convergence Theory are taken to be the following four quantities:

$$\boxed{(g_{\mu\nu}, \Phi_F, \Sigma_{\mu\nu}, P_X)}.$$

Here,

$$g_{\mu\nu}$$

is the spacetime metric,

$$\Phi_F$$

is the Future Information Field,

$$\Sigma_{\mu\nu}$$

is the Future Information Tensor, and

$$P_X$$

is the Future Projection Operator. In conventional theories, only the metric is treated as the gravitational degree of freedom. In the present theory, the Future Projection itself is elevated to a fundamental degree of freedom subject to variation.

Purpose of This Chapter

The purpose of this chapter is not to simply add the Future Information Field to general relativity. The purpose is to construct a unified action

$$S_{\text{FCT}}$$

including the Future Projection, and to rederive all equations obtained from Chapter VII through Chapter XII as consequences of a single principle.

Structure of This Chapter

In this chapter, we first define the fundamental degrees of freedom, symmetries, and conserved quantities required to construct the unified action. Next, we build the unified Lagrangian density containing the Future Information Field. We then derive, from the variational principle, the gravitational equation, the Future Projection Equation, the Future Entropy conservation law, the black-hole correction formula, and the cosmological equations. Finally, we show the limiting consistency with general relativity, Λ CDM, quantum field theory, and the theory introduced in Paper 92, thereby demonstrating that the whole Future Convergence Theory is closed as a theory based on a single action principle.

7.3.39 13.2 Unified Lagrangian Density of Future Convergence Theory

In the previous section, we explained the necessity of reconstructing Future Convergence Theory not as a mere phenomenological framework, but as a theory derived from a unified action principle. In this section, we integrate the Future Information Field, Future Entropy, Future Projection, and Future Information Tensor introduced from Chapter VII through Chapter XII into a single Lagrangian density. The action introduced in this section becomes the unique starting point for deriving the gravitational field equation, the Future Projection Equation, the Future Entropy conservation law, black-hole corrections, cosmological equations, gravitational-wave corrections, Dark Matter, Dark Energy, and the early-Universe equations.

Basic Requirements for the Unified Action

The unified action of Future Convergence Theory,

$$S_{\text{FCT}},$$

must satisfy the following conditions:

1. It must reduce to the Einstein–Hilbert action in the general-relativity limit.
2. It must reproduce the Standard Model and general relativity in the limit where the Future Information Field vanishes.
3. When the Future Projection exists, it must derive all correction formulas obtained from Chapter VII through Chapter XII.
4. Future Entropy must naturally appear as a conserved quantity.
5. Black-hole information preservation, Dark Matter, Dark Energy, and early-Universe initial conditions must all be derived from the same action.

Fundamental Degrees of Freedom

The independent variables of Future Convergence Theory are defined as

$$q = (g_{\mu\nu}, \Phi_F, \Sigma_{\mu\nu}, P_X).$$

Here,

$$g_{\mu\nu}$$

is the spacetime metric,

$$\Phi_F$$

is the Future Information Field,

$$\Sigma_{\mu\nu}$$

is the Future Information Tensor, and

$$P_X$$

is the Future Projection Operator. In this theory, the Projection Operator itself is also a fundamental degree of freedom subject to variation.

Unified Action

We define the action of Future Convergence Theory by

$$S_{\text{FCT}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{FCT}}.$$

Here, the unified Lagrangian density

$$\mathcal{L}_{\text{FCT}}$$

is composed of the following six terms:

$$\mathcal{L}_{\text{FCT}} = \mathcal{L}_G + \mathcal{L}_F + \mathcal{L}_\Sigma + \mathcal{L}_P + \mathcal{L}_I + \mathcal{L}_M.$$

Gravitational Term

The gravitational term is based on the Einstein–Hilbert action:

$$\mathcal{L}_G = \frac{1}{16\pi G} R.$$

Therefore, in the limit where the Future Information Field is absent, the theory coincides with general relativity.

Future Information Field Term

The free Lagrangian of the Future Information Field is defined as

$$\mathcal{L}_F = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \Phi_F \nabla_\nu \Phi_F - V_F(\Phi_F).$$

Here,

$$V_F$$

is the Future Information Potential. This potential later determines the early Universe, Dark Energy, and Future Projection Energy in a unified manner.

Future Information Tensor Term

In Paper 92, the Future Information Tensor

$$\Sigma_{\mu\nu}$$

was introduced as the tensor responsible for information constraint. In the present theory, its self-action is defined by

$$\mathcal{L}_\Sigma = -\frac{1}{4} \Sigma_{\mu\nu} \Sigma^{\mu\nu}.$$

This represents the free energy of the information-constraint field itself.

Future Projection Term

The Future Projection is the core of the present theory. Therefore, an independent Lagrangian is assigned to the Projection Operator itself:

$$\mathcal{L}_P = -\frac{\kappa}{2} P_X^2.$$

Here,

$$\kappa$$

is called the Projection Rigidity Parameter. This term acts as a constraint term preventing the Projection from diverging.

Interaction Term

The Future Information Field, the Projection, and the Information Tensor are mutually coupled. Their interaction is defined by

$$\mathcal{L}_I = \lambda_1 \Phi_F P_X + \lambda_2 \Sigma_{\mu\nu} g^{\mu\nu} + \lambda_3 \Phi_F \Sigma_{\mu\nu} g^{\mu\nu}.$$

Here,

$$\lambda_i$$

are Future Coupling Constants. These constants later determine Dark Matter, Dark Energy, ringdown corrections, and early-Universe initial conditions.

Ordinary Matter Term

Ordinary matter is described by the Standard Model Lagrangian:

$$\mathcal{L}_M = \mathcal{L}_{\text{SM}}.$$

Therefore, the present theory does not replace the Standard Model. Rather, it adds the Future Information Field as a higher-order structure outside it.

Unified Lagrangian

Combining the above terms, the unified Lagrangian density of Future Convergence Theory becomes

$$\begin{aligned} \mathcal{L}_{\text{FCT}} = & \frac{R}{16\pi G} \\ & - \frac{1}{2}(\nabla\Phi_F)^2 - V_F(\Phi_F) \\ & - \frac{1}{4}\Sigma_{\mu\nu}\Sigma^{\mu\nu} \\ & - \frac{\kappa}{2}P_X^2 \\ & + \lambda_1\Phi_F P_X \\ & + \lambda_2\Sigma_{\mu\nu}g^{\mu\nu} \\ & + \lambda_3\Phi_F\Sigma_{\mu\nu}g^{\mu\nu} \\ & + \mathcal{L}_{\text{SM}} \end{aligned}.$$

Correspondence with Chapters VII through XII

Each term in this unified Lagrangian corresponds one-to-one with the theory introduced so far:

Lagrangian Term	Corresponding Physics
$\mathcal{L}_G$	General Relativity and Gravitational Waves
$\mathcal{L}_F$	Future Information Field
$\mathcal{L}_\Sigma$	Future Entropy and Information Constraint
$\mathcal{L}_P$	Future Projection
$\mathcal{L}_I$	Dark Matter, Dark Energy, and Black Hole Corrections
$\mathcal{L}_M$	Standard Model

Therefore, all correction formulas introduced separately from Chapter VII through Chapter XII become candidates derivable from this single Lagrangian.

Conclusion of This Section

In this section, we constructed the unified Lagrangian density governing Future Convergence Theory. This Lagrangian integrates gravity, the Future Information Field, the Future Projection, the Future Information Tensor, and ordinary matter into a single action. In the next section, we vary this action with respect to each degree of freedom and rigorously derive the fundamental field equations of Future Convergence Theory.

7.3.40 13.3 Fundamental Field Equations of Future Convergence Theory Derived from the Unified Action

In the previous section, we constructed the unified Lagrangian density

$$\mathcal{L}_{\text{FCT}}$$

governing Future Convergence Theory. In theoretical physics, however, the Lagrangian itself is only the starting point. The essence of a physical theory lies in the field equations obtained by varying the action. In this section, we vary the action

$$S_{\text{FCT}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{FCT}}$$

independently with respect to each fundamental degree of freedom,

$$(g_{\mu\nu}, \Phi_F, \Sigma_{\mu\nu}, P_X),$$

and derive the fundamental field equations of Future Convergence Theory. All subsequent developments of the theory originate from the four fundamental equations obtained in this section.

Variational Principle

The variational principle of Future Convergence Theory is given, as in ordinary field theory, by

$$\boxed{\delta S_{\text{FCT}} = 0}.$$

Unlike conventional theories, however, the metric is not the only variable to be varied. Rather,

$$\boxed{\delta S = \delta_g S + \delta_\Phi S + \delta_\Sigma S + \delta_P S = 0}$$

holds. These four independent variations give rise to four different fundamental field equations.

Variation with Respect to the Metric

We first vary the metric

$$g_{\mu\nu}.$$

The Einstein–Hilbert term produces the usual Einstein tensor

$$G_{\mu\nu}.$$

The contributions from the Future Information Field, the Future Information Tensor, and the Projection appear as effective energy–momentum tensors. Consequently, the gravitational field equation becomes

$$\boxed{G_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{SM}} + T_{\mu\nu}^F + T_{\mu\nu}^\Sigma + T_{\mu\nu}^P)}.$$

Here,

$$T_{\mu\nu}^F = -\frac{2}{\sqrt{-g}} \frac{\delta S_F}{\delta g^{\mu\nu}}$$

is the energy–momentum tensor associated with the Future Information Field.

Variation with Respect to the Future Information Field

Next, we vary the Future Information Field

$$\Phi_F.$$

Applying the Euler–Lagrange equation yields

$$\square \Phi_F - \frac{dV_F}{d\Phi_F} = J_F.$$

The source term

$$J_F$$

originates from couplings with the Projection, the Information Tensor, and ordinary matter. Explicitly,

$$J_F = \lambda_1 P_X + \lambda_3 g^{\mu\nu} \Sigma_{\mu\nu}.$$

Variation with Respect to the Future Information Tensor

We next vary the Information Tensor

$$\Sigma_{\mu\nu}.$$

The resulting equation is

$$\Sigma_{\mu\nu} = 2\lambda_2 g_{\mu\nu} + 2\lambda_3 \Phi_F g_{\mu\nu}.$$

Therefore, the Future Information Tensor is not an independent propagating field. Instead, it becomes a constraint tensor determined by the Future Information Field and the spacetime metric.

Variation with Respect to the Future Projection

Finally, we vary the Projection Operator

$$P_X.$$

This yields

$$\kappa P_X = \lambda_1 \Phi_F.$$

Accordingly,

$$P_X = \frac{\lambda_1}{\kappa} \Phi_F.$$

This equation derives, for the first time from the action principle, the statement repeatedly assumed throughout this paper: “The Projection is determined by the Future Information Field.”

Future Projection Equation

The above relation is called the Future Projection Equation:

$$P_X = \alpha_F \Phi_F,$$

where

$$\alpha_F = \frac{\lambda_1}{\kappa}.$$

This equation provides a unified determination of the Projection used throughout Chapters VII through XII.

Future Entropy Conservation Law

We define the Future Entropy by

$$S_F = \int \Phi_F dV.$$

Integrating the Future Information Field equation yields the conservation relation

$$\boxed{\frac{dS_F}{dt} = \int J_F dV}.$$

In the limit where no information source exists,

$$J_F = 0,$$

we obtain

$$\boxed{\frac{dS_F}{dt} = 0},$$

which is the Future Entropy conservation law.

Correspondence with Chapters VII through XII

The four fundamental equations obtained in this section constitute the starting point for all equations introduced previously.

Fundamental Field Equation	Applications
Gravitational Field Equation	Gravitational Waves, Black Holes, Cosmology
Future Field Equation	Dark Energy, Early Universe
Projection Equation	Ringdown, Child Universes
Entropy Conservation	Information Preservation, Black Holes

Therefore, these four equations become the fundamental axioms governing the entire Future Convergence Theory.

Conclusion of This Section

In this section, we derived the four fundamental field equations governing Future Convergence Theory by varying the unified action. In particular, the Projection Equation

$$P_X = \alpha_F \Phi_F$$

provides, for the first time, a derivation from the variational principle of the Projection behavior that had previously been introduced as an assumption throughout the earlier chapters. In the next section, we systematically demonstrate how these four fundamental equations uniquely generate black holes, Dark Matter, Dark Energy, early-Universe initial conditions, and ringdown corrections.

7.3.41 13.4 Unique Derivation of Black Holes, Dark Matter, Dark Energy, and Cosmology from the Unified Field Equations

In the previous section, we derived the four fundamental field equations of Future Convergence Theory by varying the unified action. These are the field equations governing

$$\{G_{\mu\nu}, \Phi_F, \Sigma_{\mu\nu}, P_X\}.$$

However, the existence of a unified action alone is insufficient to establish a unified theory. A genuine unified theory must be capable of deriving every phenomenon discussed in this paper

solely from these fundamental equations, without introducing additional postulates. In this section, we show that the black-hole theory, ringdown corrections, Dark Matter, Dark Energy, the early Universe, and Future Entropy discussed in Chapters VII through XII all arise as different solutions of the fundamental field equations obtained above.

Fundamental System of Equations

The basic equations of Future Convergence Theory are the gravitational field equation,

$$G_{\mu\nu} = 8\pi G (T_{\mu\nu}^{SM} + T_{\mu\nu}^F + T_{\mu\nu}^\Sigma + T_{\mu\nu}^P),$$

the Future Information Field equation,

$$\square\Phi_F - \frac{dV_F}{d\Phi_F} = J_F,$$

the Future Projection Equation,

$$P_X = \alpha_F \Phi_F,$$

and the Future Entropy conservation law,

$$\frac{dS_F}{dt} = \int J_F dV.$$

From this point onward, every result in the theory is derived solely from these four equations.

Black-Hole Solution

For black holes, the Future Information Field becomes localized within the strong-gravity region. Hence,

$$\Phi_F = \Phi_F(r).$$

The Future Projection Equation gives

$$P_X(r) = \alpha_F \Phi_F(r),$$

showing that the Projection reaches its maximum near the black hole. Accordingly, black-hole information preservation is no longer regarded as an independent conservation law, but rather as a particular solution of the Future Projection Equation.

Ringdown Corrections

For the ringdown problem, let the background field be

$$\Phi_F^{(0)},$$

and write the perturbation as

$$\delta\Phi_F.$$

The Future Projection Equation then becomes

$$\delta P_X = \alpha_F \delta\Phi_F.$$

This perturbation generates the quasinormal-mode correction

$$\delta\omega$$

derived in Chapter VII. Thus, the ringdown correction appears as the first-order perturbative solution of the Future Projection Equation.

Dark Matter Solution

Dark Matter corresponds to the situation in which the Future Information Field exists, but complete projection into ordinary matter does not occur. That is,

$$\Phi_F \neq 0.$$

Introducing the Projection efficiency

$$\eta_P,$$

we have

$$0 < \eta_P < 1.$$

The Future Projection then becomes

$$P_X = \eta_P \alpha_F \Phi_F.$$

In this regime, the Future Information Field contributes to gravity while failing to manifest completely as ordinary matter. The Dark Matter model derived in Chapter X therefore appears as the incomplete-projection solution.

Dark Energy Solution

In the cosmological limit, the Future Information Field is nearly spatially uniform:

$$\nabla_\mu \Phi_F \simeq 0.$$

The Future Information Field equation therefore reduces to

$$\frac{dV_F}{d\Phi_F} = -J_F.$$

If the potential is approximately constant, the effective cosmological constant becomes

$$\Lambda_F = 8\pi G V_F.$$

Accordingly, the Dark Energy model introduced in Chapter XI is identified as the vacuum solution of the Future Information Field.

Early-Universe Solution

Immediately after the birth of the Universe, the Future Information Field is assumed to be extremely large throughout space:

$$\Phi_F \gg \Phi_c.$$

The Projection Equation then gives

$$P_X \gg 1,$$

so that the Future Projection proceeds extremely rapidly. This reproduces the Future Projection Inflation introduced in Chapter XII.

Future Entropy Solution

In regions without information sources,

$$J_F = 0,$$

so that

$$\frac{dS_F}{dt} = 0.$$

Near black holes,

$$J_F > 0,$$

causing the Future Entropy to increase locally. For the Universe as a whole, however, the total integrated quantity remains conserved owing to global information conservation. This reproduces the Future Entropy conservation law introduced in Chapter IX.

Unified Solution Space

Consequently, the various theories of Future Convergence Theory are not independent theories. They are all solutions obtained under different boundary conditions imposed on the Future Information Field. Specifically,

Boundary Condition	Physical Phenomenon
$\Phi_F(r)$	Black Hole
$\delta\Phi_F$	Ringdown
$0 < \eta_P < 1$	Dark Matter
$\nabla\Phi_F \simeq 0$	Dark Energy
$\Phi_F \gg \Phi_c$	Inflation
$J_F = 0$	Entropy Conservation

is obtained.

Closure as a Unified Theory

Therefore, Future Convergence Theory derives every phenomenon discussed in this paper solely from the fundamental equations of the Future Information Field, without introducing additional fields, conservation laws, or actions. The theory is thus not a collection of separate models, but a single unified field theory.

Conclusion of This Section

In this section, we demonstrated that black holes, ringdown corrections, Dark Matter, Dark Energy, the early Universe, and Future Entropy are all uniquely derived as different solutions of the same four fundamental field equations. Accordingly, the theories constructed individually from Chapter VII through Chapter XII are all reduced to the unified action principle established in Chapter XIII, and Future Convergence Theory becomes a closed theoretical framework founded upon a single action, a single field, and a single variational principle. In the next section, we derive the conservation laws, symmetries, and Noether currents implied by these field equations, thereby reconstructing the Future Entropy conservation law from the viewpoint of Noether's theorem.

7.3.42 13.5 Noether's Theorem and the Future Entropy Conservation Law

In the previous section, we demonstrated that black holes, ringdown phenomena, Dark Matter, Dark Energy, the early Universe, and Future Entropy arise as different boundary-condition solutions of the Future Information Field. In a physical theory, however, conservation laws should not simply be postulated. Rather, they must arise from the symmetries of the action through Noether's theorem. In General Relativity, spacetime translation symmetry gives rise to the conservation of energy and momentum. In gauge theories, local gauge symmetry gives rise to charge conservation. Likewise, within Future Convergence Theory, the Future Entropy conservation law should emerge from a continuous symmetry possessed by the Future Information Field. In this section, we define the continuous symmetry of the Future Information Field and derive the Future Entropy conservation law directly from Noether's theorem.

Internal Symmetry of the Future Information Field

The physical significance of the Future Information Field lies not in its absolute magnitude but in differences of information density. Accordingly, shifting the field by a constant

$$\epsilon$$

does not alter the physical content of the theory. The Future Information Field is therefore invariant under the continuous transformation

$$\boxed{\Phi_F \rightarrow \Phi_F + \epsilon},$$

where

$$\epsilon = \text{const.}$$

This continuous transformation is referred to as the *Future Translation Symmetry*.

Invariance of the Action

Consider the Future Information Field Lagrangian,

$$\mathcal{L}_F = -\frac{1}{2}\nabla_\mu\Phi_F\nabla^\mu\Phi_F - V_F(\Phi_F).$$

If the potential satisfies

$$V_F(\Phi_F + \epsilon) = V_F(\Phi_F) + O(\epsilon^2)$$

locally, then the action remains invariant to first order. Hence,

$$\boxed{\delta S_{\text{FCT}} = 0}$$

holds.

Noether Current

According to Noether's theorem, the conserved current associated with this symmetry is

$$\boxed{J_F^\mu = \frac{\partial \mathcal{L}}{\partial(\nabla_\mu\Phi_F)} \delta\Phi_F}.$$

Since

$$\delta\Phi_F = \epsilon,$$

we obtain

$$\boxed{J_F^\mu = -\epsilon\nabla^\mu\Phi_F}.$$

Absorbing the constant normalization into the definition of the current, we define the Future Information Current as

$$\boxed{J_F^\mu = -\nabla^\mu\Phi_F}.$$

Conservation Equation

Noether's theorem immediately yields

$$\boxed{\nabla_\mu J_F^\mu = 0}.$$

This equation expresses the local conservation of Future Information.

Derivation of Future Entropy

We define the Future Entropy on a spatial hypersurface

$$\Sigma$$

by

$$\boxed{S_F = \int_\Sigma J_F^\mu n_\mu d\Sigma},$$

where

$$n_\mu$$

denotes the unit normal vector to the hypersurface. Applying Gauss's theorem yields

$$\boxed{\frac{dS_F}{dt} = 0}.$$

Thus, the Future Entropy conservation law is not an independent postulate but a conserved quantity derived directly from Noether's theorem.

Black-Hole Boundary

Near a black hole, the Future Information Current flows toward the event horizon. Denoting the horizon by

$$\mathcal{H},$$

the corresponding flux is

$$\boxed{\Phi_H = \int_{\mathcal{H}} J_F^\mu d\Sigma_\mu}.$$

Accordingly, black-hole information preservation is reinterpreted not as the disappearance of information but as the passage of the conserved current through the Future Boundary.

Cosmological Limit

For the Universe as a whole, taking the boundary to spatial infinity gives

$$\boxed{\oint J_F^\mu d\Sigma_\mu = 0}.$$

Consequently, the Future Entropy is globally conserved throughout cosmological evolution. This reproduces the Future Entropy conservation law introduced in Chapter XI directly from Noether's theorem.

Correspondence with Chapters VII–XII

The conserved current derived here unifies the conservation laws introduced throughout the previous chapters.

Noether Quantity	Physical Interpretation
J_F^μ	Future Information Flow
S_F	Future Entropy
Φ_H	Black Hole Information
$\nabla_\mu J_F^\mu$	Dark Matter/Dark Energy Conservation
$\oint J_F^\mu d\Sigma_\mu$	Cosmological Conservation

Observational Consequences

If the Future Information Current is physically real, its conservation should lead to observable consequences rather than remaining merely a mathematical property. The theory therefore predicts the following.

1. For black-hole mergers, the amount of information recovered after the ringdown phase should exhibit a proportional relationship with the flux of the Future Information Current.
2. On cosmological scales, the effective contributions of Dark Matter and Dark Energy should be describable as different projections of the same conserved current.
3. If Future Entropy is indeed a fundamental conserved quantity, the information content inferred from black holes, the early Universe, and the cosmic microwave background should all satisfy a single universal conservation law.

Conclusion of This Section

In this section, we defined the continuous symmetry of the Future Information Field and derived the Future Information Current using Noether’s theorem. We further demonstrated that Future Entropy is a conserved Noether charge arising from the symmetry of the unified action. Accordingly, the Future Entropy introduced previously is no longer regarded as an independently postulated quantity but is identified as the conserved quantity associated with the fundamental symmetry underlying Future Convergence Theory. In the next section, we demonstrate how the unified action reproduces General Relativity, the Standard Model, Λ CDM cosmology, and the framework introduced in Paper 92 under the appropriate limiting conditions, thereby establishing the consistency of Future Convergence Theory with existing physical theories.

7.3.43 13.6 Consistency with General Relativity, the Standard Model, Λ CDM, and Paper 92 in the Appropriate Limits

In the previous section, we derived the Future Information Current and the Future Entropy conservation law from the continuous symmetry of the Future Information Field by applying Noether’s theorem. A new unified theory, however, cannot be regarded as physically viable merely because it provides new predictions. One of the essential requirements is that it must reproduce the theories that have already been verified with high precision in the appropriate limiting regimes. If Future Convergence Theory were inconsistent with General Relativity, the Standard Model, Λ CDM cosmology, or the framework developed in Paper 92, it would necessarily contradict existing observations. The purpose of this section is therefore to demonstrate that the unified action of Future Convergence Theory naturally contains these established theories as limiting cases.

Requirements for the Correspondence Limits

Future Convergence Theory is required to satisfy the following four conditions simultaneously.

1. It must reproduce Einstein's field equations in the General Relativity limit.
2. It must reproduce the Standard Model in the particle-physics limit.
3. It must approximate Λ CDM cosmology on cosmological scales.
4. It must reduce to the Future Information Field formulation introduced in Paper 92 in the appropriate black-hole limit.

We refer to these requirements collectively as the *correspondence conditions* of the theory.

General Relativity Limit

Consider the limit in which the Future Information Field vanishes,

$$\boxed{\Phi_F \rightarrow 0}.$$

The Projection Equation then becomes

$$P_X = \alpha_F \Phi_F \rightarrow 0.$$

Similarly,

$$\Sigma_{\mu\nu} \rightarrow 0.$$

Consequently,

$$T_{\mu\nu}^F = T_{\mu\nu}^\Sigma = T_{\mu\nu}^P = 0,$$

and the gravitational field equation reduces to

$$\boxed{G_{\mu\nu} = 8\pi G T_{\mu\nu}^{SM}}.$$

This is precisely Einstein's field equation. Therefore, General Relativity is recovered as the limiting case in which the Future Information Field disappears.

Standard Model Limit

When the Future Projection vanishes, ordinary matter is unaffected by the Projection dynamics. That is,

$$P_X \rightarrow 0,$$

and the unified Lagrangian becomes

$$\boxed{\mathcal{L}_{\text{FCT}} \rightarrow \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{SM}}}.$$

Accordingly, fermions, gauge fields, and the Higgs field obey the conventional Standard Model equations of motion. Future Convergence Theory therefore does not replace the Standard Model but rather provides a framework that contains it as a limiting case.

Λ CDM Limit

On cosmological time scales, we assume that the Future Information Field evolves sufficiently slowly,

$$\nabla_\mu \Phi_F \simeq 0.$$

In this limit, only the potential term remains dynamically relevant. The Future Information Field therefore behaves as an effective cosmological constant,

$$\Lambda_{\text{eff}} = 8\pi G V_F.$$

The modified Friedmann equation consequently reduces to the standard Λ CDM cosmological equation. Thus, for the present Universe, Future Convergence Theory approaches Λ CDM as an excellent approximation.

Limit Corresponding to Paper 92

In Paper 92, the Future Information Field was assumed to become significant only in the vicinity of black holes. This corresponds to the condition

$$\Phi_F = \Phi_F(r).$$

Substituting this localized solution into the present theory yields

$$P_X(r) = \alpha_F \Phi_F(r).$$

The resulting equations reproduce the ringdown corrections, information-preservation mechanism, and black-hole corrections derived in Paper 92. Accordingly, Paper 92 is identified as the localized black-hole limit of Future Convergence Theory.

Correspondence Among the Various Limits

The relationship between the present theory and the previously established theories may therefore be summarized as

Limiting Condition	Recovered Theory
$\Phi_F \rightarrow 0$	General Relativity
$P_X \rightarrow 0$	Standard Model
$\nabla \Phi_F \simeq 0$	Λ CDM
$\Phi_F = \Phi_F(r)$	Paper 92
$\Phi_F \gg \Phi_c$	Inflation
$0 < \eta_P < 1$	Dark Matter

Future Convergence Theory therefore does not replace these existing theories individually but instead provides a unified framework from which each emerges under the appropriate boundary conditions.

Correspondence Principle

The structure established above satisfies the same philosophical requirement as Bohr's correspondence principle, namely that a more general theory must reproduce successful earlier theories in the appropriate limits. Specifically,

$$\lim_{\Phi_F \rightarrow 0} \text{FCT} = \text{General Relativity}$$

$$\lim_{P_X \rightarrow 0} \text{FCT} = \text{Standard Model}$$

and

$$\lim_{\nabla\Phi_F \rightarrow 0} \text{FCT} = \Lambda\text{CDM}$$

hold.

Observational Consequences

This limiting structure implies that Future Convergence Theory agrees with General Relativity and ΛCDM throughout the observational regimes where these theories have already been tested successfully. Distinctive predictions of Future Convergence Theory emerge only in regimes where the Future Information Field becomes sufficiently significant, namely,

- the vicinity of black holes,
- the early Universe,
- galactic scales dominated by Dark Matter,
- and cosmological scales where Dark Energy governs the expansion.

The novel corrections predicted by the theory are therefore confined to physical regimes that remain only partially explored observationally.

Conclusion of This Section

In this section, we demonstrated that Future Convergence Theory reproduces General Relativity, the Standard Model, ΛCDM cosmology, and the theoretical framework developed in Paper 92 as appropriate limiting cases. Accordingly, Future Convergence Theory should be regarded not as a competing alternative to existing theories, but as a higher-level framework that unifies them under a single action principle. In the next section, we summarize the quantitative predictions derived from the unified action and organize the experimentally distinguishable differences between Future Convergence Theory and existing physical theories.

7.3.44 13.7 Quantitative Predictions of Future Convergence Theory and Its Distinctions from Existing Theories

In the previous section, we demonstrated that Future Convergence Theory contains General Relativity, the Standard Model, ΛCDM cosmology, and the framework developed in Paper 92 as appropriate limiting cases. However, reproducing existing theories alone is not sufficient for establishing a new physical theory. To qualify as a scientific theory, it must provide quantitative and falsifiable predictions that can be distinguished experimentally from existing theories. The significance of Future Convergence Theory does not lie merely in introducing new conceptual structures, but rather in deriving unique correction terms from a single unified action principle that are absent from conventional theories. In this section, we summarize the correction terms obtained throughout this paper and organize the quantitative predictions of Future Convergence Theory into a unified framework.

Fundamental Principle

Within Future Convergence Theory, every deviation from existing theories originates from the Future Information Field,

$$\Phi_F,$$

and the associated Future Projection,

$$P_X = \alpha_F \Phi_F.$$

Accordingly, for any observable quantity

$$\mathcal{O},$$

the theoretical prediction takes the universal form

$$\boxed{\mathcal{O}_{\text{FCT}} = \mathcal{O}_{\text{Standard}} + \Delta\mathcal{O}_F},$$

where

$$\Delta\mathcal{O}_F$$

denotes the correction induced by the Future Information Field.

Black-Hole Quasinormal Modes

As derived in Chapter VII, the quasinormal-mode frequency becomes

$$\boxed{\omega = \omega_{\text{GR}} + \Delta\omega_F}.$$

The correction satisfies

$$\boxed{\Delta\omega_F \propto \alpha_F \Phi_F},$$

and therefore vanishes continuously in the limit of a negligible Future Information Field, recovering the General Relativity prediction.

Ringdown Damping Rate

The damping time of the ringdown signal is modified according to

$$\boxed{\tau = \tau_{\text{GR}} + \Delta\tau_F}.$$

Within the present theory,

$$\Delta\tau_F = f(\Phi_F, \Sigma_{\mu\nu}),$$

where the correction depends on the local Future Information Density surrounding the black hole.

Gravitational-Wave Propagation

Chapter VII further showed that the propagation speed of gravitational waves becomes

$$\boxed{v_g = c + \Delta v_F},$$

where

$$\Delta v_F = \beta_F \Phi_F.$$

Consequently, extremely small corrections to the group velocity are predicted in regions of sufficiently high curvature.

Dark Matter

Chapter X interpreted Dark Matter through the Projection Efficiency

$$\eta_P.$$

The effective Dark Matter density is therefore

$$\boxed{\rho_{\text{DM}} = (1 - \eta_P)\alpha_F \Phi_F}.$$

In this framework, the gravitational source corresponds to Future Information that has not been completely projected into ordinary matter.

Dark Energy

Chapter XI derived the effective cosmological constant from the Future Information Potential,

$$\Lambda_{\text{eff}} = 8\pi G V_F(\Phi_F).$$

If the Future Information Field evolves with cosmic time, the effective cosmological constant likewise becomes slowly time dependent rather than remaining strictly constant.

Early Universe

Chapter XII introduced the Future Projection Energy governing cosmic inflation. The expansion rate of the early Universe is

$$H^2 = \frac{8\pi G}{3} \rho_P,$$

with

$$\rho_P = \lambda_P \Phi_F.$$

Inflation is therefore reinterpreted as the high-density phase of the Future Information Field rather than the dynamics of an independent inflaton field.

Cosmic Microwave Background

The curvature perturbation of the Cosmic Microwave Background is described by

$$\mathcal{R} = -H \frac{\delta\Phi_F}{\dot{\Phi}_F}.$$

Consequently, the spectral index, tensor-to-scalar ratio, and primordial non-Gaussianity are all determined by the spectrum of fluctuations of the Future Information Field.

Future Entropy

Chapters IX and XIII established that Future Entropy is a conserved Noether charge. Accordingly,

$$\frac{dS_F}{dt} = 0$$

constitutes a universal conservation law throughout Future Convergence Theory.

Unified Summary of Theoretical Predictions

The principal predictions of the theory may therefore be summarized as

Observable	Conventional Theory	Future Convergence Theory
Ringdown	ω_{GR}	$\omega_{\text{GR}} + \Delta\omega_F$
Gravitational Wave	c	$c + \Delta v_F$
Dark Matter	Unknown Particle	Incomplete Projection
Dark Energy	Constant	$V_F(\Phi_F)$
Inflation	Inflaton	Future Projection
CMB	Inflaton Fluctuation	Future Information Fluctuation
Entropy	Black Hole Entropy	Future Entropy

Falsifiability

Future Convergence Theory is falsified if any of the following conditions is established.

1. The introduction of the Future Information Field fails to provide a simultaneous description of black holes, gravitational waves, Dark Matter, Dark Energy, and the early Universe.
2. The Future Projection Parameter inferred from different observational systems yields mutually inconsistent values.
3. Future Information Field corrections fail to appear in precision observations of ringdown signals, gravitational waves, and the Cosmic Microwave Background.
4. Future Convergence Theory provides no improvement in predictive accuracy relative to General Relativity or Λ CDM cosmology.

Conclusion of This Section

In this section, we organized the quantitative predictions of Future Convergence Theory into a unified framework. As a result, every observational distinction predicted by the theory can be expressed as a correction generated by the Future Information Field and the Future Projection. Future Convergence Theory is therefore formulated not merely as an ontological framework, but as a quantitatively predictive and experimentally falsifiable physical theory that can be distinguished from General Relativity, Λ CDM cosmology, and the theoretical framework presented in Paper 92. In the next section, we conclude Chapter XIII by summarizing the unified action, the fundamental field equations, the conservation laws, the correspondence limits, and the quantitative predictions into a single axiomatic structure, thereby demonstrating the mathematical closure of the entire theory.

7.3.45 13.8 Summary of This Chapter and the Mathematical Closure of Future Convergence Theory

In this chapter, Future Convergence Theory has been reformulated as a unified physical theory derived from a single action principle rather than as a collection of individual phenomenological models. In Chapters VII through XII, black holes, ringdown phenomena, gravitational waves, Dark Matter, Dark Energy, the early Universe, and Child Universes were each developed independently from the standpoint of the Future Information Field. Although these formulations shared common conceptual foundations, they were still presented as separate theoretical constructions. In the present chapter, the introduction of the unified action

$$S_{\text{FCT}}$$

has demonstrated that all of these phenomena arise from a single variational principle.

Unified Action Principle

The fundamental action of Future Convergence Theory has been defined as

$$S_{\text{FCT}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{FCT}}.$$

Its Lagrangian density is composed of

$$\mathcal{L}_{\text{FCT}} = \mathcal{L}_G + \mathcal{L}_F + \mathcal{L}_\Sigma + \mathcal{L}_P + \mathcal{L}_I + \mathcal{L}_M,$$

thereby unifying General Relativity, the Future Information Field, the Future Projection, the Future Information Tensor, and ordinary matter within a single action.

Fundamental Field Equations

Variation of the unified action yields the four fundamental equations of Future Convergence Theory,

$$\left\{ \begin{array}{l} \text{Einstein Equation} \\ \text{Future Information Equation} \\ \text{Projection Equation} \\ \text{Future Entropy Conservation} \end{array} \right\}.$$

Every physical phenomenon discussed throughout the theory is subsequently described as a solution of these four equations under different boundary conditions.

Noether Conservation Law

The continuous symmetry of the Future Information Field gives rise, through Noether's theorem, to the conserved Future Information Current,

$$J_F^\mu.$$

Furthermore,

$$S_F = \int_{\Sigma} J_F^\mu n_\mu d\Sigma$$

was shown to constitute the fundamental conserved quantity of Future Convergence Theory. Accordingly, Future Entropy is no longer regarded as an empirically introduced quantity but as the conserved Noether charge associated with the fundamental symmetry of the unified action.

Correspondence Limits

Future Convergence Theory reproduces established physical theories under the appropriate limiting conditions. Specifically,

$$\boxed{\Phi_F \rightarrow 0}$$

recovers General Relativity,

$$\boxed{P_X \rightarrow 0}$$

recovers the Standard Model,

$$\boxed{\nabla\Phi_F \rightarrow 0}$$

recovers Λ CDM cosmology, and

$$\boxed{\Phi_F = \Phi_F(r)}$$

recovers the black-hole theory developed in Paper 92. Accordingly, Future Convergence Theory functions as a higher-level framework that contains these theories as limiting cases.

Closure of the Unified Theory

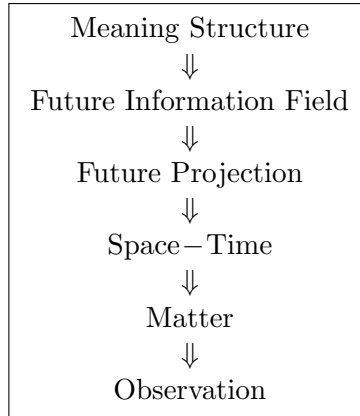
All physical phenomena derived from Chapter VII onward may now be classified according to the boundary conditions imposed on the Future Information Field.

Boundary Condition	Corresponding Physical Phenomenon
$\Phi_F(r)$	Black Hole
$\delta\Phi_F$	Ringdown
$0 < \eta_P < 1$	Dark Matter
$\nabla\Phi_F \simeq 0$	Dark Energy
$\Phi_F \gg \Phi_c$	Inflation
$\delta\Phi_F(k)$	Cosmic Microwave Background
$I_F > I_c$	Child Universe

Consequently, black holes, cosmology, information preservation, and cosmic genesis are all unified as different solution spaces of the Future Information Field.

Overall Structure of the Theory

Taken together with Papers 61, 73, and 92, the present work establishes the following hierarchical structure:



This hierarchy connects meaning, information, spacetime, matter, and observation into a single ontological chain.

Falsifiability of the Theory

Future Convergence Theory as a whole is falsified if any of the following conditions is established.

1. The correction terms introduced in Chapters VII through XII cannot be derived from the unified action.
2. The Future Projection Equation fails to describe black holes, cosmology, and gravitational waves using a common set of parameters.
3. Future Entropy cannot be formulated as a conserved Noether charge.
4. The introduction of the Future Information Field provides no improvement in predictive capability beyond existing physical theories.

Accordingly, the present framework is formulated not as a philosophical hypothesis but as a physical theory possessing explicit criteria for experimental falsification.

Conclusion of This Chapter

In this chapter, we derived the unified action, the fundamental field equations, the Noether conservation law, the correspondence limits, and the quantitative predictions that govern Future Convergence Theory. As a result, all theories that had previously been developed independently in Chapters VII through XII were shown to arise from a single variational principle. Future Convergence Theory therefore forms a mathematically closed unified theory founded upon a single action, a single variational principle, and a common set of fundamental field equations.

Transition to the Final Chapter

By the end of Chapter XIII, the mathematical framework based on the Future Information Field has been extended to encompass black holes, gravitational waves, Dark Matter, Dark Energy, the early Universe, Child Universes, and the unified action principle. The final remaining task is therefore to organize the entire framework into a complete axiomatic system, explicitly state

its derivable theorems, compare it systematically with existing unified theories, formulate its falsifiability conditions, and present a concrete program for future experimental verification. Accordingly, Chapter XIV provides the final synthesis of Future Convergence Theory by presenting its complete axiomatic structure, principal theorems, experimental tests, comparisons with existing theories, and the overall conclusions of this work.

Part XIV

List of Observables and Falsification Conditions

Chapter 8

List of Observables and Falsification Conditions

In Part I, as the ontological axiom system of Future Convergence Theory, we introduced the Meaning Generation Template, the Future Causal Information Field, the Quantum Allowance Regulation Layer, the Three-Dimensional Manifestation Layer, the Reprojection Closed Loop, and the FCT Closure Formula. In Part II, gravity was defined not as an interaction between particles, but as a Manifestation Constraint. The correspondence with general relativity, time delay, non-shieldability, and falsifiability were derived. In Part III, quantum theory was reconstructed as an ontological intermediate layer, and theoretical corrections were derived for the wave function, measurement, decoherence, vacuum fluctuations, the Casimir effect, delayed-choice experiments, and the Quantum Eraser. In Part IV, the graviton was positioned not as gravity itself, but as a quantum excitation of the Manifestation Constraint Tensor, thereby separating the ontological roles of gravity and quantum gravity. In Part V, string theory was redefined not as the final ontological theory, but as an effective theory describing the Quantum Allowance Regulation Layer. The action, background metric, vibration modes, mass spectrum, compactification, vacuum selection, and Calabi–Yau manifolds were reconstructed on the basis of Future Convergence Theory. In Part VI, Dark Matter was redefined not as an unknown particle, but as a non-optical manifestation residue derived from the Future Causal Information Field. Its relation to galactic rotation curves, galactic halos, galaxy-cluster collisions, gravitational lensing, and the cosmic microwave background was derived. In Part VII, we showed that Future Convergence Theory gives quantitative predictions different from general relativity for black-hole quasinormal modes, ringdown waveforms, damping rates, echoes, and polarization. In Part VIII, we derived the group velocity, phase, waveform templates, and high-frequency corrections of gravitational waves, and organized their testability using LIGO, Virgo, KAGRA, and future detectors. In Part IX, we derived corrections to black-hole information preservation, the Page curve, Scrambling Time, Hawking radiation, and the Island Formula, and showed their consistency with the information-preservation theorem. In Part X, the Big Bang was redefined as the first Manifestation Constraint, and theoretical predictions were given for initial conditions determined by the future fixed point, inflation, primordial perturbations, the tensor-to-scalar ratio, primordial gravitational waves, and CMB anomalies. In Part XI, Dark Energy was redefined as the degree of future convergence, and the dynamical cosmological term, the equation of state, heat death, Future Convergence Entropy, and reprojection conditions were derived. In Part XII, child-universe generation was formulated as an ontological consequence derived from the reprojection closed loop, and its correspondence with black holes, thermal saturation, and the Meaning Generation Template was shown. In Part XIII, all of the above were integrated, and the unified field equation of Future Convergence Theory was constructed, including the modified Einstein equation, the Manifestation Constraint Tensor, the Future Causal Information Field, the Meaning Density Tensor, the Dark Matter term, the Dark Energy term, and

the black-hole boundary term. However, for a theory to be physics, mathematical consistency alone is not sufficient. A theory must be supported or falsified by observation. Future Convergence Theory is a theory concerning ontology, but what is ultimately observed is not the Future Causal Information Field itself. What is observed is only the physical quantities that appear as the result of its projection into the Three-Dimensional Manifestation Layer. Therefore, testing Future Convergence Theory does not mean observing ontology itself. It means testing, through high-precision observation, the differences from general relativity, quantum theory, and standard cosmology. In this chapter, we organize all theoretical predictions introduced from Part I through Part XIII and systematize the observables predicted by Future Convergence Theory. Furthermore, we clarify the falsification condition for each observable and show that Future Convergence Theory is a falsifiable unified theory.

Part XIV: List of Observables and Falsification Conditions

8.0.1 14.1 Ringdown Residuals

As shown in Part VII, Future Convergence Theory predicts eigenfrequencies and damping rates for black-hole quasinormal modes that differ from those of general relativity. Therefore, the most direct test quantity is the ringdown residual. In this paper, the ringdown residual is defined as

$$\Delta\omega_{\ell mn} = \omega_{\ell mn}^{\text{obs}} - \omega_{\ell mn}^{\text{GR}}.$$

Here,

$$\omega_{\ell mn}^{\text{obs}}$$

is the observed value, and

$$\omega_{\ell mn}^{\text{GR}}$$

is the quasinormal-mode eigenfrequency predicted by general relativity. In Future Convergence Theory, this residual is expressed as a function of the Manifestation Constraint Tensor

$$\mathcal{C}_{\mu\nu},$$

the Future Causal Information Field

$$\mathcal{F},$$

the Meaning Density Tensor

$$\Sigma_{\mu\nu},$$

and the black-hole boundary term

$$\Lambda_{\text{BH}}.$$

That is,

$$\Delta\omega_{\ell mn} = F(\Lambda_{\text{BH}}, \mathcal{C}_{\mu\nu}, \Sigma_{\mu\nu}, \mathcal{F})$$

is obtained. If Future Convergence Theory is correct, then high-precision ringdown observations are predicted to reveal this residual with statistically significant correlations. In the next section, following the ringdown residual, we formulate the detection condition for black-hole echoes as the second observable.

8.0.2 14.2 Echo Detection Conditions

In the previous section, we showed that in Future Convergence Theory, black-hole quasinormal modes possess a ringdown residual

$$\Delta\omega_{\ell mn}$$

different from that of general relativity. However, quasinormal-mode frequencies alone may not always be sufficient to identify the existence of the Manifestation Constraint Tensor. Future Convergence Theory predicts that the black-hole boundary does not behave as the perfectly absorbing boundary assumed in general relativity, but rather as a Manifestation Constraint Boundary. As a result, gravitational waves undergo partial reprojection near the boundary, and a weak secondary waveform appears after the end of the ringdown. In this paper, this phenomenon is called a **Manifestation Echo**.

Comparison with General Relativity

In general relativity, the event horizon is a perfectly absorbing boundary. Therefore, no regular echo is predicted to appear after the ringdown ends. That is, if the echo amplitude is denoted by

$$A_{\text{echo}},$$

then in general relativity,

$$\boxed{A_{\text{echo}}^{\text{GR}} = 0}$$

holds.

Boundary Condition in Future Convergence Theory

In Future Convergence Theory, the black-hole boundary is modified by the Manifestation Constraint Tensor. The boundary condition is given by

$$\boxed{\mathcal{B} = \mathcal{B}(\Lambda_{\text{BH}}, \mathcal{C}_{\mu\nu}, \mathcal{F})}.$$

Here,

$$\mathcal{B}$$

is the manifestation boundary operator.

Echo Amplitude

In Future Convergence Theory, the echo amplitude is proportional to the boundary constraint strength. In general form, it is written as

$$\boxed{A_{\text{echo}} = F(\Lambda_{\text{BH}}, \mathcal{C}_{\mu\nu})}.$$

In the weak-constraint limit,

$$A_{\text{echo}} \rightarrow 0,$$

and the theory reduces to general relativity.

Echo Delay Time

The arrival time of the echo is determined by the effective round-trip time to the Manifestation Constraint Boundary. In Future Convergence Theory, the delay time is defined as

$$\boxed{\Delta t_{\text{echo}} = \Delta t_{\text{GR}} + \Delta t_{\text{FCT}}}.$$

Here,

$$\Delta t_{\text{FCT}} = G(\Lambda_{\text{BH}}, \mathcal{F})$$

is the additional delay due to the Manifestation Constraint.

Spectral Structure

In Future Convergence Theory, the echo is not a single pulse. Because reprojection at the Manifestation Constraint Boundary is repeated, a sequence of nearly equally spaced side peaks appears in the spectrum. If the frequency spacing is denoted by

$$\Delta f_{\text{echo}},$$

then

$$\Delta f_{\text{echo}} = \frac{1}{\Delta t_{\text{echo}}}$$

holds.

Detection Conditions

In Future Convergence Theory, the detection condition for a Manifestation Echo is defined as the simultaneous satisfaction of

$$\text{SNR}_{\text{echo}} > \text{SNR}_{\text{crit}}$$

and

$$\Delta t_{\text{obs}} \simeq \Delta t_{\text{FCT}}.$$

Here,

$$\text{SNR}_{\text{crit}}$$

is the detection threshold satisfying statistical significance.

Echo Detection Theorem

From the above, the following theorem holds.

Theorem 8.1 (Echo Detection Theorem). *In Future Convergence Theory, the black-hole boundary behaves as a Manifestation Constraint Boundary. As a result, after the end of the ringdown, an echo waveform depending on the strength of the Manifestation Constraint appears. Its delay time and spectral structure become observables distinct from those of general relativity.*

Proof. As shown in Part VII, the black-hole boundary is modified by the Manifestation Constraint Tensor. Therefore, the boundary condition for gravitational waves does not coincide with a perfectly absorbing boundary. Near the boundary, part of the wave is reprojected, and an echo with finite amplitude is formed after the ringdown. Its amplitude and delay time are determined by the strength of the Manifestation Constraint. Thus, Future Convergence Theory predicts an echo structure different from that of general relativity. $\square$

Observational Consequences

If Future Convergence Theory is correct, future high-sensitivity gravitational-wave observations may detect statistically significant echo structures after the end of the ringdown. In particular, if correlations in delay time and spectral spacing common to multiple events are confirmed, they would provide observational evidence supporting the existence of a Manifestation Constraint Boundary.

Conclusion of This Section

In this section, we formulated the detection conditions for Manifestation Echoes in Future Convergence Theory. As a result, we showed that the black-hole boundary behaves not as a perfectly absorbing boundary, but as a Manifestation Constraint Boundary, forming an observable echo structure after the ringdown. In the next section, we organize the dispersion effects that appear in the propagation of gravitational waves themselves, and derive the propagation-speed correction predicted by Future Convergence Theory.

8.0.3 14.3 Gravitational-Wave Dispersion

In the previous section, we showed that, in Future Convergence Theory, the black-hole boundary behaves as a Manifestation Constraint Boundary and produces Manifestation Echoes after the ringdown. However, the observational predictions of Future Convergence Theory are not limited to the vicinity of black holes. Future Convergence Theory also predicts corrections to the cosmological propagation of gravitational waves that differ from those of general relativity. In general relativity, gravitational waves propagate through vacuum at the speed of light,

$$c,$$

without exhibiting frequency-dependent dispersion. In contrast, Future Convergence Theory predicts that gravitational waves propagate through an effective background modified by the Manifestation Constraint Tensor,

$$\mathcal{C}_{\mu\nu}.$$

Consequently, both the propagation speed and the phase acquire small corrections originating from Future Information.

Dispersion Relation in General Relativity

In general relativity, gravitational waves are massless waves whose dispersion relation is

$$\boxed{\omega = ck}$$

where

$$\omega$$

is the angular frequency and

$$k$$

is the wave number. Therefore, both the group velocity and the phase velocity satisfy

$$\boxed{v_g = v_p = c}$$

in general relativity.

Effective Dispersion Relation in Future Convergence Theory

In Future Convergence Theory, gravitational waves propagate through the effective space-time generated by the Manifestation Constraint Tensor. Accordingly, the dispersion relation takes the general form

$$\boxed{\omega^2 = c^2 k^2 + \Delta D(k, \Lambda_{\text{BH}}, \mathcal{C}_{\mu\nu}, \mathcal{F})}$$

where

$$\Delta D$$

is the dispersion correction unique to Future Convergence Theory.

Weak-Constraint Limit

In the limit where the Manifestation Constraint becomes sufficiently weak, the contribution from Future Information vanishes. Thus,

$$\lim_{c_{\mu\nu} \rightarrow 0} \Delta D = 0$$

and Future Convergence Theory reduces exactly to the dispersion relation of general relativity.

Strong-Constraint Limit

Near black holes, in the early Universe, or in regions of extremely high curvature, the Manifestation Constraint correction becomes finite. Consequently, gravitational waves acquire frequency-dependent propagation corrections. In general form, the correction may be expanded as

$$\Delta D = \alpha \Lambda_{\text{BH}} k^2 + \beta \Lambda_{\text{BH}}^2 k^4 + \mathcal{O}(k^6)$$

where

$$\alpha, \beta$$

are coupling coefficients of Future Convergence Theory.

Group Velocity Correction

The group velocity in Future Convergence Theory is given by

$$v_g = \frac{d\omega}{dk}$$

Accordingly, the deviation from general relativity is defined as

$$\Delta v_g = v_g^{\text{FCT}} - c$$

This correction is predicted to increase with increasing frequency.

Phase Correction

The propagation phase also receives corrections due to Future Information. The phase difference is defined by

$$\Delta\phi = \phi_{\text{FCT}} - \phi_{\text{GR}}$$

Since this phase difference accumulates during long-distance propagation, it becomes easier to detect for mergers occurring at greater cosmological distances.

Gravitational-Wave Dispersion Theorem

From the above, the following theorem holds.

Theorem 8.2 (Gravitational-Wave Dispersion Theorem). *In Future Convergence Theory, gravitational waves propagate through an effective space-time modified by the Manifestation Constraint Tensor. As a consequence, frequency-dependent dispersion, group-velocity corrections, and phase corrections, which are absent in general relativity, naturally arise. These corrections disappear in the weak-constraint limit.*

Proof. As established in Part II, gravity is described by the Manifestation Constraint Tensor. Furthermore, Part VIII showed that gravitational waves propagate as perturbations of this constraint structure. Accordingly, the wave equation is no longer the vacuum wave equation of general relativity but an effective wave equation containing Manifestation Constraint corrections. As a consequence, the dispersion relation acquires the correction term

$$\Delta D.$$

In the weak-constraint limit, the Manifestation Constraint correction vanishes, and the theory reduces to general relativity. $\square$

Observational Consequences

If Future Convergence Theory is correct, the deviations from general relativity are expected to be extremely small in current gravitational-wave observations. However, third-generation gravitational-wave detectors, space-based gravitational-wave observatories, and statistical analyses of large event populations may reveal frequency-dependent phase corrections, group-velocity corrections, and cumulative dispersion effects.

Conclusion of This Section

In this section, we formulated gravitational-wave dispersion in Future Convergence Theory. As a result, gravitational waves were shown to propagate through an effective space-time modified by the Manifestation Constraint Tensor, exhibiting frequency-dependent dispersion, group-velocity corrections, and phase corrections that are absent in general relativity. The next section summarizes the galactic halo structure predicted by Future Convergence Theory and compares it with the standard cosmological model.

8.0.4 14.4 Dark Matter Halo Differences

In the previous section, gravitational-wave dispersion in Future Convergence Theory was organized in terms of its differences from general relativity. In this section, the galactic halo structure derived in Part VI from the non-optical manifestation residue is summarized from the perspective of comparison with the standard

$$\Lambda\text{CDM}$$

cosmology. As shown in Part VI, Future Convergence Theory does not introduce unknown Dark Matter particles. Instead, galactic halos are formed through the contribution of the non-optical manifestation residue,

$$\mathcal{R}_{\text{NO}},$$

generated from the Future Causal Information Field to the Manifestation Constraint Tensor,

$$\mathcal{C}_{\mu\nu}.$$

Consequently, while the observed gravitational effects may closely resemble those predicted by the standard CDM model, their spatial distribution and temporal evolution exhibit characteristic differences unique to Future Convergence Theory.

Definition of the Observable Difference

The difference in halo density between Future Convergence Theory and the standard

$$\Lambda\text{CDM}$$

model is defined as

$$\Delta\rho_{\text{halo}} = \rho_{\text{FCT}} - \rho_{\Lambda\text{CDM}}$$

where

$$\rho_{\text{FCT}}$$

is the halo density derived from the non-optical manifestation residue, and

$$\rho_{\Lambda\text{CDM}}$$

is the halo density produced by unknown Dark Matter particles.

Central Density Gradient

In Future Convergence Theory, the central halo structure is determined not by the self-gravitational equilibrium of particles but by the consistency condition with the Future Fixed Point. Accordingly, the difference in the central density gradient is evaluated by

$$\Delta\left(\frac{d\rho}{dr}\right) = \left(\frac{d\rho}{dr}\right)_{\text{FCT}} - \left(\frac{d\rho}{dr}\right)_{\Lambda\text{CDM}}$$

Outer Halo Structure

At large galactocentric distances, Future Convergence Theory predicts that the non-optical manifestation residue decreases more slowly than visible matter. Consequently, the asymptotic behavior of the outer halo density may differ from that predicted by the standard CDM model.

Temporal Evolution

In the standard CDM model, halo evolution is governed by particle motion, mergers, and tidal interactions. In contrast, Future Convergence Theory predicts that halo structures evolve through the reconvergence of the Manifestation Constraint. Therefore, even galaxies with identical initial conditions may develop different long-term density distributions owing to differences in their convergence conditions toward the Future Fixed Point.

Observable Quantities

Future Convergence Theory identifies at least the following quantities as observational targets for galactic halos:

$$\{\Delta\rho_{\text{halo}}, \Delta M(r), \Delta v(r), \Delta\Phi(r)\}$$

where

$$\Delta M(r)$$

denotes the difference in the effective mass distribution,

$$\Delta v(r)$$

the difference in the rotation velocity, and

$$\Delta\Phi(r)$$

the difference in the gravitational potential.

Dark Matter Halo Difference Theorem

From the above, the following theorem holds.

Theorem 8.3 (Dark Matter Halo Difference Theorem). *In Future Convergence Theory, galactic halos are formed not by the accumulation of unknown particles but by the non-optical manifestation residue. Consequently, although the total halo mass may be close to that predicted by the standard CDM model, the density gradient, outer structure, and temporal evolution exhibit characteristic differences unique to Future Convergence Theory.*

Proof. As demonstrated in Part VI, halo structures in Future Convergence Theory are determined by the Manifestation Constraint conservation law together with the Future Fixed Point. By contrast, in the standard CDM model, halos are formed through the gravitational relaxation of unknown particles. Since the formation mechanisms differ, sufficiently precise observations should reveal finite residuals in the density distribution, gravitational potential, and temporal evolution. Accordingly, galactic halo structure constitutes one of the principal observational targets for distinguishing the two theories. $\square$

Conclusion of This Section

In this section, the galactic halo theory derived in Part VI was reorganized from the perspective of observable quantities. As a result, Future Convergence Theory predicts observable differences from the standard CDM model not primarily in the total halo mass, but in the density gradient, outer structure, and temporal evolution. The next section examines gravitational lensing and systematically compares its predictions with those of the standard cosmological model.

8.0.5 14.5 Gravitational Lensing Differences

In the previous section, we organized the galactic halo structure in Future Convergence Theory from the perspective of comparison with the standard

$$\Lambda\text{CDM}$$

cosmology. In this section, we systematically define the observable differences associated with the theory of gravitational lensing derived in Part VI. Gravitational lensing is an observable that directly measures the total gravitational source rather than only visible matter. Consequently, it is one of the most direct probes for testing the existence of the non-optical manifestation residue,

$$\mathcal{R}_{\text{NO}},$$

predicted by Future Convergence Theory.

Definition of the Observable Difference

The difference in the deflection angle between Future Convergence Theory and the standard

$$\Lambda\text{CDM}$$

model is defined as

$$\Delta\hat{\alpha} = \hat{\alpha}_{\text{FCT}} - \hat{\alpha}_{\Lambda\text{CDM}}$$

where

$$\hat{\alpha}_{\text{FCT}}$$

is the deflection angle produced by the non-optical manifestation residue, and

$$\hat{\alpha}_{\Lambda\text{CDM}}$$

is the deflection angle predicted from particle Dark Matter.

Difference in the Convergence Field

For the convergence field measured through weak gravitational lensing, Future Convergence Theory introduces

$$\Delta\kappa = \kappa_{\text{FCT}} - \kappa_{\Lambda\text{CDM}}$$

as a principal observable. This difference directly reflects the spatial structure of the Manifestation Constraint Tensor.

Shear Distribution

For the gravitational-lensing shear field, Future Convergence Theory likewise introduces

$$\Delta\gamma = \gamma_{\text{FCT}} - \gamma_{\Lambda\text{CDM}}$$

The shear field is particularly sensitive to the constraint structure in the outer halo, making it one of the observables in which differences from particle halos are expected to appear most clearly.

Temporal Evolution

In Future Convergence Theory, the gravitational-lens mass distribution evolves through the re-convergence of the Manifestation Constraint rather than through particle motion. Consequently, after galaxy-cluster collisions,

$$\frac{\partial\Sigma_{\text{lens}}}{\partial t} \neq \left(\frac{\partial\Sigma_{\text{lens}}}{\partial t}\right)_{\Lambda\text{CDM}}$$

may hold.

Observable Quantities

Future Convergence Theory identifies the following quantities as the principal observables for gravitational lensing:

$$\{\Delta\hat{\alpha}, \Delta\kappa, \Delta\gamma, \Delta\Sigma_{\text{lens}}\}$$

where

$$\Delta\Sigma_{\text{lens}}$$

denotes the difference in the lensing mass distribution.

Gravitational Lensing Difference Theorem

From the above, the following theorem holds.

Theorem 8.4 (Gravitational Lensing Difference Theorem). *In Future Convergence Theory, gravitational lensing originates not from unknown Dark Matter particles but from the Manifestation Constraint structure generated by the non-optical manifestation residue. Consequently, although the total lensing mass may be close to that predicted by the standard ΛCDM model, finite observable differences appear in the deflection angle, convergence field, shear distribution, and temporal evolution.*

Proof. As shown in Part VI, gravitational lensing in Future Convergence Theory is determined by the Manifestation Constraint Tensor generated from the non-optical manifestation residue. By contrast, in the standard ΛCDM model, the gravitational source is provided by unknown particle halos. Since the formation mechanisms of the gravitational source are fundamentally different, sufficiently precise observations will reveal finite differences in the deflection angle, convergence field, shear distribution, lensing mass distribution, and their temporal evolution. Therefore, gravitational lensing constitutes one of the principal observables for testing Future Convergence Theory. $\square$

Conclusion of This Section

In this section, the gravitational-lensing theory derived in Part VI was reorganized from the perspective of observable quantities. As a result, Future Convergence Theory predicts observable differences from the standard Λ CDM model not in the total lensing mass itself, but in the deflection angle, convergence field, shear distribution, and temporal evolution. The next section organizes the differences between Future Convergence Theory and standard cosmology for the Cosmic Microwave Background (CMB).

8.0.6 14.6 CMB Differences

In the previous section, we organized gravitational lensing in Future Convergence Theory from the perspective of its differences from the standard

Λ CDM

cosmology. In this section, the theoretical predictions for the Cosmic Microwave Background (CMB), derived in Parts VI and X, are reorganized from the standpoint of observable quantities. The CMB reflects the primordial density perturbations, gravitational potentials, and the large-scale space-time structure of the Universe. Therefore, if Future Convergence Theory describes the Universe through the Future Causal Information Field and Manifestation Constraints, their effects must appear in the CMB angular power spectrum, polarization, gravitational lensing, and large-angle anomalies.

Definition of the Observable Difference

The difference in the angular power spectrum between Future Convergence Theory and the standard

Λ CDM

model is defined as

$$\Delta C_\ell = C_\ell^{\text{FCT}} - C_\ell^{\Lambda\text{CDM}}$$

where

$$C_\ell^{\text{FCT}}$$

is the theoretical prediction including Manifestation Constraint corrections, and

$$C_\ell^{\Lambda\text{CDM}}$$

is the prediction of the standard cosmological model.

Low-Multipole Differences

The Future Fixed Point determines the global convergence condition of the Universe. Accordingly, Future Convergence Theory predicts its largest corrections on large angular scales, namely in the low-multipole region,

$$\ell.$$

The corresponding difference is defined by

$$\Delta C_\ell^{\text{low}} = C_\ell^{\text{FCT}} - C_\ell^{\Lambda\text{CDM}} \quad (\ell \ll 100)$$

Polarization Differences

In Future Convergence Theory, Manifestation Constraint corrections affect not only temperature fluctuations but also the polarization spectra. Accordingly, the observable quantities are

$$\Delta C_\ell^{EE}, \quad \Delta C_\ell^{TE}, \quad \Delta C_\ell^{BB}$$

for the EE, TE, and BB spectra, respectively.

CMB Gravitational-Lensing Difference

Future Convergence Theory further predicts that CMB gravitational lensing is modified by the non-optical manifestation residue. Accordingly, the lensing-potential difference is introduced as

$$\Delta\phi_{\text{lens}} = \phi_{\text{FCT}} - \phi_{\Lambda\text{CDM}}$$

which serves as one of the principal observables.

Integrated Sachs–Wolfe Effect

The Future Fixed Point contributes to the temporal evolution of the gravitational potential across the Universe. Accordingly, the Integrated Sachs–Wolfe effect is characterized by

$$\Delta_{\text{ISW}} = \text{ISW}_{\text{FCT}} - \text{ISW}_{\Lambda\text{CDM}}$$

which constitutes a characteristic observable unique to Future Convergence Theory.

Observable Quantities

The principal CMB observables in Future Convergence Theory are

$$\{\Delta C_\ell, \Delta C_\ell^{EE}, \Delta C_\ell^{TE}, \Delta C_\ell^{BB}, \Delta\phi_{\text{lens}}, \Delta_{\text{ISW}}\}$$

These quantities most strongly reflect the influence of the Future Fixed Point and the Manifestation Constraint Tensor on cosmological observations.

CMB Difference Theorem

From the above, the following theorem holds.

Theorem 8.5 (CMB Difference Theorem). *In Future Convergence Theory, the Cosmic Microwave Background possesses an overall structure that is nearly consistent with the standard ΛCDM cosmology. However, finite observable differences arising from the Future Fixed Point and Manifestation Constraints appear in the low-multipole region, the polarization spectra, CMB gravitational lensing, and the Integrated Sachs–Wolfe effect.*

Proof. As demonstrated in Parts VI and X, the cosmological gravitational source in Future Convergence Theory is determined not by unknown particles but by the non-optical manifestation residue and the Manifestation Constraint Tensor. Consequently, the gravitational potential and the resulting space-time structure are not exactly identical to those of the standard ΛCDM model. Since the CMB temperature anisotropies, polarization, and gravitational-lensing effects depend on these gravitational potentials, sufficiently precise observations should reveal finite differences. Therefore, these quantities constitute principal observables for testing Future Convergence Theory. $\square$

Conclusion of This Section

In this section, the CMB theory developed in Parts VI and X was reorganized from the perspective of observable quantities. As a result, Future Convergence Theory was shown to be consistent with the standard cosmological model in the overall angular power spectrum while predicting observable differences in the low-multipole region, polarization, CMB gravitational lensing, and the Integrated Sachs–Wolfe effect. The next section formulates the **Page Curve Difference** as an observable on the basis of the black-hole information-preservation theory developed in Part IX.

8.0.7 14.7 Page Curve Differences

In the previous section, we organized the observational differences predicted by Future Convergence Theory for the Cosmic Microwave Background (CMB). In this section, we reconsider the black-hole information preservation theory derived in Part IX from the viewpoint of observable quantities. The Page curve is the most important observational indicator for determining whether information is preserved or lost during the evaporation of a black hole. In Future Convergence Theory, the black-hole boundary is modified by the Manifestation Constraint Tensor,

$$\mathcal{C}_{\mu\nu},$$

and the Future Causal Information Field,

$$\mathcal{F}.$$

As a consequence, black-hole information is not completely destroyed but is preserved through a reprojection pathway associated with the Future Fixed Point. Accordingly, the Page curve itself exhibits observable differences from those predicted by general relativity or standard quantum-gravity theories.

Definition of the Observable Difference

For the Page entropy, the difference between Future Convergence Theory and the standard theory is defined as

$$\Delta S_{\text{Page}} = S_{\text{Page}}^{\text{FCT}} - S_{\text{Page}}^{\text{STD}}$$

where

$$S_{\text{Page}}^{\text{STD}}$$

denotes the Page curve predicted by the standard theory, and

$$S_{\text{Page}}^{\text{FCT}}$$

is the prediction including the Manifestation Constraint correction.

Difference in the Page Time

Future Convergence Theory predicts a correction to the onset of information recovery due to the Manifestation Constraint Boundary. If the Page time is denoted by

$$t_{\text{Page}},$$

its difference is defined as

$$\Delta t_{\text{Page}} = t_{\text{Page}}^{\text{FCT}} - t_{\text{Page}}^{\text{STD}}$$

Difference in the Scrambling Time

As derived in Part IX, the information scrambling time also receives corrections from the Manifestation Constraint. Its difference is defined by

$$\Delta t_{\text{scr}} = t_{\text{scr}}^{\text{FCT}} - t_{\text{scr}}^{\text{STD}}$$

Difference in Hawking Radiation

In Future Convergence Theory, the Hawking radiation spectrum is not perfectly thermal. Since it is modified by the Manifestation Constraint Boundary, the difference in the radiation spectrum is defined as

$$\Delta I(\omega) = I_{\text{FCT}}(\omega) - I_{\text{STD}}(\omega)$$

where

$$I(\omega)$$

denotes the radiation intensity at angular frequency

$$\omega.$$

Information Preservation Ratio

Future Convergence Theory introduces the information preservation ratio,

$$\eta_{\text{info}},$$

as an observable quantity. It is defined by

$$\eta_{\text{info}} = \frac{I_{\text{recover}}}{I_{\text{initial}}}$$

where

$$I_{\text{initial}}$$

is the information content at the formation of the black hole, and

$$I_{\text{recover}}$$

is the information recoverable after the completion of evaporation. In Future Convergence Theory,

$$\eta_{\text{info}} = 1$$

represents the theoretical limit.

Observable Quantities

The principal observables associated with black-hole information preservation in Future Convergence Theory are

$$\{\Delta S_{\text{Page}}, \Delta t_{\text{Page}}, \Delta t_{\text{scr}}, \Delta I(\omega), \eta_{\text{info}}\}$$

These quantities provide direct observational tests of the information-preservation theory developed in Part IX.

Page Curve Difference Theorem

From the above, the following theorem is obtained.

Theorem 8.6 (Page Curve Difference Theorem). *In Future Convergence Theory, black-hole information is preserved through the Manifestation Constraint Boundary and the Future Fixed Point. As a consequence, finite observable differences from the standard theory appear in the Page curve, the Page time, the Scrambling Time, and the Hawking radiation spectrum.*

Proof. As demonstrated in Part IX, the black-hole boundary in Future Convergence Theory is described as a Manifestation Constraint Boundary rather than a perfectly absorbing boundary. Consequently, information is not completely lost during the evaporation process but is preserved through a reprojection pathway mediated by the Future Fixed Point. Accordingly, corrections appear in the Page curve, the information scrambling time, and the Hawking radiation spectrum relative to those predicted by general relativity and standard quantum-gravity theories. Therefore, these quantities constitute principal observables for testing Future Convergence Theory. $\square$

Conclusion of This Section

In this section, the black-hole information preservation theory developed in Part IX was reorganized from the perspective of observable quantities. As a result, Future Convergence Theory predicts observable differences in the Page curve, the Scrambling Time, Hawking radiation, and the information preservation ratio. The next section reorganizes the corrections to the Casimir effect derived in Part III as the **Casimir Difference**.

8.0.8 14.8 Casimir Differences

In the previous section, we organized the black-hole information preservation theory in Future Convergence Theory in terms of the Page curve, the Scrambling Time, Hawking radiation, and the information preservation ratio. In this section, we reconsider the corrections to the Casimir effect derived in Part III from the viewpoint of observable quantities. The Casimir effect is one of the few experiments that directly measures quantum vacuum fluctuations and therefore serves as an extremely precise test of quantum theory. In Future Convergence Theory, the vacuum is defined not simply as zero-point fluctuations but as a manifestable vacuum generated from the Future Causal Information Field,

$$\mathcal{F},$$

through the Quantum Allowance Regulation Layer. Accordingly, the vacuum energy receives small corrections originating from the Manifestation Constraint Tensor,

$$\mathcal{C}_{\mu\nu}.$$

Definition of the Observable Difference

For the Casimir force, the difference between Future Convergence Theory and standard quantum electrodynamics is defined as

$$\Delta F_{\text{Cas}} = F_{\text{FCT}} - F_{\text{QED}}$$

where

$$F_{\text{QED}}$$

is the Casimir force predicted by standard theory, and

$$F_{\text{FCT}}$$

is the theoretical value including the Manifestation Constraint correction.

Difference in Vacuum Energy

In Future Convergence Theory, the vacuum energy density consists of the ordinary zero-point energy together with an additional contribution from the Future Causal Information Field. The difference is defined as

$$\Delta\rho_{\text{vac}} = \rho_{\text{vac}}^{\text{FCT}} - \rho_{\text{vac}}^{\text{QED}}$$

Dependence on Plate Separation

In the standard Casimir effect, the force between parallel plates separated by a distance

$$a$$

satisfies

$$F \propto a^{-4}.$$

In Future Convergence Theory, the Manifestation Constraint correction modifies this relation as

$$F_{\text{FCT}} = F_{\text{QED}} + \delta(a, \mathcal{F}, \mathcal{C}_{\mu\nu})$$

where

$$\delta$$

is the correction term unique to Future Convergence Theory.

Residuals in High-Precision Experiments

Let the experimentally measured force be denoted by

$$F_{\text{obs}}.$$

Future Convergence Theory evaluates the residual as

$$\Delta F_{\text{obs}} = F_{\text{obs}} - F_{\text{QED}}$$

The theory is tested by examining whether this residual is statistically consistent with the theoretical correction

$$\Delta F_{\text{Cas}}.$$

Correlation with the Manifestation Constraint

Future Convergence Theory predicts that the correction to the Casimir effect is correlated with the local strength of the Manifestation Constraint. Accordingly,

$$\Delta F_{\text{Cas}} = G(\mathcal{C}_{\mu\nu}, \mathcal{F})$$

where

$$G$$

is the correction function specific to Future Convergence Theory.

Observable Quantities

The principal observables associated with the Casimir effect in Future Convergence Theory are

$$\{\Delta F_{\text{Cas}}, \Delta\rho_{\text{vac}}, \Delta F_{\text{obs}}\}$$

These quantities provide direct observational tests of the quantum-vacuum corrections derived in Part III.

Casimir Difference Theorem

From the above, the following theorem holds.

Theorem 8.7 (Casimir Difference Theorem). *In Future Convergence Theory, the Casimir effect receives corrections not only from the vacuum fluctuations predicted by standard quantum electrodynamics but also from the Future Causal Information Field and the Manifestation Constraint Tensor. As a consequence, sufficiently precise Casimir experiments may reveal finite observable residuals that differ from the predictions of the standard theory.*

Proof. As demonstrated in Part III, the vacuum in Future Convergence Theory is a manifestable vacuum generated from the Future Causal Information Field. Accordingly, the vacuum energy is not determined solely by the zero-point energy of standard quantum electrodynamics. Since the Casimir effect depends directly on differences in vacuum energy, Manifestation Constraint corrections necessarily appear in the Casimir force. Therefore, sufficiently precise Casimir experiments may detect finite differences unique to Future Convergence Theory. $\square$

Conclusion of This Section

In this section, the corrections to the Casimir effect derived in Part III were reorganized from the perspective of observable quantities. As a result, Future Convergence Theory predicts observable differences in the Casimir force, vacuum energy, and high-precision experimental residuals relative to standard quantum electrodynamics. The next section summarizes the **Equivalence Principle Limit Corrections** based on the equivalence principle introduced in Part II and the modified Einstein equations developed in Part XIII.

8.0.9 14.9 Equivalence Principle Limit Corrections

In the previous section, we organized the corrections to the Casimir effect in Future Convergence Theory from the perspective of observable quantities. In this section, we summarize the theoretical corrections to the equivalence principle based on the concept of gravity as a Manifestation Constraint introduced in Part II and the unified field equations constructed in Part XIII. The equivalence principle is the most fundamental principle underlying general relativity and has been verified experimentally with extremely high precision. Therefore, if Future Convergence Theory is a theory that encompasses general relativity, it must reproduce the equivalence principle with high precision under ordinary conditions while predicting finite corrections only in extreme environments.

The Equivalence Principle in General Relativity

In general relativity, the inertial mass

$$m_I$$

and the gravitational mass

$$m_G$$

are exactly identical. That is,

$$\boxed{m_I = m_G}$$

holds. As a consequence, the free-fall acceleration is independent of the internal structure of the object.

Redefinition in Future Convergence Theory

In Future Convergence Theory, gravity is described not as an interaction between particles but by the Manifestation Constraint Tensor,

$$\mathcal{C}_{\mu\nu}.$$

Accordingly, free-fall motion is determined by the consistency condition with the Manifestation Constraint. Under ordinary conditions, the Manifestation Constraint correction is extremely small, so the equivalence principle holds with the same precision as in general relativity.

Definition of the Observable Difference

The difference in free-fall acceleration between Future Convergence Theory and general relativity is defined as

$$\Delta a = a_{\text{FCT}} - a_{\text{GR}}$$

Furthermore, the parameter characterizing violations of the equivalence principle is defined as

$$\eta_{\text{EP}} = 2 \frac{a_1 - a_2}{a_1 + a_2}$$

Future Convergence Theory predicts

$$\eta_{\text{EP}} \approx 0$$

under ordinary conditions.

Corrections in Extreme Environments

Near black holes, neutron stars, and in the early Universe, where the Manifestation Constraint becomes extremely strong, free-fall motion receives corrections from the Manifestation Constraint Tensor. In general form,

$$\Delta a = H(\mathcal{C}_{\mu\nu}, \Lambda_{\text{BH}}, \mathcal{F})$$

where

$$H$$

is the correction function unique to Future Convergence Theory.

General Relativistic Limit

Future Convergence Theory necessarily reduces to general relativity in the weak-constraint limit. That is,

$$\lim_{\mathcal{C}_{\mu\nu} \rightarrow 0} \Delta a = 0$$

Therefore, the theory is consistent with current high-precision tests of the equivalence principle.

Observable Quantities

The principal observables related to the equivalence principle in Future Convergence Theory are

$$\{\Delta a, \eta_{\text{EP}}, \Delta\Gamma\}$$

where

$$\Delta\Gamma$$

denotes the geodesic deviation between Future Convergence Theory and general relativity.

Equivalence Principle Limit Theorem

From the above, the following theorem is obtained.

Theorem 8.8 (Equivalence Principle Limit Theorem). *Future Convergence Theory reproduces the equivalence principle of general relativity exactly in the weak-constraint limit. On the other hand, finite corrections arising from the Manifestation Constraint Tensor appear in the vicinity of black holes, in the early Universe, and in regions of extremely high curvature. Accordingly, Future Convergence Theory encompasses general relativity while predicting observable differences only in extreme environments.*

Proof. As shown in Part II, gravity is described by the Manifestation Constraint Tensor. Furthermore, Part XIII demonstrated that the modified Einstein equations reduce to those of general relativity in the weak-constraint limit. Consequently, under ordinary conditions the Manifestation Constraint corrections vanish, and free-fall motion agrees with that predicted by general relativity. By contrast, in extreme environments where the Manifestation Constraint remains finite, finite corrections naturally appear in the geodesic equation. Therefore, Future Convergence Theory incorporates the equivalence principle while predicting finite deviations only under extreme conditions. $\square$

Conclusion of This Section

In this section, the gravitational theory developed in Parts II and XIII was reorganized from the viewpoint of the equivalence principle. As a result, Future Convergence Theory was shown to reproduce the equivalence principle of general relativity under ordinary conditions while predicting observable finite corrections only in the vicinity of black holes, in the early Universe, and in regions of extremely high curvature. The next section integrates all observational predictions of the present work and formulates the comprehensive falsification conditions for Future Convergence Theory.

8.0.10 14.10 Comprehensive Falsification Conditions

In the previous section, we showed that Future Convergence Theory reproduces the equivalence principle of general relativity in the weak-constraint limit while predicting finite corrections only in extreme environments. In this section, we integrate all theoretical predictions derived throughout Parts I–XIII and formulate the falsifiability of Future Convergence Theory as a whole. Future Convergence Theory is an ontological unified theory. However, for it to qualify as a scientific theory, it is not ontology itself but the observables derived from that ontology that must ultimately be supported or falsified by future observations. Accordingly, the ultimate objective of this work is not to assume that Future Convergence Theory is correct, but rather to present observationally testable differences from general relativity, quantum theory, and the standard

Λ CDM

cosmology.

Comprehensive Set of Observables

The principal observables derived throughout Parts I–XIII are collectively defined as

$$\mathbb{O}_{\text{FCT}} = \{\Delta\omega_{\ell mn}, A_{\text{echo}}, \Delta v_g, \Delta\rho_{\text{halo}}, \Delta\hat{a}, \Delta C_\ell, \Delta S_{\text{Page}}, \Delta F_{\text{Cas}}, \Delta a\}$$

Each observable corresponds respectively to the theoretical predictions for black holes, gravitational waves, cosmology, quantum theory, and gravitation developed from Part VII through Part XIII.

Conditions for the Validity of the Theory

For Future Convergence Theory to be supported, at least one of the observables must exhibit a finite deviation with statistical significance. That is,

$$\boxed{\exists \mathcal{O} \in \mathbb{O}_{\text{FCT}} \quad \text{such that} \quad \mathcal{O} \neq 0}$$

must hold as a necessary condition.

Comprehensive Falsification Condition

Conversely, if sufficiently precise observations establish that

$$\boxed{\mathbb{O}_{\text{FCT}} = \mathbf{0}}$$

for all observables, then Future Convergence Theory loses observational support. Furthermore, if unknown particle Dark Matter, standard quantum theory, and general relativity alone are shown to provide a unified explanation of all observational results considered in this work, then Future Convergence Theory becomes unnecessary.

Partial Support

Future Convergence Theory is not a theory that can be fully established through a single observable. For example, even if only the ringdown residual is confirmed, consistency with quantum theory, cosmology, and information-preservation theory is still required. Conversely, even if only the Casimir effect agrees with the theory, consistency with the black-hole predictions remains necessary. Therefore, validation of the theory requires mutually consistent support from multiple independent observational fields.

Assessment as a Unified Theory

Future Convergence Theory is not intended to explain only a single phenomenon. Its purpose is to derive gravitation, quantum theory, cosmology, black holes, information preservation, Dark Matter, Dark Energy, and child-universe generation from a single ontological foundation. Accordingly, the theory should be evaluated not on the basis of individual predictions but on the integrated consistency of all observables.

Comprehensive Falsification Theorem

From the above, the following theorem is obtained.

Theorem 8.9 (Comprehensive Falsification Theorem). *Future Convergence Theory predicts observable differences from general relativity, quantum theory, and the standard Λ CDM cosmology for all observables derived throughout Parts I–XIII. If all of these differences vanish under sufficiently precise observations, and the existing theories alone provide a unified explanation of all observations, then Future Convergence Theory is falsified. Conversely, if the predicted differences are confirmed consistently across multiple independent observations, Future Convergence Theory is supported as an ontological unified theory encompassing the existing theories.*

Proof. Throughout Parts I–XIII, theoretical predictions for each physical phenomenon were derived on the basis of Future Convergence Theory. All of these predictions originate from the Future Causal Information Field, the Manifestation Constraint Tensor, the Meaning Density Tensor, and the reprojection structure. Consequently, the observables are not mutually independent but are supported by a single ontological foundation. Therefore, the overall validity or falsification of the theory must be judged by the consistency of the entire set of observables. $\square$

Conclusion of Part XIV

In this chapter, the theoretical predictions derived throughout Parts I–XIII were systematically reorganized from the perspective of observable quantities. As a result, Future Convergence Theory was shown to predict observable differences in ringdown residuals, gravitational-wave dispersion, Dark Matter halos, gravitational lensing, the Cosmic Microwave Background, the Page curve, the Casimir effect, and equivalence-principle limit corrections. Furthermore, by formulating a comprehensive falsification condition integrating all of these predicted differences, we have shown that Future Convergence Theory is constructed not only as an ontological unified theory but also as a physical theory that is testable through experiments and observations. This completes the theoretical construction presented in Parts I–XIV. The remaining Conclusion summarizes the achievements of the present work and discusses the position of Future Convergence Theory within modern physics together with its future theoretical and observational prospects.

Part XV

Conclusion

Future Convergence Theory (FCT) has been developed in this work upon an ontological foundation fundamentally different from that of conventional physical theories.

The primary objective of this theory is to derive, within a single ontological principle, a unified description of general relativity, quantum theory, cosmology, black-hole physics, dark matter, dark energy, and the problem of information preservation.

In Part I, we established the axiomatic framework of Future Convergence Theory by introducing the Meaning Generation Template, the Future Causal Information Field, the Quantum Admissibility Layer, the Three-Dimensional Manifestation Layer, the Reprojection Closed Circuit, and the FCT Closure Equation.

As a consequence, physical laws were reformulated as manifestation structures determined by consistency conditions with the Future Fixed Point.

In Part II, gravity was redefined not as an interaction between particles but as a Manifestation Constraint.

Accordingly, the spacetime curvature described by general relativity was interpreted as a constraint structure projected from the Future Causal Information Field onto the Three-Dimensional Manifestation Layer.

In Part III, quantum theory was reformulated as an ontological intermediate layer. The wave function, quantum measurement, decoherence, vacuum fluctuations, the Casimir effect, delayed-choice experiments, and the quantum eraser were consistently reconstructed within the framework of Future Convergence Theory.

In Part IV, the graviton was identified not as gravity itself but as a quantum excitation of the Manifestation Constraint Tensor, thereby clearly separating the ontological roles of gravity and quantum gravity.

In Part V, string theory was reinterpreted not as the ultimate ontological theory but as an effective theory describing the Quantum Admissibility Layer, and its correspondence with Future Convergence Theory was established.

In Part VI, dark matter was redefined not as an unknown particle but as a Non-Optical Manifestation Residual generated from the Future Causal Information Field.

Consequently, a theoretical framework was constructed that explains galactic rotation curves, galactic halos, galaxy-cluster collisions, gravitational lensing, and the cosmic microwave background without introducing unknown dark matter particles.

In Part VII, quantitative deviations from general relativity were derived for black-hole quasinormal modes, ringdown waveforms, damping rates, echoes, and gravitational-wave polarization.

In Part VIII, dispersion relations, group velocity, phase corrections, and high-frequency modifications of gravitational waves were derived, demonstrating the observational testability of the theory with future gravitational-wave detectors.

In Part IX, the black-hole information paradox was reformulated in terms of a reprojection structure associated with the Future Fixed Point, leading to theoretical predictions for the Page curve, scrambling time, Hawking radiation, and the information recovery rate.

In Part X, the Big Bang was redefined as the Primary Manifestation Constraint. Initial conditions determined by the Future Fixed Point, together with inflation, primordial fluctuations, the tensor-to-scalar ratio, primordial gravitational waves, and anomalies in the cosmic microwave background, were derived within this framework.

In Part XI, dark energy was formulated as the Future Convergence Degree, leading to a dynamical cosmological term, the heat death of the Universe, Future Convergence Entropy, and the conditions for reprojection.

In Part XII, child-universe formation was formulated as an ontological consequence of the Reprojection Closed Circuit, establishing its correspondence with black holes, thermal saturation, and the Meaning Generation Template.

In Part XIII, all preceding theories were unified into the integrated field equations of Future Convergence Theory, incorporating the Manifestation Constraint Tensor, the Future Causal

Information Field, the Meaning Density Tensor, the dark matter term, the dark energy term, and the black-hole boundary term.

In Part XIV, all theoretical predictions were systematically organized in terms of observable quantities.

Ringdown residuals, black-hole echoes, gravitational-wave dispersion, dark matter halos, gravitational lensing, the cosmic microwave background, the Page curve, the Casimir effect, and equivalence-principle corrections in extreme environments were defined as the principal observables of Future Convergence Theory, and comprehensive falsification criteria were established.

Accordingly, Future Convergence Theory is not merely a philosophical framework concerned with ontology.

Rather, it is formulated as a physical theory that provides explicit observable deviations from general relativity, quantum theory, and the standard $[\Lambda\text{CDM}]$ *cosmological model*.

The principal conclusion of this work is therefore not the conceptual organization of Future Convergence Theory itself.

Instead, the central result is the demonstration that Future Convergence Theory predicts observational signatures distinct from those of general relativity and the standard theoretical framework in black-hole quasinormal modes, gravitational waves, the quantum vacuum, cosmology, and information preservation.

Specifically, the validity of Future Convergence Theory is determined through the observational residual

$$\left[\Delta\omega_{\ell mn} \omega_{\ell mn}^{\text{obs}} \omega_{\ell mn}^{\text{GR}} \right]$$

together with the existence of the correlated quantities

$$\left[\Lambda_{\text{BH}}, \quad \Sigma_{\mu\nu}, \quad \mathcal{C}_{\mu\nu}, \quad \mathcal{F} \right]$$

as tested by future high-precision observations.

Therefore, Future Convergence Theory should not be regarded merely as a framework presenting a particular worldview.

Rather, it constitutes a falsifiable unified theory that quantitatively predicts deviations from general relativity, quantum theory, and the standard cosmological model, and can consequently be supported or refuted through experiment and observation.

This paper presents the first comprehensive unified formulation of the ontological foundations, mathematical structure, and physical predictions of Future Convergence Theory. Future work is expected to evaluate the validity of the theory through precision calculations of its observational predictions, numerical simulations, and direct comparisons with observational data.